16 Component Dirac Fermion Condensate Singlet Superfluid on 4+4 Deflating Spacetime

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A 16-component Dirac field with a mass term and a quartic self-interaction is studied on a spacetime with four space-like and four time-like dimensions in which three of the time-like coordinates deflate exponentially as the ordinary time advances. The Dirac equation is solved exactly: every admissible mode that depends on a deflating time is extinguished, and fields that do not depend on the deflating times obey the flat equations with couplings that grow with the deflation. Of the nine couplings of the self-interaction only four are independent. In the sector that can be quantized the self-interaction has two forms, which differ by a sign between the two halves of the field; the covariant form is the definition, and the other is given beside it. Matter and vacuum are analyzed with the Kohn-Sham method and with the effective action of Cornwall, Jackiw and Tomboulis, which agree. In the covariant form a gas polarizes and becomes self-bound matter with one half of the field filled, a massless field acquires masses of opposite sign in the two halves, and the self-interaction distinguishes the fermion pair along the hidden axis from the pairs along the additional times, each at a definite moment of the deflation. With equal rates the spacetime solves Einstein-Gauss-Bonnet gravity without matter; the solution is unstable, and matter accelerates the deflation. Bounds on the rate from cosmology and from atomic clocks are given.

I. INTRODUCTION

A. The question

This paper studies a single fermion field with sixteen complex components on a spacetime of eight dimensions. Two such spacetimes are of interest. One has four space-like and four time-like dimensions, and is written 4+4; the other has seven space-like dimensions and one time-like dimension, and is written 7+1. On both, the eight Dirac matrices can be chosen real [1–3], and the field supports a superfluid of fermion pairs of spin zero whose order parameter points along one of the four coordinates that are not the coordinates of ordinary spacetime. This paper treats the 4+4 spacetime. Spacetimes with more than one time have been studied for their own sake [4]; what a second time does to the evolution of a field is taken up in Sec. V.

The eight coordinates are written $x0, x1, \ldots, x7$. Throughout, $x1, x2, x3$ are the coordinates of ordinary space and $x4$ is the ordinary time. The coordinate $x0$ is an additional space-like coordinate, called the hidden coordinate. The remaining three, $x5, x6, x7$, are three additional *time-like* coordinates.

On a flat spacetime nothing explains why the fields of ordinary physics do not depend on $x5$, $x6$ and $x7$. On the 4+4 spacetime this is more than a gap in the explanation. A field that depends on an additional time does not evolve unitarily in the ordinary time, and on flat spacetime one has to *impose* that the fields do not depend on the additional times. The present paper examines a definite proposal for what happens to these three coordinates, and it does so for a field with a mass and with a self-interaction.

B. The hypothesis of deflation

The hypothesis is that *the coordinates $x5$, $x6$ and $x7$ deflate exponentially*: as the ordinary time $x4$ advances, the proper extent of every interval along each of these coordinates shrinks as an exponential of $x4$. Flat spacetime is replaced by a curved spacetime with a diagonal metric, in which the three components of the metric along the deflating coordinates are

$$\begin{aligned} g_{55}^{(4+4)} &= -\mathrm{const5}^2\, e^{-2\,\mathrm{g55}\,x4}, \\ g_{66}^{(4+4)} &= -\mathrm{const6}^2\, e^{-2\,\mathrm{g66}\,x4}, \\ g_{77}^{(4+4)} &= -\mathrm{const7}^2\, e^{-2\,\mathrm{g77}\,x4}. \end{aligned} \tag{1}$$

Here g55, g66 and g77 are three positive constants, the *rates* of deflation; const5, const6 and const7 are three positive constants that set the scale at $x4 = 0$ and may differ from one another. The names of the rates are written upright and without a subscript, to distinguish them from the components $g_{55}^{(4+4)}$, $g_{66}^{(4+4)}$, $g_{77}^{(4+4)}$ of the metric, which carry subscripts and the superscript $(4+4)$. The other five components of the metric keep their flat values, $+1$ for $x0, \ldots, x3$ and -1 for the time $x4$. The sign convention is that space-like components of the metric are positive and time-like components negative: the minus signs in Eq. (1) say that $x5, x6, x7$ are time-like.

C. Mass term, self-interaction, and the two methods

The Lagrangian of the fermion field has three parts (Sec. III): the kinetic term; a *mass term*, bilinear in the field; and a *self-interaction*, quartic in the field. The self-interaction is the most general one without derivatives: nine invariants, one for each number $r = 0, \ldots, 8$

of Dirac matrices in the bilinears from which it is built, with nine couplings g_r. The field equation is a nonlinear Dirac equation. The paper answers the following questions about it. How many of the nine couplings are independent? What do the mass term and the self-interaction do to matter made of these fermions, on flat spacetime and while three coordinates deflate? What do they do to the pairing of the fermions, which in this model is mediated by a second field, the Vorton-Scalar? And what does such matter do to the spacetime itself, if it is the source of gravity?

Two methods are used throughout, and each checks the other.

- The *Kohn-Sham method*, known from density-functional theory: the interacting fermions are replaced by independent fermions that move in fields which are determined self-consistently by their own densities. For uniform matter the fields are a shift of the mass and a shift of the energy.
- The *effective potential for composite operators* of Cornwall, Jackiw and Tomboulis (CJT): a function of the full propagator of the fermions whose stationary points are the possible states of matter and whose value there is minus the pressure.

The two methods are equivalent at the level of approximation used here, and the paper verifies this numerically (Sec. XIII).

The paper also examines three hypotheses about the superfluid.

H1. The fermion field forms a superfluid state of spin 0 in which the fermions are paired by the exchange of a new scalar boson, the *Vorton-Scalar*, through a neutral current of the kind known from the weak interaction.

H2. This superfluid state is the Higgs field.

H3. This superfluid state explains the observations of dark matter, or the Hubble tension, or both.

The purpose is to carry out every calculation that these hypotheses call for under the hypothesis of deflation, with the mass term and the self-interaction, and to report what follows for H1, H2 and H3, favorable or not.

D. Summary of results

Geometry (Sec. II). The deflating spacetime is homogeneous and has 14 symmetries, or 15 or 20 if two or three of the rates are equal, instead of the 36 of flat spacetime. It is not a solution of Einstein's equations without matter. With three equal rates it *is* an exact solution of the equations of Einstein-Gauss-Bonnet gravity without matter, for one definite relation between the rate and the two constants of that theory.

The Dirac field (Secs. III, V). A field that does not depend on $x5, x6, x7$ obeys, after multiplication by the square root of the volume factor of the deflating coordinates, exactly the flat equation with the mass unchanged and with every coupling of the self-interaction growing as the exponential of the time. A field that does depend on the deflating coordinates is affected directly. The linear equation is solved exactly. Of its sixteen solutions eight grow as the exponential of an exponential of the time and are not admissible. The other eight are standing waves, which are extinguished as the exponential of an exponential. *The deflation removes the dependence of the field on the additional times*, and with it the sector in which the 4+4 theory is not unitary.

The self-interaction (Sec. IV). For anticommuting fields the nine invariants obey five linear relations: only four couplings are independent, a scalar, a vector, an axial-vector and a pseudoscalar one. Uniform matter feels two combinations of them, pairs of particles at rest feel a third. On the 4+4 spacetime the self-interaction has two forms in the sector that can be quantized, which agree for particles in the same half of the field and differ in sign for particles in different halves. The covariant form is the definition of the model in this paper; the other is recorded beside it, and the two are discussed in Sec. XVIII D.

Matter (Secs. VII, XIII). The Kohn-Sham method and the CJT effective potential give the same states. In the sector of fields that do not depend on the deflating coordinates a scalar attraction grows with the deflation and turns a gas into self-bound matter at a definite moment. With the filled states of negative energy taken into account, the same growth gives a massless field a mass at a critical coupling. In the covariant form matter polarizes: the self-bound state fills one half of the field and forms at two to three times the strength that the alternative form needs, and the mass that a massless field acquires has opposite signs in the two halves. A bare mass together with the Vorton-Scalar gives the phase mode of that field a mass.

The superfluid (Secs. VI to XIV). For fields that do not depend on the extra coordinates the pairing calculation of flat spacetime holds at each moment with the couplings of that moment. The masses do not change. Every threshold is passed at a definite time. The gap of the paired state grows much faster than the coupling; in the pairs along the additional times a scalar contact attraction raises it and a vector contact repulsion lowers it, and in the pair along the hidden axis the covariant form reverses both. The gap equation has a solution that does not depend on the upper limit of its momentum integral only below an ultraviolet criterion. Two fermions become bound at a scattering resonance. The paired state can follow the deflation only once its gap exceeds a few times the rate divided by the pairing strength; before that it lags behind and breaks pairs. Matter of a given density loses its fermion mass at a definite time. A vortex line carries four fermion zero modes.

Gravity (Sec. XV). With Kohn-Sham matter as the source of the Einstein-Lovelock equations the deflating spacetime is not stationary: the three coordinates deflate faster than exponentially and the four flat coordinates expand. Even without matter the exactly exponential deflation is an unstable solution of the gravity in which it is a solution.

Cosmology (Sec. XVI). If gravity is the Einstein theory in eight dimensions, Newton's constant grows at the total rate of deflation; lunar laser ranging then bounds that rate to 3×10^{-3} of the Hubble rate, and the Hubble constant inferred from the microwave background falls. The deflation does not relieve the Hubble tension. Whatever keeps the rates constant must carry a tension and must take part in the perturbations of ordinary space; the spectra of the microwave background at all but the largest angular scales do not depend on how it does so, and the growth of structure does.

The Higgs hypothesis (Sec. XVII). The symmetry under which the pair condensate has the pattern of the Higgs vacuum is a hidden symmetry on the 4+4 spacetime, flat or deflating. A contact self-interaction of the covariant form breaks it explicitly, down to the rotations of the additional times. If the condensate were the Higgs field, the electroweak scale would follow the growing coupling, and atomic clocks would limit the rate of deflation to about 10^{-6} of the Hubble rate. The hypothesis H2 and a deflation of any consequence today exclude each other.

E. Method, and the limits of what is claimed

Every statement of this paper about geometry, matrices and operators is an identity that a program has verified in exact rational arithmetic, and that an independent program has verified again by computer algebra or with Grassmann-valued fields. Every numerical statement is the output of a program, with a stated accuracy, and the central ones were computed twice by independent programs. The programs and their output are public [5]; Appendix E says how each number of this paper is reproduced. The program performs 310 checks and the independent verification 88, all of which pass. The metric of Eq. (1) is a hypothesis. The paper derives its consequences; it does not claim that the hypothesis is true. Section XVIII lists what has been established and what has not.

Units are natural, $\hbar = c = k_B = 1$. Symbols that carry the tag $1644d$ refer to the deflating 4+4 spacetime. A general coordinate among the three deflating ones is written xt with $t \in \{5, 6, 7\}$ and its rate rate_t; a general coordinate among $x0, x1, x2, x3$ is written xj with $j \in \{0, 1, 2, 3\}$. The letter a is kept for the four extra axes, $a \in \{0, 5, 6, 7\}$ (Sec. VIII). Every symbol is defined where it first appears, and Appendix A lists all of them.

II. THE DEFLATING SPACETIME

This section derives the geometry that the hypothesis implies: the curvature, the field equations that the metric does and does not solve, its symmetries, and the motion of a classical particle. Every formula of this section was obtained twice by exact computer algebra, by the program of this paper and by an independent script (Appendix E); the steps are short enough to be repeated by hand, and the text says how.

A. Metric, frame and volume

Throughout, the metric is written in the sign convention of the hypothesis (1): *space-like components are positive and time-like components are negative*. The flat metric is

$$\eta^{(4+4)} = \text{diag}(+1, +1, +1, +1, -1, -1, -1, -1), \qquad (2)$$

with the coordinates in the order $x0, \ldots, x7$: four space-like coordinates $x0, \ldots, x3$ and four time-like ones, $x4, \ldots, x7$. Define three *scale factors*, one for each deflating coordinate,

$$b_t(x4) = \text{const}_t \, e^{-\text{rate}_t \, x4}, \qquad t = 5, 6, 7, \qquad (3)$$

with $\text{rate}_5 = \text{g55}$, $\text{rate}_6 = \text{g66}$, $\text{rate}_7 = \text{g77}$. The metric of the deflating spacetime is

$$g^{(4+4)}_{\mu\nu} = \text{diag}\big(+1, +1, +1, +1, -1, -b_5^2, -b_6^2, -b_7^2\big), \quad (4)$$

whose last three components are those of Eq. (1). The superscript $(4+4)$ marks the metric of this spacetime in this sign convention; a component of the metric always carries it, so that $g^{(4+4)}_{55}$ cannot be taken for the rate g55. Greek indices label the coordinates. A coordinate interval dxt has the proper extent $b_t \, |dxt|$, which shrinks by the factor e in the time $1/\text{rate}_t$. Because $x5, x6, x7$ are time-like, what shrinks is a duration measured along an additional time.

Capital indices A, B label an orthonormal frame, eight vector fields $e_A = e_A{}^\mu \partial_\mu$ with $g^{(4+4)}_{\mu\nu} e_A{}^\mu e_B{}^\nu = \eta^{(4+4)}_{AB}$. We take the frame along the coordinates, $e_A{}^\mu = \text{diag}(1, 1, 1, 1, 1, b_5^{-1}, b_6^{-1}, b_7^{-1})$. The volume element is $V_d \, d^8x$ with

$$V_d(x4) = b_5 b_6 b_7 \propto e^{-\text{gsum}\, x4}, \quad \text{gsum} = \text{g55} + \text{g66} + \text{g77}. \qquad (5)$$

V_d is the proper volume of a unit coordinate cell of the deflating coordinates, and the *total rate* gsum is the rate at which it shrinks. Most results below depend on the three rates only through gsum.

B. Curvature

The only components of the metric that depend on a coordinate are the three $g^{(4+4)}_{tt}$, and they depend on $x4$

alone. The Christoffel symbols $\Gamma^\rho{}_{\mu\nu} = \frac{1}{2}g^{\rho\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu})$ that do not vanish are therefore

$$\Gamma^t{}_{4t} = -\mathrm{rate}_t, \qquad \Gamma^4{}_{tt} = +\mathrm{rate}_t\, b_t^2, \tag{6}$$

without summation over t. The Riemann tensor has, in the orthonormal frame, the components

$$R^t{}_{4t4} = -\mathrm{rate}_t^2, \quad R^t{}_{sts} = -\mathrm{rate}_t\,\mathrm{rate}_s \quad (t \neq s), \tag{7}$$

and those related to them by symmetry. No component carries an index of $x0, \ldots, x3$: these four coordinates span a flat space, and the curvature lives in the four coordinates $x4, \ldots, x7$. With $\mathrm{gsq} = \mathrm{g55}^2 + \mathrm{g66}^2 + \mathrm{g77}^2$ and $\mathrm{gprod} = \mathrm{g55g66} + \mathrm{g55g77} + \mathrm{g66g77}$, the curvature scalar is the constant $R = \mathrm{gsq} + \mathrm{gsum}^2$, and the Einstein tensor $\mathcal{G}^\mu{}_\nu = R^\mu{}_\nu - \frac{1}{2}R\,\delta^\mu_\nu$ is diagonal with

$$\begin{aligned} \mathcal{G}^4{}_4 &= -\mathrm{gprod}, \qquad \mathcal{G}^j{}_j = -\tfrac{1}{2}(\mathrm{gsq} + \mathrm{gsum}^2), \\ \mathcal{G}^t{}_t &= \mathrm{rate}_t\,\mathrm{gsum} - \tfrac{1}{2}(\mathrm{gsq} + \mathrm{gsum}^2), \end{aligned} \tag{8}$$

with $j = 0, 1, 2, 3$. In the sign convention used here these mixed components are the same on the 4+4 and on the 7+1 spacetime. (They follow from the sectional curvatures of Appendix C: $\mathcal{G}^A{}_A$ is minus the sum of the sectional curvatures of the planes that do not contain the direction A.) The Einstein tensor does not vanish, and it is not a multiple of the unit matrix unless all rates vanish. *The deflating spacetime is not a solution of Einstein's equations without matter, with or without a cosmological constant.*

C. A solution of Einstein-Gauss-Bonnet gravity

In more than four dimensions the general field equations of second order for the metric are those of Lovelock [6]. Their first term beyond Einstein's is the Lanczos tensor $\mathcal{H}^\mu{}_\nu$ of the Gauss-Bonnet term [7], and the equations without matter read

$$\mathcal{G}^\mu{}_\nu + \Lambda_8\,\delta^\mu_\nu + \alpha_{\mathrm{GB}}\,\mathcal{H}^\mu{}_\nu = 0, \tag{9}$$

with a cosmological constant Λ_8 and a coupling α_{GB}. For the metric (4) we find

$$\begin{aligned} \mathcal{H}^j{}_j &= -4\,\mathrm{g55}\,\mathrm{g66}\,\mathrm{g77}\,\mathrm{gsum} \quad (j = 0, 1, 2, 3), \\ \mathcal{H}^4{}_4 &= \mathcal{H}^t{}_t = 0, \end{aligned} \tag{10}$$

and the Lovelock tensor of the third order, the last that can differ from zero in eight dimensions, vanishes identically. The components of Eq. (9) along $x4, \ldots, x7$ therefore contain no Gauss-Bonnet term and require the four corresponding components of the Einstein tensor to be equal, which holds if and only if the three rates are equal. The components along $x0, \ldots, x3$ then fix the coupling. The result:

$$\mathrm{g55} = \mathrm{g66} = \mathrm{g77}: \qquad \Lambda_8 = 3\,\mathrm{g55}^2, \qquad \alpha_{\mathrm{GB}} = -\frac{1}{4\,\mathrm{g55}^2}. \tag{11}$$

With three equal rates the deflating spacetime is an exact solution of Einstein-Gauss-Bonnet gravity without matter, in a theory whose two constants satisfy $4\alpha_{\mathrm{GB}}\Lambda_8 = -3$; the rate is then $\mathrm{g55} = \sqrt{\Lambda_8/3}$. With unequal rates it is not a solution for any Λ_8 and α_{GB}. Solutions of this theory in which every scale factor is an exponential of the time have been classified in five and six dimensions [8]; the present one, with four static and three contracting directions in eight dimensions, is of that kind. In the notation of the Einstein-Lovelock equations of Sec. XV, Eq. (11) reads $\bar{\Lambda} = 3\,\mathrm{g55}^2$, $\alpha_2 = 1/(4\,\mathrm{g55}^2)$. Section XV examines the stability of this solution, with and without matter: it is unstable.

D. Symmetries

A Killing vector is a vector field along which the metric does not change. Flat spacetime in eight dimensions has 36. The deflating spacetime has, for any rates, 14: the seven translations along all coordinates but $x4$; the six rotations of $x0, x1, x2, x3$; and the scaling

$$\mathcal{X}_{\mathrm{scale}} = \partial_4 + \mathrm{g55}\,x5\,\partial_5 + \mathrm{g66}\,x6\,\partial_6 + \mathrm{g77}\,x7\,\partial_7, \tag{12}$$

which shifts the time and stretches the deflating coordinates so as to undo the shrinking. Two equal rates add the rotation of the two coordinates concerned. Three equal rates add the three rotations of $x5, x6, x7$ and three further vectors, 20 in all; the four coordinates $x4, \ldots, x7$ then form a space of constant curvature of radius $1/\mathrm{g55}$: minus a four-dimensional hyperbolic space. These lists are complete. (A Killing vector that vanishes at a point is a Lorentz transformation of the tangent space that leaves the Riemann tensor (7) unchanged; the dimension of the space of such transformations is 6, 7 or 12, which bounds the number of Killing vectors by 14, 15 or 20.)

Three consequences are used below. The spacetime is homogeneous: no place and no moment is distinguished. The constants const_t are not observable by themselves, since a rescaling of xt changes them; the deflation is observable only through an object that fixes a coordinate scale along xt, such as a wave of given coordinate wavelength. And there is no conserved energy, because a translation in time alone is not a symmetry.

Of the 28 rotations and boosts of flat spacetime, only the six rotations of $x0, \ldots, x3$ survive for general rates (Fig. 1, Table LXXI). Nothing that involves the time survives, in particular not the boosts of ordinary special relativity: the deflation defines a preferred state of rest, as the expansion of the universe does.

E. Geodesics

The translations give seven conserved momenta P_μ ($\mu \neq 4$) of a classical particle. Let $E_\perp = (m^2 + \sum_j P_j^2)^{1/2}$, with the sum over $j = 0, 1, 2, 3$, be the energy

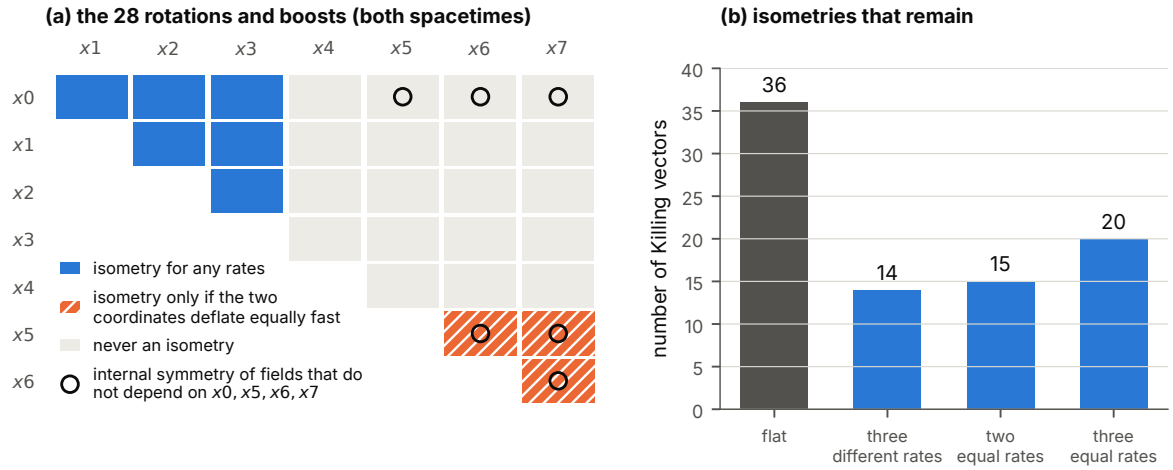


FIG. 1. Symmetries of the deflating spacetime. (a) The 28 rotations and boosts of flat spacetime, one square for each pair of coordinates: those that remain symmetries for any rates, those that remain symmetries only if the two coordinates deflate equally fast, and those that are never symmetries. Circles mark the six pairs whose rotations act as internal symmetries on fields that do not depend on $x0, x5, x6, x7$ (Sec. XVII). (b) The number of Killing vectors.

that a particle of mass m would have without momentum along the deflating coordinates. The conservation of the norm of the momentum gives the velocity along the time, with an affine parameter ς along the path:

$$\left(\frac{dx4}{d\varsigma}\right)^2 = E_\perp^2 - \sum_t \frac{P_t^2}{b_t(x4)^2}. \tag{13}$$

The last term grows as $e^{2\,\mathrm{rate}_t\,x4}$: the momentum P_t belongs to a fixed coordinate wavelength, and the proper wavelength shrinks. With the minus sign the right-hand side vanishes at a *turning value* of the time; for momentum along $x5$ alone,

$$x4_{\mathrm{turn}} = \frac{1}{\mathrm{g55}} \ln \frac{E_\perp\,\mathrm{const5}}{|P_5|}. \tag{14}$$

The path reaches this value and returns to smaller $x4$ [Fig. 2(b)]. A classical particle with momentum along a deflating time cannot pass the turning value of the ordinary time. Section V shows the quantum form of this result. (On the 7+1 spacetime the sign in Eq. (13) is the opposite, and the energy of such a particle grows without bound.)

III. THE DIRAC FIELD WITH MASS TERM AND SELF-INTERACTION

A. Matrices and spin connection

At each point of a curved spacetime the tangent space is flat. The Dirac matrices of the orthonormal frame are therefore eight constant matrices γ^A, $A = 0, \ldots, 7$. They are real 16×16 matrices with entries 0 and ±1, printed in full in Appendix B. The four matrices of the space-like coordinates are symmetric and square to $+1$, the four matrices of the time-like coordinates are antisymmetric and square to -1, and different matrices anticommute:

$$\gamma^A\gamma^B + \gamma^B\gamma^A = 2\,\eta^{AB}_{(4+4)}. \tag{15}$$

All of these properties were verified by exact integer arithmetic for this work. The matrices with a coordinate index are $\gamma^\mu = e_A{}^\mu\gamma^A$: five of them are the constant matrices, and the three along the deflating coordinates are the constant matrices divided by the scale factors, $\gamma^{A=t}/b_t$. They satisfy $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\,g^{\mu\nu}_{(4+4)}$ with the inverse metric $g^{\mu\nu}_{(4+4)}$.

A spinor is defined relative to a frame, and its derivative must account for the turning of the frame from point to point [9]. The covariant derivative of a spinor is $\nabla_\mu\psi = (\partial_\mu + \Omega_\mu)\psi$ with the spin connection

$$\begin{aligned} \Omega_\mu &= \tfrac{1}{4}\,\omega_{\mu AB}\,\gamma^A\gamma^B, \\ \omega_{\mu AB} &= \eta^{(4+4)}_{AC}\,e^C{}_\nu\big(\partial_\mu e_B{}^\nu + \Gamma^\nu{}_{\mu\lambda}e_B{}^\lambda\big), \end{aligned} \tag{16}$$

whose coefficients $\omega_{\mu AB} = -\omega_{\mu BA}$ follow from the metric ($e^C{}_\nu$ is the inverse of the frame). For the metric (4) the only coefficients that do not vanish are $\omega_{t\,t4} = -\omega_{t\,4t} = \mathrm{rate}_t\,b_t$, so that

$$\begin{aligned} &\Omega_t = \tfrac{1}{2}\,\mathrm{rate}_t\,b_t\,\gamma^t\gamma^4 \quad (t = 5, 6, 7), \\ &\Omega_0 = \cdots = \Omega_4 = 0, \end{aligned} \tag{17}$$

with the constant matrices of the frame on the right. Each Ω_t generates the Lorentz transformation in the plane of xt and $x4$. The spin connection is correct if and only if the Dirac matrices are covariantly constant, $\partial_\mu\gamma^\nu + \Gamma^\nu{}_{\mu\lambda}\gamma^\lambda + [\Omega_\mu, \gamma^\nu] = 0$, which holds for all 64 pairs of indices. Its curvature is the Riemann tensor, $[\nabla_\mu, \nabla_\nu]\psi = \frac{1}{4}R_{AB\mu\nu}\gamma^A\gamma^B\psi$, a third route to Eq. (7).

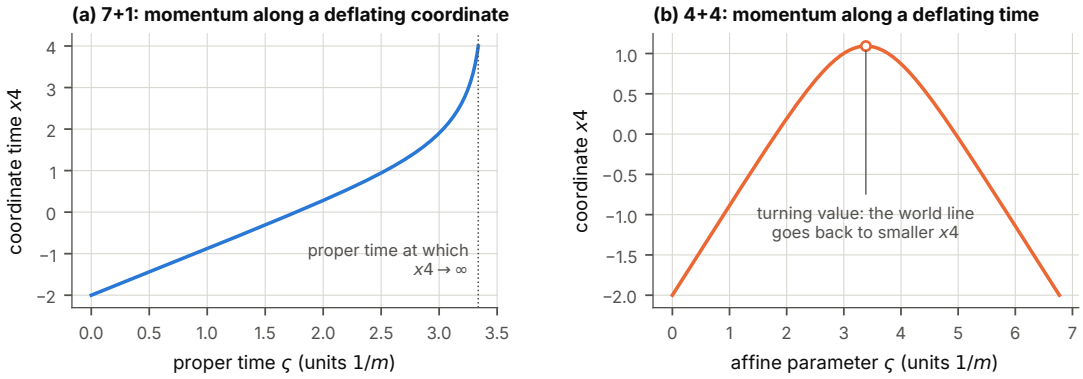


FIG. 2. A classical particle of the mass $m = 1$ with the momenta $P_1 = 0.5$ along $x1$ and $P_5 = 0.3$ along the deflating coordinate $x5$, for g55 = 1 and const5 = 0.8, started at $x4 = -2$. (a) On the 7+1 spacetime the time $x4$ grows without bound within the finite proper time 3.386. (b) On the 4+4 spacetime the path reaches the turning value $x4 = 1.0924$ and returns to smaller $x4$.

One combination enters the Dirac equation:

$$\gamma^\mu \Omega_\mu = -\tfrac{1}{2}\,\mathrm{gsum}\,\gamma^4. \tag{18}$$

The scale factors cancel, and only the total rate remains.

B. The Lagrangian: kinetic term, mass term, self-interaction

The field of this paper, $\psi_{1644d}(x)$, is a column of sixteen complex-valued Grassmann operators. Its Lagrangian has three parts,

$$\mathcal{L}_{1644d} = V_d \left[\mathcal{L}_{\mathrm{kin}} + \mathcal{L}_{\mathrm{mass}} + \mathcal{L}_{\mathrm{self}}\right], \tag{19}$$

where $V_d = \sqrt{|\det g|}$ is the factor that makes $\mathcal{L}_{1644d}\, d^8x$ invariant. A bosonic field theory would call the second part the "mass times ϕ^2" term and the third the "ϕ^4" term; for a fermion field they are bilinear and quartic in ψ.

Kinetic and mass terms. With the adjoint $\tilde\psi = \psi^\dagger C^-$, where $C^- = \gamma^0\gamma^1\gamma^2\gamma^3$ (a real symmetric matrix that turns each Dirac matrix into minus its transpose),

$$\mathcal{L}_{\mathrm{kin}} + \mathcal{L}_{\mathrm{mass}} = \tilde\psi\left(\gamma^\mu \nabla_\mu - m\right)\psi. \tag{20}$$

It contains no factor i: with the real matrices of 4+4 the operator $\gamma^\mu\nabla_\mu - m$ gives the mass shell of a particle of mass m. The mass term $-m\tilde\psi\psi$ is real because C^- is symmetric, and it is invariant under the transformations of $SO(4,4)$ of the frame.

Self-interaction. The most general interaction without derivatives that is quartic in the field and invariant under the transformations of the frame is built from the products of the Dirac matrices. Let $K = (A_1 < A_2 < \cdots < A_r)$ be a strictly increasing list of r indices out of $0,\ldots,7$; $r = |K|$ is its *rank*, and there are $2^8 = 256$ lists, with $1, 8, 28, 56, 70, 56, 28, 8, 1$ of the ranks $0,\ldots,8$.

Define

$$\Theta^K = \zeta_r\, \gamma^{A_1}\gamma^{A_2}\cdots\gamma^{A_r}, \qquad \zeta_r = \begin{cases} i, & r = 1,2,5,6, \\ 1, & \text{otherwise}, \end{cases} \tag{21}$$

($\Theta^{()} = 1$ for the empty list) and $\Theta_K = (\Theta^K)^{-1} = \mathrm{sgn}_K\,\Theta^K$, where the sign $\mathrm{sgn}_K = \pm 1$ is $(-1)^r$ times the product of the squares of the matrices in K. The phase ζ_r makes $C^-\Theta^K$ Hermitian, so that every bilinear $\tilde\psi\Theta^K\psi$ is real. The *invariant of rank r* and the self-interaction are

$$I_r = \sum_{|K|=r} (\tilde\psi\Theta_K\psi)(\tilde\psi\Theta^K\psi), \qquad \mathcal{L}_{\mathrm{self}} = -\frac{1}{2}\sum_{r=0}^{8} g_r\, I_r. \tag{22}$$

The nine real constants g_r are the couplings: g_0 multiplies the square of the scalar $\tilde\psi\psi$, g_1 the square of the current $\tilde\psi\gamma^A\psi$, g_7 and g_8 the squares of the axial current and of the pseudoscalar; in the traditional names $g_S = g_0$, $g_V = g_1$, $g_A = -g_7$, $g_P = -g_8$. A coupling g_r carries one digit as its subscript; it must not be confused with a rate (g55) or with a component of the metric ($g^{(4+4)}_{55}$). In a gravitational field each I_r is written with the frame matrices exactly as in flat spacetime, because it is invariant under the transformations of the frame. Equation (22) is the *covariant form* of the self-interaction, called *form C* below; it is the definition of the self-interaction of the 4+4 model in this paper. Section IV shows that for anticommuting fields only four of the nine invariants are independent. Section IV A shows that the kinetic term (20) is indefinite in the sector of the theory that can be quantized, defines the model that is quantized (the positive kinetic term with the self-interaction of form C), and records a second form of the self-interaction, *form Q*, as the alternative.

C. The field equation

The variation of the action with respect to $\tilde{\psi}$ gives

$$\big(\gamma^\mu \nabla_\mu - m\big)\psi_{1644d} = \sum_{r=0}^{8} g_r \sum_{|K|=r} \big(\tilde{\psi}\Theta_K\psi\big)\,\Theta^K\psi. \quad (23)$$

(The bilinears are even elements of the Grassmann algebra, so that the derivative with respect to $\tilde{\psi}$ acts on $\mathcal{L}_{\text{self}}$ as on a product of ordinary numbers.) The left-hand side is the Dirac operator with the mass term; the right-hand side is the self-interaction, cubic in the field. Without it the equation is linear. With Eq. (18),

$$\begin{aligned}\gamma^4\Big(\partial_4 - \frac{\text{gsum}}{2}\Big)\psi + \sum_{j=0}^{3}\gamma^j\partial_j\psi + \sum_{t=5}^{7}\frac{\gamma^t}{b_t}\,\partial_t\psi - m\,\psi \\ = \sum_{r=0}^{8} g_r \sum_{|K|=r} \big(\tilde{\psi}\Theta_K\psi\big)\,\Theta^K\psi,\end{aligned} \quad (24)$$

with the constant matrices. Three things distinguish the left-hand side and the interaction from their forms on flat spacetime: the derivative along the time is shifted by gsum/2, each derivative along a deflating coordinate is divided by its scale factor, and, as the next subsection shows, the self-interaction is strengthened by the shrinking of the volume.

D. The weighted field

The shift is removed by a change of normalization. Define the *weighted field*

$$\hat{\psi} = \sqrt{V_d}\,\psi. \quad (25)$$

Because $\partial_4\sqrt{V_d} = -\frac{1}{2}\text{gsum}\sqrt{V_d}$, the left-hand side of Eq. (24) applied to ψ equals $V_d^{-1/2}$ times the same expression without the term gsum/2 applied to $\hat{\psi}$. The right-hand side is cubic in the field and therefore equals $V_d^{-3/2}$ times the same expression in $\hat{\psi}$. The weighted field thus obeys Eq. (24) *without the term* gsum/2 *and with every coupling* g_r *replaced by*

$$\frac{g_r}{V_d(x4)} \;\propto\; g_r\, e^{\text{gsum}\,x4}. \quad (26)$$

The quantity $\psi^\dagger\psi$ is a density per unit of proper volume and grows as the deflating coordinates shrink; $\hat{\psi}^\dagger\hat{\psi}$ is the density per unit of coordinate volume, from which this trivial growth is removed.

The central observation of this paper follows at once. *For a field that does not depend on* $x5$, $x6$, $x7$, *the equation of the weighted field is exactly the flat equation with a self-interaction whose couplings grow as* $e^{\text{gsum}\,x4}$. Every result that holds on flat spacetime for such fields, in particular everything about the pairing of the fermions, holds for the weighted field at each moment with the couplings of that moment. The deflation enters those results in one way only, through the dependence of the couplings on V_d (Sec. VI): the mass term is not changed, and every quartic coupling grows at the same rate.

A field that does depend on a deflating coordinate feels the deflation directly. The rest of this section and Sec. V treat the equation without the self-interaction, which is linear and can be solved exactly; it describes a field so dilute that the cubic term is negligible, and the single-particle states from which the many-body treatment of Secs. IV–VII starts. Because the coefficients depend on $x4$ only, a solution of the linear equation can be taken proportional to $\exp(i\sum_j k_j\,xj + i\sum_t k_t\,xt)$ with seven constant wavenumbers, four along the flat coordinates xj, $j = 0,1,2,3$, and three along the deflating ones; such a solution is a *mode*. Multiplying the equation of the weighted field by γ^4 gives an ordinary differential equation in the time:

$$\partial_4\hat{\psi} = K(x4)\,\hat{\psi}, \qquad K = K_\perp + i\,k_d(x4)\,\gamma^4\gamma^d, \quad (27)$$

where $K_\perp = -m\gamma^4 + i\sum_j k_j\gamma^4\gamma^j$ is the operator of the modes that do not depend on the deflating times,

$$k_d(x4) = \Big(\sum_t \frac{k_t^2}{b_t(x4)^2}\Big)^{1/2} \quad (28)$$

is the *physical* wavenumber along the deflating coordinates, and $\gamma^d = k_d^{-1}\sum_t (k_t/b_t)\gamma^t$ is the frame matrix along its direction. With $E_\perp = (m^2 + \sum_j k_j^2)^{1/2}$,

$$K^2 = \big(k_d^2 - E_\perp^2\big)\,1. \quad (29)$$

$K_\perp$ is anti-Hermitian and $i\gamma^4\gamma^d$ is Hermitian. The evolution in the time is unitary, and $\hat{\psi}^\dagger\hat{\psi}$ is conserved, if and only if $k_5 = k_6 = k_7 = 0$. For the other modes the square of K is negative while $k_d < E_\perp$ and positive afterwards: the mode oscillates until its physical wavenumber, which grows as $e^{\text{rate}_t\,x4}$, has reached $E_\perp$, and its solutions grow or decay exponentially thereafter. A conserved charge exists, $\hat{\psi}^\dagger W_{1644}\hat{\psi}$ with the Hermitian matrix $W_{1644} = iC^-\gamma^4$ ($W_{1644}^2 = 1$), but it is not positive.

IV. THE SELF-INTERACTION: NINE CHANNELS, FOUR COUPLINGS

This section analyzes the quartic term $\mathcal{L}_{\text{self}}$ of Eq. (22) by exact algebra. Its results are used in every later section: the number of independent couplings, the couplings that act in uniform matter, the couplings that act between two particles, and the symmetries that the self-interaction keeps or breaks. On the 4+4 spacetime the quartic term can be completed in two ways once the theory is quantized. Section IV A defines them: *form C*, the covariant form, which is the definition of the model

in this paper, and *form Q*, the quantized-sector form, which is the alternative and is recorded beside it. The later subsections give the algebra for both forms, and Sec. XVIII D discusses the two.

A. The quantized sector and the two forms of the self-interaction

On the 4+4 spacetime only fields that do not depend on the three additional times $x5, x6, x7$ have a unitary evolution (Sec. III); they form the *quantized sector*. For such fields the kinetic and mass terms (20) become, with $\beta = -i\gamma^4$, $\alpha^j = -\gamma^4\gamma^j$ $(j = 0, 1, 2, 3)$, the Hamiltonian matrix $\mathsf{h} = \sum_j \alpha^j(-i\partial_j) + \beta m$, the matrix $W_{1644} = i\,C^-\gamma^4$ and the identity $C^- = -W_{1644}\beta$,

$$\tilde\psi\big(\gamma^A\partial_A - m\big)\psi = -\psi^\dagger\, W_{1644}\,\big(i\partial_4 - \mathsf{h}\big)\psi. \tag{30}$$

The matrix W_{1644} is Hermitian with $W_{1644}^2 = 1$; it has the eigenvalues $+1$ and -1, eight times each, and it commutes with h. The field therefore consists of two *halves*,

$$\psi = \psi_+ + \psi_-, \qquad W_{1644}\,\psi_\pm = \pm\,\psi_\pm, \tag{31}$$

and h acts inside each half: of the eight states of positive energy of a given momentum, four lie in each half. By Eq. (30) the covariant Lagrangian is the ordinary Lagrangian $\psi^\dagger(i\partial_4 - \mathsf{h})\psi$ of a Dirac field on the half $W_{1644} = -1$, and *minus* that Lagrangian on the half $W_{1644} = +1$. For a free field the sign does not change the equation of motion, and the quantum theory is defined by the positive Lagrangian

$$\begin{aligned}\mathcal{L}_{\rm kin} + \mathcal{L}_{\rm mass} &= \psi^\dagger\big(i\partial_4 - \mathsf{h}\big)\psi \\ &= \bar\psi\,(i\beta\partial_4 - \beta\mathsf{h})\,\psi, \qquad \bar\psi = \psi^\dagger\beta,\end{aligned} \tag{32}$$

for all sixteen components, which gives a particle eight internal states of positive norm. Its mass term is $-m\,\bar\psi\psi$, the same mass for both halves. The mass term of the covariant Lagrangian is different:

$$-m\,\tilde\psi\psi = m\,\big(\psi_+^\dagger\beta\,\psi_+ - \psi_-^\dagger\beta\,\psi_-\big), \tag{33}$$

a mass term of opposite sign in the two halves. For a free field this difference is invisible, because the spectrum of h does not depend on the sign of m. With a self-interaction the two halves are no longer equivalent, and the quantized theory can be completed in two ways. Both keep the positive Lagrangian (32); they differ in the adjoint with which the nine invariants are built.

Form C, the covariant form: the definition. The quartic term is that of the covariant Lagrangian, Eq. (22), with the adjoint $\tilde\psi = \psi^\dagger C^-$ and the matrices (21):

$$\begin{aligned}\mathcal{L}_{\rm kin} + \mathcal{L}_{\rm mass} + \mathcal{L}^{\rm C}_{\rm self} &= \psi^\dagger\big(i\partial_4 - \mathsf{h}\big)\psi - \frac{1}{2}\sum_{r=0}^{8} g_r\, I^{\rm C}_r, \\ I^{\rm C}_r &= \sum_{|K|=r} \mathrm{sgn}_K\,\big(\tilde\psi\,\Theta^K\psi\big)^2.\end{aligned} \tag{34}$$

($I^{\rm C}_r$ is the invariant I_r of Eq. (22); the superscript distinguishes it from the invariant of the second form.) *This is the definition of the* 4+4 *model in this paper.* Its quartic term is invariant under all 28 rotations and boosts of the frame, as the hypothesis requires of the invariants of the spacetime. Its kinetic and mass terms are invariant under the 13 of them that commute with W_{1644}: the ten rotations and boosts of $x0, \ldots, x4$ and the three rotations of $x5, x6, x7$. The other fifteen mix one of $x0, \ldots, x4$ with an additional time and do not preserve the quantized sector. The field equation is

$$i\partial_4\psi = \mathsf{h}\,\psi + \sum_r g_r \sum_{|K|=r} \big(\tilde\psi\Theta_K\psi\big)\,C^-\Theta^K\psi; \tag{35}$$

it conserves $\int \psi^\dagger\psi$, the number of particles of both halves together.

Form Q, the quantized-sector form: the alternative. The bilinears are formed with the adjoint $\bar\psi = \psi^\dagger\beta$ of Eq. (32) and with the matrices

$$\gamma^A_{44q} = -i\gamma^A \ (A = 0, \ldots, 4), \qquad \gamma^A_{44q} = \gamma^A \ (A = 5, 6, 7), \tag{36}$$

which satisfy $\gamma^4_{44q} = \beta$ and in terms of which the positive Lagrangian reads $\bar\psi(i\gamma^A_{44q}\partial_A - m)\psi$ for fields of the quantized sector. They have the squares of the Dirac matrices of a spacetime with one time: $+1$ for $A = 4$ and -1 otherwise. With

$$\begin{aligned}\Theta^K_Q &= \zeta^Q_r\,\gamma^{A_1}_{44q}\cdots\gamma^{A_r}_{44q}, \\ \zeta^Q_r &= \begin{cases} i, & r = 2, 3, 6, 7, \\ 1, & \text{otherwise,}\end{cases}\end{aligned} \tag{37}$$

and with the sign $\mathrm{sgn}^Q_K = \pm 1$, the product of the squares of the matrices γ^A_{44q} in K, so that $(\Theta^K_Q)^2 = \mathrm{sgn}^Q_K$, the invariants and the self-interaction of this form are

$$I^{\rm Q}_r = \sum_{|K|=r} \mathrm{sgn}^Q_K\,\big(\bar\psi\,\Theta^K_Q\psi\big)^2, \qquad \mathcal{L}^{\rm Q}_{\rm self} = -\frac{1}{2}\sum_r g_r\, I^{\rm Q}_r. \tag{38}$$

The field equation,

$$i\partial_4\psi = \mathsf{h}\,\psi + \sum_r g_r \sum_{|K|=r} \mathrm{sgn}^Q_K\,\big(\bar\psi\,\Theta^K_Q\psi\big)\,\beta\,\Theta^K_Q\psi, \tag{39}$$

is term by term that of the 7+1 spacetime for fields that do not depend on $x5, x6, x7$, because the algebra of the matrices (36) is that of the 7+1 Dirac matrices: every result for such fields on 7+1 holds for form Q without change. Its invariants are invariant under the group that the 28 products $\frac{1}{2}\gamma^A_{44q}\gamma^B_{44q}$ generate. Thirteen of these are the transformations of the coordinates named above; the other fifteen mix the two halves of the field and are not transformations of the coordinates of the 4+4 spacetime.

The exact relation between the two forms. For every one of the 256 lists K, $C^-\Theta^K = z_K\,W_{1644}\,\beta\,\Theta^K_Q$ with a phase $z_K = \pm 1$ if K contains an even number of the

indices 5, 6, 7 and $z_K = \pm i$ if it contains an odd number. Consequently

$$I_r^{\mathrm{C}} = \sum_{|K|=r} \mathrm{sgn}_K^Q \left(\psi^\dagger W_{1644}\, \beta\, \Theta_Q^K \psi\right)^2, \tag{40}$$

to be compared with $I_r^{\mathrm{Q}} = \sum_{|K|=r} \mathrm{sgn}_K^Q\, (\psi^\dagger \beta\, \Theta_Q^K \psi)^2$: *form C is form Q with the matrix W_{1644} inserted in every bilinear, and nothing else.* What the insertion does is seen with the bilinears of the halves (31), $\mathcal{J}_K^{++} = \psi_+^\dagger \beta\, \Theta_Q^K \psi_+$, $\mathcal{J}_K^{--} = \psi_-^\dagger \beta\, \Theta_Q^K \psi_-$, $\mathcal{J}_K^{+-} = \psi_+^\dagger \beta\, \Theta_Q^K \psi_-$ and $\mathcal{J}_K^{-+} = (\mathcal{J}_K^{+-})^\dagger$. If K contains an even number of the indices 5, 6, 7, the Hermitian matrix $\beta\, \Theta_Q^K$ commutes with W_{1644} and acts inside each half: form Q contains $(\mathcal{J}_K^{++} + \mathcal{J}_K^{--})^2$ and form C contains $(\mathcal{J}_K^{++} - \mathcal{J}_K^{--})^2$. The squares, which act between two particles of the same half, are equal; the product $2\, \mathcal{J}_K^{++} \mathcal{J}_K^{--}$, which acts between a particle of one half and a particle of the other, has opposite signs. If K contains an odd number, $\beta\, \Theta_Q^K$ anticommutes with W_{1644} and connects the halves: form Q contains $(\mathcal{J}_K^{+-} + \mathcal{J}_K^{-+})^2$ and form C contains $(\mathcal{J}_K^{+-} - \mathcal{J}_K^{-+})^2$. The squares $(\mathcal{J}_K^{+-})^2$ and $(\mathcal{J}_K^{-+})^2$, which move two particles from one half to the other, are equal; the products of $\mathcal{J}_K^{+-}$ and $\mathcal{J}_K^{-+}$, which exchange a particle of one half with a particle of the other, have opposite signs. *Two particles of the same half interact identically in the two forms; a particle of the half $W_{1644} = +1$ and a particle of the half $W_{1644} = -1$ interact with opposite signs.* Two further consequences are used below. In both forms the interaction can move two particles together from one half to the other, so that only the total number of particles is conserved, and the difference of the numbers of the two halves modulo 2. And both forms are symmetric under the exchange of the two halves, which for fields in ordinary space is the transformation $\psi \to \gamma^0\gamma^5\psi$ (a Hermitian matrix with the square 1 that commutes with $\alpha^1, \alpha^2, \alpha^3$ and β and anticommutes with W_{1644}).

What the covariant field equation itself says. If the quartic term of form C is combined with the covariant kinetic and mass terms (20) in place of the positive ones, the field equation is Eq. (23). Restricted to the quantized sector it reads

$$i\partial_4 \psi = \mathsf{h}\, \psi + W_{1644}\, \frac{\partial \mathcal{L}_{\mathrm{self}}^{\mathrm{C}}}{\partial \psi^\dagger}, \tag{41}$$

whereas Eq. (35) has $-\partial \mathcal{L}_{\mathrm{self}}^{\mathrm{C}}/\partial\psi^\dagger$ in place of $+W_{1644}\, \partial \mathcal{L}_{\mathrm{self}}^{\mathrm{C}}/\partial\psi^\dagger$. On the half $W_{1644} = -1$ the two agree. On the half $W_{1644} = +1$ the covariant equation has the opposite sign of the whole interaction, the same change of sign as in Eq. (30); it conserves $\psi^\dagger W_{1644} \psi$ rather than $\psi^\dagger\psi$ and is not the equation of a Hamiltonian with states of positive norm when both halves are occupied. It is a consistent classical field equation; no quantum theory of particles is built on it here.

Table I places the three options side by side. The hypotheses fix the nine invariants as the invariants of the spacetime. On the 7+1 spacetime this determines the model. On the 4+4 spacetime it does not, because the adjoint that makes the invariants covariant, C^-, is not the adjoint that makes the energy positive, β: the two differ by the matrix W_{1644}. This paper takes form C as the definition, by the decision of the author of the hypotheses, and reports every result of form Q beside it.

TABLE I. The three ways to complete the self-interaction on the 4+4 spacetime in the quantized sector.

option	kinetic and mass term	bilinears of the quartic term	status
form C	positive, Eq. (32)	$\bar\psi\, \Theta^K \psi$, Eq. (34)	the definition
form Q	positive, Eq. (32)	$\bar\psi\, \Theta_Q^K \psi$, Eq. (38)	the alternative, reported beside the definition
covariant classical equation	indefinite, Eq. (30)	as in form C	Eq. (41); not quantized

B. The Fierz matrix

The 256 matrices Θ^K are a basis of all 16×16 matrices, $\mathrm{Tr}(\Theta_K \Theta^J) = 16\, \delta_{KJ}$ for two lists K and J, so that $\sum_K (\Theta_K)_{cd} (\Theta^K)_{ab} = 16\, \delta_{ad}\delta_{bc}$, where the indices $a, b, c, d = 1, \ldots, 16$ number the components. This completeness relation rearranges a product of two matrices of the basis, as in the rearrangement of Fierz for four Dirac matrices [10]:

$$\sum_{|K|=r} (\Theta_K)_{ab}\, (\Theta^K)_{cd} = \sum_{r'=0}^{8} \mathsf{F}_{rr'} \sum_{|J|=r'} (\Theta_J)_{ad}\, (\Theta^J)_{cb}, \tag{42}$$

with the *Fierz matrix*

$$\mathsf{F}_{rr'} = \frac{1}{16} \sum_l (-1)^{rr'-l} \binom{r'}{l} \binom{8-r'}{r-l}, \tag{43}$$

in which r and r' are ranks and l is a summation index. The derivation takes three lines. Expanding the left-hand side of Eq. (42) in the basis gives the coefficient $\frac{1}{256} \sum_{|K|=r} \mathrm{Tr}(\Theta_K \Theta_J \Theta^K \Theta^J)$ for each list J of rank r'. Two products of Dirac matrices either commute or anticommute: $\Theta_K \Theta_J \Theta^K = (-1)^{rr'-l} \Theta_J$ if the lists K and J have l indices in common. And $\binom{r'}{l}\binom{8-r'}{r-l}$ lists K of rank r share exactly l indices with a given J of rank r'. Table II prints the matrix. It satisfies $\mathsf{F}^2 = 1$; its eigenvalues are $+1$ (five times) and -1 (four times). It is the same matrix for both forms (with Θ_Q^K for Θ^K in form Q).

TABLE II. The Fierz matrix: the integers $16\,\mathsf{F}_{rr'}$ of Eq. (43), rows $r = 0, \ldots, 8$, columns $r' = 0, \ldots, 8$.

$r\backslash r'$	0	1	2	3	4	5	6	7	8
0	1	1	1	1	1	1	1	1	1
1	8	−6	4	−2	0	2	−4	6	−8
2	28	14	4	−2	−4	−2	4	14	28
3	56	−14	−4	6	0	−6	4	14	−56
4	70	0	−10	0	6	0	−10	0	70
5	56	14	−4	−6	0	6	4	−14	−56
6	28	−14	4	2	−4	2	4	−14	28
7	8	6	4	2	0	−2	−4	−6	−8
8	1	−1	1	−1	1	−1	1	−1	1

C. Anticommuting fields: only four invariants are independent

Contract Eq. (42) with two adjoint fields (indices a and c) and two fields (indices b and d). The left-hand side is the invariant I_r. The right-hand side contains the two fields ψ in the opposite order, and the components of ψ anticommute, which costs a sign. The nine invariants therefore obey

$$I_r = -\sum_{r'} \mathsf{F}_{rr'}\, I_{r'}, \qquad \text{that is} \qquad (1+\mathsf{F})\, I = 0. \quad (44)$$

Because F has the eigenvalue $+1$ five times, these are five independent linear relations: *of the nine invariants only four are independent.* Solved for the tensor invariants,

$$\begin{aligned} I_2 &= -3I_0 - I_1 - I_7 - I_8, & I_3 &= -4I_0 - I_7 + 4I_8, \\ I_4 &= -5I_0 - 5I_8, & I_5 &= -4I_0 - I_1 + 4I_8, \\ I_6 &= -I_0 + I_1 + I_7 - 3I_8. \end{aligned} \quad (45)$$

Inserted into Eq. (22),

$$\mathcal{L}_{\text{self}} = -\tfrac{1}{2}\big(g_0^{\text{eff}}\, I_0 + g_1^{\text{eff}}\, I_1 + g_7^{\text{eff}}\, I_7 + g_8^{\text{eff}}\, I_8\big) \quad (46)$$

with the four *effective couplings*

$$\begin{aligned} g_0^{\text{eff}} &= g_0 - 3g_2 - 4g_3 - 5g_4 - 4g_5 - g_6, \\ g_1^{\text{eff}} &= g_1 - g_2 - g_5 + g_6, \\ g_7^{\text{eff}} &= g_7 - g_2 - g_3 + g_6, \\ g_8^{\text{eff}} &= g_8 - g_2 + 4g_3 - 5g_4 + 4g_5 - 3g_6. \end{aligned} \quad (47)$$

A tensor coupling $g_2, \ldots, g_6$ is not a new parameter of the theory: it shifts the scalar, vector, axial-vector and pseudoscalar couplings. Every physical quantity computed below depends on the nine g_r only through the four combinations (47), and this is verified in each case. The relations (45) and the combinations (47) are the same in both forms, with I_r^{C} or I_r^{Q} for I_r: a model is specified, in either form, by the four numbers $g_0^{\text{eff}}, g_1^{\text{eff}}, g_7^{\text{eff}}, g_8^{\text{eff}}$, and the choice of the form decides what these four numbers mean. The theorem (44) was verified in two independent ways: by exact integer arithmetic on the 16^4 coefficients of the quartic form, antisymmetrized as anticommuting fields require, and by evaluating the nine invariants on Grassmann-valued fields with an implementation of the Grassmann algebra; in both, the rank of the nine invariants is exactly four.

D. Uniform matter: the mean-field couplings

In uniform matter at rest the expectation value of a product of two fields is the density matrix $\mathsf{n}_{ab} = \langle \psi_b^\dagger \psi_a \rangle$. When the two halves (31) of the field are filled with particles up to the Fermi momenta k_{F+} and k_{F-}, and no coherence between the halves is present, it has four terms,

$$\mathsf{n} = \tfrac{1}{16}\big(n + n_S\,\beta + n_W\, W_{1644} + n_{SW}\, W_{1644}\beta\big), \quad (48)$$

with $n = n_+ + n_-$, $n_W = n_+ - n_-$, $n_S = n_{S+} + n_{S-}$ and $n_{SW} = n_{S+} - n_{S-}$, where $n_\pm = \langle \psi_\pm^\dagger \psi_\pm \rangle$ are the number densities of the two halves and $n_{S\pm} = \langle \psi_\pm^\dagger \beta\, \psi_\pm \rangle$ their scalar densities. By Wick's theorem the expectation value of a normal-ordered product of two bilinears is a direct (Hartree) term minus an exchange (Fock) term. With it the energy density of the self-interaction is, in both forms, a sum of four squares,

$$\varepsilon_{\text{int}} = \tfrac{1}{2}\big(G_V^{\text{tot}}\, n^2 + G_S^{\text{tot}}\, n_S^2 + G_V^{\text{dif}}\, n_W^2 + G_S^{\text{dif}}\, n_{SW}^2\big). \quad (49)$$

$G_V^{\text{tot}}, G_S^{\text{tot}}$ are the couplings of the squares of the *total* densities, $G_V^{\text{dif}}, G_S^{\text{dif}}$ those of the squares of the *differences* between the halves. They are four combinations of the effective couplings,

$$\begin{aligned} G_S^{\text{HF}} &= \tfrac{1}{16}\big(15\, g_0^{\text{eff}} - 8\, g_1^{\text{eff}} - 8\, g_7^{\text{eff}} - g_8^{\text{eff}}\big), \\ G_V^{\text{HF}} &= \tfrac{1}{16}\big(-\, g_0^{\text{eff}} + 22\, g_1^{\text{eff}} - 6\, g_7^{\text{eff}} + g_8^{\text{eff}}\big), \\ G_S^{\text{c}} &= \tfrac{1}{16}\big(-\, g_0^{\text{eff}} - 2\, g_1^{\text{eff}} + 2\, g_7^{\text{eff}} + g_8^{\text{eff}}\big), \\ G_V^{\text{c}} &= -\tfrac{1}{16}\big(g_0^{\text{eff}} + g_8^{\text{eff}}\big), \end{aligned} \quad (50)$$

the same four in both forms, with the roles of the totals and of the differences exchanged:

$$\begin{aligned} \text{form C:}\quad & G_S^{\text{tot}} = G_S^{\text{c}}, \quad G_V^{\text{tot}} = G_V^{\text{c}}, \\ & G_S^{\text{dif}} = G_S^{\text{HF}}, \quad G_V^{\text{dif}} = G_V^{\text{HF}}; \\ \text{form Q:}\quad & G_S^{\text{tot}} = G_S^{\text{HF}}, \quad G_V^{\text{tot}} = G_V^{\text{HF}}, \\ & G_S^{\text{dif}} = G_S^{\text{c}}, \quad G_V^{\text{dif}} = G_V^{\text{c}}. \end{aligned} \quad (51)$$

In terms of the nine couplings, $G_S^{\text{HF}} = \frac{1}{16}(15g_0 - 8g_1 - 28g_2 - 56g_3 - 70g_4 - 56g_5 - 28g_6 - 8g_7 - g_8)$, $G_V^{\text{HF}} = \frac{1}{16}(-g_0 + 22g_1 - 14g_2 + 14g_3 - 14g_5 + 14g_6 - 6g_7 + g_8)$, $G_S^{\text{c}} = \frac{1}{16}(-g_0 - 2g_1 + 2g_2 + 6g_3 + 10g_5 - 2g_6 + 2g_7 + g_8)$ and $G_V^{\text{c}} = \frac{1}{16}(-g_0 + 4g_2 + 10g_4 + 4g_6 - g_8)$. G_S^{HF} and G_V^{HF} are the Hartree-Fock couplings of the 7+1 model: the direct terms alone would give g_0 and g_1, and the exchange terms reduce the scalar coupling by one sixteenth and add to the vector coupling. An equivalent way to write Eq. (49) shows what happens between particles:

$$\begin{aligned} \varepsilon_{\text{int}} = {} & \tfrac{1}{2}\, G_V^{\text{same}}\, (n_+^2 + n_-^2) + \tfrac{1}{2}\, G_S^{\text{same}}\, (n_{S+}^2 + n_{S-}^2) \\ & + G_V^{\text{opp}}\, n_+ n_- + G_S^{\text{opp}}\, n_{S+} n_{S-}, \end{aligned} \quad (52)$$

with $G^{\rm same} = G^{\rm tot} + G^{\rm dif}$ and $G^{\rm opp} = G^{\rm tot} - G^{\rm dif}$ for the scalar and for the vector couplings alike:

$$\begin{aligned} G_S^{\rm same} &= \tfrac{1}{8}\big(7\,g_0^{\rm eff} - 5\,g_1^{\rm eff} - 3\,g_7^{\rm eff}\big), \\ G_V^{\rm same} &= \tfrac{1}{8}\big(-\,g_0^{\rm eff} + 11\,g_1^{\rm eff} - 3\,g_7^{\rm eff}\big), \\ G_S^{\rm opp} &= \pm\tfrac{1}{8}\big(8\,g_0^{\rm eff} - 3\,g_1^{\rm eff} - 5\,g_7^{\rm eff} - g_8^{\rm eff}\big), \\ G_V^{\rm opp} &= \pm\tfrac{1}{8}\big(11\,g_1^{\rm eff} - 3\,g_7^{\rm eff} + g_8^{\rm eff}\big), \end{aligned} \tag{53}$$

with the upper sign in form Q and the lower sign in form C. $G_S^{\rm same}, G_V^{\rm same}$ are the couplings between two particles of the same half; they are the same in both forms. $G_S^{\rm opp}, G_V^{\rm opp}$ are the couplings between a particle of one half and a particle of the other; they have opposite signs in the two forms. This is the statement of Sec. IV A for the mean field.

For an attractive scalar coupling alone ($g_0 < 0$) the consequence is this. In form Q the attraction stands in $G_S^{\rm tot} = \frac{15}{16} g_0$: it lowers the energy of any scalar density, as on 7+1. In form C it stands in $G_S^{\rm dif} = \frac{15}{16} g_0$: it lowers the energy only if the scalar densities of the two halves differ, and the total scalar density is weakly repelled ($G_S^{\rm tot} = -\frac{1}{16} g_0 > 0$). Between two particles of the same half the coupling attracts in both forms ($G_S^{\rm same} = \frac{7}{8} g_0$); between particles of opposite halves it attracts in form Q ($G_S^{\rm opp} = g_0$) and repels in form C ($G_S^{\rm opp} = -g_0$). Matter in which only one half is filled is matter of one kind of particle with four internal states and the couplings $G_S^{\rm same}$, $G_V^{\rm same}$; matter of form Q with equal halves is matter of one kind with eight internal states and the couplings $G_S^{\rm HF}$, $G_V^{\rm HF}$. Section VII sets out the method for one kind of particle with a scalar coupling G_S and a vector coupling G_V, and then treats the two halves together.

E. Two particles: the pair couplings

Two particles at rest can be in $8 \cdot 7/2 = 28$ antisymmetric states, since a particle at rest has eight internal states (the eigenvalue $+1$ of β). The self-interaction acts on a pair state Φ, an antisymmetric 16×16 matrix, as $\Phi \mapsto \sum_r g_r \sum_{|K|=r} {\rm sgn}_K\, (C^-\Theta^K)\,\Phi\,(C^-\Theta^K)^T$ in form C (in form Q with $\beta\,\Theta_Q^K$ and ${\rm sgn}_K^Q$), projected on the states of positive energy. Its eigenvalue is the *pair coupling* $G_{\rm pair}$, the interaction energy of the pair times the volume; a negative value means attraction. Diagonalizing this map (exactly in the program, numerically in the independent verification) gives, in form Q and on the 7+1 spacetime, two values:

$$G_{\rm pair} = \begin{cases} g_0^{\rm eff} + g_1^{\rm eff} + 5\,g_7^{\rm eff} & \text{for 7 pairs,} \\ g_0^{\rm eff} + g_1^{\rm eff} - 3\,g_7^{\rm eff} & \text{for 21 pairs.} \end{cases} \tag{54}$$

The seven pairs are those created by the vector pair operators $\psi^T C^- \gamma^A \psi$ along the seven axes other than the time ($A \neq 4$); among them is the pair along the hidden axis $x0$, the pair of the superfluid of this paper. A scalar attraction ($g_0^{\rm eff} < 0$) binds all 28 pairs equally; a vector repulsion ($g_1^{\rm eff} > 0$) opposes all of them equally; the axial coupling distinguishes the seven from the twenty-one; and the pseudoscalar coupling does not act between particles at rest.

TABLE III. The pair couplings of the 28 antisymmetric states of two particles at rest on the 4+4 spacetime, in the two forms of the self-interaction (exact).

form	pairs	the two particles are in	$G_{\rm pair}$
C	4	different halves	$-g_0^{\rm eff} - g_1^{\rm eff} - 5\,g_7^{\rm eff}$
C	12	different halves	$-g_0^{\rm eff} - g_1^{\rm eff} + 3\,g_7^{\rm eff}$
C	9	same half	$g_0^{\rm eff} + g_1^{\rm eff} - 3\,g_7^{\rm eff}$
C	3	same half	$g_0^{\rm eff} + g_1^{\rm eff} + 5\,g_7^{\rm eff}$
Q	12	different halves	$g_0^{\rm eff} + g_1^{\rm eff} - 3\,g_7^{\rm eff}$
Q	4	different halves	$g_0^{\rm eff} + g_1^{\rm eff} + 5\,g_7^{\rm eff}$
Q	9	same half	$g_0^{\rm eff} + g_1^{\rm eff} - 3\,g_7^{\rm eff}$
Q	3	same half	$g_0^{\rm eff} + g_1^{\rm eff} + 5\,g_7^{\rm eff}$

In form C the result is different. The state that a vector pair operator creates has a definite behavior under W_{1644}: $W_{1644}\,\Phi\,W_{1644}^T = -\Phi$ for the pairs along $x0, x1, x2, x3$ and $W_{1644}\,\Phi\,W_{1644}^T = +\Phi$ for the pairs along $x5, x6, x7$. *A pair along one of the four space-like axes consists of one particle of each half; a pair along one of the three additional time-like axes consists of two particles of the same half.* By Sec. IV A the two forms agree on the pairs of two particles of the same half and have opposite signs on the pairs of one particle of each half (Table III). In form C the seven vector pairs split into the four pairs along the space-like axes, with $G_{\rm pair} = -(g_0^{\rm eff} + g_1^{\rm eff} + 5\,g_7^{\rm eff})$, and the three pairs along the additional axes, with $G_{\rm pair} = +(g_0^{\rm eff} + g_1^{\rm eff} + 5\,g_7^{\rm eff})$: *a scalar coupling that attracts in form Q, and on* 7+1*, repels in the hidden-axis pair of form C.*

F. Symmetries

Chiral symmetry. The transformation $\psi \to e^{i\epsilon\chi}\psi$ with the chirality matrix χ (Hermitian, $\chi^2 = 1$, anticommuting with every γ^A) leaves the self-interaction invariant if and only if

$$g_1^{\rm eff} - g_7^{\rm eff} = 0 \text{ (form C)}, \qquad g_0^{\rm eff} + g_8^{\rm eff} = 0 \text{ (form Q)}. \tag{55}$$

In form Q, as on the 7+1 spacetime, the transformation leaves I_1 and I_7 unchanged and rotates I_0 and I_8 into each other. In form C the roles of the even and of the odd ranks are exchanged, because the bilinears of the two forms differ by the matrix W_{1644}, which anticommutes with χ: a scalar coupling alone is chirally invariant in form C and not in form Q, and a vector coupling alone is chirally invariant in form Q and not in form C. The mass term breaks this symmetry in any case; it is the symmetry $U(1)_{\rm VS}$ of the model with the Vorton-Scalar (Sec. VI), in which the mass arises from the vacuum value of a scalar field that transforms as well.

The internal symmetry of fields in ordinary space. A field that does not depend on the four extra coordinates $x0, x5, x6, x7$ has an internal symmetry $[SU(2)\times SU(2)]\times U(1)_E$ in addition to the phase symmetry $U(1)_F$ of the fermion number (Sec. XVII): six generators proportional to $\gamma^a\gamma^b$ ($a<b$ out of $0,5,6,7$), which generate the rotations $SO(4)$ of the four extra axes, and a generator P proportional to $\gamma^0\gamma^5\gamma^6\gamma^7$. The three rotations of $x5, x6, x7$ among themselves commute with W_{1644}; the three generators $\gamma^0\gamma^t$ ($t=5,6,7$) and P anticommute with W_{1644} and connect the two halves. Table IV gives the couplings for which the self-interaction is invariant under the unitary transformations that each generator generates. *In form C every self-interaction that does not vanish breaks the three transformations that mix the hidden axis with $x5, x6, x7$, and breaks $U(1)_E$*: the internal symmetry is reduced from $SO(4)\times U(1)_F\times U(1)_E$ to $SO(3)_T\times U(1)_F$, the rotations of the three additional times among themselves and the fermion number. In form Q, as on 7+1, the self-interaction keeps the six rotations and the phase for all couplings and keeps $U(1)_E$ only if $g_1^{\rm eff}=g_7^{\rm eff}=0$. The reason for the difference is this. In the frame group of the 4+4 spacetime $\gamma^0\gamma^t$ generates a boost, which mixes a space-like with a time-like coordinate and is not a unitary transformation of the field. The unitary transformation generated by the same matrix is a different transformation: it is a symmetry of the kinetic term of fields in ordinary space by accident, and it is not an element of the frame group. The quartic term of form C is invariant under the boost and has no reason to be invariant under the unitary transformation. In form Q the same matrices generate rotations, as on the 7+1 spacetime, where all four extra axes are space-like.

V. MODES THAT DEPEND ON THE DEFLATING COORDINATES

A. Three kinds of modes

Modes fall into three kinds. A mode that does not depend on the deflating coordinates, $k_5=k_6=k_7=0$, obeys the flat equation; its solutions are the plane waves of flat spacetime, with the energies $\pm E_\perp$, eight of each sign. A mode with some $k_t\neq 0$ behaves in the same way as long as its physical wavenumber is negligible, $k_d \ll E_\perp$. But k_d grows without bound, and every such mode sooner or later reaches $k_d \gg E_\perp$. The moment at which $k_d = E_\perp$ is its *crossing*. It is the turning value (14) of the classical particle, with the wavenumbers in place of the momenta. The crossing is where the deflation acts on the field, and this section solves the equation of the weighted field across it.

The solution is exact when *one rate governs*: when only one of the k_t differs from zero, or when the three rates are equal. In both cases the matrix γ^d does not depend on the time, and k_d is proportional to $e^{{\rm rate}_t\, x4}$ with the rate that governs. The natural variable is then the physical wavenumber in units of the rate,

$$u_d = \frac{k_d(x4)}{{\rm rate}_t}, \qquad \frac{d}{dx4} = {\rm rate}_t\, u_d\, \frac{d}{du_d}. \tag{56}$$

The crossing is at $u_d = E_\perp/{\rm rate}_t$, and this ratio of the energy to the rate is the one parameter on which the results depend.

B. The exact solution

In $K = K_\perp + i\,k_d\,\gamma^4\gamma^d$ the first matrix is anti-Hermitian with $K_\perp^2 = -E_\perp^2$, the matrix $i\gamma^4\gamma^d$ is Hermitian with the square one, and the two anticommute. Let w be one of the eight eigenvectors of $K_\perp$ with the eigenvalue $+iE_\perp$; then $i\gamma^4\gamma^d w$ is an eigenvector with the eigenvalue $-iE_\perp$ and is orthogonal to w. The equation is solved by the modified Bessel functions that decay at large argument, of the orders $\frac{1}{2} \mp iE_\perp/{\rm rate}_t$. Such a function is written here with its order as a subscript and with its argument; the letter K without subscript and argument is the matrix of Eq. (27). With the recurrence relations of these functions one verifies by substitution that

$$\hat\psi = \sqrt{u_d}\,\Big[K_{\frac{1}{2}-iE_\perp/{\rm rate}_t}(u_d)\,w - K_{\frac{1}{2}+iE_\perp/{\rm rate}_t}(u_d)\,i\gamma^4\gamma^d w\Big] \tag{57}$$

solves Eq. (27) for each of the eight w; the relative residual, computed with the Bessel functions of an arbitrary-precision library, is 2×10^{-15}. The other eight solutions are built in the same way from the modified Bessel functions that *grow* as e^{+u_d}.

C. Standing waves, and their extinction

Since u_d itself grows exponentially with the time, a solution of the second kind grows as the exponential of an exponential. Such a solution is not admissible; it is the counterpart of the growing exponential under a potential barrier. Admissible are the eight solutions (57), which stay bounded. Their properties are exact.

Magnitude. Because w and $i\gamma^4\gamma^d w$ are orthogonal, and because the two functions in Eq. (57) are complex conjugates of each other,

$$|\hat\psi|^2 = 2\,u_d\,\big|K_{\frac{1}{2}+iE_\perp/{\rm rate}_t}(u_d)\big|^2\,|w|^2. \tag{58}$$

Early times, $u_d\to 0$: $|\hat\psi|^2 \to \pi\,|w|^2/\cosh(\pi E_\perp/{\rm rate}_t)$, a constant. The solution is then a combination of $e^{+iE_\perp x4}\,w$ and $e^{-iE_\perp x4}\,i\gamma^4\gamma^d w$ with coefficients of *equal* magnitude: the two signs of the frequency enter with equal weight. A bounded solution is a *standing wave* in the ordinary time, not a traveling wave. Its conserved charge $\hat\psi^\dagger W_{1644}\hat\psi$ vanishes.

TABLE IV. The couplings for which the self-interaction is invariant under the internal symmetry of fields in ordinary space, on the 7+1 spacetime and in the two forms of the 4+4 spacetime (exact).

generators	number	7+1	4+4, form Q	4+4, form C
phase of the field (fermion number)	1	all couplings	all couplings	all couplings
rotations of $x5, x6, x7$ among themselves	3	all couplings	all couplings	all couplings
rotations that mix $x0$ with $x5$, $x6$ or $x7$	3	all couplings	all couplings	no coupling ($g_0^{\text{eff}} = g_1^{\text{eff}} = g_7^{\text{eff}} = g_8^{\text{eff}} = 0$)
P, the generator of $U(1)_E$	1	$g_1^{\text{eff}} = g_7^{\text{eff}} = 0$	$g_1^{\text{eff}} = g_7^{\text{eff}} = 0$	no coupling ($g_0^{\text{eff}} = g_1^{\text{eff}} = g_7^{\text{eff}} = g_8^{\text{eff}} = 0$)

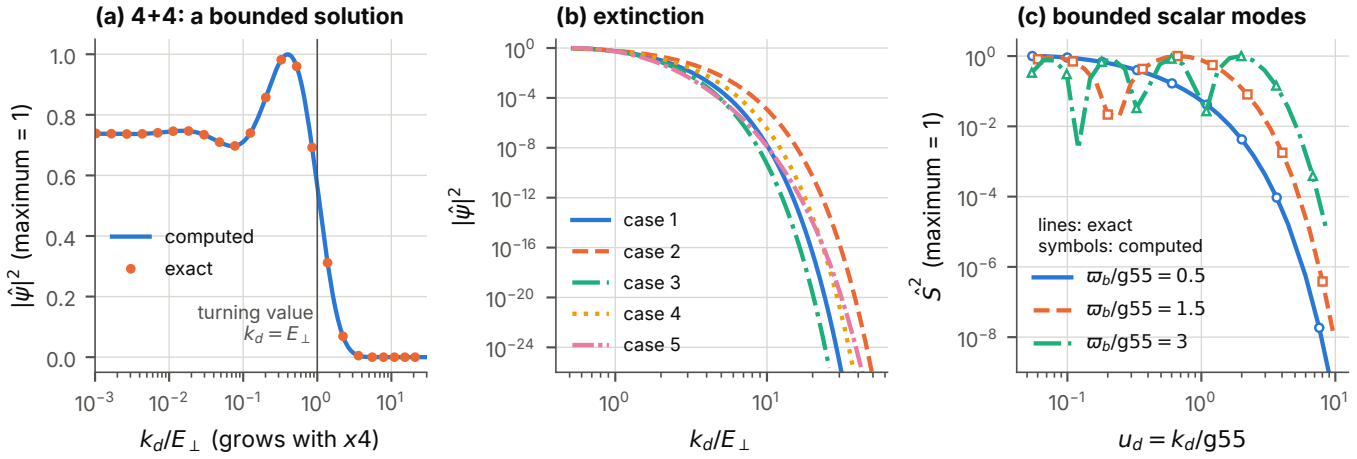


FIG. 3. Extinction on the deflating 4+4 spacetime. (a) A bounded solution: $|\hat\psi|^2$ against the physical wavenumber $k_d/E_\perp$, which grows with the time; computed (line) and the exact form (58) (symbols). It is constant long before the crossing at $k_d = E_\perp$ and falls after it. (b) Five cases, with one, two or three wavenumbers k_t different from zero and with equal and unequal rates, on a logarithmic scale. (c) The bounded modes of the Vorton-Scalar on the same spacetime (Sec. VI D), which are extinguished in the same way; lines: exact, symbols: computed.

Late times, $u_d \to \infty$: $|\hat\psi|^2 \to \pi\,|w|^2 e^{-2u_d}$. The mode dies out as the exponential of an exponential of the time.

The law of the extinction. In the semiclassical (WKB) approximation the bounded solution falls beyond the turning value, at which $k_d = E_\perp$, as $e^{-S_{\text{WKB}}}$ with

$$S_{\text{WKB}} = \int_{x4_{\text{turn}}}^{x4} \sqrt{k_d^2 - E_\perp^2}\; dx4' = \frac{1}{\text{rate}_t}\Big[\sqrt{k_d^2 - E_\perp^2} - E_\perp \arccos\frac{E_\perp}{k_d}\Big], \qquad (59)$$

which approaches $(k_d - \frac{\pi}{2}E_\perp)/\text{rate}_t$ for $k_d \gg E_\perp$. The exact solution fixes the constant that the approximation leaves open:

$$\ln\frac{|\hat\psi|_{\text{early}}}{|\hat\psi|_{\text{late}}} - S_{\text{WKB}} \;\longrightarrow\; \tfrac{1}{2}\ln 2 - \tfrac{1}{2}\ln\big(1 + e^{-2\pi E_\perp/\text{rate}_t}\big). \qquad (60)$$

The numerical solution of Eq. (27) for all sixteen components confirms these statements (Fig. 3). Beyond the turning value exactly eight of the sixteen solutions stay bounded; followed backwards to early times they contain parts of positive and of negative frequency of equal weight, to 2×10^{-14}. Where one rate governs, the computed magnitude equals Eq. (58) over the range $0.02 < u_d < 12$ to a relative deviation of 3.6×10^{-12}, and Eq. (60) holds to 3.6×10^{-4} at $S_{\text{WKB}} = 30$. With three unequal rates no exact solution is known; the bounded solutions fall as $e^{-S_{\text{WKB}}}$ times a factor that depends on the internal state, with a spread of 0.40 in its logarithm at $S_{\text{WKB}} = 30$.

D. What the result means

On the flat 4+4 spacetime a field that depends on the additional times evolves non-unitarily in the ordinary time. This is the principal difficulty of four timelike dimensions [11, 12]; on flat spacetime it can only be met by imposing that the fields do not depend on those times. On the deflating 4+4 spacetime every admissible mode that depends on a deflating time is a standing wave that is extinguished after its crossing, with an exponent S_{WKB} that grows as $e^{\text{rate}_t\, x4}$. What remains at late times is the sector of fields that do not depend on $x5, x6, x7$, and in that sector the evolution is unitary and $\hat\psi^\dagger\hat\psi$ is conserved. *The deflation of the three additional times removes, dynamically, the sector that causes the difficulty.* The classical counterpart is the turning value of Sec. II E.

The limits of this conclusion must be stated with equal clarity. The eight growing solutions are discarded be-

cause they are unbounded. That is a condition of admissibility, imposed as in the theory of a potential barrier; it is not derived from a principle. The theory still has no positive conserved charge for modes that depend on the deflating times, and a consistent quantization exists, as before, only for the sector that does not depend on them. What has been shown is that the admissible modes outside that sector die out, not that the difficulty of four time-like dimensions has been solved.

E. The numerical method

The equations of this section are linear, with coefficients that grow exponentially, and an integrator with a fixed step fails on them. The program uses a commutator-free Magnus method of the fourth order: over each step the solution is advanced by the product of two matrix exponentials of linear combinations of the coefficient matrix at two Gauss points. Because the square of K is a multiple of the unit matrix at any one time, each exponential is evaluated in closed form. The length of a step is chosen so that neither the phase accumulated in it nor the relative change of the coefficients exceeds a fixed bound. An independent program, which uses an explicit Runge-Kutta method of the eighth order and the Bessel functions of another library, confirms Eq. (57) and, to the accuracy of the semiclassical form at the exponent that it reaches, Eq. (60) (Appendix E).

VI. THE VORTON-SCALAR AND THE DEFLATION CLOCK

A. The model

The model contains, besides the fermion field, a complex scalar field, the *Vorton-Scalar* $\Phi_{\rm VS}$, whose vacuum value gives the fermions a mass and whose exchange binds two fermions into a pair. This subsection states the complete model on the deflating spacetime: the fermion field with a mass term and with the self-interaction of Sec. III, and the Vorton-Scalar. The Vorton-Scalar brings three real, positive constants: its self-coupling $\lambda_{\rm VS}$, the constant $v_{\rm VS}$ that fixes the magnitude $v_{\rm VS}/\sqrt{2}$ at which the potential of the scalar is lowest, and its Yukawa coupling $y_{\rm VS}$ to the fermions. The fermion field brings the mass m_D of its Lagrangian, called the *bare Dirac mass* from here on, and the couplings g_r of its self-interaction. The chirality matrix is denoted by χ, with $\chi = \gamma^0\gamma^1\cdots\gamma^7$, which is real on the 4+4 spacetime, and the chiral projectors are $P^{L,R} = \frac{1}{2}(1 \mp \chi)$.

On a curved spacetime a Lagrangian is multiplied by the volume factor V_d, derivatives of spinors become covariant derivatives, and the flat metric is replaced by the curved one. On the 4+4 spacetime the Yukawa coupling can be written only in the sector of fields that do not depend on the additional times, the quantized sector of Sec. IV A, because only there does the adjoint $\bar\psi = \psi^\dagger\beta$ exist with which the chiral projectors combine into a real, invariant term; the model is defined there. (The covariant adjoint cannot serve in the Yukawa coupling: C^- commutes with χ, so that $\tilde\psi\psi$ and $\tilde\psi\chi\psi$ are each invariant under the chiral transformation and do not rotate into each other, whereas a complex scalar field whose phase is the Goldstone mode of that transformation must couple to two bilinears that do, $\bar\psi\psi$ and $\bar\psi\, i\chi\,\psi$. The Yukawa coupling is therefore written with $\bar\psi$ whatever the form of the self-interaction, and an expectation value of the scalar shifts the masses of both halves of the field by the same amount.) On the deflating spacetime Sec. V gives that sector a dynamical reason. In it the model is

$$\begin{aligned}\mathcal{L}^{\rm VS}_{1644d} = \hat\psi^\dagger\big(i\partial_4 + i\sum_{j=0}^{3}\alpha^j\partial_j - \beta\, m_D\big)\hat\psi + V_d\,\mathcal{L}_{\rm self}\\ + V_d\left[|\partial_4\Phi_{\rm VS}|^2 - \sum_{j=0}^{3}|\partial_j\Phi_{\rm VS}|^2\right]\\ - V_d\,\lambda_{\rm VS}\Big(|\Phi_{\rm VS}|^2 - \frac{v_{\rm VS}^2}{2}\Big)^2\\ - V_d\,y_{\rm VS}\,\bar\psi\big(\Phi_{\rm VS}\,P^R + \Phi^*_{\rm VS}\,P^L\big)\psi,\end{aligned} \tag{61}$$

with the Hermitian matrices $\beta = -i\gamma^4$ and $\alpha^j = -\gamma^4\gamma^j$, $\bar\psi = \psi^\dagger\beta$, and the self-interaction $\mathcal{L}_{\rm self} = \mathcal{L}^{\rm C}_{\rm self}$ of form C, Eq. (34), the definition (with $\mathcal{L}^{\rm Q}_{\rm self}$ of Eq. (38) in its place the model is that of the alternative form Q). The first line is the positive Lagrangian (32) for the weighted field $\hat\psi = \sqrt{V_d}\,\psi$, with the bare Dirac mass, and the self-interaction; without the self-interaction its equation of motion is the covariant Dirac equation (23) in this sector. The other terms carry the volume factor. The Lagrangian is invariant under the phase transformation $\psi \to e^{i\epsilon}\psi$ of the fermion number, $U(1)_F$, whatever its constants are. Without the bare Dirac mass, and with a self-interaction that satisfies the condition (55), it is also invariant under the chiral transformation $U(1)_{\rm VS}$: $\psi \to e^{i\epsilon\chi}\psi$ with $\Phi_{\rm VS} \to e^{-2i\epsilon}\Phi_{\rm VS}$. These statements hold on the deflating spacetime as on flat spacetime, because the transformations do not involve the coordinates. *The case $m_D = 0$, $g_r = 0$ is the model in which the whole mass of the fermions comes from the Vorton-Scalar and the whole interaction from its exchange;* Secs. VIII to XIV treat it first, and Secs. VII, IX and XIII then add the bare mass and the self-interaction and say what each changes.

A constant field $\Phi_{\rm VS} = v_{\rm VS}/\sqrt{2}$ with $\psi = 0$ solves the field equations on the deflating spacetime, because a constant has no derivatives: the vacuum value is the same at all times. It breaks $U(1)_{\rm VS}$. Around it the Vorton-Scalar has two modes: the *radial mode*, a real field of the mass m_s, and the *phase mode*, a real massless field, the Goldstone boson of the broken symmetry (massless if $U(1)_{\rm VS}$ is exact; Sec. XIII computes the mass that a bare Dirac mass gives it). The fermions acquire the mass $m = m_D + m_Y$, of which $m_Y = y_{\rm VS}v_{\rm VS}/\sqrt{2}$ comes from

the Vorton-Scalar. Both modes couple to the fermions with the same strength $g_{\rm VS}$, the radial mode to the scalar density $\bar\psi\psi$ and the phase mode to the pseudoscalar density $\bar\psi\, i\chi\,\psi$:

$$\begin{aligned} m_Y &= \frac{y_{\rm VS}\, v_{\rm VS}}{\sqrt{2}}, \qquad & m_s &= \sqrt{2\lambda_{\rm VS}}\, v_{\rm VS}, \\ g_{\rm VS} &= \frac{m_Y}{v_{\rm VS}}, & \alpha_{\rm VS} &= \frac{g_{\rm VS}^2}{4\pi}. \end{aligned} \tag{62}$$

B. Fields that do not depend on the extra coordinates

Consider fields that do not depend on the four extra coordinates $x0, x5, x6, x7$, and introduce the weighted fields

$$\hat\psi = \sqrt{V_d}\,\psi, \qquad \hat\Phi_{\rm VS} = \sqrt{V_d}\,\Phi_{\rm VS}. \tag{63}$$

Each term of Eq. (61) is rewritten in turn. The kinetic term and the mass term of the fermions are the flat ones for $\hat\psi$ (Sec. III). The self-interaction contains four fermion fields and therefore the factor $V_d \cdot V_d^{-2}$, as in Eq. (26). The Yukawa term contains one scalar and two fermion fields and therefore the factor $V_d \cdot V_d^{-3/2}$. The potential becomes $(\lambda_{\rm VS}/V_d)\,(|\hat\Phi_{\rm VS}|^2 - V_d v_{\rm VS}^2/2)^2$. In the kinetic term of the scalar the derivative along the time acts on the weight as well, $\sqrt{V_d}\,\partial_4\Phi_{\rm VS} = \partial_4\hat\Phi_{\rm VS} + \frac{1}{2}{\rm gsum}\,\hat\Phi_{\rm VS}$, which gives a term $\frac{1}{4}{\rm gsum}^2|\hat\Phi_{\rm VS}|^2$ and a total derivative. The result is the flat Lagrangian for the weighted fields, with four replacements and one additional term:

$$\begin{aligned} &y_{\rm VS} \to \frac{y_{\rm VS}}{\sqrt{V_d(x4)}}, \qquad \lambda_{\rm VS} \to \frac{\lambda_{\rm VS}}{V_d(x4)}, \qquad g_r \to \frac{g_r}{V_d(x4)}, \\ &v_{\rm VS} \to v_{\rm VS}\sqrt{V_d(x4)}, \qquad \text{and } + \frac{{\rm gsum}^2}{4}\,|\hat\Phi_{\rm VS}|^2. \end{aligned} \tag{64}$$

The bare Dirac mass m_D is not replaced. With a constant volume of the extra coordinates the same replacements define the couplings of a four-dimensional description on flat spacetime. Here that volume is the coordinate volume of a cell times $V_d(x4)$, and it shrinks. The additional term lowers the square of the mass of each mode of the scalar by ${\rm gsum}^2/4$ in the equation of the weighted field; Section VI D shows what it does. From here on the couplings $y_{\rm VS}$, $\lambda_{\rm VS}$, $v_{\rm VS}$, $g_{\rm VS}$, $\alpha_{\rm VS}$ and g_r are those of the four-dimensional description at the time considered.

C. The deflation clock

Define the *deflation clock*

$$\tau_d = {\rm gsum}\,(x4 - x4_{\rm ref}), \tag{65}$$

the number of e-foldings by which the volume V_d has shrunk since a reference time $x4_{\rm ref}$, which is chosen in each application. By Eq. (64) every quantity of the four-dimensional description is proportional to $e^{n\tau_d}$ with a definite exponent n (Table V). Three observations organize what follows.

The masses do not change. The bare Dirac mass is not affected at all, and the part m_Y of the fermion mass and the mass m_s of the radial mode contain the couplings in combinations from which the volume cancels. So do the Fermi momentum and the Fermi energy of matter with a given number of particles per unit of ordinary volume.

The couplings grow. The coupling $g_{\rm VS} = m_Y/v_{\rm VS}$ grows as $e^{\tau_d/2}$, because the four-dimensional vacuum value falls. The strength of the attraction, $\alpha_{\rm VS}$, grows as e^{τ_d}: it doubles every time the volume of the deflating coordinates halves. Every coupling g_r of the self-interaction, and with them the couplings G_S, G_V of Sec. VII, grows as e^{τ_d} as well: *the contact interaction and the exchange of the Vorton-Scalar keep their ratio.*

The two numbers that decide the state of matter grow at the same rate. These are

$$D_{\rm VS} = \alpha_{\rm VS}\,\frac{m}{m_s}, \qquad C = \frac{g_{\rm VS}^2 m^2}{m_s^2}. \tag{66}$$

The first decides whether two fermions form a bound pair (Sec. X), the second whether matter keeps its fermion mass (Sec. XII). Their ratio $C/D_{\rm VS} = 4\pi\, m/m_s$ is fixed.

On flat spacetime the model is specified by the masses and the couplings. On a deflating spacetime it is specified by the masses and the values of the couplings *at one time*; the deflation then carries it through all larger couplings. The constants of the flat theory become functions of the time, and its thresholds are passed one after another (Fig. 4).

D. Modes of the Vorton-Scalar

Let $\varphi_{\rm bos}$ stand for the radial mode or the phase mode, with the mass $m_{\rm bos} = m_s$ or $m_{\rm bos} = 0$. To first order in the field, a mode with a wavenumber of the magnitude k along the flat coordinates and the wavenumbers k_t along the deflating ones obeys

$$\begin{aligned} &\frac{1}{V_d}\,\partial_4\big(V_d\,\partial_4\varphi_{\rm bos}\big) \\ &\quad + \Big(k^2 - \sum_t \frac{k_t^2}{b_t^2} + m_{\rm bos}^2\Big)\varphi_{\rm bos} = 0. \end{aligned} \tag{67}$$

The wavenumbers along the deflating coordinates, which are time-like, enter with the sign opposite to that of k^2, as for the fermions.

Modes that do not depend on the deflating coordinates. With $k_t = 0$ the equation has constant coefficients, and its solutions are the exponentials

$$\varphi_{\rm bos} \propto \exp\Big\{\Big[\frac{\rm gsum}{2} \pm \sqrt{\frac{{\rm gsum}^2}{4} - k^2 - m_{\rm bos}^2}\Big]x4\Big\}. \tag{68}$$

If $k^2 + m_{\rm bos}^2 > {\rm gsum}^2/4$, the mode oscillates with the frequency $\varpi_b = (k^2 + m_{\rm bos}^2 - {\rm gsum}^2/4)^{1/2}$, and its amplitude grows as $e^{{\rm gsum}\,x4/2}$. The growth is that of the weight

TABLE V. How the quantities of the four-dimensional description follow the deflation clock: each is proportional to $e^{n\tau_d}$. The last two rows hold under additional assumptions, which Secs. XVI and XVII state and examine.

quantity	meaning	exponent n
V_d	$b_5 b_6 b_7$: the factor $\sqrt{\lvert\det g\rvert}$ of the measure	-1
y_{VS} of the four-dimensional description	Yukawa coupling of the weighted fields	$\frac{1}{2}$
λ_{VS} of the four-dimensional description	quartic coupling of the weighted Vorton-Scalar	1
v_{VS} of the four-dimensional description	vacuum value of the weighted Vorton-Scalar	$-\frac{1}{2}$
m_D	bare Dirac mass (the mass term of the Lagrangian of the fermion field)	0
m_Y	part of the fermion mass that the Vorton-Scalar gives, $y_{\mathrm{VS}} v_{\mathrm{VS}}/\sqrt{2}$; the mass in vacuum is $m = m_D + m_Y$	0
m_s	mass of the radial mode $\sqrt{2\lambda_{\mathrm{VS}}}\, v_{\mathrm{VS}}$	0
g_r of the four-dimensional description	each of the nine couplings of the self-interaction of the weighted field, and with them the mean-field couplings G_S, G_V	1
g_{VS}	coupling of both modes of the Vorton-Scalar to the fermions, m_Y/v_{VS}	$\frac{1}{2}$
α_{VS}	$g_{\mathrm{VS}}^2/4\pi$	1
G_{VS}	constant of the Fermi type of the neutral-current interaction, $1/(\sqrt{2}\, v_{\mathrm{VS}}^2)$	1
D_{VS}	$\alpha_{\mathrm{VS}}\, m/m_s$: decides whether two fermions bind	1
C	$g_{\mathrm{VS}}^2 m^2/m_s^2$: decides the state of matter	1
m_ϑ	mass of the phase mode of the Vorton-Scalar when $m_D \neq 0$, $m_\vartheta^2 v_{\mathrm{VS}}^2 = \lvert\langle\bar\psi\psi\rangle_{\mathrm{vac}}\rvert\, m_D m_Y/m$	$\frac{1}{2}$
vortex energy per length	proportional to v_{VS}^2 (four-dimensional)	-1
G_{Fermi} (Fermi constant of the weak interaction) if the vacuum value of the Vorton-Scalar sets the electroweak scale (assumption HV)	$1/(\sqrt{2}\, v_{\mathrm{VS}}^2)$, four-dimensional vacuum value	1
G_N (Newton's constant) if gravity is eight-dimensional (A2)	G_8 divided by the volume of $x5, x6, x7$	1

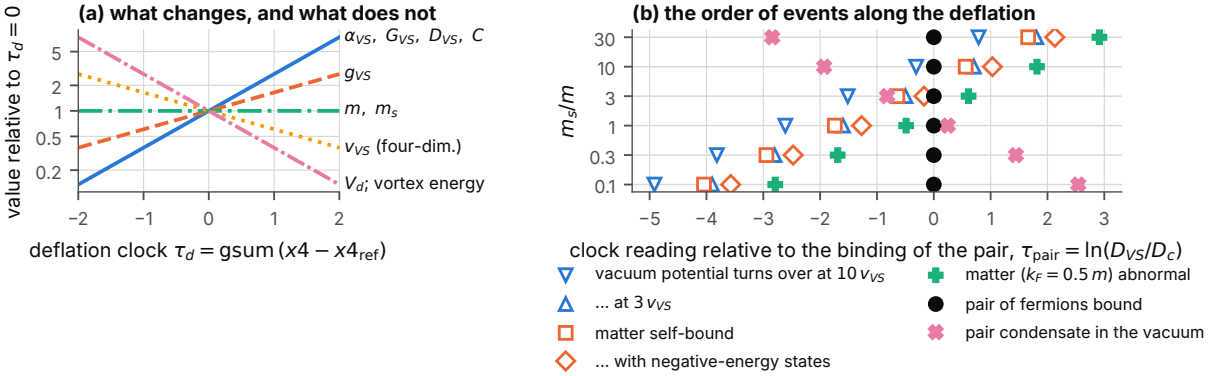


FIG. 4. The deflation clock. (a) How the quantities of the four-dimensional description change with the clock τ_d (logarithmic scale; values relative to $\tau_d = 0$). (b) The order of events along the deflation for six ratios m_s/m: the clock reading of each event relative to the moment at which two fermions form a bound pair (Table XIX).

alone: the weighted field has a constant amplitude, and the number of quanta in a cell of fixed coordinates is conserved while the cell shrinks. Otherwise the mode does not oscillate. For the radial mode this never happens if $m_s > \mathrm{gsum}/2$, which holds by many orders of magnitude for any rate that observation allows (Sec. XVI). For the phase mode it happens at wavelengths longer than $4\pi/\mathrm{gsum}$. At $k = 0$ the two exponents of the phase mode are 0 and gsum: a constant phase, which costs nothing, and a phase that advances at a rate proportional to $1/V_d$, which is the conservation of the charge of $U(1)_{\mathrm{VS}}$ in a shrinking volume. Neither is an instability of the vacuum. Figure 5 shows the exponents.

The potential between two fermions. Two fermions at rest attract each other by exchanging the radial mode. The density of a fixed number of fermions grows as $1/V_d$, so that the radial mode obeys Eq. (67) with a source proportional to $e^{\mathrm{gsum}\, x4}$. The solution that follows the

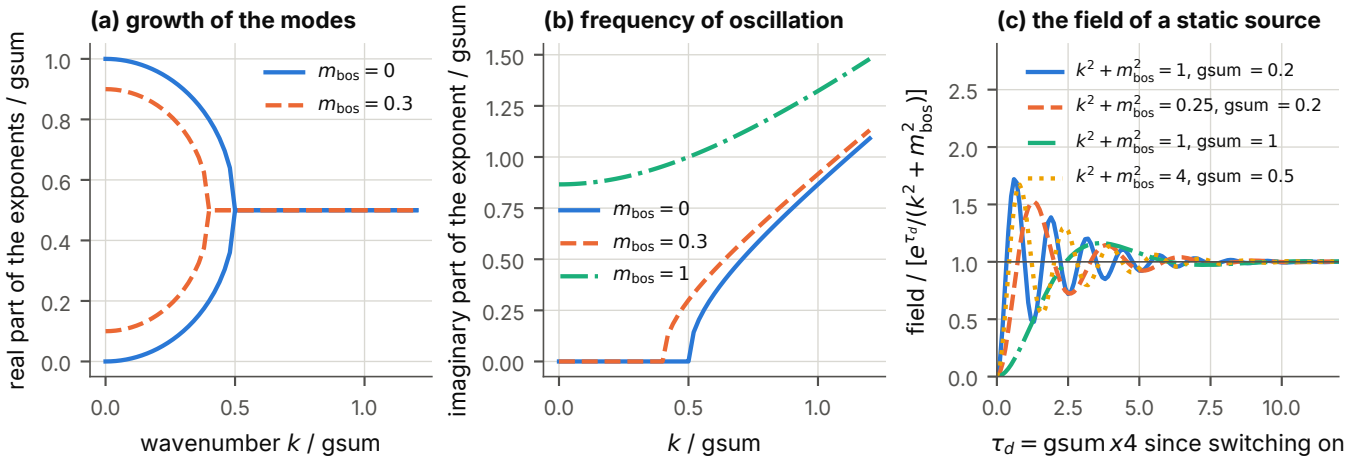


FIG. 5. Modes of the Vorton-Scalar that do not depend on the deflating coordinates. (a) Real parts of the two exponents of Eq. (68) against the wavenumber, for the phase mode and for a radial mode, in units of gsum. (b) Their imaginary part, the frequency of oscillation. (c) The field that a static source builds up from zero, divided by the form $e^{\tau_d}/(k^2+m_{\rm bos}^2)$ that leads to Eq. (69).

source is $e^{\mathrm{gsum}\,x4}/(k^2+m_{\rm bos}^2)$ times the source, as substitution shows: the terms $\mathrm{gsum}^2 - \mathrm{gsum}\cdot\mathrm{gsum}$ cancel, and the denominator contains the *unshifted* mass. The potential between two fermions at rest is therefore, at every moment, a Yukawa potential,

$$V_s(r) = -\alpha_{\rm VS}(x4)\,\frac{e^{-m_s r}}{r}, \tag{69}$$

with the range $1/m_s$ unchanged and the strength of the moment. The numerical solution from rest approaches this form with a transient that decays as $e^{-\tau_d/2}$ [Fig. 5(c)]; after $\tau_d = 20$ the relative difference is below 4×10^{-5} in the four cases computed.

Modes that depend on a deflating coordinate. For $k_t \neq 0$ the weighted field obeys $\partial_4^2\hat\varphi_{\rm bos} + (\varpi_b^2 - k_d^2)\,\hat\varphi_{\rm bos} = 0$ with the physical wavenumber k_d of Eq. (28), and the scalar behaves as the fermion field does. Beyond the turning value, at which $k_d = \varpi_b$, one solution decays and one grows. The bounded solution is the modified Bessel function of the imaginary order $i\varpi_b/\mathrm{rate}_t$ that decays at large argument. At early times it is a standing wave; at late times it falls as e^{-u_d} (Fig. 3). The scalar field, like the fermion field, loses its dependence on the deflating times.

VII. MATTER: THE KOHN-SHAM ANALYSIS

This section determines the states of uniform matter made of the fermions of the model when the self-interaction acts between them. The Vorton-Scalar is set aside here ($y_{\rm VS} = 0$, so that the mass m of the fermions in vacuum is the bare Dirac mass); Sec. XIII restores it.

A. The method

In the method of Kohn and Sham [13] a system of interacting fermions is replaced by independent fermions that move in fields which their own densities determine. The energy is written as a function of the densities; the fields are its derivatives; and the densities are computed from the occupied states in those fields, until both agree. The method is set out here for matter of one kind of particle, whose densities are two numbers: the number density $n = \langle\psi^\dagger\psi\rangle$ and the scalar density $n_S = \langle\bar\psi\psi\rangle$. (Section VII B treats the two halves of the field together.) The interaction energy is $\frac{1}{2}G_V n^2 + \frac{1}{2}G_S n_S^2$ with a scalar coupling G_S and a vector coupling G_V, and each particle obeys the Dirac equation with the Hamiltonian matrix

$$\begin{aligned} \mathsf{h}_{\rm KS} &= \sum_j \alpha^j k_j + \beta\, m^* + V_0, \\ m^* &= m + G_S\, n_S, \qquad V_0 = G_V\, n. \end{aligned} \tag{70}$$

The scalar field changes the mass of the particles in matter from m to m^*; the vector field shifts all energies by V_0. A negative G_S is an attraction (it lowers the mass and the energy) and a positive G_V a repulsion; this is the structure of the mean-field theory of dense matter with a scalar and a vector field [14]. With the couplings of Eq. (50) the exchange terms are contained exactly, because for a contact interaction they are a function of the same two densities.

A particle of the momentum $\vec k$ has the energy $E^*(k) + V_0$ with $E^*(k) = (k^2 + m^{*2})^{1/2}$. The occupied states are those with $|\vec k| < k_F$, ν for each momentum ($\nu = 8$ if all eight internal states of a particle are filled, $\nu = 4$ if the particles of one half of the field are present alone), and averages over them are written $\langle f\rangle = \nu\int d^D k\, f/(2\pi)^D$, where D is the number of space dimensions in which the

particles have momentum. Then

$$n = \langle 1 \rangle = \frac{\nu\, V_D\, k_F^D}{(2\pi)^D}, \qquad n_S = \Big\langle \frac{m^*}{E^*} \Big\rangle, \tag{71}$$

with the volume $V_D = \pi^{D/2}/\Gamma_{\rm E}(D/2+1)$ of the unit ball ($\Gamma_{\rm E}$ is Euler's gamma function). The second of Eqs. (70) with the second of Eqs. (71) is the *gap equation* for m^*. The energy density is

$$\varepsilon = \langle E^* \rangle + \tfrac{1}{2} G_V\, n^2 - \frac{(m^*-m)^2}{2G_S}, \tag{72}$$

which is the kinetic and mass energy of the particles plus the interaction energy [the first term counts the mass as m^* instead of m, which the last term corrects]; it is stationary with respect to m^* exactly when the gap equation holds. The pressure along the coordinate xj is

$$p_j = \Big\langle \frac{k_j^2}{E^*} \Big\rangle + p_{\rm int}, \qquad p_{\rm int} = \tfrac{1}{2} G_V\, n^2 + \frac{(m^*-m)^2}{2G_S}. \tag{73}$$

The first term is the momentum that the particles carry across a surface; the *interaction pressure* $p_{\rm int}$ is the same in all directions. *Along a coordinate on which the fields do not depend the particles have no momentum, and the pressure is* $p_{\rm int}$ *alone*; this remark is used for the deflating coordinates below. The chemical potential is $\mu = E^*(k_F) + V_0$, and for an isotropic state $p = \mu n - \varepsilon$. Matter is *self-bound* if its energy per particle, ε/n, is below the mass m at some density; the density at which ε/n is smallest is the *saturation* density, where the pressure vanishes and $\mu = \varepsilon/n$ [15].

Two limitations are stated here once. The filled states of negative energy are not counted (the *no-sea approximation*); Sec. XIII counts them. And the state is the one of lowest energy among the solutions of the gap equation: without the filled negative-energy states the energy as a function of an unrestricted trial mass has no lowest value, because a large negative trial mass places particles in states of negative rest energy.

B. Fields that do not depend on the deflating coordinates

On the 4+4 spacetime matter exists only in the quantized sector: a mode that depends on a deflating time is extinguished (Sec. V). The particles then have momentum along $x0, \ldots, x3$, so that $D = 4$, or, if the fields do not depend on the hidden coordinate $x0$ either, along ordinary space, $D = 3$. By Eq. (64) the equations of flat spacetime hold for the weighted field at each moment with the couplings of that moment: every coupling g_r, and with it every mean-field coupling of Sec. IV D, grows as e^{τ_d}, while the mass m and the density of particles per unit of the volume of $x0, \ldots, x3$ do not change. The ratios of the couplings are fixed. The deflation therefore carries the matter through a one-parameter sequence of interactions of growing strength.

The equations of two halves. The field has two halves (Sec. IV A), and the interaction energy (49) depends on the densities of both. The particles of the half $W_{1644} = \pm 1$ have the mass $m^*_\pm$ and the energy shift $V_{0\pm}$,

$$\begin{aligned} m^*_\pm &= m + G_S^{\rm tot}\, n_S \pm G_S^{\rm dif}\, n_{SW}, \\ V_{0\pm} &= G_V^{\rm tot}\, n \pm G_V^{\rm dif}\, n_W, \end{aligned} \tag{74}$$

and fill the states up to the Fermi momenta $k_{F\pm}$, four for each momentum. With $E^*_\pm(k) = (k^2 + m^{*2}_\pm)^{1/2}$,

$$\begin{aligned} n_\pm &= 4 \int_{|\vec{k}|<k_{F\pm}} \frac{d^D k}{(2\pi)^D}, \\ n_{S\pm} &= 4 \int_{|\vec{k}|<k_{F\pm}} \frac{d^D k}{(2\pi)^D}\, \frac{m^*_\pm}{E^*_\pm}, \end{aligned} \tag{75}$$

$$\begin{aligned} \varepsilon = \sum_\pm 4 \int_{|\vec{k}|<k_{F\pm}} \frac{d^D k}{(2\pi)^D}\, E^*_\pm + \tfrac{1}{2}\big(G_V^{\rm tot}\, n^2 + G_V^{\rm dif}\, n_W^2\big) \\ - \tfrac{1}{2}\big(G_S^{\rm tot}\, n_S^2 + G_S^{\rm dif}\, n_{SW}^2\big), \end{aligned} \tag{76}$$

and the pressure along $x0, \ldots, x3$ is the sum of the kinetic terms of Eq. (73) for the two halves and of the interaction pressure $p_{\rm int} = \frac{1}{2}(G_V^{\rm tot} n^2 + G_V^{\rm dif} n_W^2) + \frac{1}{2}(G_S^{\rm tot} n_S^2 + G_S^{\rm dif} n_{SW}^2)$, which is the whole pressure along the deflating coordinates. The first of Eqs. (74) with the second of Eqs. (75) are two coupled gap equations for m^*_+ and m^*_-. If one half is empty, or if both halves are equally filled, these equations are those of Sec. VII A for one kind of particle: with $\nu = 4$ and the couplings $G_S^{\rm same}$, $G_V^{\rm same}$ for one filled half, and with $\nu = 8$ and the couplings $G_S^{\rm tot}$, $G_V^{\rm tot}$ for equal halves. Because the interaction can move two particles from one half to the other (Sec. IV A), the numbers of the two halves are not fixed separately: at a given total density n the state is the one of lowest energy with respect to the *polarization*

$$\mathrm{pol}_W = \frac{n_+ - n_-}{n}, \tag{77}$$

between 0 (equal halves) and 1 (one half filled; the sign is free by the symmetry between the halves). Among the solutions of the gap equations those with non-negative masses are taken, which continue the dilute gas. Such a solution is a regular function of the density and of the couplings while the eigenvalues of the matrix $\mathsf{R}_{\rm gap}$ of the derivatives of the right-hand sides of the two gap equations with respect to the two masses stay below 1; they are $2\, G_S^{\rm tot}\, \partial n_{S+}/\partial m^*_+$ and $2\, G_S^{\rm dif}\, \partial n_{S+}/\partial m^*_+$ for equal halves, and $G_S^{\rm same}\, \partial n_{S+}/\partial m^*_+$ and 0 for one filled half.

The numbers below are given for two sets of couplings of the Lagrangian, each with one strength $g > 0$ that grows as e^{τ_d}: *set A*, a scalar attraction alone, $g_0 = -g$; and *set B*, $g_0 = -g$ with a vector repulsion one tenth as large, $g_1 = +0.1\, g$; all other couplings zero. For these Table VI gives the mean-field couplings of Eqs. (49) and (52).

TABLE VI. The mean-field couplings of the two halves of the field for the sets A and B of couplings of the Lagrangian, in the two forms, in units of the strength g (exact).

set	form	$G_V^{\rm tot}$	$G_S^{\rm tot}$	$G_V^{\rm dif}$	$G_S^{\rm dif}$	$G_V^{\rm same}$	$G_S^{\rm same}$	$G_V^{\rm opp}$	$G_S^{\rm opp}$
A	C	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$-\frac{15}{16}$	$\frac{1}{8}$	$-\frac{7}{8}$	0	1
A	Q	$\frac{1}{16}$	$-\frac{15}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$-\frac{7}{8}$	0	-1
B	C	$\frac{1}{16}$	$\frac{1}{20}$	$\frac{1}{5}$	$-\frac{79}{80}$	$\frac{21}{80}$	$-\frac{15}{16}$	$-\frac{11}{80}$	$\frac{83}{80}$
B	Q	$\frac{1}{5}$	$-\frac{79}{80}$	$\frac{1}{16}$	$\frac{1}{20}$	$\frac{21}{80}$	$-\frac{15}{16}$	$\frac{11}{80}$	$-\frac{83}{80}$

Form C polarizes matter; form Q does not. Table VII and Fig. 6 give the energy per particle as a function of the polarization, for the same couplings of the Lagrangian in the two forms. In form C the energy falls with the polarization, in form Q it rises. Over 48 cases (two dimensions, two sets, three strengths, four densities) the state of lowest energy is polarized in 48 cases in form C and in 0 cases in form Q. The reason is Eq. (52): in form C a particle is attracted by the particles of its own half ($G_S^{\rm same} < 0$) and repelled by those of the other half ($G_S^{\rm opp} > 0$), so that the particles gather in one half and break the symmetry between the halves spontaneously; in form Q a particle is attracted by both. (Of the 3936 states examined, 108 states of form Q with unequal halves do not exist as solutions with non-negative masses, at three and six times the threshold and high density; the states with equal halves and with one filled half exist in all cases, and in form C every state exists.)

A gas becomes self-bound matter at a definite moment: when the strength reaches the threshold of Table VIII. In form C the bound state has one half filled, in form Q equal halves, in all eight cases; each threshold agrees with that of matter of one kind (four states with $G_S^{\rm same}, G_V^{\rm same}$, or eight states with $G_S^{\rm tot}, G_V^{\rm tot}$) computed by the routines of Sec. VII A, to 5.7×10^{-14}. Matter of form C needs 2.74 times (set A) and 2.74 times (set B) the strength of form Q in $D = 4$, and 2.51 and 2.52 times in $D = 3$: on the clock it binds later by $\Delta\tau_d = 1.01$ and 1.01 ($D = 4$), 0.92 and 0.92 ($D = 3$). Two things make the difference. Only four of the eight internal states take part. And the attraction inside one half, $G_S^{\rm same} = \frac{7}{8} g_0$ for set A, is weaker than the attraction $G_S^{\rm tot} = \frac{15}{16} g_0$ that matter of form Q with equal halves feels.

Figure 7 and Table IX follow the saturated matter from the moment of binding on. Its energy per particle falls monotonically and the mass of its particles falls; the kind of the state does not change (one half filled in form C, equal halves in form Q) at any of the 101 clock readings examined up to five units of the clock after the threshold. Counted from its own threshold, matter of form C is somewhat less strongly bound than matter of form Q at the same clock reading. Along the table the largest eigenvalue of $\mathsf{R}_{\rm gap}$ is 0.881. Later the saturated state of form Q reaches the eigenvalue 1, at which its two masses can separate ($D = 4$, set A: $\tau_d = 6.416$; $D = 4$, set B: not before $\tau_d = 9$; $D = 3$, set A: $\tau_d = 5.296$; $D = 3$, set B: $\tau_d = 7.323$); that of form C does not.

Matter of a *fixed* density is carried through the same sequence (Fig. 8, Table X). In form C the two halves become unequal at a definite clock reading before the binding threshold, and one half is empty at most 0.35 units of the clock later: the gas polarizes. Its pressure along the flat coordinates becomes negative after the polarization, or, in 3 of the 8 cases (the higher density), not at all, because the repulsion $G_V^{\rm same}$ between the particles of the filled half grows with the couplings while the attraction saturates as the mass falls. In form Q the halves stay equal and the pressure becomes negative in all cases: a uniform gas is then unstable against breaking up into drops of saturated matter. Along the deflating coordinates the pressure is the interaction pressure $p_{\rm int}$ alone, which is negative as long as the scalar attraction dominates: *the matter pulls on the deflating coordinates.* Section XV uses this.

C. Form Q for other ratios of the couplings

Matter of form Q with equal halves is matter of one kind of particle with eight internal states, whose two couplings are the Hartree-Fock couplings, $G_S = G_S^{\rm HF}$ and $G_V = G_V^{\rm HF}$; it is also the matter of the 7+1 spacetime whose fields do not depend on the deflating coordinates. For it the equations of Sec. VII A hold with $G_S(x4) = G_S^{\rm ref} e^{\tau_d}$ and $G_V(x4) = G_V^{\rm ref} e^{\tau_d}$, and Table XI gives the smallest scalar attraction that binds it for three ratios of the two couplings (the ratio 0.2025 is that of set B). The thresholds were computed twice, by the program and by the independent script, and agree to 10^{-9}.

VIII. PAIRING AND THE GAP ALONG THE CLOCK

A. The pair channels

A pair of fermions is created by an operator $\psi^T X \psi$ with a 16×16 matrix X; the choice of X is the *channel* of the pair. For each channel one computes the *exchange amplitudes* F_S and F_P of the radial mode and of the phase mode: the matrix elements of the exchange of one boson between two fermions of opposite momenta in the internal state of the channel. The radial mode attracts in every channel, and the phase mode attracts in the four

TABLE VII. Energy per particle minus m of matter of the total density of the Fermi momentum $k_{F,\mathrm{tot}} = 0.6\,m$ (the Fermi momentum that the density would have with equal halves), at three polarizations, in the two forms of the 4+4 model (units m). Strength: in units of the strength at which form Q first binds matter.

D	set	strength	C, $\mathrm{pol}_W = 0$	C, 0.5	C, 1	Q, $\mathrm{pol}_W = 0$	Q, 0.5	Q, 1
4	A	1.5	+0.1358	+0.1155	+0.0598	−0.0159	+0.0016	+0.0598
4	A	3	+0.1588	+0.1121	−0.0099	−0.1137	−0.0897	−0.0099
4	B	1.5	+0.1436	+0.1187	+0.0543	−0.0343	−0.0138	+0.0543
4	B	3	+0.1745	+0.1239	+0.0179	−0.1044	−0.0764	+0.0179
3	A	1.5	+0.1264	+0.1074	+0.0553	−0.0380	−0.0158	+0.0553
3	A	3	+0.1511	+0.1039	−0.0167	−0.1430	−0.1132	−0.0167
3	B	1.5	+0.1301	+0.1107	+0.0612	−0.0406	−0.0164	+0.0612
3	B	3	+0.1583	+0.1140	+0.0197	−0.1188	−0.0864	+0.0197

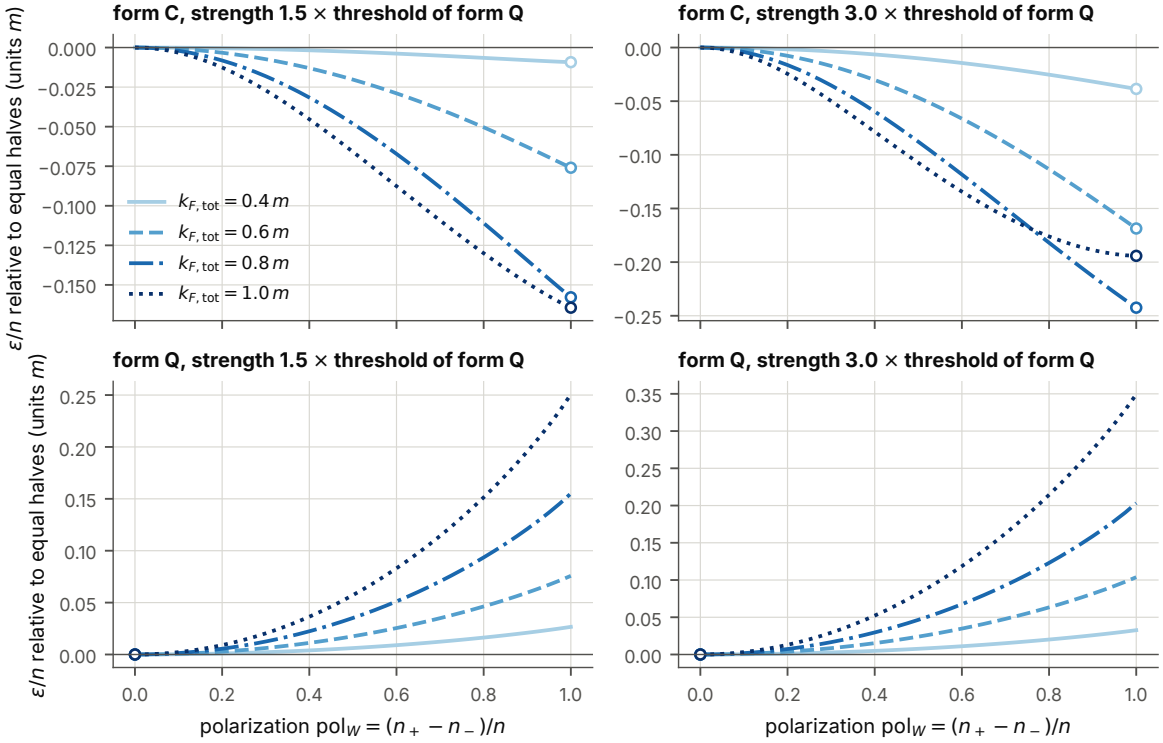


FIG. 6. Energy per particle of matter of a given total density against the polarization of the two halves of the field, relative to its value for equal halves ($D = 4$, set A). Upper row: form C; lower row: form Q; left: 1.5 times, right: 3 times the strength at which form Q first binds matter. The four curves are four total densities; the circle marks the lowest point of each curve.

TABLE VIII. The smallest strength g of the couplings of the Lagrangian at which uniform matter is self-bound, in the two forms of the 4+4 model (units $m = 1$), with the state just above the threshold: its kind, the Fermi momentum $k_{F,\mathrm{tot}}$ that its density would have with equal halves, and the masses of the particles of the two halves (for an empty half the mass that a particle of that half would have). Last column: the largest cutoff for which the vacuum is stable at that strength (Sec. XIII B).

D	set	form	$g\,m^{D-1}$	state	$k_{F,\mathrm{tot}}/m$	m_+^*/m	m_-^*/m	$\Lambda_{\max}/m$
4	A	C	224.7386	one half	0.6140	0.5239	1.5441	0.520
4	A	Q	82.1116	equal halves	0.8054	0.4946	0.4946	0.727
4	B	C	332.2917	one half	0.5211	0.5625	1.4842	0.449
4	B	Q	121.0858	equal halves	0.6792	0.5424	0.5424	0.628
3	A	C	24.5717	one half	0.5374	0.6454	1.4053	0.463
3	A	Q	9.8090	equal halves	0.7356	0.6276	0.6276	0.733
3	B	C	31.1674	one half	0.4635	0.6712	1.3638	0.400
3	B	Q	12.3819	equal halves	0.6351	0.6575	0.6575	0.635

TABLE IX. Self-bound matter in the two forms along the deflation clock, counted for each case from its own binding threshold: energy per particle of the saturated state minus m at four clock readings, and the mass of the particles of the filled half or halves and the Fermi momentum $k_{F,\mathrm{tot}}$ at the clock readings 1 and 3 (units m).

D	set	form	$\tau_d = 1$	2	3	4	m^* at 1	m^* at 3	$k_{F,\mathrm{tot}}$ at 1	$k_{F,\mathrm{tot}}$ at 3
4	A	C	−0.1092	−0.2079	−0.2913	−0.3597	0.351	0.183	0.538	0.361
4	A	Q	−0.1284	−0.2449	−0.3438	−0.4254	0.319	0.156	0.713	0.486
4	B	C	−0.0829	−0.1577	−0.2209	−0.2727	0.392	0.215	0.452	0.301
4	B	Q	−0.0968	−0.1842	−0.2580	−0.3185	0.371	0.199	0.592	0.395
3	A	C	−0.1226	−0.2400	−0.3365	−0.4116	0.373	0.157	0.502	0.308
3	A	Q	−0.1424	−0.2801	−0.3945	−0.4844	0.347	0.137	0.699	0.442
3	B	C	−0.0940	−0.1835	−0.2567	−0.3133	0.407	0.183	0.426	0.256
3	B	Q	−0.1093	−0.2136	−0.2990	−0.3652	0.389	0.170	0.588	0.357

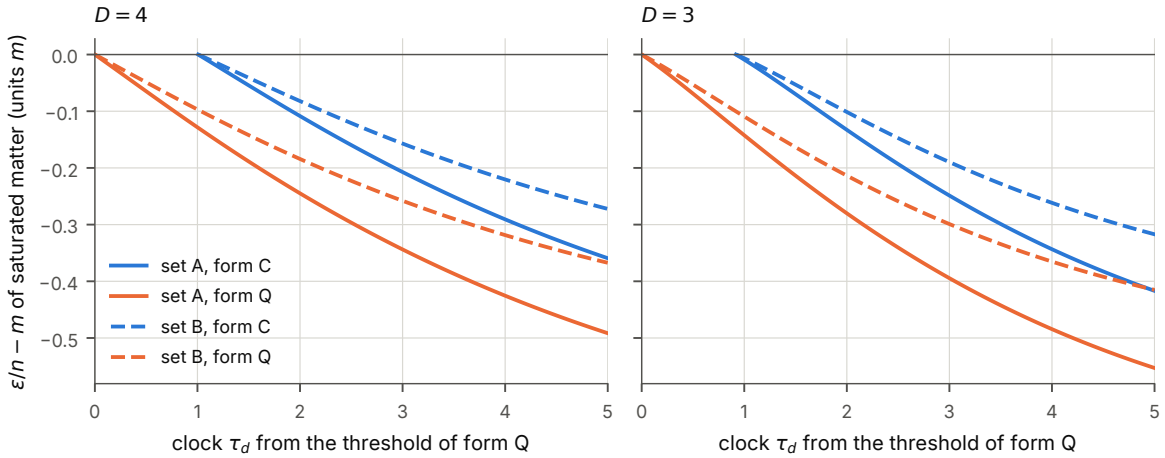


FIG. 7. Energy per particle minus the mass of self-bound matter along the deflation clock, for the same couplings of the Lagrangian in both forms. The clock is counted here from the moment at which form Q first binds matter of the same set, so that the curves of form C begin later, where form C first binds. Solid lines: set A; dashed lines: set B; blue: form C (one half filled); orange: form Q (equal halves). Left: $D = 4$; right: $D = 3$.

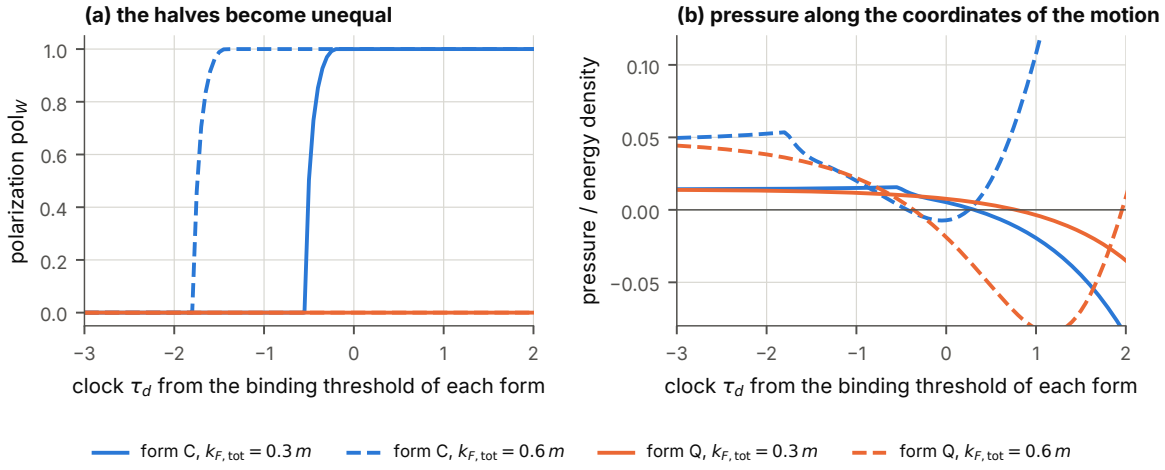


FIG. 8. Matter of a fixed total density while the couplings grow ($D = 4$, set A; the clock of each form is counted from its own binding threshold). (a) The polarization: in form C the two halves become unequal at a definite clock reading and one half empties soon after; in form Q they stay equal. (b) The pressure along the coordinates of the motion, divided by the energy density. Solid lines: $k_{F,\mathrm{tot}} = 0.3\,m$; dashed lines: $0.6\,m$.

TABLE X. A gas of fixed total density in the two forms: the clock readings, counted from the binding threshold of each case, at which the two halves become unequal, at which one half is empty, and at which the pressure along the flat coordinates becomes negative. A dash: the event does not happen for $-3 \le \tau_d \le 5$.

D	set	form	$k_{F,\mathrm{tot}}/m$	unequal	one half	$p<0$
4	A	C	0.3	-0.53	-0.17	$+0.31$
4	A	C	0.6	-1.77	-1.43	-0.43
4	A	Q	0.3	—	—	$+0.78$
4	A	Q	0.6	—	—	-0.36
4	B	C	0.3	-0.82	-0.47	$+0.06$
4	B	C	0.6	-1.98	-1.62	—
4	B	Q	0.3	—	—	$+0.51$
4	B	Q	0.6	—	—	-0.49
3	A	C	0.3	-0.93	-0.72	-0.03
3	A	C	0.6	-1.48	-1.27	—
3	A	Q	0.3	—	—	$+0.21$
3	A	Q	0.6	—	—	-0.21
3	B	C	0.3	-1.02	-0.88	-0.12
3	B	C	0.6	-1.52	-1.32	—
3	B	Q	0.3	—	—	$+0.10$
3	B	Q	0.6	—	—	-0.14

TABLE XI. Form Q with equal halves (matter of one kind with eight internal states): the smallest scalar attraction at which uniform matter is self-bound, for $D=4$ and $D=3$ and three ratios of the vector repulsion to the scalar attraction, with the Fermi momentum of the bound state at the threshold (units $m=1$). Last column: the critical coupling of the vacuum for a cutoff equal to m (Sec. XIII B).

D	$G_V/\lvert G_S\rvert$	$\lvert G_S\rvert\, m^{D-1}$	k_F/m	$\lvert G_c\rvert\, m^{D-1}$
4	0	62.0767	0.8879	29.6088
4	0.1000	85.6504	0.7707	29.6088
4	0.2025	119.5723	0.6792	29.6088
3	0	8.0671	0.7932	4.9348
3	0.1000	9.8364	0.7092	4.9348
3	0.2025	12.2272	0.6351	4.9348

vector pairs along the extra axes,

$$\psi^T C^- \gamma^a \psi, \qquad a \in \{0,5,6,7\}, \tag{78}$$

and repels in their partners that are built with two Dirac matrices. Here $C^- = \gamma^0\gamma^1\gamma^2\gamma^3$ is the matrix that turns each Dirac matrix into minus its transpose (Sec. III), and the index a labels the four extra axes. The pair along the hidden axis, $a=0$, has spin 0 in ordinary spacetime; it is the *hidden-axis singlet* of hypothesis H1.

The amplitudes are computed from the Dirac matrices of the frame, which the deflation does not change, and each is proportional to the square of the coupling. For fields that do not depend on $x5, x6, x7$ the amplitude of a channel at the time $x4$ is therefore

$$F_{S,P}(x4) = g_{\mathrm{VS}}(x4)^2 \times \big(F_{S,P} \text{ of the flat spacetime at } g_{\mathrm{VS}}=1\big). \tag{79}$$

The antisymmetric matrices X fall into classes under the symmetries of the theory, with 15 representative channels. The program computes the amplitudes of these 15 channels from the matrices of the deflating spacetime, at three momenta and 13 scattering angles; the numbers agree with those computed with the matrices of flat spacetime to every digit, and the amplitudes of the hidden-axis channel reproduce the closed forms of Eq. (83) to 1.1×10^{-15}. The conclusion about the channels therefore holds at every moment (Fig. 9): the most attractive channel is the vector pair along an extra axis.

B. The gap grows much faster than the coupling

In matter of a given density the attraction opens a gap Δ_F at the Fermi momentum k_F: the energy that it costs to break a pair. The gap equation [Eq. (88) below, where it is written out with all its terms], which generalizes that of Bardeen, Cooper and Schrieffer [16], holds at each moment with the couplings of that moment. Its solution as a function of the clock is the *adiabatic history* of the paired state: the state that matter would have if it were always in equilibrium. (Section XI examines when it can be.) The strength of the pairing is measured by the dimensionless number $\lambda = N(0)\,W(k_F,k_F)$, the density of states at the Fermi surface times the kernel of the gap equation there; in the vector channel it is the sum $\lambda_S+\lambda_P$ of the contributions of the two modes. Since the kernel is proportional to g_{VS}^2,

$$\lambda \;\propto\; e^{\tau_d}. \tag{80}$$

The program solves the gap equation at 34 moments for each of four sets of the mass m_s and the Fermi momentum (Table XII, Fig. 10). There are three results.

The order of the channels does not change. At every moment the gap of the vector channel is at least as large as the gap that the radial mode alone would give, and that at least as large as the gap of the channel in which the phase mode repels. The gap never decreases.

The critical temperature follows the gap. Wherever the gap is below 5 per cent of the Fermi kinetic energy E_F, the critical temperature is $T_c = 0.567\,\Delta_F$, the ratio of weak coupling, to within 0.002.

The law of growth. In weak coupling the gap is proportional to $e^{-1/\lambda}$. With $\lambda \propto e^{\tau_d}$,

$$\frac{d\ln\Delta_F}{d\tau_d} = \frac{1}{\lambda}. \tag{81}$$

In the computed histories the product $\lambda\, d\ln\Delta_F/d\tau_d$ lies between 1.00 and 1.03 wherever the gap is between 10^{-6} and 0.02 of E_F.

Equation (81) is the central quantitative statement about the superfluid on a deflating spacetime. While the coupling grows by one e-folding, the gap of a weakly paired state with $\lambda = 0.1$ grows by ten e-foldings, a factor of twenty thousand. A paired state that is unobservably

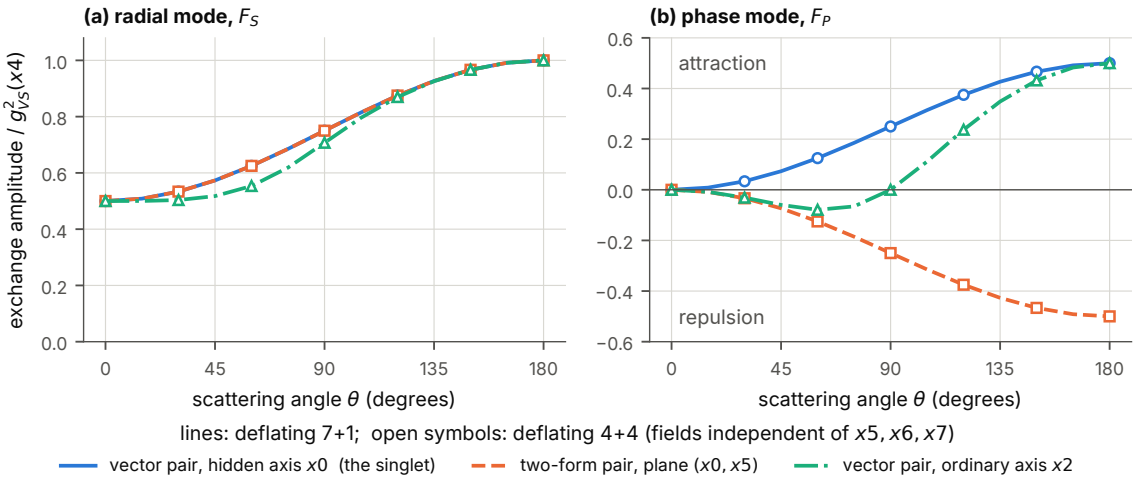


FIG. 9. Exchange amplitudes in three pair channels, divided by the square of the coupling of the moment, against the scattering angle, for momenta of the magnitude m in the plane of $x1$ and $x2$. Lines: deflating 7+1 spacetime; open symbols: deflating 4+4 spacetime. (a) The radial mode attracts in all three channels. (b) The phase mode attracts in the vector pair along the hidden axis and repels in its partner built with two Dirac matrices. The amplitudes are those of flat spacetime: the deflation changes their overall strength and nothing else.

TABLE XII. The gap along the deflation clock for two of the four sets computed. $g_{\rm VS}$ and λ: coupling and pairing strength of the moment. E_F: Fermi kinetic energy. T_c: critical temperature. Rate of growth: $d\ln\Delta_F/d\tau_d$; the last column, its product with λ, is given where the gap is between 10^{-6} and 0.02 of E_F.

m_s/m, k_F/m	τ_d	$g_{\rm VS}$	λ	Δ_F/E_F	T_c/Δ_F	rate of growth	$\lambda\times$ rate
0.10, 0.50	−0.63	0.730	0.062	1.17×10^{-7}	0.567	—	—
0.10, 0.50	−0.13	0.939	0.102	7.40×10^{-5}	0.567	9.88	1.01
0.10, 0.50	0.38	1.208	0.169	3.70×10^{-3}	0.567	6.01	1.01
0.10, 0.50	0.88	1.554	0.279	0.0407	0.565	3.74	—
0.10, 0.50	1.39	2.000	0.463	0.189	0.555	2.50	—
0.10, 0.50	1.89	2.573	0.766	0.563	0.528	1.96	—
0.10, 0.50	2.39	3.311	1.268	1.49	0.480	1.96	—
1.0, 0.50	0.88	1.554	0.056	1.02×10^{-7}	0.567	—	—
1.0, 0.50	1.13	1.763	0.072	5.62×10^{-6}	0.567	14.02	1.01
1.0, 0.50	1.39	2.000	0.092	1.27×10^{-4}	0.567	10.93	1.01
1.0, 0.50	1.64	2.269	0.119	1.45×10^{-3}	0.567	8.54	1.02
1.0, 0.50	1.89	2.573	0.153	9.77×10^{-3}	0.567	6.71	1.03
1.0, 0.50	2.14	2.919	0.197	0.0440	0.566	5.33	—
1.0, 0.50	2.39	3.311	0.253	0.148	0.563	4.38	—

weak at one time is strong a few e-foldings of the volume later.

IX. PAIRING WITH THE SELF-INTERACTION

Section VIII treated the pairing by the exchange of the Vorton-Scalar alone. This section adds the self-interaction of the fermion field. It writes out the amplitudes and the gap equation in full, shows that the exchange of the Vorton-Scalar is itself a self-interaction of finite range, gives the condition under which the gap equation has a solution that does not depend on the upper limit of its momentum integral, and gives the gap along the deflation clock when both interactions act. On the 4+4 spacetime the contact couplings of form C enter the pair along the hidden axis with the sign opposite to that of form Q, and the two forms select different pairs.

A. Amplitudes of the four contact channels

Two particles of the mass m^* with the momenta $\vec{p}$ and $-\vec{p}$ in ordinary space scatter into $\vec{p}'$ and $-\vec{p}'$. Let $E=(m^{*2}+p^2)^{1/2}$ and E' likewise, and let $\mathsf{h}(\vec{p})=\sum_j\alpha^j p_j+\beta\,m^*$ be the Hamiltonian matrix of one particle, so that $\frac{1}{2}(1+\mathsf{h}/E)$ projects on its states of positive energy. The

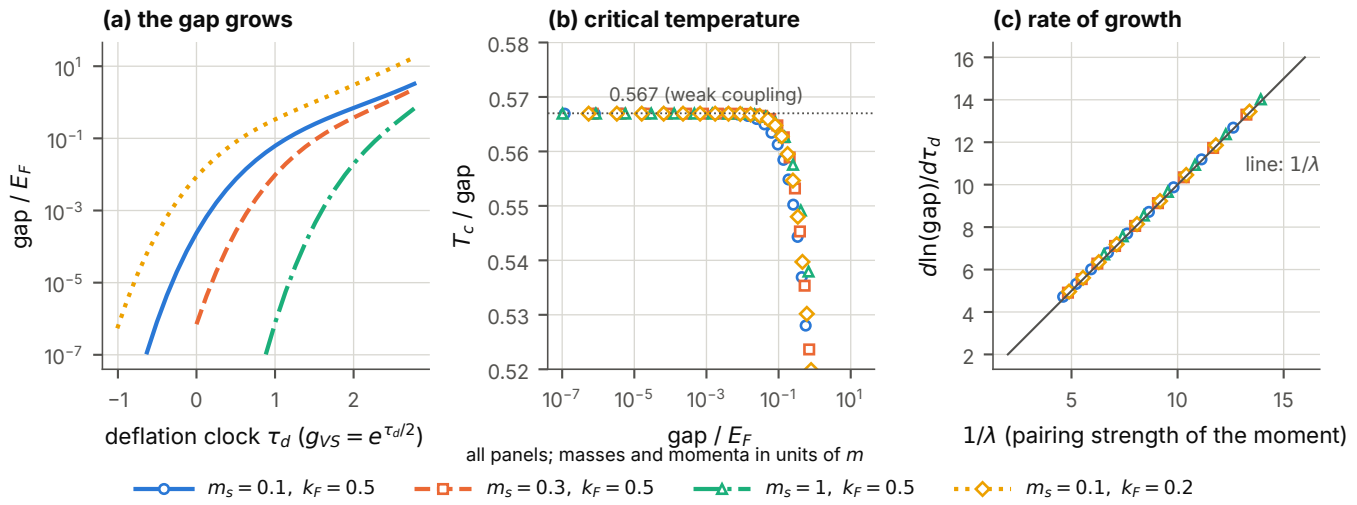


FIG. 10. The gap along the deflation clock, for four sets of the mass of the radial mode and of the Fermi momentum (in units of m). (a) The gap of the vector channel in units of the Fermi kinetic energy. (b) The ratio of the critical temperature to the gap. (c) The rate of growth of the gap against $1/\lambda$; the line is Eq. (81).

pair along the hidden axis has the wave function

$$\Phi(\vec{p}) = \tfrac{1}{2}\Big(1+\frac{\mathsf{h}(\vec{p})}{E}\Big)\, X^*\, \tfrac{1}{2}\Big(1+\frac{\mathsf{h}(-\vec{p})}{E}\Big)^T, \qquad X = C^-\gamma^0, \tag{82}$$

a 16×16 matrix. A vertex $\psi^\dagger \,\mathrm{Adj}\,\Theta^K \psi$ acting on both particles, with the adjoint matrix Adj of the model, gives the amplitude $\mathrm{Tr}[\Phi(\vec{p}')^\dagger \,\mathrm{Adj}\,\Theta^K\,\Phi(\vec{p})\,(\mathrm{Adj}\,\Theta^K)^T]$ divided by the norms of the two wave functions. Summed over the lists K of one rank with the signs sgn_K, this defines the amplitude of the rank r. In form Q ($\mathrm{Adj} = \beta$, matrices Θ^K_Q, signs sgn^Q_K), as on the 7+1 spacetime, the result for the four independent channels of Sec. IV C is

$$\begin{aligned} F_0 &= \frac{EE' + m^{*2} - \vec{p}\cdot\vec{p}'}{2EE'}, \\ F_1 &= \frac{3EE' - 2m^{*2} - \vec{p}\cdot\vec{p}'}{EE'}, \\ F_7 &= \frac{3EE' + 2m^{*2} - \vec{p}\cdot\vec{p}'}{EE'}, \\ F_8 &= -\frac{EE' - m^{*2} - \vec{p}\cdot\vec{p}'}{2EE'}. \end{aligned} \tag{83}$$

F_0 is the amplitude F_S of the exchange of the radial mode of the Vorton-Scalar, which couples to the scalar density, and $F_8 = -F_P$ is minus the amplitude of the exchange of its phase mode, which couples to the pseudoscalar density. At rest $(F_0, F_1, F_7, F_8) = (1, 1, 5, 0)$, the pair coupling (54). The closed forms were verified against the direct evaluation with the 256 matrices at random momenta, by the program to 1.8×10^{-15} and by the independent script with projectors built from numerical eigenvectors; the 7+1 spacetime gives the same four functions. They hold for momenta in ordinary space; a momentum along the hidden axis itself changes them, because the pair operator points along that axis.

The sign of the contact couplings in the two forms. The same evaluation for the pairs along $x5, x6, x7$, $\psi^T C^- \gamma^t \psi$, and for both forms gives one rule. In form Q the amplitudes of all four vector pairs along the extra axes are the functions (83). In form C ($\mathrm{Adj} = C^-$) the amplitudes of the pairs along $x5, x6, x7$ are the same functions, and those of the pair along the hidden axis are *minus* these functions, at all momenta (direct evaluation at random momenta, to 2.2×10^{-15}). This is Sec. IV E at general momenta: the pair along the hidden axis consists of one particle of each half of the field, the pairs along $x5, x6, x7$ of two particles of the same half. The contact amplitudes therefore enter the pairing potential with a sign

$$s_{\mathrm{ct}} = \begin{cases} -1 & \text{form C, pair along the hidden axis,} \\ +1 & \text{form C, pairs along } x5, x6, x7, \\ +1 & \text{form Q, all four pairs.} \end{cases} \tag{84}$$

The relations (45) between the nine invariants hold for the amplitudes at rest, and at general momenta for the part of each amplitude that is even under $\vec{p}' \to -\vec{p}'$, which is the sum of the direct and of the exchange scattering of two identical particles; they do not hold for the odd part, which does not contribute to a contact interaction in a pair state without orbital angular momentum. For a strictly local interaction the four couplings (47) therefore suffice. When the interaction is given a range, as below, the model is *defined* by giving that range to the four channels.

B. The pairing potential and the gap equation

The pairing potential of the contact interaction in each of these four pair channels is

$$V_{\rm ct}(\vec{p}',\vec{p}) = s_{\rm ct}\left(g_0^{\rm eff}F_0+g_1^{\rm eff}F_1+g_7^{\rm eff}F_7+g_8^{\rm eff}F_8\right)\frac{m_{\rm ct}^2}{q^2+m_{\rm ct}^2}, \tag{85}$$

attractive where it is negative, with $q=|\vec{p}'-\vec{p}|$ and the sign (84). A contact interaction in a gap equation needs a regularization at large momentum transfer. The last factor gives each coupling the range $1/m_{\rm ct}$ of an exchanged particle of the mass $m_{\rm ct}$, the *range parameter*; it is the form that the exchange of the Vorton-Scalar itself has:

$$V_{\rm VS}(\vec{p}',\vec{p}) = -g_{\rm VS}^2\Big[\frac{F_S}{q^2+m_s^2}+\frac{F_P}{q^2+m_\vartheta^2}\Big], \tag{86}$$

with the mass m_s of the radial mode and the mass m_ϑ of the phase mode, which vanishes when the chiral symmetry is exact; it is the same in all four pair channels and in both forms, because the Yukawa coupling contains the adjoint $\bar\psi$, in which the two halves of the field enter alike. Comparing Eqs. (85) and (86): *the exchange of the Vorton-Scalar is a self-interaction of finite range* of form Q, with the momentum-dependent couplings

$$g_0^{\rm eff}(q) = -\frac{g_{\rm VS}^2}{q^2+m_s^2}, \qquad g_8^{\rm eff}(q) = +\frac{g_{\rm VS}^2}{q^2+m_\vartheta^2}. \tag{87}$$

At momenta far below m_s the radial mode acts as the scalar contact coupling $g_0=-g_{\rm VS}^2/m_s^2$ of form Q. In the phase with unbroken symmetry the two masses are equal and $g_0^{\rm eff}(q)+g_8^{\rm eff}(q)=0$, the condition (55) of chiral invariance in form Q. In the 4+4 model of this paper the contact terms are of form C and the exchange is not: in the pair along the hidden axis the two enter with opposite relative sign, and in the pairs along $x5,x6,x7$ with the same. The program confirms that the scalar contact coupling $-g_{\rm VS}^2/m_s^2$ with $m_{\rm ct}=m_s$ reproduces the kernel of the exchange of the radial mode to 3×10^{-16}.

The gap function $\Delta(k)$ of the paired state obeys, at the temperature T,

$$\Delta(k) = \frac{1}{4\pi^2}\int_0^{k_{\rm max}} dk'\,k'^2\,W(k,k')\,\frac{\Delta(k')}{E_{k'}}\tanh\frac{E_{k'}}{2T},$$
$$W(k,k') = \frac{1}{2}\int_{-1}^{1} d\cos\theta\,\big[-V_{\rm VS}-V_{\rm ct}\big], \tag{88}$$

with $\xi_k=(k^2+m^2)^{1/2}-\mu$, the quasi-particle energy $E_k=(\xi_k^2+\Delta(k)^2)^{1/2}$, the chemical potential $\mu=(k_F^2+m^2)^{1/2}$, and the angle θ between $\vec{p}$ and $\vec{p}'$ ($k=|\vec{p}|$, $k'=|\vec{p}'|$); $\Delta_F=\Delta(k_F)$; $k_{\rm max}$ is the upper limit of the momentum integral, which the next subsection examines. It is the condition that the CJT effective action be stationary with respect to the pair part of the propagator, in the truncation of Sec. XII, and equally the form that the Kohn-Sham equations take with a pairing field. The angular integral is elementary, because each amplitude is linear in $\cos\theta$.

C. The upper limit of the momentum integral: an ultraviolet criterion

At momenta far above all masses every term of the pairing potential has the form of a coupling times a function of $\cos\theta$ divided by q^2, with $F_0,F_S,F_P\to\frac{1}{2}(1-\cos\theta)$, $F_8\to-\frac{1}{2}(1-\cos\theta)$ and $F_1,F_7\to3-\cos\theta$. The kernel is then homogeneous, $W(k,k')\to w_\infty(k'/k)/k^2$ with a function w_∞ of the ratio of the two momenta, and above the gap Eq. (88) at zero temperature reads $\Delta(k)=\frac{1}{4\pi^2}\int_0^\infty dt\,t\,w_\infty(t)\,\Delta(tk)$. A power $\Delta(k)\propto k^{-s_\Delta}$ solves it if the exponent s_Δ obeys

$$\int_0^\infty dt\;t^{1-s_\Delta}\,w_\infty(t) = 4\pi^2. \tag{89}$$

Because $W(k,k')$ is symmetric in its arguments, $w_\infty(1/t)=t^2\,w_\infty(t)$; the left-hand side of Eq. (89) is therefore unchanged when s_Δ is replaced by $2-s_\Delta$, and it is smallest at $s_\Delta=1$. Its value there is the *ultraviolet strength* of the kernel,

$$U_{\rm uv} = \int_0^\infty dt\;w_\infty(t). \tag{90}$$

If $U_{\rm uv}<4\pi^2$, Eq. (89) has two real roots, $s_\Delta>1$ and $2-s_\Delta<1$; the gap function falls as k^{-s_Δ} with the larger root, and the gap equation has a solution that does not depend on $k_{\rm max}$, which is approached as $k_{\rm max}^{-2(s_\Delta-1)}$, slowly when $U_{\rm uv}$ is close to $4\pi^2$. If $U_{\rm uv}>4\pi^2$ the roots are complex, no such solution exists, and with an upper limit the gap grows in proportion to $k_{\rm max}$: the pair collapses to short distances. This is the fall to the center of relativistic particles in an attractive Coulomb potential whose strength exceeds a critical value; the exchange of a boson of the mass m_s gives a Yukawa potential, which is a Coulomb potential at distances below $1/m_s$.

The integrals over t and over the angle are elementary. With $\int_0^\infty dt/(1+t^2-2t\cos\theta)=(\pi-\theta)/\sin\theta$ the two functions of the angle give

$$u_S \equiv \int_0^\infty dt\,\frac{1}{2}\int_{-1}^{1}d\cos\theta\,\frac{\frac{1}{2}(1-\cos\theta)}{1+t^2-2t\cos\theta} = \frac{\pi^2}{8}-\frac{1}{2},$$
$$u_V \equiv \int_0^\infty dt\,\frac{1}{2}\int_{-1}^{1}d\cos\theta\,\frac{3-\cos\theta}{1+t^2-2t\cos\theta} = \frac{3\pi^2}{4}-1, \tag{91}$$

that is $u_S=0.7337$ and $u_V=6.4022$, and therefore

$$U_{\rm uv} = 2\,u_S\,g_{\rm VS}^2 - s_{\rm ct}\,m_{\rm ct}^2\Big[\big(g_0^{\rm eff}-g_8^{\rm eff}\big)\,u_S + \big(g_1^{\rm eff}+g_7^{\rm eff}\big)\,u_V\Big] \tag{92}$$

when both modes of the Vorton-Scalar are exchanged (half the first term for the radial mode alone). The formula was verified against the numerical integral of the kernel by the program and by the independent script. *The exchange of the two modes alone reaches* $U_{\rm uv}=4\pi^2$ *at* $g_{\rm VS}^2=4\pi^2/(2u_S)=26.90$, the clock reading $\tau_d=3.29$

of the histories of Sec. VIII, beyond their last reading 2.77. A scalar contact attraction with $s_{\rm ct} = +1$ brings the criterion nearer, a vector repulsion moves it away (Table XIII); with $s_{\rm ct} = -1$ it is the other way round, and the vector contact coupling $+0.3\, g_{\rm VS}^2/m^2$ in the hidden-axis pair of form C reaches the criterion at $g_{\rm VS}^2 = 11.65$, the clock reading 2.46, from which on no gap is given.

The criterion was tested numerically: the gap was computed with the upper limits $k_{\rm max} = 60$, 2×10^3, 2×10^4 and 2×10^5, in units of the larger of m and k_F, for 192 kernels (four sets of m_s and k_F, four clock readings, seven combinations of couplings, and both signs of the contact part). Below the criterion the gap converges (Table XIV); above it (8 kernels) it grows by a factor between 11.49 and 14.56 from $k_{\rm max} = 2\times10^4$ to 2×10^5, so that its value is not a property of the model. *All gaps of this paper are computed with $k_{\rm max} = 2\times10^4$, and none is given above the criterion.* A limit of 60, with which gaps of this model were computed before, gives gaps of the exchange alone that are too small by up to 1.08 per cent at the strongest coupling of the histories, and gaps with contact couplings that are too small by the percentages of the last column of the table.

D. The gap along the clock with contact couplings

All couplings of the four-dimensional description grow as e^{τ_d} (Sec. VI), so that the ratio of a contact coupling to $g_{\rm VS}^2$ is fixed. Table XV and Fig. 11 give the gap, for the sign $s_{\rm ct} = +1$, for seven choices: the exchange alone; the exchange with a scalar contact attraction $g_0^{\rm eff} = -g_{\rm VS}^2/m^2$; with a scalar attraction and a vector repulsion one tenth as large, $g_1^{\rm eff} = +0.1\, g_{\rm VS}^2/m^2$; with a vector repulsion alone, $g_1^{\rm eff} = +0.3\, g_{\rm VS}^2/m^2$; with an axial coupling $g_7^{\rm eff} = -0.1\, g_{\rm VS}^2/m^2$; the contact couplings without the Vorton-Scalar; and the exchange alone with a phase mode of the mass $m_\vartheta = 0.3\, m$. The range parameter is $m_{\rm ct} = m$.

Three statements summarize the result for $s_{\rm ct} = +1$; each was verified at every clock reading at which the gap exceeds 10^{-6} of the Fermi kinetic energy.

A scalar contact attraction raises the gap, strongly so where the exchange alone is weak: for $m_s = m$ at the clock reading of Table XV by the factor 12.1.

A vector contact repulsion lowers it. A vector coupling of three tenths of $g_{\rm VS}^2/m^2$ removes most of the gap of a weakly paired state (Table XV, $m_s = m$) and reduces the gap of a strongly paired one.

A mass of the phase mode lowers it, because it weakens the attraction of that mode at small momentum transfer.

Without contact couplings the program reproduces the gaps of Sec. VIII to 7×10^{-16}. For small gaps the critical temperature remains 0.567 times the gap. An independent solution of Eq. (88) on another grid of momenta, by plain iteration, gives the gaps of Table XV for the first three rows to 8.2×10^{-7} (relative). The law of growth (81) holds with contact couplings as it does without, because every part of the kernel grows as e^{τ_d}.

E. The pair along the hidden axis in form C

In the definition of the model, form C, the contact couplings enter the pair along the hidden axis with $s_{\rm ct} = -1$. Table XVI and Fig. 12 compare the two signs for the same couplings. With $s_{\rm ct} = -1$ the couplings act in the opposite way: *a scalar contact attraction lowers the gap* (for $m_s = m$ it removes it at the clock reading of the table), an axial coupling $g_7^{\rm eff} < 0$ lowers it, and *a vector contact repulsion raises it*; the contact couplings of the signs of set B alone give no gap in this pair at any clock reading. The gap with $s_{\rm ct} = -1$ and the vector coupling agrees with an independent solution of the gap equation to 7.8×10^{-5}.

In form C the self-interaction removes the degeneracy of the pair along the hidden axis with the pairs along $x5, x6, x7$, and so selects the direction of the order parameter (Table XVII). With an attractive scalar coupling the pairs along the additional time-like axes have the larger gap, not the pair along the hidden axis; with a repulsive vector coupling alone the pair along the hidden axis has the larger gap. In form Q, as on the 7+1 spacetime, the four pairs have the same gap, and the direction is free (Sec. XIV).

X. THE BOUND PAIR AND THE ORDER OF EVENTS

A. The moment of binding

Two fermions in the Yukawa potential (69) form a bound state if and only if the number $D_{\rm VS}$ of Eq. (66) exceeds a critical value, $D_c = 1.67981$. On a deflating spacetime $D_{\rm VS} \propto e^{\tau_d}$, so that the pair becomes bound at a definite time. Measure the clock from that moment:

$$\tau_{\rm pair} = \ln \frac{D_{\rm VS}}{D_c} = {\rm gsum}\left(x4 - x4_{\rm binding}\right). \qquad (93)$$

Table XVIII and Fig. 13 show the scattering length $a_{\rm sc}$ of two fermions and the binding energy E_b of the pair as the clock passes the binding. The linearized gap equation in vacuum, whose kernel is relativistic, gives nearly the same critical value: 1.7347, 1.6902 and 1.6810 for $m_s/m = 0.3$, 0.1 and 0.03 (Table LII).

The scattering length. Just before the binding the scattering length is large and negative, just after it large and positive: $-134.0/m_s$ at $0.99\, D_c$ and $+135.9/m_s$ at $1.01\, D_c$. At the moment of binding it is infinite: the two fermions are at a scattering resonance. *Every pair of fermions of the model passes through this resonance once, at the same time.* Section XVI examines what that means if the fermions are the dark matter.

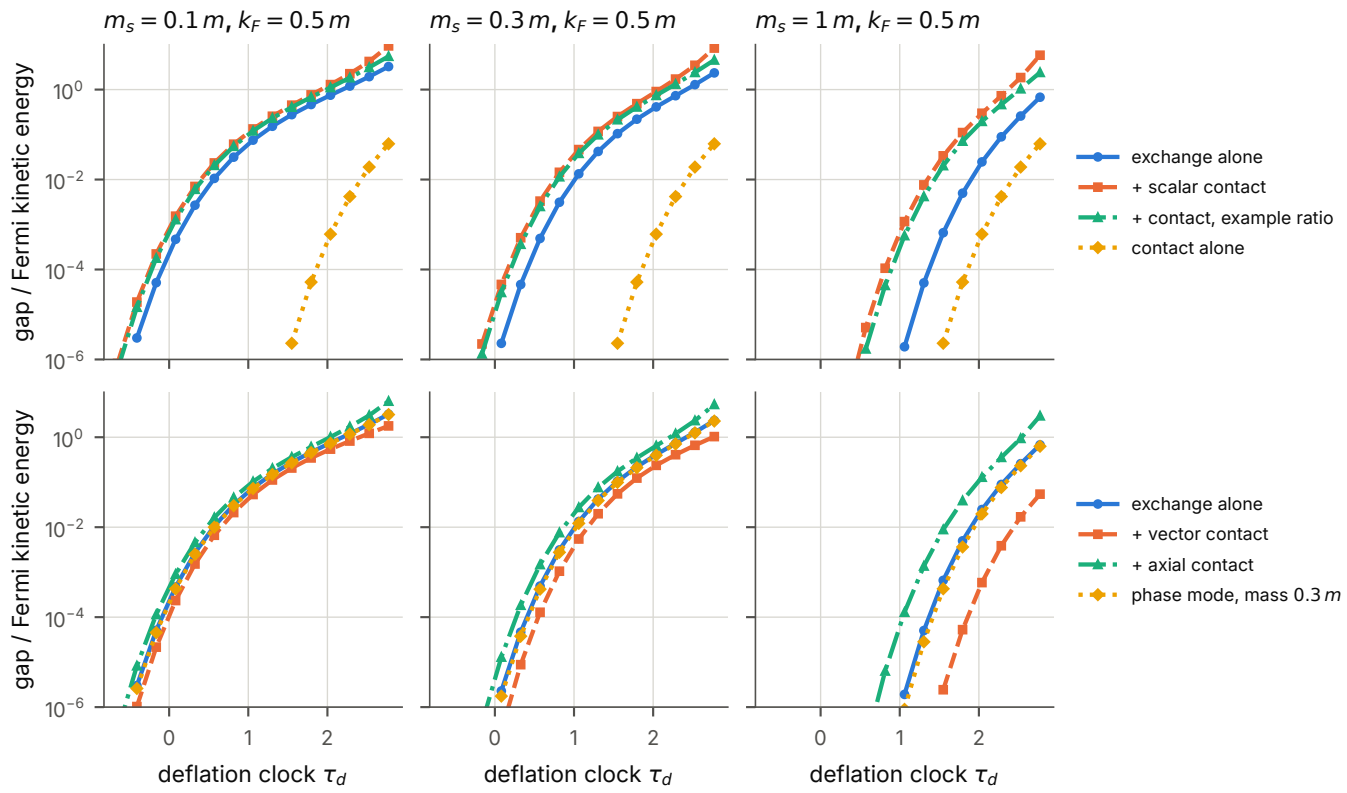


FIG. 11. The gap with contact couplings that enter with the sign $s_{\rm ct} = +1$ (form Q; form C in the pairs along $x5, x6, x7$) along the deflation clock, for the Fermi momentum $0.5\,m$ and three masses m_s of the radial mode of the Vorton-Scalar. Upper row: the exchange of the Vorton-Scalar alone; with a scalar contact attraction; with a scalar attraction and a vector repulsion one tenth as large; the contact couplings alone. Lower row: the exchange alone; with a vector contact repulsion; with an axial contact coupling; with a phase mode of the mass $0.3\,m$. Couplings and range parameter as in Table XV.

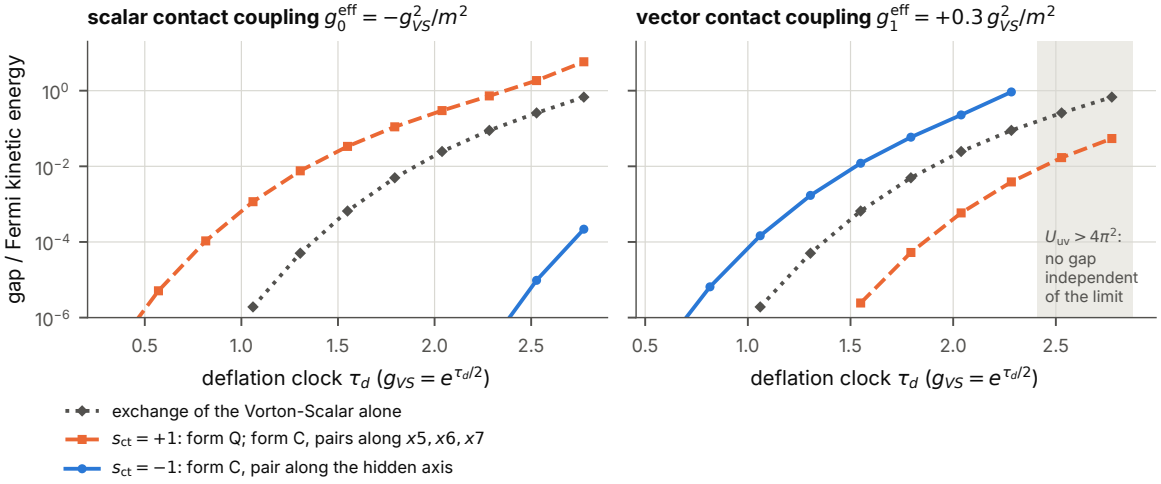


FIG. 12. The gap at the Fermi surface, in units of the Fermi kinetic energy, along the deflation clock, for $m_s = m$ and $k_F = 0.5\,m$. Left: with the scalar contact coupling $g_0^{\rm eff} = -g_{\rm VS}^2/m^2$; right: with the vector contact coupling $g_1^{\rm eff} = +0.3\,g_{\rm VS}^2/m^2$. Dotted line: the exchange of the Vorton-Scalar alone. Dashed line (squares): the contact coupling with the sign $s_{\rm ct} = +1$ (form Q; form C in the pairs along $x5, x6, x7$). Solid line (circles): with the sign $s_{\rm ct} = -1$ (form C in the pair along the hidden axis). In the shaded band the kernel with $s_{\rm ct} = -1$ exceeds the ultraviolet criterion of Sec. IX C, and no gap is given.

TABLE XIII. The ultraviolet strength $U_{\rm uv}$ of the pairing kernel per unit $g_{\rm VS}^2$, the value of $g_{\rm VS}^2$ at which it reaches $4\pi^2$, and the clock reading of that moment ($g_{\rm VS} = e^{\tau_d/2}$), for the exchange of the two modes of the Vorton-Scalar alone and with contact couplings ($m_{\rm ct} = m$), for the two signs (84). A dash: the criterion is never reached.

interaction	$s_{\rm ct} = +1$: $U_{\rm uv}/g_{\rm VS}^2$	critical $g_{\rm VS}^2$	τ_d	$s_{\rm ct} = -1$: $U_{\rm uv}/g_{\rm VS}^2$	critical $g_{\rm VS}^2$	τ_d
exchange of the Vorton-Scalar alone	1.467	26.90	3.29	1.467	26.90	3.29
with $g_0^{\rm eff} = -g_{\rm VS}^2/m^2$	2.201	17.94	2.89	0.734	53.81	3.99
with $g_0^{\rm eff} = -g_{\rm VS}^2/m^2$, $g_1^{\rm eff} = +0.1\, g_{\rm VS}^2/m^2$	1.561	25.29	3.23	1.374	28.73	3.36
with $g_1^{\rm eff} = +0.3\, g_{\rm VS}^2/m^2$	−0.453	—	—	3.388	11.65	2.46
with $g_7^{\rm eff} = -0.1\, g_{\rm VS}^2/m^2$	2.108	18.73	2.93	0.827	47.73	3.87
contact couplings alone ($g_0^{\rm eff} = -g_{\rm VS}^2/m^2$, $g_1^{\rm eff} = +0.1\, g_{\rm VS}^2/m^2$)	0.093	422.32	6.05	−0.093	—	—

TABLE XIV. Convergence of the gap with the upper limit $k_{\max}$ of the momentum integral, by classes of the ultraviolet strength (kernels below the criterion with a gap above 10^{-4} of the Fermi kinetic energy): the largest relative change of the gap from $k_{\max} = 2\times10^4$ to 2×10^5, and the largest error of the limit $k_{\max} = 60$.

$U_{\rm uv}/4\pi^2$	kernels	change	error of 60 (per cent)
$U_{\rm uv}/4\pi^2 < 0.5$	118	2.5×10^{-6}	0.75
0.5 to 0.7	28	6.4×10^{-5}	2.37
0.7 to 0.9	8	3.3×10^{-3}	12.59

The binding energy. Just after the binding the binding energy starts quadratically,

$$E_b \;\propto\; \left(e^{\tau_{\rm pair}}-1\right)^2 \frac{m_s^2}{m}, \tag{94}$$

with the fitted exponent 2.01 between 2 and 8 per cent above D_c. Because E_b vanishes at the threshold, the time $1/E_b$ that the pair needs to adjust is infinite there. However slow the deflation is, the pair cannot follow it adiabatically at first.

The same threshold from the gap equation. In vacuum the linearized gap equation is the equation of a bound state of two fermions, with a relativistic kernel. It reaches the eigenvalue 1 at $D_{\rm VS} = 1.7347$, 1.6902 and 1.6810 for $m_s/m = 0.3$, 0.1 and 0.03, approaching the value D_c of the Schrödinger equation as the radial mode becomes light and the binding non-relativistic.

B. The order of events

The other thresholds of the model are thresholds in the number C of Eq. (66), which grows at the same rate as $D_{\rm VS}$. Because $C/D_{\rm VS} = 4\pi m/m_s$ is fixed, the order in which the events happen depends on the ratio m_s/m alone. Table XIX gives the clock reading $\tau_{\rm pair}$ of each event; a negative number means that the event precedes the binding of the pair. The events other than the binding are derived in Sec. XII.

For a *light radial mode*, $m_s/m \le 3$ in the table, the pair becomes bound only *after* the one-loop potential of the Vorton-Scalar has fallen below its vacuum value at a field of three times the vacuum value: the deflation endangers the vacuum of the model before it binds the pair. The condition for the opposite order is $4\pi D_c\, m/m_s < 4.27$, that is, $m_s/m > 4.9$. For a *heavy radial mode*, $m_s/m = 10$ or 30, the pair becomes bound first, and matter becomes self-bound and loses its fermion mass later. Nothing in the model decides between the two orders; the table states what follows from each.

XI. CAN THE PAIRED STATE FOLLOW THE DEFLATION?

A. The question, and the parameter that decides it

Section VIII gave the gap that the paired state would have in equilibrium at each moment. But a state can change at most at the rate of its own energy scale, which for a paired state is the gap. By Eq. (81) the equilibrium gap, written $\Delta_{\rm eq}$ in this section, changes at the rate gsum/λ. The paired state can follow only if that rate is small compared with the gap itself:

$$\Xi_d = \frac{\mathrm{gsum}}{\lambda\,\Delta_{\rm eq}} \;\ll\; 1. \tag{95}$$

Early in the deflation the gap is exponentially small, and the condition fails however small the rate is. A system that is carried through a change faster than it can respond falls out of equilibrium, as in the formation of defects at a phase transition [17, 18]. The question is what the paired state does then, and when it catches up.

B. The model

The question is answered with the standard reduced model of a superfluid, whose dynamics without collisions is known in detail [19]. Pairs of states of opposite momenta, with the single-particle energy ξ_k measured from the Fermi energy, fill a shell $-E_{\rm sh} < \xi_k < E_{\rm sh}$ with a constant density of states, and all attract one another with the dimensionless strength λ. In the mean-field approximation each pair of states is described by its pair correlation κ, a complex number, and its occupation n_k; the order parameter is $\Delta = \lambda\int \kappa\, d\xi_k$. The equations of motion are

$$\begin{aligned}\frac{d\kappa}{dx4} &= -i\left[2\,\xi_k\,\kappa - \Delta\,(1-2n_k)\right],\\ \frac{dn_k}{dx4} &= 2\,\mathrm{Im}\big(\Delta^*\kappa\big).\end{aligned} \tag{96}$$

They conserve $|2\kappa|^2 + (1-2n_k)^2 = 1$ for every pair of states. In equilibrium $\kappa = \Delta/2E_k$ and $n_k = \frac{1}{2}(1-\xi_k/E_k)$ with the quasi-particle energy $E_k = (\xi_k^2+\Delta^2)^{1/2}$, and the gap equation gives

$$\Delta_{\rm eq}(\lambda) = \frac{E_{\rm sh}}{\sinh(1/\lambda)} \;\approx\; 2\,E_{\rm sh}\,e^{-1/\lambda} \quad (\lambda \ll 1). \tag{97}$$

The model keeps what matters for the question, a gap that depends exponentially on a coupling that grows exponentially, and omits the dependence of the gap on the momentum; $E_{\rm sh}$ stands for the range of energies over which the attraction acts. The program starts the model in equilibrium at $\lambda = 0.15$, lets the coupling grow as $\lambda = 0.15\,e^{\tau_d}$ up to $\lambda = 0.5$, and integrates Eqs. (96) for 2000 energies. Doubling the number of energies and

TABLE XV. The gap at the Fermi surface in units of the Fermi kinetic energy, with contact couplings that enter with the sign $s_{\rm ct} = +1$ (form Q, all four vector pairs along the extra axes; form C, the pairs along $x5, x6, x7$), at the clock reading 2.04 [$g_{\rm VS} = 2.771$], for the Fermi momentum $k_F = 0.5\,m$ and three masses of the radial mode. Contact couplings in units of $g_{\rm VS}^2/m^2$; $m_{\rm ct} = m$.

interaction	$m_s = 0.1\,m$	$m_s = 0.3\,m$	$m_s = m$
exchange of the Vorton-Scalar alone	0.7504	0.4118	0.0247
with a scalar contact attraction, $g_0^{\rm eff} = -1$	1.2895	0.9035	0.2975
with $g_0^{\rm eff} = -1$ and $g_1^{\rm eff} = +0.1$	1.1105	0.7407	0.1963
with a vector contact repulsion, $g_1^{\rm eff} = +0.3$	0.5438	0.2370	0.0006
with an axial-vector contact coupling, $g_7^{\rm eff} = -0.1$	1.0114	0.6459	0.1290
contact couplings alone, $g_0^{\rm eff} = -1$, $g_1^{\rm eff} = +0.1$	0.0006	0.0006	0.0006
exchange alone, phase mode of the mass $0.3\,m$	0.7341	0.3967	0.0197

TABLE XVI. The gap at the Fermi surface in units of the Fermi kinetic energy at the clock reading 2.04, for $k_F = 0.5\,m$ and two masses of the radial mode, for the two signs with which the contact couplings enter the pairing potential: $s_{\rm ct} = +1$ (form Q; form C in the pairs along $x5, x6, x7$) and $s_{\rm ct} = -1$ (form C in the pair along the hidden axis). Last two columns: the ultraviolet strength of the kernel for $m_s = m$. Contact couplings in units of $g_{\rm VS}^2/m^2$; $m_{\rm ct} = m$.

interaction	$m_s = 0.3\,m$, $s_{\rm ct} = +1$	-1	$m_s = m$, $s_{\rm ct} = +1$	-1	$U_{\rm uv}/4\pi^2$, $s_{\rm ct} = +1$	-1
exchange of the Vorton-Scalar alone	0.4118	0.4118	0.0247	0.0247	0.285	0.285
with $g_0^{\rm eff} = -g_{\rm VS}^2/m^2$	0.9035	0.1481	0.2975	0.0000	0.428	0.143
with $g_0^{\rm eff} = -g_{\rm VS}^2/m^2$, $g_1^{\rm eff} = +0.1\,g_{\rm VS}^2/m^2$	0.7407	0.1875	0.1963	0.0000	0.304	0.267
with $g_1^{\rm eff} = +0.3\,g_{\rm VS}^2/m^2$	0.2370	0.8481	0.0006	0.2285	-0.088	0.659
with $g_7^{\rm eff} = -0.1\,g_{\rm VS}^2/m^2$	0.6459	0.2509	0.1290	0.0007	0.410	0.161
contact couplings alone ($g_0^{\rm eff} = -g_{\rm VS}^2/m^2$, $g_1^{\rm eff} = +0.1\,g_{\rm VS}^2/m^2$)	0.0006	0.0000	0.0006	0.0000	0.018	-0.018

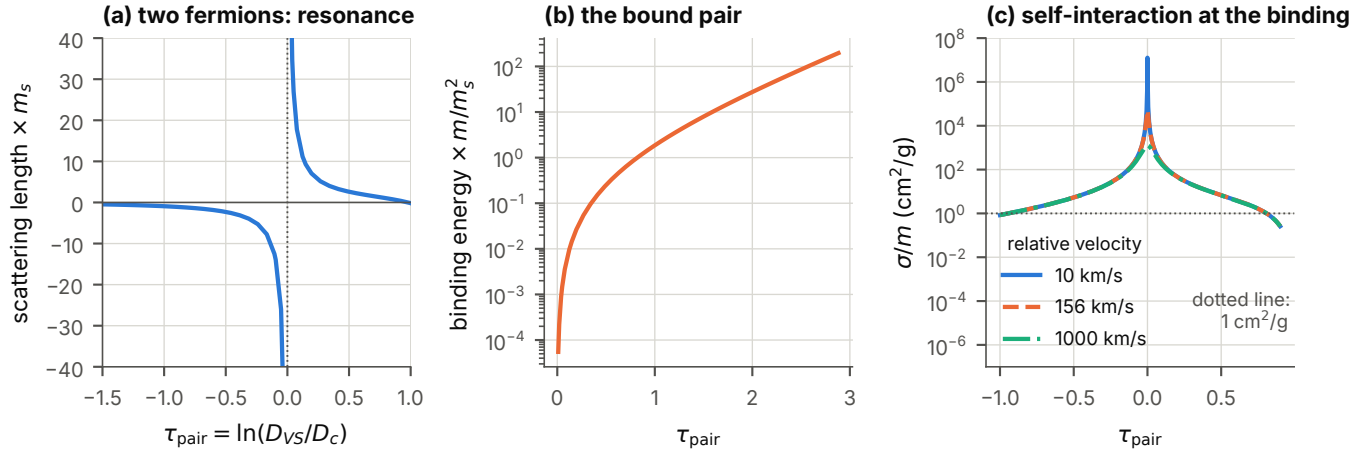


FIG. 13. The bound pair along the clock. (a) Scattering length of two fermions in units of $1/m_s$ against $\tau_{\rm pair}$: the resonance at the moment of binding. (b) Binding energy of the pair. (c) Cross-section per mass of two fermions at three relative velocities while the clock passes the binding, for the example particle of Sec. XVII.

halving the step changes no digit of the results quoted. Besides $|\Delta|$ the program records the density $n_{\rm ex}$ of *broken pairs*: the number of pairs of states, per unit of $E_{\rm sh}$ times the density of states, that are not in the ground state that belongs to the order parameter of the moment. In equilibrium it is zero.

C. Results

Table XX and Fig. 14 show three results.

The faster the deflation, the further the order parameter falls behind. For a rate of $10^{-4}\,E_{\rm sh}$ the order parameter drops to 0.958 of the equilibrium gap of the moment before it recovers; for $10^{-1}\,E_{\rm sh}$ it drops to 0.098.

TABLE XVII. Form C: the number of clock readings, among those at which both gaps are defined, at which the pairs along $x5, x6, x7$ ($s_{\rm ct} = +1$) or the pair along the hidden axis ($s_{\rm ct} = -1$) have the larger gap (four sets of m_s and k_F).

interaction	$x5, x6, x7$	hidden axis
with $g_0^{\rm eff} = -g_{\rm VS}^2/m^2$	53	0
with $g_0^{\rm eff} = -g_{\rm VS}^2/m^2$, $g_1^{\rm eff} = +0.1\, g_{\rm VS}^2/m^2$	53	0
with $g_1^{\rm eff} = +0.3\, g_{\rm VS}^2/m^2$	0	43
with $g_7^{\rm eff} = -0.1\, g_{\rm VS}^2/m^2$	51	0

TABLE XVIII. Two fermions as the clock passes the moment of binding: the scattering length in units of $1/m_s$, the number of bound states, and the binding energy in units of m_s^2/m.

$\tau_{\rm pair}$	$D_{\rm VS}$	$a_{\rm sc}\, m_s$	bound states	$E_b\, m/m_s^2$
-0.105	1.5118	-12.50	0	0
-0.095	1.5279	-13.93	0	0
-0.010	1.6630	-134.0	0	0
0.010	1.6966	135.9	1	5.50×10^{-5}
0.486	2.7314	2.698	1	0.232
0.994	4.5409	-0.1288	1	1.82

The state catches up when Ξ_d has fallen to a value between 0.23 and 0.52. In six of the seven runs the order parameter returns to within 10 per cent of the equilibrium gap at such a value. In the fastest run it has not caught up when the coupling reaches 0.5. This confirms the estimate (95) and fixes the number that "small" means.

The lag leaves broken pairs behind. Their density at the end grows with the rate, from 6.0×10^{-6} to 4.1×10^{-2} in the six slower runs. They are excitations of the superfluid that the deflation itself has made; in a system without dissipation they remain.

Because $\Delta_{\rm eq}$ grows as $e^{-1/\lambda}$ while λ grows only as e^{τ_d}, the parameter Ξ_d falls steeply with time. For any rate there is a moment at which Ξ_d passes through the window of the second result. Before it the paired state is out of equilibrium, with an order parameter below the equilibrium value; after it the adiabatic history of Sec. VIII applies. The slower the deflation, the earlier that moment and the smaller the gap at which it happens: it is the moment at which the equilibrium gap has grown to between two and four times gsum/λ.

XII. THE EFFECTIVE ACTION ALONG THE CLOCK

A. The method, and why it carries over

The results on the paired state, on matter, on temperature and on the vacuum follow from the effective action for composite operators of Cornwall, Jackiw and Tomboulis (CJT) [20]: a functional of the expectation value ϕ of the Vorton-Scalar and of the full propagators, whose two-particle-irreducible part is truncated at two loops. Its stationarity conditions are the equation for the fermion mass in matter and the gap equation. On the deflating spacetime, and for fields that do not depend on the extra coordinates, nothing in this construction changes, for two reasons established above. The weighted fermion field obeys the flat equations, so that its propagator is the flat one. And the interaction between the fermions is, at each moment, the exchange of the two modes of the scalar with the coupling of that moment and with the unshifted range (Sec. VI D).

The one new term, the shift $-{\rm gsum}^2/4$ of the squares of the boson masses in Eq. (64), is smaller than every other scale of the calculation by the square of the ratio of the rate of deflation to a particle mass. For the largest rate that observation allows (Sec. XVI) and a mass of 1 keV that ratio is below 10^{-36}. The term is neglected in this section; this is the only approximation made here beyond the truncation itself. The test of the truncation on a solvable model in zero dimensions, which does not involve the spacetime, was executed again: the two-loop truncation has a solution for every coupling from 0.01 to 30 and stays within 10.7 per cent of the exact propagator.

Every result of this section is a statement about a *sequence of equilibrium states.* Whether a system can follow the sequence was examined, for the paired state, in Sec. XI.

B. The paired state

The CJT potential $V[\Delta]$ of the paired state is the effective action per unit volume, measured from the unpaired state, as a functional of the gap function $\Delta(k)$. The program evaluates it at 13 moments, for two choices of the mass m_s at the Fermi momentum $k_F = 0.5\, m$, in the vector channel (Table XXI, Fig. 15).

The solution of the gap equation is a minimum at every moment. Newton's method applied to the potential converges to the gap function that the iteration of the gap equation finds, to 7.7×10^{-12}, and the smallest eigenvalue of the second variation is positive throughout. The temperature at which the second variation at $\Delta = 0$ loses its positivity equals the temperature at which the linearized gap equation has the eigenvalue 1, to 2.6×10^{-12}: two routes to the critical temperature agree.

The condensation energy deepens. It is within 15 per cent of the weak-coupling value wherever the gap is below $0.1\, E_F$, and in weak coupling its logarithmic slope along the clock is twice that of the gap, by Eq. (81):

$$\frac{d \ln |V|}{d\tau_d} = \frac{2}{\lambda}. \tag{98}$$

The pairs shrink. The coherence length of the first set falls from 25 to 1.0 in units of $1/k_F$. At its last five moments the gap exceeds $0.1\, E_F$, and at the last the pair is as small as the distance between the particles. *The*

TABLE XIX. The order of events along the deflation: the clock reading τ_{pair} of each event, relative to the moment at which two fermions form a bound pair, for six ratios of the mass m_s of the radial mode to the fermion mass m.

event	threshold	0.1	0.3	1	3	10	30
the one-loop potential of the Vorton-Scalar falls below its vacuum value at the field $10\,v_{\mathrm{VS}}$	$C = 1.5477$	−4.92	−3.82	−2.61	−1.51	−0.31	0.79
matter is self-bound (Fermi sea only)	$C = 3.7011$	−4.04	−2.94	−1.74	−0.64	0.56	1.66
the one-loop potential falls below its vacuum value at the field $3\,v_{\mathrm{VS}}$	$C = 4.2694$	−3.90	−2.80	−1.60	−0.50	0.70	1.80
matter is self-bound (with the negative-energy states)	$C = 5.9218$	−3.57	−2.48	−1.27	−0.17	1.03	2.13
matter of Fermi momentum $0.5\,m$ becomes abnormal (massless fermions)	$C = 12.9713$	−2.79	−1.69	−0.49	0.61	1.82	2.91
two fermions form a bound pair	$D_{\mathrm{VS}} = 1.6798$	0.00	0.00	0.00	0.00	0.00	0.00
pair condensate in the vacuum (ladder equation, cut-off $100\,m$)	$\alpha_{\mathrm{VS}} = 2.13$ to 2.93	2.54	1.44	0.25	−0.82	−1.93	−2.84

TABLE XX. The paired state while the pairing strength grows from 0.15 to 0.5, for seven rates of deflation: the parameter Ξ_d at the start; the smallest ratio of the order parameter to the equilibrium gap of the moment; the clock reading at which the order parameter has returned to within 10 per cent of the equilibrium gap, and Ξ_d at that moment; the ratio at the end; and the density of broken pairs at the end.

$\mathrm{gsum}/E_{\mathrm{sh}}$	Ξ_d at the start	smallest $\lvert\Delta\rvert/\Delta_{\mathrm{eq}}$	τ_d of the catch-up	Ξ_d there	final $\lvert\Delta\rvert/\Delta_{\mathrm{eq}}$	n_{ex} at the end
1×10^{-4}	0.262	0.958	0.020	0.23	1.0000	6.0×10^{-6}
3×10^{-4}	0.786	0.890	0.069	0.47	0.9998	5.1×10^{-5}
1×10^{-3}	2.62	0.737	0.234	0.52	0.9980	4.4×10^{-4}
3×10^{-3}	7.86	0.539	0.454	0.44	0.9886	2.4×10^{-3}
0.01	26.2	0.328	0.780	0.32	0.9347	0.012
0.03	78.6	0.189	1.172	0.24	0.8076	0.041
0.1	262	0.098	—	—	0.1189	0.022

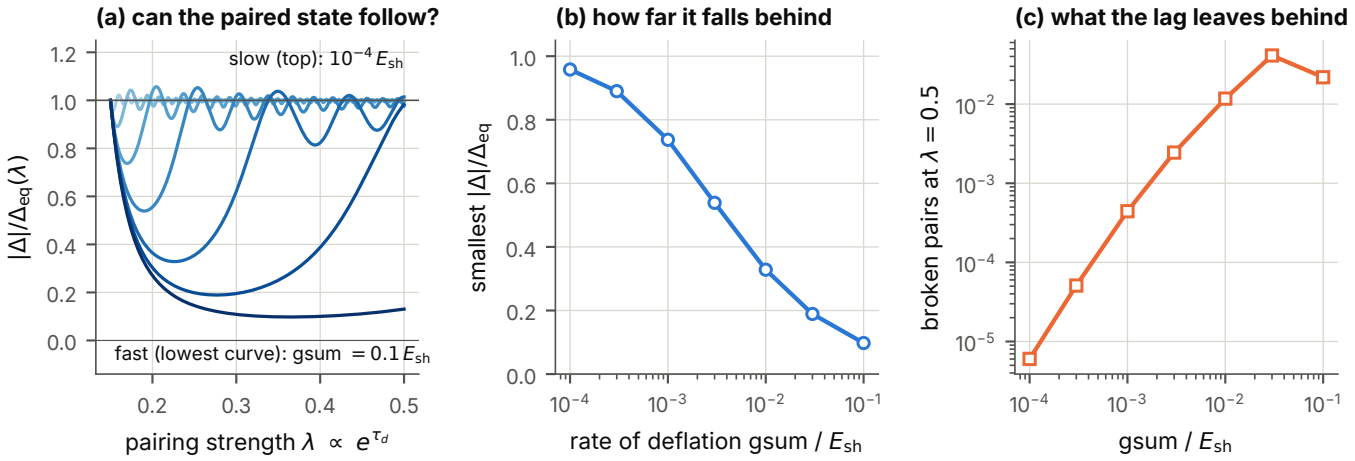


FIG. 14. The paired state while the coupling grows. (a) The order parameter divided by the equilibrium gap of the moment, against the pairing strength, for seven rates of deflation. (b) The smallest value of that ratio against the rate. (c) The density of broken pairs at the end of the run against the rate.

deflation carries the paired state from overlapping pairs toward bound pairs. On flat spacetime this crossover [21, 22] is reached by choosing a larger coupling; here it is reached by waiting.

The universal ratios hold while the coupling is weak. While the critical temperature rises by a factor of sixteen, from 6.3×10^{-3} to 0.10 of the Fermi energy, the ratio of the gap at zero temperature to the critical temperature stays within 0.019 of the weak-coupling value 1.7639, and the relative jump of the heat capacity at T_c within 0.048 of 1.4261. The coefficients of the Ginzburg-Landau energy agree with their weak-coupling formulas to 1.0 per cent while T_c is below $0.05\,E_F$ and differ by up to 7.9 per cent at $T_c = 0.10\,E_F$. The coherence length of

TABLE XXI. The paired state along the clock from the CJT potential, for $k_F = 0.5\,m$. λ: pairing strength. V: the potential at the solution, the condensation energy. Ratio: V divided by its weak-coupling value $-\nu N(0)\Delta_F^2/4$, with $\nu = 8$ internal states and the density of states $N(0)$. Curvature: the smallest eigenvalue of the matrix of second derivatives of the potential. Size: the coherence length of the pair in units of $1/k_F$.

m_s/m, k_F/m	τ_d	$g_{\rm VS}$	λ	Δ_F/m	V/m^4	ratio	curvature	T_c/Δ_F	size $\times k_F$
0.30, 0.50	0.00	1.80	0.216	2.81×10^{-3}	-4.46×10^{-7}	1.000	0.21	0.567	25.4
0.30, 0.50	0.21	2.00	0.266	6.89×10^{-3}	-2.68×10^{-6}	0.998	0.26	0.566	10.3
0.30, 0.50	0.40	2.20	0.322	0.0135	-1.03×10^{-5}	0.992	0.30	0.564	5.3
0.30, 0.50	0.58	2.40	0.383	0.0230	-2.93×10^{-5}	0.981	0.34	0.559	3.1
0.30, 0.50	0.74	2.60	0.450	0.0354	-6.82×10^{-5}	0.961	0.37	0.552	2.0
0.30, 0.50	0.88	2.80	0.522	0.0511	-1.38×10^{-4}	0.932	0.38	0.542	1.4
0.30, 0.50	1.02	3.00	0.599	0.0707	-2.53×10^{-4}	0.895	0.38	0.530	1.0
1.0, 0.50	0.00	2.40	0.133	4.25×10^{-4}	-1.02×10^{-8}	1.000	0.13	0.567	167.5
1.0, 0.50	0.16	2.60	0.156	1.32×10^{-3}	-9.90×10^{-8}	1.000	0.15	0.567	53.8
1.0, 0.50	0.31	2.80	0.181	3.28×10^{-3}	-6.09×10^{-7}	0.999	0.17	0.567	21.7
1.0, 0.50	0.45	3.00	0.208	6.90×10^{-3}	-2.69×10^{-6}	0.998	0.19	0.566	10.3
1.0, 0.50	0.58	3.20	0.237	0.0128	-9.29×10^{-6}	0.994	0.21	0.564	5.5
1.0, 0.50	0.70	3.40	0.267	0.0219	-2.67×10^{-5}	0.987	0.23	0.561	3.3

the Ginzburg-Landau energy, $0.1333\,v_F/T_c$ in weak coupling with the Fermi velocity v_F, shrinks from 80 to 5.1 in units of $1/m$: the cores of the vortices of the superfluid shrink as the deflation proceeds. Tables L and LI of Appendix D give the numbers at the three moments computed.

C. Matter

Matter of fixed density loses its fermion mass. In matter the expectation value of the Vorton-Scalar is reduced by the scalar density of the fermions, and with it the fermion mass in the matter, m^*. The reduction depends on the density and on the single number C, and above a critical value of C the mass vanishes: the matter is then in the *abnormal* state of Lee and Wick [23], a gas of massless fermions in a region without expectation value of the scalar. On the deflating spacetime $C \propto e^{\tau_d}$ grows at fixed density. The critical values are $C = 151.892$, 12.971, 2.556 and 0.617 for the Fermi momenta $k_F = 0.2$, 0.5, 1 and 2 in units of m; they exceed the estimate $\pi^2 m^2/4k_F^2$, which holds for dense matter, by the factors 2.46, 1.31, 1.04 and 1.00. *Every piece of matter becomes abnormal at a definite moment, the denser the earlier.*

Self-bound matter. Matter is self-bound, that is, its energy per particle at some density lies below the mass of a free fermion, if $C > 3\pi^2/8 = 3.7011$ without the filled negative-energy states and if $C > 3\pi^2/5 = 5.9218$ with them. Once the threshold is passed, the lowest energy per particle falls monotonically along the clock (Table XXII), from $-0.0192\,m$ at $C = 4$ to $-0.405\,m$ at $C = 29.6$. At every one of these moments the matter of lowest energy is in the abnormal state. The exchange energies of the two modes, which the mean field omits, are 0.44 and 0.51 per cent of the mean-field energy at the first moment and 0.90 and 1.22 per cent at the last (for $m_s/m = 0.3$): the mean field remains a good approximation along the whole history. A gas of the fermions that is dilute and unbound at early times thus has, at late times, a state of lower energy: drops of dense abnormal matter, bound by up to 40 per cent of the rest mass at the last moment of the table. Whether and how fast a gas reaches that state is a question of kinetics that is not answered here.

TABLE XXII. Self-bound matter along the clock, in the mean-field approximation without the negative-energy states, for $C = 4\,e^{\tau_d}$: the lowest energy per particle minus the fermion mass, and the Fermi momentum at which it is reached.

τ_d	C	$(E/N - m)/m$	k_F/m
0.00	4.00	−0.0192	0.981
0.25	5.14	−0.0787	0.921
0.50	6.59	−0.1345	0.866
0.75	8.47	−0.1869	0.813
1.00	10.87	−0.2362	0.764
1.25	13.96	−0.2825	0.718
1.50	17.93	−0.3259	0.674
1.75	23.02	−0.3668	0.633
2.00	29.56	−0.4051	0.595

D. Temperature

At high temperature the expectation value of the Vorton-Scalar vanishes and the fermions are massless. The temperature $T_{1/2}$ at which the expectation value has fallen to one half is of the order of the four-dimensional vacuum value $v_{\rm VS} = m/g_{\rm VS}$, which falls as $e^{-\tau_d/2}$ while the masses stay fixed. In units of the constant fermion mass, $T_{1/2}$ falls monotonically as the coupling grows, for each of the three ratios $m_s/m = 0.3$, 1 and 3 examined; for $m_s/m = 1$, from $2.33\,m$ at $g_{\rm VS} = 0.3$ to $0.37\,m$ at $g_{\rm VS} = 4$ [Fig. 16(d)]. *Matter of a fixed temperature melts*

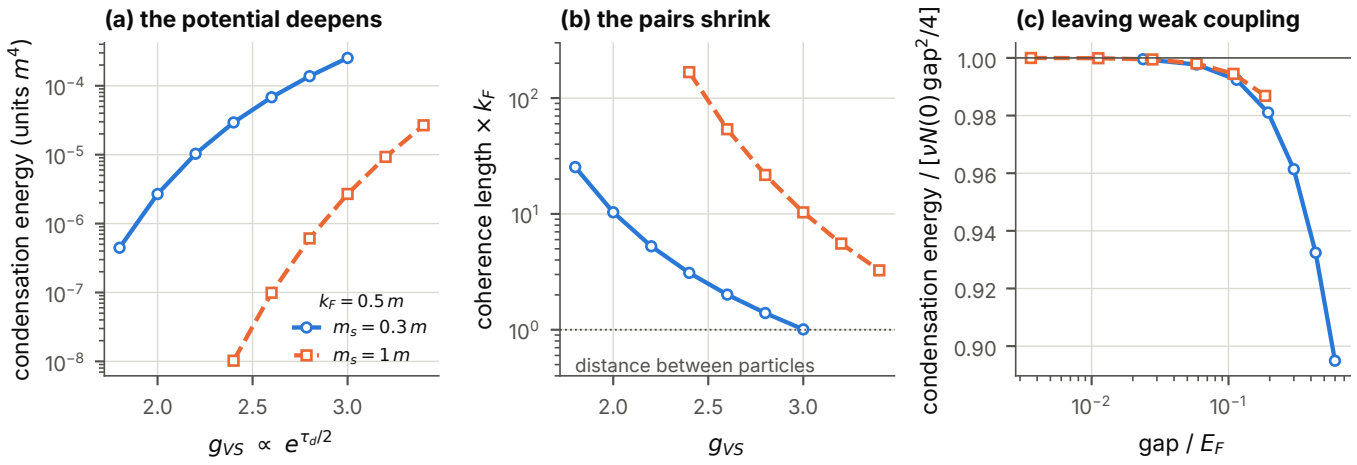


FIG. 15. The paired state along the clock, from the CJT potential, for the two sets of Table XXI. (a) The condensation energy against the coupling of the moment. (b) The coherence length of the pair in units of $1/k_F$. (c) The condensation energy divided by its weak-coupling value, against the gap in units of the Fermi kinetic energy: the departure from weak coupling.

at a definite moment of the deflation. For $m_s/m = 0.3$ no melting temperature can be given from $C = 32$ on: there the one-loop potential of the vacuum turns over so close to the vacuum value that the Hartree potential ceases to exist before the expectation value has fallen, and the program marks these moments as not defined (Table LIII).

E. The vacuum

The one-loop potential of the Vorton-Scalar in vacuum contains the negative contribution of the fermion loop [24], which grows with the fourth power of the field and eventually wins: beyond some field the potential lies below its value in the vacuum. The larger C, the smaller that field: 24.7 times the vacuum value at $C = 1$, and 1.77 times at $C = 10$ [Fig. 16(c), Table LIV]. As the deflation proceeds, the barrier between the vacuum of the model and the region where the one-loop potential is unbounded below moves in toward the vacuum.

The one-loop potential is not reliable at fields far from the vacuum value, and a potential that falls at large field signals that the effective theory needs completion there, not that the vacuum decays. What the result establishes is a trend: every statement of the model that rests on the vacuum of the Vorton-Scalar becomes less secure with time. The order of this event relative to the binding of the pair was given in Sec. X.

A pair condensate in the vacuum. The ladder equation with a momentum cutoff shows that the pair can condense without a medium only if $\alpha_{\rm VS}$ exceeds a critical value α_c of order one [25]; with the cutoff $100\,m$ the critical values lie between 2.13 and 2.93 for the six mass ratios of Table XIX. On the deflating spacetime $\alpha_{\rm VS}$ reaches such a value at a definite moment. But at that moment the coupling is strong, the exchange of one boson is no longer a controlled approximation, and the vacuum of the model is no longer protected. A pair condensate in the vacuum is not reached by the deflation within the range of validity of the model.

XIII. THE EFFECTIVE POTENTIAL WITH THE SELF-INTERACTION

Section XII applied the effective action of Cornwall, Jackiw and Tomboulis to the model in which the Vorton-Scalar alone provides mass and interaction. This section applies the same method to the self-interaction of the fermion field: first without the Vorton-Scalar, where it must reproduce the Kohn-Sham results of Sec. VII and goes beyond them by counting the filled states of negative energy; then with the bare Dirac mass, the self-interaction and the Vorton-Scalar together.

A. The effective potential of uniform matter

The CJT effective action is a functional of the full propagator S of the fermions,

$$\Gamma[S] = -i\,\mathrm{Tr}\ln S^{-1} - i\,\mathrm{Tr}\big(S_0^{-1}S - 1\big) + \Gamma_2[S], \qquad (99)$$

where S_0 is the propagator of free particles of the mass m and Γ_2 is the sum of the vacuum diagrams that cannot be cut in two by cutting two lines, drawn with the full propagator. For the contact interaction $\mathcal{L}_{\rm self}$ the lowest order of Γ_2 consists of two diagrams, each with two loops that meet at one vertex: the direct (Hartree) and the exchange (Fock) diagram. At this order the stationarity condition $\delta\Gamma/\delta S = 0$ says that the self-energy $S_0^{-1} - S^{-1}$ does not depend on the momentum. For uniform matter at rest it is a combination of the unit matrix and of β (Sec. IV D): the full propagator is that of free particles

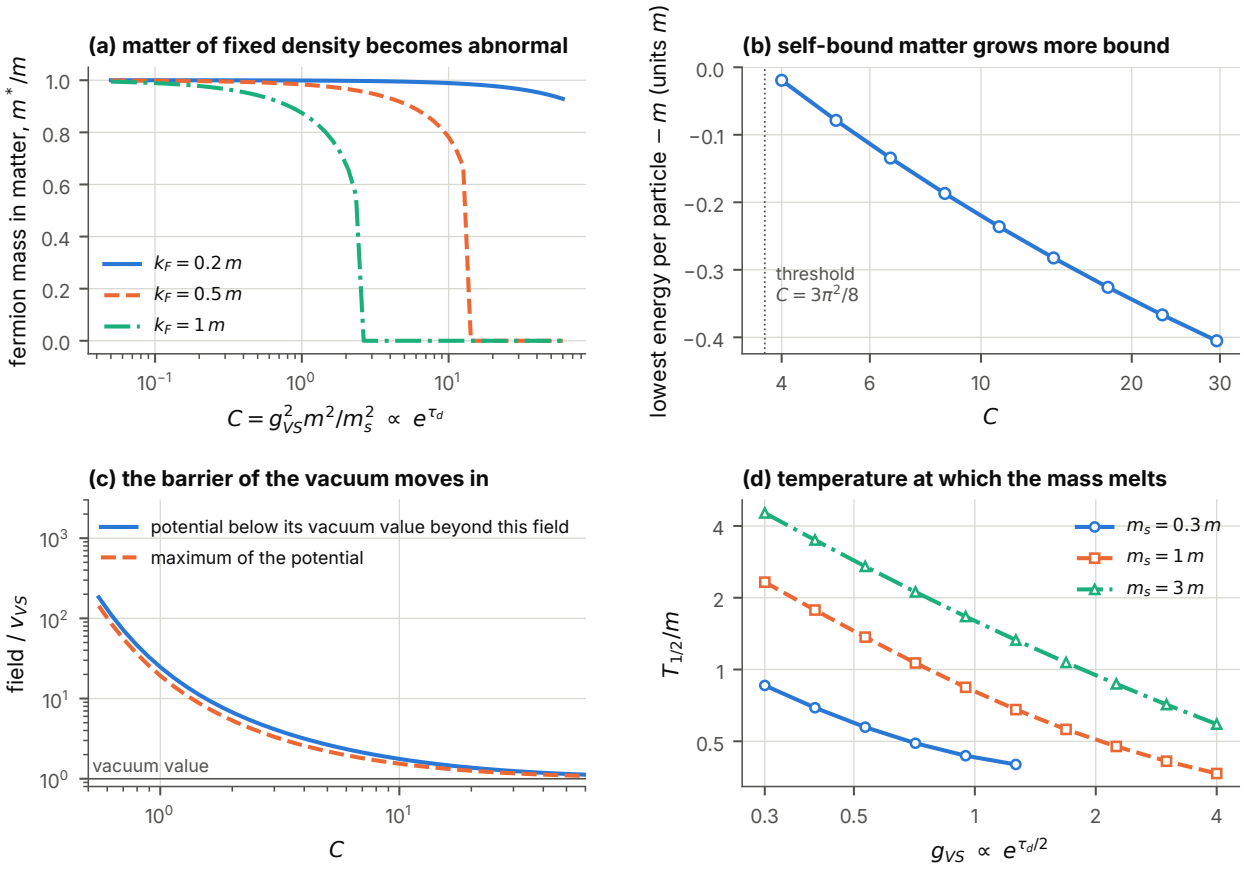


FIG. 16. Matter, vacuum and temperature along the clock. (a) The fermion mass in matter of three fixed densities against $C \propto e^{\tau_d}$: it drops to zero at the critical value. (b) The lowest energy per particle of self-bound matter, minus m. (c) The field at which the one-loop potential of the vacuum has its maximum, and the field beyond which it lies below the vacuum value, in units of the vacuum value. (d) The temperature at which the fermions have lost half of their mass, against the coupling, for three ratios m_s/m.

of a mass m^* whose energies are shifted by a constant V_0. The effective action per unit volume and time then becomes an ordinary function of the two numbers m^* and V_0, the *effective potential*. At zero temperature and at the chemical potential μ,

$$\Omega_{\rm CJT}(m^*, V_0; \mu) = -\frac{(m^*-m)^2}{2\,G_S} - \frac{V_0^2}{2\,G_V} - \nu \int \frac{d^D k}{(2\pi)^D} \left(\mu - V_0 - E^*\right) \theta\left(\mu - V_0 - E^*\right), \tag{100}$$

with $E^* = (k^2 + m^{*2})^{1/2}$, the step function θ, and the scalar and vector coupling G_S, G_V of matter of one kind of particle with ν internal states (Sec. VII A). The integral is the grand potential of free particles of the mass m^* at the chemical potential $\mu - V_0$; the two other terms remove the interaction energy that the integral counts twice. The two stationarity conditions are

$$\begin{aligned} \frac{\partial \Omega_{\rm CJT}}{\partial V_0} = 0: &\quad V_0 = G_V\, n, \\ \frac{\partial \Omega_{\rm CJT}}{\partial m^*} = 0: &\quad m^* = m + G_S\, n_S, \end{aligned} \tag{101}$$

the two equations (70) of the Kohn-Sham method. At the stationary point $-\Omega_{\rm CJT}$ is the pressure and $\Omega_{\rm CJT} + \mu\, n$ the energy density. *The Kohn-Sham method and the CJT effective potential at this order are two formulations of the same approximation*: the first works at a fixed density and solves the gap equation, the second works at a fixed chemical potential and searches the stationary point of a function.

The program evaluates the two with separate code (Gauss-Legendre integrals and a root search in the one, Simpson integrals and a minimization in the other) for 30 states in $D = 7$, 4 and 3 dimensions (Table XXIII). Mass, density, energy density and pressure agree to 1.6×10^{-12} (sum of the four relative deviations); the independent script repeats the comparison with its own integrals and finds the same.

TABLE XXIII. The Kohn-Sham method (KS) and the CJT effective potential compared, without the filled states of negative energy (units $m = 1$): the mass of the particles in matter by both methods, and $-\Omega_{\rm CJT}$ at its stationary point divided by the Kohn-Sham pressure p, for ten of the 30 states computed (the data files hold eleven digits).

D	G_S	G_V	k_F	m^* (KS)	m^* (CJT)	$-\Omega_{\rm CJT}/p$
7	−100000	10000	0.9	0.143888	0.143888	1.00000000
7	−100000	10000	1.4	0.011510	0.011510	1.00000000
7	−98750	20000	0.9	0.145495	0.145495	1.00000000
7	−98750	20000	1.4	0.011654	0.011654	1.00000000
4	−256.9512	25.6951	0.9	0.140053	0.140053	1.00000000
4	−256.9512	25.6951	1.4	0.040348	0.040348	1.00000000
3	−29.5093	2.9509	0.9	0.182319	0.182319	1.00000000
3	−29.5093	2.9509	1.4	0.079333	0.079333	1.00000000
3	−4	0	0.9	0.394848	0.394848	1.00000000
3	−4	0	1.4	0.240321	0.240321	1.00000000

The effective potential says more than the gap equation in one respect. At a given chemical potential the stationary point with the lowest $\Omega_{\rm CJT}$, that is, with the highest pressure, is the state of equilibrium (the vector field V_0 is taken at its stationary value, a maximum; the minimum is over m^*). Of the 23 states of the comparison with a positive pressure, 20 are that minimum. In the others a denser state has the higher pressure at the same chemical potential: the Kohn-Sham state of that density is metastable, a dilute gas that can lower its energy by condensing into saturated matter.

The two halves of the field. For the matter of Sec. VII B, whose two halves have different masses and energy shifts, the effective potential is a function of four numbers,

$$\begin{aligned}\Omega_{\rm CJT} = \sum_{\pm} \Omega_{\rm half}\big(m^*_{\pm}, \mu - V_{0\pm}\big) \\ - \frac{\big[\frac{1}{2}(m^*_+ + m^*_-) - m\big]^2}{2\,G_S^{\rm tot}} - \frac{\big[\frac{1}{2}(m^*_+ - m^*_-)\big]^2}{2\,G_S^{\rm dif}} \\ - \frac{\big[\frac{1}{2}(V_{0+} + V_{0-})\big]^2}{2\,G_V^{\rm tot}} - \frac{\big[\frac{1}{2}(V_{0+} - V_{0-})\big]^2}{2\,G_V^{\rm dif}},\end{aligned} \tag{102}$$

where $\Omega_{\rm half}(m^*, \mu) = -4\int d^D k\,(\mu - E^*)\,\theta(\mu - E^*)/(2\pi)^D$ is the grand potential of free particles of one half, with four internal states. Its four stationarity conditions are the equations (74) with the densities (75). For 80 states of both forms (40 with equal halves, 17 with unequal halves, 23 with one half filled) the stationary point, found by Newton's method from displaced start values, agrees with the Kohn-Sham state in both masses, density, energy density and pressure to 4.9×10^{-11}: the two methods are equivalent in both forms.

B. The filled states of negative energy

The Kohn-Sham calculation counts the particles of matter only. The term $-i\,{\rm Tr}\ln S^{-1}$ of Eq. (99) also contains the filled states of negative energy of the vacuum, whose energy depends on the mass of the particles. Because the contact interaction is an effective interaction, their momenta are counted up to a cutoff Λ:

$$\begin{aligned}\Omega_{\rm CJT} \to \Omega_{\rm CJT} - \nu \int_{|\vec{k}|<\Lambda} \frac{d^D k}{(2\pi)^D} \\ \times \Big[\sqrt{k^2 + m^{*2}} - \sqrt{k^2 + m^2}\Big].\end{aligned} \tag{103}$$

The filled states contribute the scalar density $-\nu\int_{|\vec{k}|<\Lambda} d^D k\,(m^*/E^*)/(2\pi)^D$ to the gap equation. The mass parameter of the Lagrangian is then no longer the mass of a particle in vacuum and is called the *bare mass* $m_{\rm bare}$; the mass in vacuum, $m_{\rm vac}$, is the solution of

$$\begin{aligned} m_{\rm vac} &= m_{\rm bare} + G_S\,\langle\bar\psi\psi\rangle_{\rm vac}, \\ \langle\bar\psi\psi\rangle_{\rm vac} &= -\nu\int_{|\vec{k}|<\Lambda} \frac{d^D k}{(2\pi)^D}\,\frac{m_{\rm vac}}{\sqrt{k^2 + m_{\rm vac}^2}}.\end{aligned} \tag{104}$$

The scalar density of the vacuum is negative, so that an attraction ($G_S < 0$) *raises* the mass above the bare mass.

A critical coupling. For $m_{\rm bare} = 0$ Eq. (104) always has the solution $m_{\rm vac} = 0$. A second solution, of lower energy, exists if the attraction exceeds a critical value, $G_S < G_c$. Expanding the integral for small $m_{\rm vac}$, with the area $A_D = D\,V_D$ of the unit sphere,

$$\begin{aligned} G_c &= -\frac{(2\pi)^D\,(D-1)}{\nu\,A_D\,\Lambda^{D-1}}, \\ G_c\,\Lambda^{D-1} &= -90\pi^4,\ -3\pi^2,\ -\frac{\pi^2}{2} \quad \text{for } D = 7,\ 4,\ 3.\end{aligned} \tag{105}$$

A field without a mass term acquires a mass by its own self-interaction when the scalar attraction exceeds G_c. This is the mechanism of Nambu and Jona-Lasinio [26]; the minimization of the effective potential gives no mass two per cent below G_c and a mass two per cent above it in all three cases.

On the deflating spacetime the critical coupling is crossed at a definite moment. In the sector of fields that do not depend on $x5, x6, x7$ the coupling grows as e^{τ_d} while the cutoff, a momentum along the flat coordinates, does not change. A field that is massless at early times therefore acquires a mass at the clock reading $\tau_d = \ln[G_c/G_S(x4_{\rm ref})]$, and from then on the mass grows [Fig. 17(a), Table XXIV]. With a small bare mass the transition is smooth. The mass reaches the cutoff about half a unit of the clock after the critical coupling is crossed; beyond that point the effective theory says nothing. (Table XXIV and Fig. 17 are for one kind of particle with eight internal states, that is, for form Q of the 4+4 model and for the 7+1 model; the two halves of the field in form C follow below.)

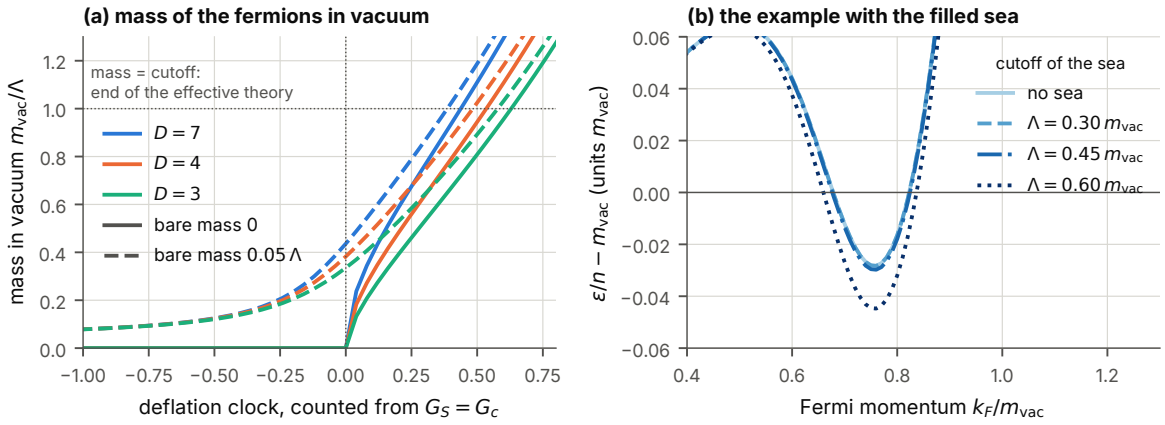


FIG. 17. The filled states of negative energy. (a) The mass of the fermions in vacuum, in units of the cutoff, along the deflation clock counted from the moment at which the scalar attraction equals the critical coupling, for $D = 7$, 4 and 3 dimensions, without bare mass and with the bare mass $0.05\,\Lambda$. (b) Energy per particle of matter relative to the vacuum, for an example on the 7+1 spacetime ($D = 7$, $G_S = -10^5$, $G_V = +10^4$ in units of the mass in vacuum) without the filled states and with them for three cutoffs; the bare mass is chosen so that the mass in vacuum is the same.

TABLE XXIV. The mass of the fermions in vacuum, $m_{\rm vac}/\Lambda$, with the filled states of negative energy, along the deflation clock counted from the moment at which $G_S = G_c$, for a field without bare mass and for the bare mass $0.05\,\Lambda$.

D	$m_{\rm bare}/\Lambda$	$\tau_d = -1$	$\tau_d = 0$	$\tau_d = 0.24$	$\tau_d = 0.48$
7	0	0	0	0.658	1.074
7	0.05	0.079	0.436	0.774	1.154
4	0	0	0	0.544	0.913
4	0.05	0.079	0.383	0.664	0.995
3	0	0	0	0.447	0.780
3	0.05	0.078	0.335	0.573	0.866

The vacuum of the two halves. For the two halves of the field the gap equations of the vacuum are

$$\begin{aligned} \tfrac{1}{2}(m^*_+ + m^*_-) - m_{\rm bare} &= G_S^{\rm tot}\left(n^{\rm vac}_{S+} + n^{\rm vac}_{S-}\right), \\ \tfrac{1}{2}(m^*_+ - m^*_-) &= G_S^{\rm dif}\left(n^{\rm vac}_{S+} - n^{\rm vac}_{S-}\right), \\ n^{\rm vac}_{S\pm} &= -4\int_{|\vec{k}|<\Lambda} \frac{d^D k}{(2\pi)^D}\,\frac{m^*_\pm}{E^*_\pm}, \end{aligned} \tag{106}$$

with the scalar densities $n^{\rm vac}_{S\pm}$ of the filled states of the two halves; where they have several solutions, the vacuum is the one of lowest energy density

$$\begin{aligned} \Omega_{\rm vac} = &-4\sum_\pm \int_{|\vec{k}|<\Lambda} \frac{d^D k}{(2\pi)^D}\left[E^*_\pm - \sqrt{k^2 + m^2_{\rm bare}}\right] \\ &- \frac{\left[\tfrac{1}{2}(m^*_+ + m^*_-) - m_{\rm bare}\right]^2}{2\,G_S^{\rm tot}} - \frac{\left[\tfrac{1}{2}(m^*_+ - m^*_-)\right]^2}{2\,G_S^{\rm dif}}. \end{aligned} \tag{107}$$

For the sets A and B the more attractive of the two scalar couplings is $G_S^{\rm HF}$ in both forms ($G_S^{\rm tot}$ in form Q, $G_S^{\rm dif}$ in form C), so that the critical strength $g_{\rm crit} = G_c\,g/G_S^{\rm HF}$ of a field without mass term is the same for both (Table XXV). Above it, *in form Q the two halves acquire the same mass, a mass term proportional to $\bar\psi\psi$, the mass term of the positive Lagrangian; in form C they acquire masses of opposite sign, by Eq. (33) a mass term proportional to $\tilde\psi\psi$, the mass term of the covariant Lagrangian.* Each form generates dynamically the mass term that belongs to its own adjoint; the magnitude of the mass is the same. With a bare mass, whose term is $-m_{\rm bare}\bar\psi\psi$ in both forms, the two forms differ in kind (Table XXVI, Fig. 18). In form Q the bare mass and the dynamical mass have the same structure, and the mass grows smoothly through the critical strength. In form C the two masses are equal and slightly smaller than the bare mass up to a definite strength, because $G_S^{\rm tot} > 0$ repels the total scalar density, and there they separate: a sharp transition at which the symmetry between the halves is broken spontaneously. With the bare mass $0.05\,\Lambda$ it lies at $D = 4$, set A: $\tau_d = 0.009$; $D = 4$, set B: $\tau_d = 0.009$; $D = 3$, set A: $\tau_d = 0.019$; $D = 3$, set B: $\tau_d = 0.020$, counted from the critical strength of the massless field, and agrees with the linear criterion $1 + 2\,G_S^{\rm dif}\cdot 4\int_{|\vec{k}|<\Lambda} d^D k\,(k^2/E^{*3})/(2\pi)^D = 0$ at the common mass to 2.1×10^{-10}.

The thresholds of Sec. VII and the vacuum. The calculation of Sec. VII without the filled states is consistent only while the vacuum is stable, that is, while the most attractive scalar coupling is weaker than G_c for the cutoff of the model. The last column of Table VIII gives the largest cutoff for which this holds at the binding threshold. It is below the mass in all eight cases, and smaller in form C than in form Q, because form C needs stronger couplings to bind matter: *in both forms, and more so in form C, self-bound matter by contact couplings alone presupposes a cutoff below the mass of the particles.* How the filled states change a saturation point when they are included was computed for an example of one kind of particle on the 7+1 spacetime ($D = 7$), with the bare mass

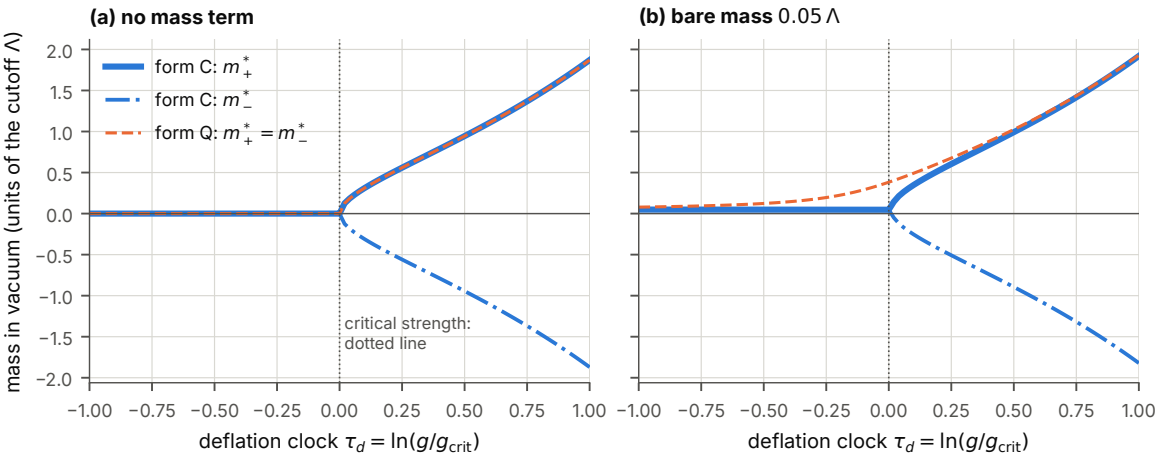


FIG. 18. The masses of the particles of the two halves in vacuum with the filled states of negative energy, along the clock counted from the critical strength of a massless field (dotted line; $D = 4$, set A; units of the cutoff). (a) A field without mass term: above the critical strength form Q gives both halves the same mass, form C gives them masses of opposite sign. (b) With the bare mass $0.05\,\Lambda$: form Q passes smoothly through the critical strength; in form C the two masses are equal up to a sharp transition and separate there.

TABLE XXV. The vacuum of a field without mass term in the two forms, with the filled states of negative energy (units of the cutoff): the critical strength of the sets A and B [for the critical coupling G_c of Eq. (105) with $\nu = 8$], and the masses (m_+^*, m_-^*) of the two halves at a strength 2 per cent above it (the representative with $m_+^* > 0$ of the degenerate solutions).

D	set	$g_{\rm crit}\Lambda^{D-1}$	form Q	form C
4	A	31.5827	(0.1243, 0.1243)	(0.1243, −0.1243)
4	B	29.9836	(0.1243, 0.1243)	(0.1243, −0.1243)
3	A	5.2638	(0.0860, 0.0860)	(0.0860, −0.0860)
3	B	4.9973	(0.0860, 0.0860)	(0.0860, −0.0860)

adjusted so that the mass in vacuum stays the same: the saturation point moves little while the cutoff stays below about half the mass in vacuum, and the binding grows for larger cutoffs [Fig. 17(b)]. The results of Sec. VII are therefore those of an effective theory with a low cutoff.

C. Bare mass, self-interaction and Vorton-Scalar together

Now let all terms of the model (61) act together, for fields in ordinary space ($D = 3$) and without the filled states. Let ϕ be the expectation value of the Vorton-Scalar, $\langle\Phi_{\rm VS}\rangle = \phi/\sqrt{2}$, and $x = \phi/v_{\rm VS}$. The mass of a fermion in a region with the expectation value ϕ is $m_D + m_Y\,x$; let $f_D = m_D/m$ be the fraction of the mass in vacuum that is a bare Dirac mass. In units $m = 1$ the energy density of matter of the Fermi momentum k_F is the stationary value of

$$\varepsilon = \frac{(1-f_D)^2\,(x^2-1)^2}{8\,C} + \langle E^*\rangle + \tfrac{1}{2}G_V\,n^2 - \frac{\left[m^* - f_D - (1-f_D)\,x\right]^2}{2\,G_S} \tag{108}$$

with respect to x and m^*: the first term is the potential of the scalar, with the number C of Eq. (66); the others are Eq. (72) with the mass $f_D + (1-f_D)x$ in place of m. Equation (108) is written for one kind of particle; for the two halves of the field the last three terms are replaced by the energy density (76) with $f_D + (1-f_D)x$ in place of m, because the Yukawa coupling shifts the masses of both halves alike (Sec. VI). For $f_D = 0$ and $G_S = G_V = 0$ this is the functional of Sec. XII, and the program reproduces its masses and energies to 2.1×10^{-8} and its binding threshold $C = 3\pi^2/8$.

At low density the Vorton-Scalar acts as a scalar contact coupling. Expanding Eq. (108) for small densities gives $m^* = m + (G_S - C/m^2)\,n_S$: the exchange of the radial mode adds $-g_{\rm VS}^2/m_s^2$ to the scalar coupling, whatever fraction of the mass is a bare mass (verified for five sets of parameters to 1.8×10^{-4} at Fermi momenta up to a twentieth of m). The scalar channel of the self-interaction and the exchange of the radial mode are the same interaction at momenta far below m_s; they differ at higher densities, where the scalar field itself changes. For the two halves this means that the Vorton-Scalar adds $-C/m^2$ to the coupling $G_S^{\rm tot}$ of the total scalar density, in both forms.

The moment at which matter becomes self-bound. All couplings grow as e^{τ_d}, so that their ratios are fixed and the deflation carries matter through a sequence of states with growing C. Table XXVII gives the smallest C at which matter is self-bound in the two forms, for contact

TABLE XXVI. The masses of the two halves in vacuum, in units of the cutoff, along the deflation clock counted from the critical strength of the massless field (set A), without bare mass and with the bare mass $0.05\,\Lambda$. A single number: both halves have this mass; a pair: the two masses, the larger first (which half has which is free by the symmetry between the halves).

D	$m_{\rm bare}/\Lambda$	form	$\tau_d=-0.5$	0.1	0.5	1
4	0	Q	0	0.311	0.944	1.871
4	0	C	0	(0.311, −0.311)	(0.944, −0.944)	(1.871, −1.871)
4	0.05	Q	0.123	0.492	1.025	1.930
4	0.05	C	0.048	(0.350, −0.255)	(0.992, −0.894)	(1.920, −1.821)
3	0	Q	0	0.241	0.809	1.643
3	0	C	0	(0.241, −0.241)	(0.809, −0.809)	(1.643, −1.643)
3	0.05	Q	0.121	0.426	0.892	1.703
3	0.05	C	0.048	(0.278, −0.183)	(0.855, −0.758)	(1.692, −1.593)

couplings $g_0<0$ and $g_1>0$ of the Lagrangian proportional to C, and the shift of that moment relative to the model with the Vorton-Scalar alone. *A bare Dirac mass makes matter bind earlier*, by 0.39 units of the clock if a quarter of the mass is a bare mass and by 0.58 if half of it is: the matter no longer has to pay for the destruction of the whole expectation value of the scalar. In all 18 cases of the table the bound state at the threshold has equal halves, in both forms: the attraction that binds it is that of the scalar field, which acts on both halves alike, and contact couplings of these strengths are too weak to polarize it. The threshold is that of matter of one kind with the couplings $G_S^{\rm tot}$, $G_V^{\rm tot}$ of the form, confirmed with the functional of two halves minimized over the density, the field and the polarization. The forms differ because the couplings of the total densities differ (Table VI). In form Q the scalar contact coupling adds its attraction $G_S^{\rm HF}$ to that of the scalar field, and a vector coupling adds the repulsion $G_V^{\rm HF}$ [Fig. 19(a,b), drawn against the mean-field couplings of matter of one kind]. In form C the couplings of the total densities are $G_S^{\rm c}$ and $G_V^{\rm c}$, both small, so that the contact couplings change the threshold less; which form binds earlier depends on the case. Without bare mass the bound state of form C at the threshold consists of massless particles ($x=0$), whose scalar densities vanish, and the threshold does not depend on the vector coupling of the table.

D. The mass of the phase mode

A bare Dirac mass breaks the chiral symmetry $U(1)_{\rm VS}$ explicitly, and the phase mode of the Vorton-Scalar is then no longer massless. Write $\Phi_{\rm VS}=(v_{\rm VS}/\sqrt{2})\,e^{i\vartheta_{\rm VS}}$ with a constant phase $\vartheta_{\rm VS}$. The mass term of the fermions becomes $m_D+m_Y\,e^{i\vartheta_{\rm VS}\chi}$, the mass of a particle

$$m(\vartheta_{\rm VS})=\sqrt{m_D^2+m_Y^2+2\,m_D m_Y\cos\vartheta_{\rm VS}}, \qquad (109)$$

largest at $\vartheta_{\rm VS}=0$. The energy of the filled states of negative energy is lowest where the mass is largest, so that the vacuum selects $\vartheta_{\rm VS}=0$, and the second derivative of the energy density with respect to the phase is $m_\vartheta^2 v_{\rm VS}^2$. With $dm/d\vartheta_{\rm VS}=0$ and $d^2m/d\vartheta_{\rm VS}^2=-m_Dm_Y/m$ at $\vartheta_{\rm VS}=0$, and with the derivative of the energy density with respect to the mass, which is the scalar density,

$$m_\vartheta^2\,v_{\rm VS}^2=\Big(\big|\langle\bar\psi\psi\rangle_{\rm vac}\big|-n_S\Big)\,\frac{m_D\,m_Y}{m}. \qquad (110)$$

In vacuum ($n_S=0$) this is the relation of Gell-Mann, Oakes and Renner [27] for the model at hand: the square of the mass of the phase mode is proportional to the bare mass and to the scalar density of the vacuum. In ordinary space $|\langle\bar\psi\psi\rangle_{\rm vac}|=(\nu m/4\pi^2)\,[\Lambda\sqrt{\Lambda^2+m^2}-m^2\,{\rm arsinh}(\Lambda/m)]$; the program confirms Eq. (110) against the numerical second derivative of the energy in 24 cases to 3.2×10^{-7} of the value in vacuum. Three consequences follow.

The phase mode becomes heavier along the clock. In the couplings of the model, $m_\vartheta^2=C\,m_s^2\,f_D\,(|\langle\bar\psi\psi\rangle_{\rm vac}|-n_S)/[(1-f_D)\,m^3]$, which grows as e^{τ_d}: $m_\vartheta\propto e^{\tau_d/2}$, while all other masses stay fixed. Its value depends on the cutoff through the scalar density of the vacuum (Table XXVIII).

In matter the phase mode is lighter. The matter subtracts its scalar density: $m_\vartheta^2(k_F)/m_\vartheta^2(0)=1-n_S/|\langle\bar\psi\psi\rangle_{\rm vac}|$ [Table XXVIII, Fig. 19(c)].

A massive phase mode binds pairs less. The exchange of the phase mode contributes to the pairing potential (86) with the range $1/m_\vartheta$; a mass of three tenths of the fermion mass reduces the gap of the example of Table XV ($m_s=m$) from 0.0247 to 0.0197 of the Fermi kinetic energy.

XIV. ORDER PARAMETER, HIDDEN SYMMETRY AND VORTICES

A. Frame components and coordinate components

The order parameter of the superfluid is the expectation value Δ^a of the vector pair operator (78) along the extra axis $a\in\{0,5,6,7\}$: four complex numbers. On the deflating spacetime the index a is a *frame* index: Δ^a is the component along the unit vector e_a. The coordinate components along the deflating coordinates are

TABLE XXVII. Bare Dirac mass, contact couplings and Vorton-Scalar together in the two forms ($D = 3$, without the filled states): the smallest C at which uniform matter is self-bound, in parentheses the shift of the moment of binding along the deflation clock relative to the first row (negative: earlier), and the value x of the scalar field in the bound state just above the threshold.

f_D	$-g_0/C$	g_1/C	form C: C ($\Delta\tau_d$)	x	form Q: C ($\Delta\tau_d$)	x
0	0	0	3.7011 (0)	0.000	3.7011 (0)	0.000
0	1	0	3.9399 (+0.063)	0.000	3.7156 (+0.004)	0.614
0	1	0.1	3.9399 (+0.063)	0.000	4.1527 (+0.115)	0.652
0.25	0	0	2.5112 (−0.388)	0.000	2.5112 (−0.388)	0.000
0.25	1	0	2.6336 (−0.340)	0.000	2.4758 (−0.402)	0.000
0.25	1	0.1	2.6307 (−0.341)	0.000	2.6992 (−0.316)	0.000
0.5	0	0	2.0643 (−0.584)	0.000	2.0643 (−0.584)	0.000
0.5	1	0	2.1617 (−0.538)	0.000	1.8252 (−0.707)	0.000
0.5	1	0.1	2.1559 (−0.540)	0.000	1.9335 (−0.649)	0.000

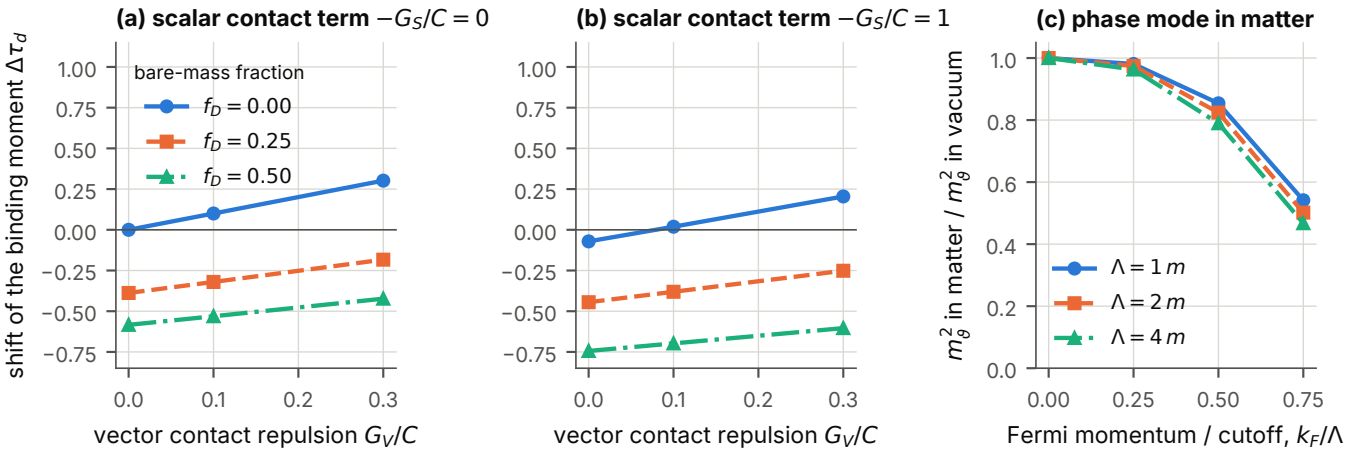


FIG. 19. Bare Dirac mass, contact couplings and Vorton-Scalar together. (a), (b) The shift of the moment at which matter becomes self-bound, in units of the deflation clock, against the vector contact coupling, for three fractions f_D of bare mass, without and with a scalar contact coupling. (c) The square of the mass of the phase mode in matter divided by its value in vacuum, against the Fermi momentum in units of the cutoff.

TABLE XXVIII. The phase mode of the Vorton-Scalar when a tenth of the fermion mass is a bare Dirac mass ($f_D = 0.1$; $D = 3$; units $m = 1$): scalar density of the vacuum, square of the mass of the phase mode in vacuum in units of $C\,m_s^2$, and its reduction in matter of the Fermi momenta $\Lambda/4$, $\Lambda/2$ and $3\Lambda/4$.

Λ	$\lvert\langle\bar\psi\psi\rangle_{\rm vac}\rvert$	$m_\vartheta^2/(C\,m_s^2)$	$k_F = \Lambda/4$	$\Lambda/2$	$3\Lambda/4$
1	0.1080	0.0120	0.981	0.854	0.541
2	0.6137	0.0682	0.974	0.824	0.502
4	2.9176	0.3242	0.963	0.790	0.467

$\Delta_t = b_t(x4)\,\Delta^t$, and they shrink with the scale factors when the frame components are constant. All statements below are about frame components.

The energy of the paired state depends on the order parameter through the 8×8 gap matrix M on the internal states of a pair. The identity

$$\tfrac{1}{8}\,\mathrm{Tr}\big[(\mathsf{M}\mathsf{M}^\dagger)^2\big] = 2\Big(\sum_a |\Delta^a|^2\Big)^2 - \Big|\sum_{a,b}\eta_{ab}\,\Delta^a\Delta^b\Big|^2, \tag{111}$$

holds on the deflating spacetime with the frame metric η_{ab} of the four axes, to 2×10^{-14}. In coordinate components the same expression contains the metric g_{tt} of the moment.

B. The ground state: no direction is preferred

Table XXIX gives the potential of the paired state for six states of the order parameter. Every *unitary* state, in which Δ^a is a real unit vector times a phase, has all eight gaps equal and the lowest potential, whichever direction the vector has. It has four flat directions, the Goldstone modes (one phase and three rotations of the direction), and four directions of positive curvature. The *half-paired* states, in which four of the eight gaps vanish, have half

TABLE XXIX. The potential V of the paired state for six states of the order parameter, in units of $N(0)\Delta_F^2$, where Δ_F is the gap of a unitary state. Each state is taken at the magnitude of the order parameter at which its potential is stationary. d_-^2 and d_+^2: the squares of the two fourfold gaps of the state, in units of Δ_F^2. Flat and positive: the numbers of eigenvalues of the matrix of second derivatives of the potential that vanish and that are positive.

state	d_-^2	d_+^2	V	flat	positive
pair along the frame axis of $x0$ (hidden-axis singlet)	1.000	1.000	−2.0000	4	4
pair along the frame axis of $x5$	1.000	1.000	−2.0000	4	4
pair along the frame axis of $x7$	1.000	1.000	−2.0000	4	4
unitary, rotated between $x0$ and $x5$	1.000	1.000	−2.0000	4	4
unitary, general direction and phase	1.000	1.000	−2.0000	4	4
half-paired: $\eta_{ab}\Delta^a\Delta^b = 0$	0.000	1.000	−1.0000	—	—

the condensation energy.

One might have expected the deflation to select the direction of the order parameter: the hidden axis $x0$, which does not deflate, against the three axes that do. *In the sector of fields that do not depend on the extra coordinates it does not.* The reason is the central observation of Sec. III: in this sector the weighted field obeys the flat equations, and their internal symmetry rotates the four frame axes into one another. The symmetry of the manifold of ground states is larger than the symmetry of the spacetime.

This holds for the model without contact self-interaction, for which Table XXIX is computed, and for a self-interaction of form Q. *A self-interaction of form C, the definition of the model, does select a direction.* It breaks the three generators that rotate the hidden axis into $x5, x6, x7$ (Sec. IV F), and it enters the pair along the hidden axis and the pairs along $x5, x6, x7$ with opposite signs, so that their gaps differ (Sec. IX E). What selects the direction is the self-interaction, through the distinction that the signature of the spacetime makes between the space-like hidden axis and the additional times; the deflation still selects nothing, because it does not act in this sector. The three flat directions of Table XXIX that rotate the hidden axis into the other three are then no longer flat. The potential of the order parameter with the contact couplings of form C has not been computed.

C. The internal symmetry of the sector, and the symmetries of the spacetime

For fields that depend on $x1, x2, x3, x4$ only, the equation of the weighted field contains five matrices: $\alpha^1, \alpha^2, \alpha^3$, β and $i\beta\chi$, the last through the coupling of the phase mode. The matrices that commute with all five generate the *internal symmetry* of the sector. The program computes it again for the deflating spacetime, item by item (Table LXXII), and finds it unchanged:

- the algebra is $su(2)_A \oplus su(2)_B \oplus u(1)_F \oplus u(1)_E$, of dimension 8;
- the eight internal states of a fermion form the two doublets $(2,1) + (1,2)$;
- the four vector pair operators (78) form the $(2,2)$, with fermion number 2, and their one invariant quadratic form is the frame metric of the four axes, with the signs $(+,-,-,-)$;
- a condensate along one axis breaks $SO(4) = [SU(2)_A \times SU(2)_B]/Z_2$ to $SO(3)$.

Here $U(1)_E$ is a second phase symmetry of the sector, with a generator P that has the square one and distinguishes the doublets of $SU(2)_A$ from those of $SU(2)_B$.

Six of the generators are the products $\gamma^A\gamma^B$ of the six pairs taken from the four axes $x0, x5, x6, x7$. On the flat 4+4 spacetime only three of them belong to transformations of the spacetime that the sector inherits: the products $\gamma^s\gamma^t$ of two additional times generate the rotations $SO(3)_T$ of $x5, x6, x7$. The other three, $\gamma^0\gamma^t$, are Hermitian; the unitary transformations that they generate in the quantized sector are not the boosts of the planes of $x0$ and xt, which are not unitary. The full $SU(2)_A \times SU(2)_B$ was therefore a *hidden symmetry* already on the flat spacetime: a symmetry of the equations of a sector that is not a transformation of the spacetime. On the deflating spacetime even $SO(3)_T$ is a symmetry of the spacetime only if the three rates are equal (Fig. 1). With two equal rates one rotation remains, and with three different rates none. The internal symmetry of the sector is nevertheless exact in all three cases.

D. Vortices and their zero modes

Because the vacuum value of the Vorton-Scalar breaks the phase symmetry $U(1)_{\rm VS}$, the field has line-like defects [28]: along a straight line, say the $x3$ axis, its magnitude vanishes, and around the line its phase winds by 2π times an integer. For unit winding, $\Phi_{\rm VS}$ is $v_{\rm VS}/\sqrt{2}$ times a profile that rises from 0 on the line to 1 at a distance of the order of $1/m_s$, times $e^{i\theta_{12}}$ with the angle θ_{12} around the line in the plane of $x1$ and $x2$.

Energy per unit length. It is proportional to the square of the four-dimensional vacuum value and so falls as $e^{-\tau_d}$: in the four-dimensional description vortex lines become lighter as the deflation proceeds, while their core, of the radius $1/m_s$, keeps its size.

Fermions on the line. On the line the fermion mass vanishes, and fermions can be trapped there [29]. In the

field of the vortex the Hamiltonian of a fermion is

$$\sum_{j=1}^{3} \alpha^j p_j + m\,(\text{profile})\,\beta\left(\cos\theta_{12} + i\chi\,\sin\theta_{12}\right), \qquad (112)$$

with the momenta p_j along $x1, x2, x3$. A state of zero energy at zero momentum along the line is a constant spinor w_z times a function that decays away from the line. Inserting this form, the Hamiltonian annihilates the state for every angle if and only if

$$\left(i\alpha^1 + \beta\right) w_z = 0, \qquad \left(i\alpha^2 + i\beta\chi\right) w_z = 0. \qquad (113)$$

The five matrices $\alpha^1, \alpha^2, \alpha^3, \beta, i\beta\chi$ are Hermitian, have the square one and anticommute with one another. The two conditions state that w_z is an eigenvector, with the eigenvalue -1, of the two commuting matrices $i\beta\alpha^1$ and $\alpha^2\beta\chi$, and each joint eigenspace of these two matrices has the dimension $16/4 = 4$. *A vortex line of unit winding carries four zero modes of the sixteen-component field.*

For a momentum p_3 along the line the energy is the eigenvalue of $\alpha^3 p_3$. On the four zero modes α^3 equals the matrix

$$P_c = i\,\alpha^1\alpha^2\alpha^3\chi, \qquad (114)$$

the product of the five matrices, which commutes with each of them, is Hermitian and traceless and has the square one; it is, up to its sign, the generator P of $U(1)_E$. Its trace on the four zero modes vanishes. *Two zero modes move along the line in one direction at the speed of light and two in the other*, and the direction of motion is tied to the charge under $U(1)_E$. For winding -1 the two directions are exchanged.

Equations (113) contain only the matrices of the frame and the fermion mass, neither of which changes. The four zero modes therefore exist at every moment of the deflation. The count also holds on flat spacetime. (The familiar count for a four-component field is one zero mode for each unit of winding; the sixteen components are equivalent to four such fields.)

Massless fermions that run along a vortex line carry a current [30]; a closed loop of the line can be held open by that current, and such loops are the *vortons* of the literature on cosmic strings [31]. Because the energy per length of the line falls as $e^{-\tau_d}$ while the fermions on it stay massless, the balance between the tension of a loop and the current on it shifts in favor of the current as the deflation proceeds. This paper establishes the ingredients of that statement; it does not compute the equilibrium radius of a loop.

XV. GRAVITY: THE DEFLATING SPACETIME WITH MATTER AS ITS SOURCE

The hypothesis (1) prescribes the metric. This section asks what the field equations of gravity make of it when the matter of Sec. VII is their source: whether such matter can produce the deflation, and what it does to a deflation that is already under way.

A. Two scale factors and the Einstein-Lovelock equations

Let the three rates be equal, and let the four flat coordinates have a scale factor of their own:

$$ds^2 = -(dx4)^2 + \mathsf{a}(x4)^2 \sum_{j=0}^{3} (dxj)^2 - \mathsf{b}(x4)^2 \sum_{t=5}^{7} (dxt)^2, \qquad (115)$$

with the rates $H_\mathsf{a} = \mathsf{a}'/\mathsf{a}$ and $H_\mathsf{b} = \mathsf{b}'/\mathsf{b}$ (a prime is a derivative with respect to $x4$). The deflating spacetime of the hypothesis is $\mathsf{a} = 1$, $\mathsf{b} = \text{const5}\, e^{-\text{g55}\, x4}$, that is, $H_\mathsf{a} = 0$ and $H_\mathsf{b} = -\text{g55}$; it is called the *undisturbed deflation* below.

For a metric of this kind the Riemann tensor is determined by one number for each pair of directions of the frame, the sectional curvature of their plane (Appendix C):

$$K_{4i} = H_i' + H_i^2, \qquad K_{ij} = H_i\, H_j \quad (i \neq j), \qquad (116)$$

where i, j run over the seven directions other than the time and H_i is the rate of the direction i. The same formulas hold for the time-like directions $x5, x6, x7$ of the 4+4 spacetime, so that everything that follows is the same on the 4+4 and on the 7+1 spacetime.The field equations of second order for the metric in eight dimensions are those of Lovelock [6], with tensors $E^{(k)\mu}{}_\nu$ of the orders $k = 1, 2, 3$ in the curvature. For the metric (115) they are diagonal, and

$$E^{(k)m}{}_m = (-1)^{k+1}\, k!\, 2^{k-1} \sum_{\substack{\text{sets of } 2k \\ \text{directions} \neq m}} \; \sum_{\substack{\text{divisions} \\ \text{into pairs}}} \; \prod_{\text{pairs}} K_{\text{pair}}, \qquad (117)$$

without summation over m: the sum runs over all sets of $2k$ directions that do not contain m, and over the ways to divide such a set into k pairs (one way for $k = 1$, three for $k = 2$, fifteen for $k = 3$). In this notation and sign $E^{(1)\mu}{}_\nu = -\mathcal{G}^\mu{}_\nu$ is minus the Einstein tensor and $E^{(2)\mu}{}_\nu = \mathcal{H}^\mu{}_\nu$ the Lanczos tensor of Sec. II C. The *Einstein-Lovelock equations* are

$$\sum_{k=1}^{3} \alpha_k\, E^{(k)\mu}{}_\nu - \bar{\Lambda}\, \delta^\mu_\nu = \kappa_8\, T^\mu{}_\nu, \qquad \alpha_1 = 1, \qquad (118)$$

with the couplings α_2, α_3 of the two higher terms, the cosmological term $\bar{\Lambda}$, and the gravitational coupling κ_8 of the eight-dimensional theory. The energy-momentum tensor of uniform matter has the components $T^4{}_4 = \varepsilon$ and $T^m{}_m = -p_m$, with the energy density and the pressures of Sec. VII. [In terms of Eq. (9): $\bar{\Lambda} = \Lambda_8$ and $\alpha_2 = -\alpha_{\text{GB}}$.]

Written out for four directions with the rate H_a and three with the rate H_b, the first two orders are

$$\begin{aligned} E^{(1)4}{}_4 &= 6H_\mathsf{a}^2 + 12H_\mathsf{a}H_\mathsf{b} + 3H_\mathsf{b}^2, \\ E^{(1)\mathsf{a}}{}_\mathsf{a} &= 3H_\mathsf{a}' + 3H_\mathsf{b}' + 6H_\mathsf{a}^2 + 9H_\mathsf{a}H_\mathsf{b} + 6H_\mathsf{b}^2, \\ E^{(1)\mathsf{b}}{}_\mathsf{b} &= 4H_\mathsf{a}' + 2H_\mathsf{b}' + 10H_\mathsf{a}^2 + 8H_\mathsf{a}H_\mathsf{b} + 3H_\mathsf{b}^2, \end{aligned} \qquad (119)$$

$$
\begin{aligned}
E^{(2)4}{}_4 = & -12H_{\mathsf{a}}^4 - 144H_{\mathsf{a}}^3H_{\mathsf{b}} - 216H_{\mathsf{a}}^2H_{\mathsf{b}}^2 \\
& - 48H_{\mathsf{a}}H_{\mathsf{b}}^3, \\
E^{(2)\mathsf{a}}{}_{\mathsf{a}} = & -12\big(H_{\mathsf{a}}^2 + 6H_{\mathsf{a}}H_{\mathsf{b}} + 3H_{\mathsf{b}}^2\big)H'_{\mathsf{a}} \\
& - 12\big(3H_{\mathsf{a}}^2 + 6H_{\mathsf{a}}H_{\mathsf{b}} + H_{\mathsf{b}}^2\big)H'_{\mathsf{b}} \\
& - 12H_{\mathsf{a}}^4 - 108H_{\mathsf{a}}^3H_{\mathsf{b}} - 180H_{\mathsf{a}}^2H_{\mathsf{b}}^2 \\
& - 108H_{\mathsf{a}}H_{\mathsf{b}}^3 - 12H_{\mathsf{b}}^4, \\
E^{(2)\mathsf{b}}{}_{\mathsf{b}} = & -16\big(3H_{\mathsf{a}}^2 + 6H_{\mathsf{a}}H_{\mathsf{b}} + H_{\mathsf{b}}^2\big)H'_{\mathsf{a}} \\
& - 16\big(3H_{\mathsf{a}}^2 + 2H_{\mathsf{a}}H_{\mathsf{b}}\big)H'_{\mathsf{b}} - 60H_{\mathsf{a}}^4 \\
& - 192H_{\mathsf{a}}^3H_{\mathsf{b}} - 136H_{\mathsf{a}}^2H_{\mathsf{b}}^2 - 32H_{\mathsf{a}}H_{\mathsf{b}}^3,
\end{aligned}
\tag{120}
$$

where the index a (b) stands for any one of the four (three) directions. Every term of the third order contains at least two factors of H_{a} or of its derivative, for example $E^{(3)4}{}_4 = 1080H_{\mathsf{a}}^4H_{\mathsf{b}}^2 + 1440H_{\mathsf{a}}^3H_{\mathsf{b}}^3$. The component along the time contains no derivative of a rate: it is a *constraint* on the initial data. The two other components are linear in H'_{a} and H'_{b} and are solved for them at every step of an integration. The polynomials (119) and (120) were obtained by the program from Eq. (117) and, independently, by computer algebra in two ways: from the Riemann tensor of the metric, and as the Euler-Lagrange equations of the reduced action for the lapse and the two scale factors.

B. A reference calculation

The program was tested on a calculation done before with other tools: gravity with three coordinates of ordinary space (scale factor a) and four further space coordinates (scale factor b), driven by the matter of the reference example of the 7+1 spacetime ($D = 7$, $G_S = -10^5$, $G_V = +10^4$ in units $m = 1$, at its saturation density), with $\kappa_8 = 1000$, $\bar\Lambda = 0$, $H_{\mathsf{a}}(0) = 0.05$ and $\mathsf{a}(0) = \mathsf{b}(0) = 1$, integrated to $x4 = 20$ in Einstein gravity and in Einstein-Lovelock gravity with $\alpha_2 = 20$, $\alpha_3 = 500$. The constraint gives $H_{\mathsf{b}}(0)$. The ten numbers of the first calculation are reproduced to 4.8×10^{-8} by the program and by the independent script (Table XXX, Fig. 20); the constraint is preserved along the integration to 1.3×10^{-14}.

TABLE XXX. The reference calculation: three space coordinates with the scale factor a and four with the scale factor b, with matter of seven space dimensions as the source (units $m = 1$). EL: Einstein-Lovelock gravity with $\alpha_2 = 20$, $\alpha_3 = 500$.

gravity	quantity	this program	first calculation
Einstein	$H_{\mathsf{b}}(0)$	0.00914371	0.00914371
Einstein	$\mathsf{a}(20)$	1.58173868	1.58173870
Einstein	$\mathsf{b}(20)$	1.19755856	1.19755860
Einstein	$H_{\mathsf{a}}(20)$	0.01187800	0.01187800
Einstein	$H_{\mathsf{b}}(20)$	0.00741911	0.00741911
EL	$H_{\mathsf{b}}(0)$	0.01498070	0.01498070
EL	$\mathsf{a}(20)$	1.47735148	1.47735150
EL	$\mathsf{b}(20)$	1.27608405	1.27608410
EL	$H_{\mathsf{a}}(20)$	0.01017591	0.01017591
EL	$H_{\mathsf{b}}(20)$	0.00848876	0.00848876

C. The undisturbed deflation as a solution

Without matter. For $H_{\mathsf{a}} = 0$, $H_{\mathsf{b}} = -\mathrm{g55}$ and constant rates, Eqs. (119) and (120) give $E^{(1)4}{}_4 = 3\mathrm{g55}^2$, $E^{(1)\mathsf{a}}{}_{\mathsf{a}} = 6\mathrm{g55}^2$, $E^{(1)\mathsf{b}}{}_{\mathsf{b}} = 3\mathrm{g55}^2$, $E^{(2)\mathsf{a}}{}_{\mathsf{a}} = -12\mathrm{g55}^4$, and zero for the other components. The three equations (118) without matter hold if and only if

$$
\bar\Lambda = 3\,\mathrm{g55}^2, \qquad \alpha_2 = \frac{1}{4\,\mathrm{g55}^2}, \tag{121}
$$

which is Eq. (11); α_3 is arbitrary. The gravity with these two constants is called the *tuned* Einstein-Lovelock gravity below. At this solution the coefficients of $(H'_{\mathsf{a}}, H'_{\mathsf{b}})$ in the two evolution equations form the matrix $\mathrm{diag}(-6, 2)$, which can be inverted: the evolution of a perturbed solution is well defined.

In Einstein gravity. Without the Gauss-Bonnet term and without $\bar\Lambda$ the same spacetime requires a source with

$$
\kappa_8\,\varepsilon = 3\,\mathrm{g55}^2, \qquad p_{\mathsf{a}} = -2\,\varepsilon, \qquad p_{\mathsf{b}} = -\varepsilon, \tag{122}
$$

a tension along the flat coordinates twice as large as the energy density, and a tension along the deflating coordinates equal to it. *The matter of Sec. VII cannot be this source.* Its pressure along a coordinate on which the fields do not depend is the interaction pressure p_{int} of Eq. (73), and $\varepsilon + p_{\mathrm{int}} = \langle E^* \rangle + G_V\, n^2$, which is positive for a repulsive or vanishing vector coupling (for the two halves of the field, $\varepsilon + p_{\mathrm{int}} = \sum_{\pm} \langle E^*_{\pm} \rangle + G_V^{\mathrm{tot}}\, n^2 + G_V^{\mathrm{dif}}\, n_W^2$, which is positive for the sets A and B in both forms); $p_{\mathsf{b}} = -\varepsilon$ is not reached. Something else must hold the deflation: an *agent* with the equation of state (122), or the tuned Gauss-Bonnet term.

D. Matter on the deflating spacetime

Now let matter be present, with the initial data $\mathsf{a} = \mathsf{b} = 1$, $H_{\mathsf{a}} = 0$, and H_{b} from the constraint. Its amount is given by the *loading*, $\kappa_8\,\varepsilon(0)/(3\,\mathrm{g55}^2)$, the energy density at the start in units of the energy density that would make three coordinates deflate at the rate g55 in Einstein gravity. The rate is $\mathrm{g55} = 0.01\,m$. The matter is that of the quantized sector, whose fields do not depend on $x5, x6, x7$ ($D = 4$): self-bound matter of set B at three times its binding threshold, at its saturation density, in the two forms (Table XXXI). In form C one half of the field is filled, in form Q both halves equally. In an evolution without collisions every particle keeps its state, so that a half that is empty stays empty: the matter of form C is then matter of one kind of particle with four internal states and the couplings G_S^{same}, G_V^{same} throughout the evolution, and the matter of form Q is matter of one kind

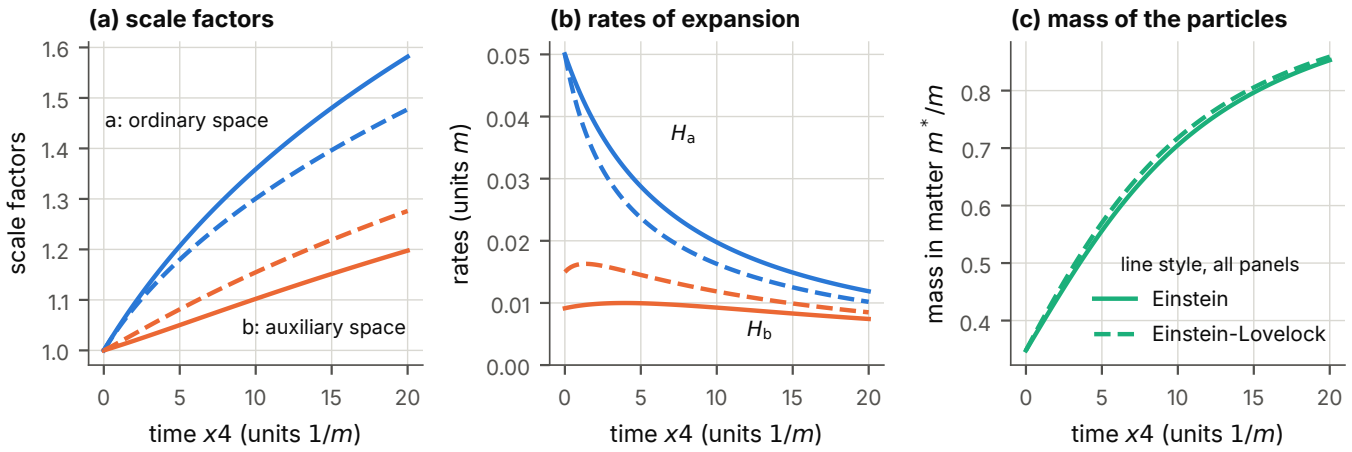


FIG. 20. The reference calculation of Table XXX: (a) the two scale factors, (b) their rates, (c) the mass of the particles in the matter, which rises as the matter is diluted. Solid lines: Einstein gravity; dashed lines: Einstein-Lovelock gravity.

TABLE XXXI. The matter of the evolutions of Table XXXII: self-bound matter of set B at three times the binding threshold of its form ($D = 4$; units $m = 1$). Second column: the strength g at the binding threshold. The Fermi momentum k_{F+} and the mass m^*_+ are those of the filled half in form C (taken as the half $W_{1644} = +1$) and of each of the two halves in form Q.

form	$g\,m^3$	state	k_{F+}/m	m^*_+/m	$\varepsilon/n - m$
C	332.2917	one half filled	0.52859	0.37920	−0.09072
Q	121.0858	equal halves	0.58171	0.35844	−0.10601

with eight states and the couplings $G_S^{\rm HF}$, $G_V^{\rm HF}$ (both were computed again with the routines for matter of one kind, with agreement to 1.3×10^{-15}). The pressure along the deflating coordinates is $p_{\rm int}$. The integration runs until the clock of the undisturbed deflation, $\tau_d^0 = 3\,\mathrm{g55}\,x4$, reads 6, or until a rate exceeds $50\,\mathrm{g55}$ (Table XXXII, Fig. 21).

Matter makes the deflation faster than exponential and makes the flat coordinates expand. In the tuned gravity the smaller loading leaves the deflation close to the undisturbed one for about four units of the clock; then the rates grow, and at $\tau_d^0 = 6$ the three coordinates deflate 34.1 times faster than at the start for the matter of form C, and 15.6 times for that of form Q, while the four flat coordinates expand. With the larger loading the rates exceed $50\,\mathrm{g55}$ before the clock reads 4: they grow without bound in a finite time. In Einstein gravity with the matter as the only source the initial rate is $-H_{\rm b}(0) = \mathrm{g55} \times (\mathrm{loading})^{1/2}$, far below g55 for a small loading, and it grows as the matter is compressed. In four of the eight evolutions of the table the rates run away. The two forms give matter of different composition and behave alike in kind: the conclusions of this section do not depend on the form (Fig. 22 shows the evolutions with the matter of form Q). The constraint is preserved along every evolution to 6.7×10^{-10}.

E. The undisturbed deflation is unstable

The behavior just described is not caused by the matter alone. Near the undisturbed deflation, with $H_{\rm b} = -\mathrm{g55} + \delta H_{\rm b}$ and to the lowest order in $\delta H_{\rm b}$, $H_{\rm a}$ and the energy density, the constraint of the tuned gravity reads

$$-6\,\mathrm{g55}\,\delta H_{\rm b} - 48\,H_{\rm a}^2 = \kappa_8\,\varepsilon. \qquad (123)$$

The terms linear in $H_{\rm a}$ cancel between the Einstein and the Gauss-Bonnet part, which is why $H_{\rm a}$ enters quadratically. Since $\varepsilon \geq 0$, the left-hand side must not be negative: *initial data exist only for a deflation that is at least as fast as the undisturbed one*, $\delta H_{\rm b} \leq 0$. For a deflation exactly at the rate g55 and without matter the constraint is $H_{\rm a}^2\,(H_{\rm a}^2 - 12\,\mathrm{g55}H_{\rm a} + 16\,\mathrm{g55}^2) = 0$, with the roots 0, 0 and $(6 \pm \sqrt{20})\,\mathrm{g55}$: the undisturbed deflation is a double root, on the edge of the allowed initial data.

For a deflation faster by one part in a thousand and no matter, the constraint has the two roots $H_{\rm a}(0) = \pm\mathrm{g55}\,(0.001/8)^{1/2}$ to the lowest order. Starting from either, the deflation does not return to the rate g55: it becomes faster and faster [Table XXXIII, Fig. 21(c)]. *The exactly exponential deflation is an unstable solution of the gravity in which it is a solution without matter*; every allowed perturbation, with or without matter, grows.

Three conclusions follow for the hypothesis. First, the matter of the model does not produce the deflation; an agent with the equation of state (122), or a tuned Gauss-Bonnet term, is needed. Second, with the Gauss-Bonnet term the deflation is not stable, so that the Gauss-Bonnet term alone does not explain a deflation that lasts. Third, whatever holds the deflation must also absorb the reaction of the matter, which pushes toward a faster deflation and an expansion of the flat coordinates; the sections on cosmology assume such an agent and say so (assumption A2 of Sec. XVI).

TABLE XXXII. The deflating spacetime with matter as a source of gravity. Tuned EL: Einstein-Lovelock gravity with the constants (121), in which the undisturbed deflation is a solution without matter; Einstein: Einstein gravity with the matter as the only source. The last five columns give the state at the end of the integration; "runaway" marks an evolution in which a rate exceeded 50 g55 before $\tau_d^0 = 6$.

matter	gravity	loading	$H_\mathrm{b}(0)/\mathrm{g55}$	end	τ_d^0	a	b	$H_\mathrm{a}/\mathrm{g55}$	$H_\mathrm{b}/\mathrm{g55}$
4+4, form C	tuned EL	10^{-3}	-1.0005	reached	6.00	1.5963	0.01521	$+6.2500$	-34.104
4+4, form C	tuned EL	10^{-1}	-1.0488	runaway	3.51	1.6307	0.02463	$+9.1436$	-50.090
4+4, form C	Einstein	10^{-3}	-0.0316	reached	6.00	1.0014	0.9332	$+0.0015$	-0.038
4+4, form C	Einstein	10^{-1}	-0.3162	runaway	4.01	1.5871	0.09687	$+11.7156$	-50.030
4+4, form Q	tuned EL	10^{-3}	-1.0005	reached	6.00	1.4087	0.02928	$+2.8489$	-15.606
4+4, form Q	tuned EL	10^{-1}	-1.0488	runaway	3.57	1.6321	0.02386	$+9.1359$	-50.050
4+4, form Q	Einstein	10^{-3}	-0.0316	reached	6.00	1.0015	0.9332	$+0.0016$	-0.038
4+4, form Q	Einstein	10^{-1}	-0.3162	runaway	4.02	1.6299	0.09367	$+12.2541$	-50.124

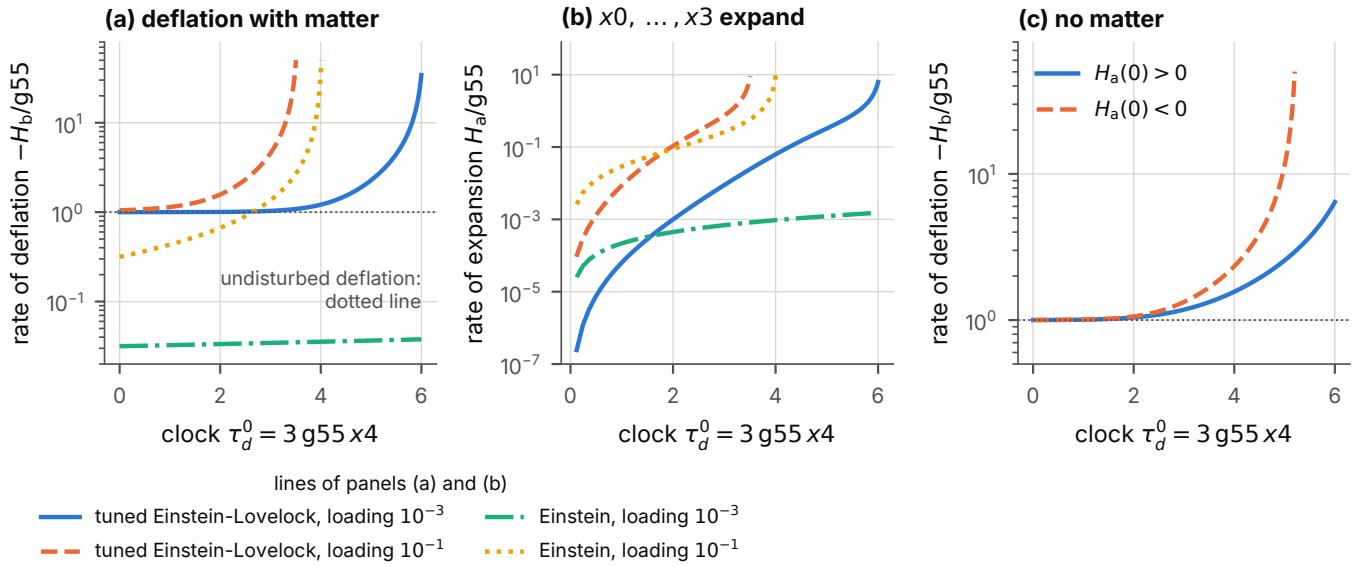


FIG. 21. The deflating spacetime with matter as a source of gravity (matter of form C, one half of the field filled; the runs of Table XXXII). (a) The rate of deflation $-H_\mathrm{b}$ in units of g55 against the clock of the undisturbed deflation. (b) The rate H_a at which the four flat coordinates expand. (c) The tuned Einstein-Lovelock gravity without matter, started with a deflation one part in a thousand faster than the undisturbed one: the two initial data that the constraint allows.

TABLE XXXIII. The tuned Einstein-Lovelock gravity without matter, started at $H_\mathrm{b}(0) = -1.001\,\mathrm{g55}$ with the two values of $H_\mathrm{a}(0)$ that the constraint allows: the state at the end of the integration (at $\tau_d^0 = 6$, or where a rate exceeded 50 g55).

$H_\mathrm{a}(0)/\mathrm{g55}$	τ_d^0 at the end	$H_\mathrm{a}/\mathrm{g55}$	$H_\mathrm{b}/\mathrm{g55}$
$+0.01147$	6.00	$+1.7334$	-6.422
-0.01088	5.20	-0.0050	-50.155

XVI. COSMOLOGY

Hypothesis H3 says that the superfluid explains the dark matter, or the Hubble tension, or both. The model offers a candidate for each. For the Hubble tension it is the phase mode of the Vorton-Scalar, a massless boson that would act as radiation in addition to photons and neutrinos (*dark radiation*) and make the early universe expand faster. For the dark matter it is the fermions themselves, as a degenerate gas or as a condensate of bound pairs. This section computes what each candidate gives when $x5, x6, x7$ deflate; the case of static extra coordinates is the case of vanishing rates, gsum = 0, and appears in every table below as the row or column without deflation. The deflation enters cosmology in several ways, and each requires an assumption beyond the hypothesis of deflation. The assumptions are stated first and are kept apart throughout, so that the reader can see which conclusion rests on which.

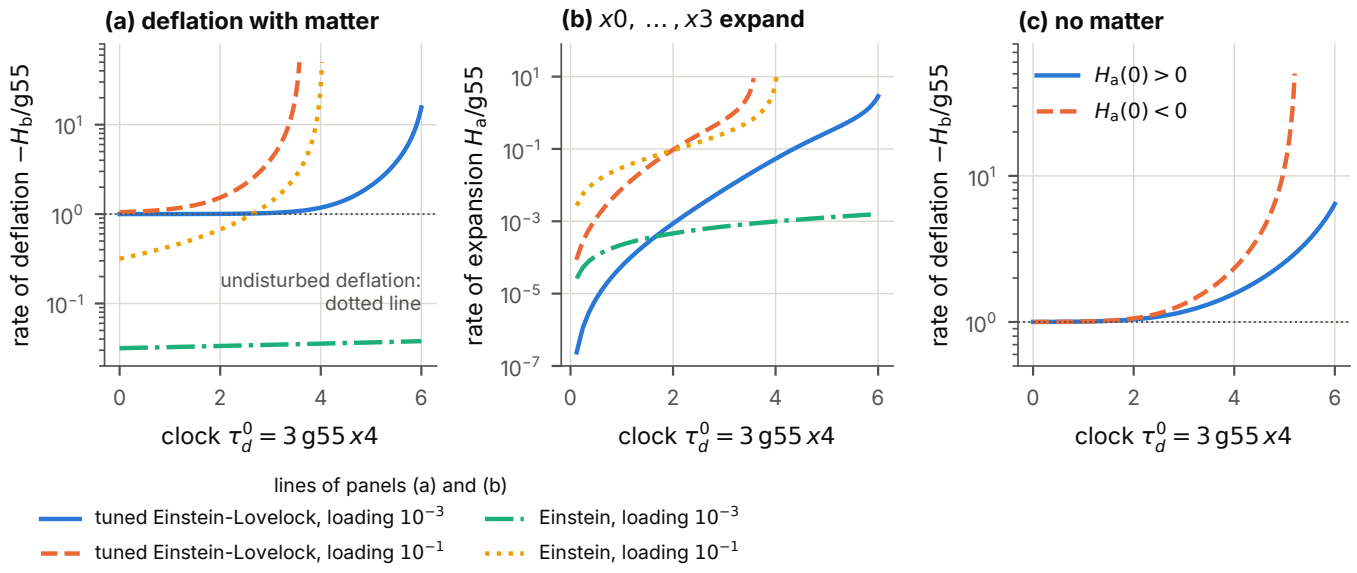


FIG. 22. As Fig. 21, with the matter of form Q (both halves of the field equally filled) as the source.

A. Assumptions, and the equation of the expansion

A1: ordinary space expands. The coordinates $x1, x2, x3$ carry the scale factor $a_{\rm sp}(x4)$ of standard cosmology, with the Hubble rate $H = d\ln a_{\rm sp}/dx4$; the hidden coordinate $x0$ is static; $x5, x6, x7$ deflate at constant rates. The metric of Sec. II is the case $a_{\rm sp} = 1$.

A2: gravity is the Einstein theory in eight dimensions, with a constant G_8. The constant of gravitation that an observer in ordinary space measures is then G_8 divided by the proper volume of the extra coordinates,

$$G_N(x4) \;\propto\; \frac{1}{V_d} \;\propto\; e^{\mathrm{gsum}\,x4}, \qquad \frac{1}{G_N}\frac{dG_N}{dx4} = \mathrm{gsum}. \tag{124}$$

Newton's constant *grows* as the extra volume shrinks: gravity was weaker in the past. That a changing volume of extra dimensions makes the constants of four dimensions change, and that nucleosynthesis bounds the change, was pointed out long ago [32]; for a review of varying constants see Ref. [33]. By Sec. II C, A2 is an assumption that the deflating spacetime itself puts in doubt: without matter the metric solves the equations of Einstein-Gauss-Bonnet gravity, not Einstein's.

For a diagonal metric whose seven directions other than the time have the rates H, H, H (ordinary space), 0 (the coordinate $x0$) and $-\mathrm{g55}, -\mathrm{g66}, -\mathrm{g77}$, the component of Einstein's equations along the time is the sum of the products of all pairs of rates, set equal to $8\pi G_8$ times the energy density in eight dimensions. Dividing the density by the extra volume turns G_8 into G_N:

$$3H^2 - 3\,\mathrm{gsum}\,H + \mathrm{gprod} = 8\pi\,G_N(x4)\,\rho, \tag{125}$$

with the density ρ of radiation, matter and the cosmological term in ordinary space. For the unperturbed universe Eq. (125) is the only one of Einstein's equations that is needed. The others determine the stresses of the unknown source that keeps the three rates constant; they enter when ordinary space is perturbed (Sec. XVI E).

The calculations take three equal rates, so that $\mathrm{gprod} = \mathrm{gsum}^2/3$, and the densities of the standard model of cosmology [34]: baryons $\omega_b = 0.02237$, cold dark matter $\omega_c = 0.1200$, and $H_0 = 67.36\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ without deflation. The cosmological term is chosen so that Eq. (125) holds today. Without deflation the program gives the two numbers on which this section rests: the angle θ_* under which the sound horizon at last scattering is seen, $100\,\theta_* = 1.03965$, and that sound horizon, $r_s = 144.449$ Mpc.

B. The Hubble constant inferred from the sound horizon

The microwave background fixes the angle θ_* to high precision. A model that changes the history of the expansion must keep that angle, and the Hubble constant that it then requires is the *inferred* Hubble constant. The tension is that the standard model infers $67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ [34] while the local distance network measures 73.50 ± 0.81 [35]. Table XXXIV gives the inferred value on the deflating spacetime under A1 and A2, at fixed θ_* and fixed densities of baryons and dark matter. The redshift z_* of last scattering and the mass fraction $Y_{\rm He}$ of helium, which the sound horizon needs, are computed on the deflating background (Sec. XVI G).

The inferred Hubble constant falls, with the slope $d\ln H_0/d(\mathrm{gsum}/H_0) = -0.969$ at small rates. The reason is this. By Eq. (124) gravity was weaker in the early universe; the universe expanded more slowly at a given density; sound had more time to travel, and the sound

TABLE XXXIV. The Hubble constant inferred from the angular scale of the sound horizon if Newton's constant grows with the deflation (assumptions A1 and A2). G_N ratio: Newton's constant in the early universe divided by its value today, $e^{-\mathrm{gsum}\, t_0}$ with the age t_0 of the model. z_*: redshift of last scattering; Y_{He}: mass fraction of helium. Last column: the Hubble constant that results if z_* and Y_{He} are held at their values without deflation. The rows at $\mathrm{gsum}/H_0 = 0.00324$ and 0.0644 are the bounds of lunar laser ranging and of nucleosynthesis (Table XXXV).

gsum/H_0	gsum (per year)	H_0 ($\mathrm{km\,s^{-1}\,Mpc^{-1}}$)	change	G_N ratio	z_*	Y_{He}	r_s (Mpc)	t_0 (Gyr)	H_0, z_* and Y_{He} held
0	0	67.36	0.00	1.0000	1089.92	0.2454	144.45	13.81	67.36
0.00100	6.89×10^{-14}	67.29	−0.07	0.9990	1089.90	0.2453	144.52	13.82	67.30
0.00300	2.07×10^{-13}	67.16	−0.20	0.9971	1089.86	0.2452	144.66	13.84	67.18
0.00324	2.23×10^{-13}	67.15	−0.21	0.9969	1089.86	0.2451	144.68	13.84	67.16
0.0100	6.89×10^{-13}	66.70	−0.66	0.9905	1089.73	0.2446	145.16	13.90	66.74
0.0300	2.07×10^{-12}	65.36	−2.00	0.9714	1089.37	0.2430	146.61	14.06	65.48
0.0644	4.43×10^{-12}	62.98	−4.38	0.9383	1088.72	0.2401	149.23	14.37	63.23
0.100	6.89×10^{-12}	60.37	−6.99	0.9037	1088.04	0.2370	152.13	14.70	60.77
0.300	2.07×10^{-11}	41.94	−25.42	0.6995	1083.78	0.2168	173.35	17.29	43.41

horizon is larger. To keep the angle, the distance to last scattering must be larger, which requires a smaller Hubble constant. The deflation moves the inferred value *away* from the local measurement [Fig. 23(a)]. Weaker gravity also delays last scattering and lowers the helium fraction, because the universe expanded more slowly during recombination and during nucleosynthesis as well; both enlarge the sound horizon further. If z_* and Y_{He} are held at their values without deflation, the slope is −0.911: that simpler calculation underestimates the fall by six per cent of itself. An independent calculation of recombination confirms both numbers (Appendix E). The term gprod of Eq. (125) plays no part in this: it is of the second order in the rate, and it acts as a constant, which the cosmological term that is adjusted to today's expansion absorbs almost exactly.

C. How fast can the coordinates deflate today?

Table XXXV and Fig. 24 collect the bounds.

Under A2, lunar laser ranging measures $(dG_N/dt)/G_N$ directly, and Eq. (124) turns its limit into $\mathrm{gsum} < 2.23 \times 10^{-13}$ per year, that is, $3.24 \times 10^{-3}\, H_0$. At this rate the deflation clock has advanced by $\mathrm{gsum}\, t_0 = 3.1 \times 10^{-3}$ over the age t_0 of the universe, and the inferred Hubble constant is lower by $0.21\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.

Under A2, nucleosynthesis bounds the ratio of Newton's constant then to its value today; for a rate that was constant since, $\mathrm{gsum} < 0.064\, H_0$. This is weaker than the first bound but does not assume that the rate today equals the rate in the past.

Under HV, nucleosynthesis bounds the change of the Fermi constant at fixed fermion masses to 10 to 20 per cent, which gives $\mathrm{gsum} < 0.111$ to $0.234\, H_0$. HV is the assumption that the Vorton-Scalar itself plays the role of the Higgs field, so that $G_{\mathrm{Fermi}} = 1/(\sqrt{2}\, v_{\mathrm{VS}}^2)$ grows as e^{τ_d} while every mass stays constant. It is *not* the hypothesis H2, and the Vorton-Scalar, which is a singlet of every non-Abelian symmetry of the model, cannot be the Higgs field of the Standard Model, which is a doublet; the bound is given only to show what the assumption would imply.

Under the hypothesis H2, atomic clocks give by far the strongest bound, and one that needs no assumption about gravity. Section XVII derives it: $\mathrm{gsum} < 1.16 \times 10^{-6}\, H_0/n_{\mathrm{EW}}$ with an exponent n_{EW} of order one or larger.

The fine-structure constant is bounded by the same clocks to parts in 10^{18} per year. This would bound the rate, to $4.6 \times 10^{-8}\, H_0$, only if an electromagnetic field propagated along the deflating coordinates. The model contains no electromagnetic field; the row shows how strongly such an extension would be constrained.

Under A2, the microwave background would bound the rate as well. Section XVI H finds, as a forecast for mock data of the quality of the Planck satellite and not as a measurement, $\mathrm{gsum} < 0.0385\, H_0$; Fig. 24 includes it for comparison.

In the model as it stands, without H2, HV or an electromagnetic extension and without A2, no mass and no coupling of ordinary matter depends on the deflation, and none of these observations bounds the rate.

The consequence for the histories of Secs. VIII to XII is severe. Under A2 the clock has advanced by at most 0.0031 during the whole of cosmic history, if the rate was constant; under H2 by at most $1.1 \times 10^{-6}/n_{\mathrm{EW}}$. A history in which pairs become bound, matter loses its fermion mass or the gap grows by orders of magnitude within the age of the universe requires that gravity does not feel the deflation, that the superfluid is not the Higgs field, or that the rate was much larger in the past than it is today.

D. Bound pairs lose rest mass

If the dark matter consists of bound pairs of the fermions, its particles have the rest mass $2m$ minus the

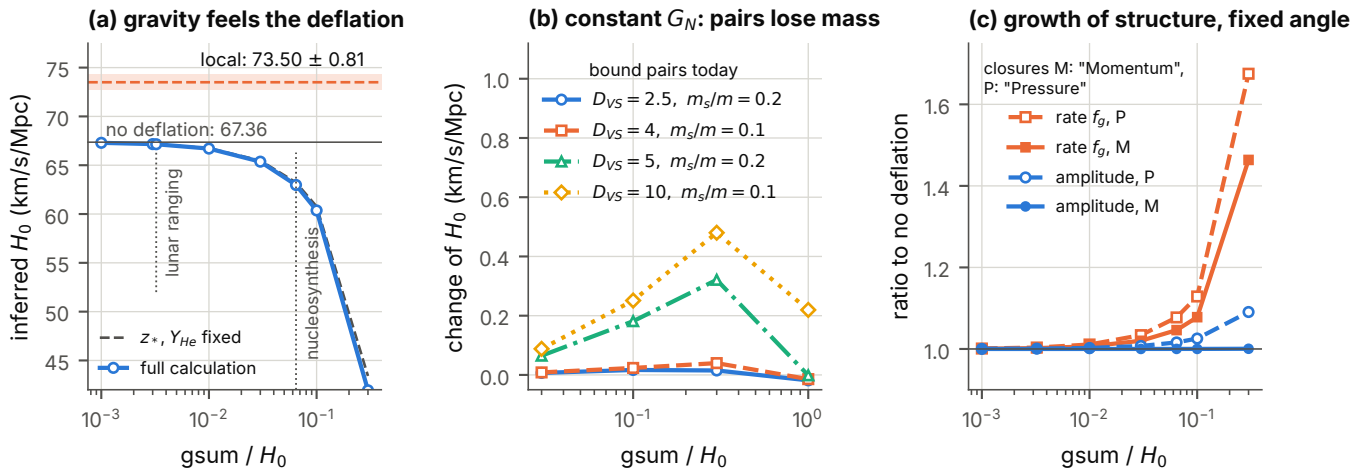


FIG. 23. The Hubble constant and the growth of structure on a deflating spacetime. (a) The Hubble constant inferred from the angular scale of the sound horizon against the rate of deflation, if gravity feels the deflation (A1, A2): with recombination computed on the deflating background, and with last scattering and the helium fraction held fixed; the bounds of Table XXXV and the local measurement are marked. (b) The change of the inferred Hubble constant if Newton's constant is held fixed and bound pairs lose rest mass. (c) Amplitude and rate of the growth of structure relative to the model without deflation, at a fixed angle of the sound horizon, for the two closures of Sec. XVI E.

TABLE XXXV. Upper bounds on the present total rate of deflation, each under the assumption named. Uncertainties of the measured values are at 68 per cent unless stated; the bounds are at two standard deviations. The bound in units of H_0 uses $H_0 = 67.36\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. A2: gravity is the Einstein theory in eight dimensions. HV: the vacuum value of the Vorton-Scalar sets the electroweak scale. H2: the superfluid state is the Higgs field; the bound is given for $n_{\mathrm{EW}} = 1$ and scales as $1/n_{\mathrm{EW}}$ (Sec. XVII).

observation	measured	assumption	bound on gsum (per year)	in units of H_0
lunar laser ranging [36]	$(dG_N/dt)/G_N = (7.1 \pm 7.6) \times 10^{-14}$ per year	A2	2.230×10^{-13}	3.237×10^{-3}
nucleosynthesis, Newton's constant [37]	G_N then, divided by G_N today: $0.99^{+0.06}_{-0.05}$ (two standard deviations)	A2; constant rate	4.434×10^{-12}	0.0644
nucleosynthesis, Fermi constant [38]	G_{Fermi} at $t = 1$ s within 20 (10) per cent of today's value, at fixed fermion masses	HV; constant rate	1.615×10^{-11} (7.627×10^{-12})	0.234 (0.111)
atomic clocks, mass ratio [39]	$(d/dt)\ln(m_p/m_e) = (-8 \pm 36) \times 10^{-18}$ per year	H2, $n_{\mathrm{EW}} = 1$	8.000×10^{-17}	1.161×10^{-6}
atomic clocks, fine-structure constant [39]	$(d\alpha_{\mathrm{em}}/dt)/\alpha_{\mathrm{em}} = (1.0 \pm 1.1) \times 10^{-18}$ per year	an electromagnetic field along $x5, x6, x7$ (not part of the model)	3.200×10^{-18}	4.645×10^{-8}

binding energy. The binding energy grows with the coupling, so that the pairs become lighter as the deflation proceeds, and the density of dark matter falls slightly faster than the inverse volume. A dark matter that was heavier in the past raises the inferred Hubble constant. The program computes the effect for four examples with α_{VS} between 0.4 and 1 today, where the binding energy is at most 8.2 per cent of the rest mass [Fig. 23(b), Table LVI]. If gravity feels the deflation, the inferred Hubble constant changes by between −4.11 and $-0.19\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ at the rates that nucleosynthesis and lunar laser ranging allow: the weaker gravity of the past outweighs the loss of mass. (These examples hold z_* and Y_{He} fixed, which by Table XXXIV underestimates a fall by about six per cent of itself.) If Newton's constant is held fixed and the deflation acts on the dark sector alone, rates between 0.03 and $1\,H_0$ raise the inferred Hubble constant by at most $+0.48\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, for a loss of up to 7.7 per cent of the rest mass. This is the only effect of the deflation found here that raises the inferred Hubble constant, and in these examples it supplies less than a tenth of the $6.1\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ that separate the model from the local measurement.

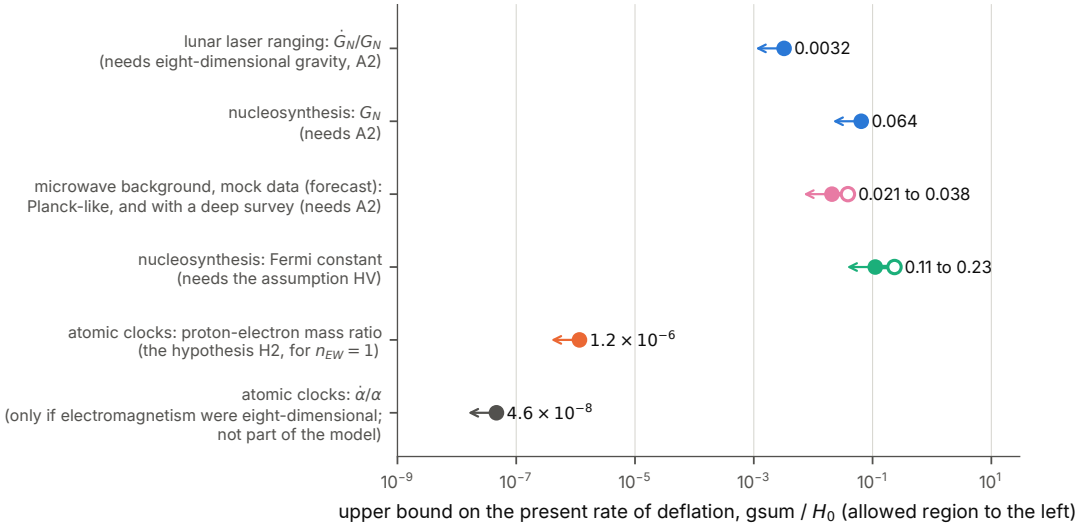


FIG. 24. Upper bounds on the present total rate of deflation in units of the Hubble constant, each under the assumption of Table XXXV. The allowed region is to the left of each bound. The row "microwave background" is not a measurement but the forecast of Sec. XVI H for mock data (open symbol: Planck-like; filled symbol: with a deep survey).

E. Perturbations of ordinary space: what must be assumed

The quantities of the preceding subsections belong to the unperturbed universe. The growth of structure and the spectra of the microwave background are built from small perturbations of ordinary space, and the equations of the perturbations contain more of Einstein's equations than Eq. (125). They therefore contain what the hypothesis of deflation leaves open.

The agent. Something must keep the three rates constant while ordinary space expands and matter dilutes. We call it the agent and assume nothing about its nature. Assumption A2 says that it has no energy density: otherwise its density would stand on the right of Eq. (125). It cannot be without stresses. The components of Einstein's equations along ordinary space, together with the conservation of matter in ordinary space and Eq. (124), require the agent to have, in ordinary space, the pressure

$$8\pi G_N\, p_{\text{agent}} = -\frac{\text{gsum}}{H}\frac{dH}{dx4} - 2\,\text{gsum}\,H + 2\,\text{gprod} - \frac{\text{gsum}\,\text{gprod}}{3H}. \tag{126}$$

In an era of matter this is $-\text{gsum}\,H/2$ to the first order in the rates. The agent is a medium without energy density and with a tension, of the relative size gsum/H. When ordinary space is perturbed, such a medium will in general be perturbed too.

A3: the perturbations. We assume (i) that in a universe with small inhomogeneities the scale factors b_t of the deflating coordinates remain the prescribed functions of the time, the time being the proper time of observers who move with the cold matter, and that $x0$ remains static; and (ii) that the agent has no energy density in the perturbed universe either. Part (i) is the plainest reading of the hypothesis of deflation in a universe that is not exactly homogeneous; part (ii) is A2 taken to first order. Both are assumptions.

The equations. We use the conformal time τ_{cf} of ordinary space, defined by $d\tau_{\text{cf}} = dx4/a_{\text{sp}}$, write a prime for the derivative with respect to it, and write $H_{\text{cf}} = a'_{\text{sp}}/a_{\text{sp}} = a_{\text{sp}}H$ for the conformal Hubble rate. A scalar perturbation of ordinary space with the comoving wavenumber k is described, in the synchronous gauge, by two functions $h_{\text{sy}}(\tau_{\text{cf}})$ and $\eta_{\text{sy}}(\tau_{\text{cf}})$ of the perturbed metric; they are the h and η of Ref. [40], written here with an index because h and η have other meanings in this paper. The combination $\alpha_{\text{sy}} = (h'_{\text{sy}} + 6\,\eta'_{\text{sy}})/2k^2$ occurs repeatedly. Matter and radiation enter through four sums over the species, each multiplied by $8\pi G_N(x4)\,a_{\text{sp}}^2$ with the Newton's constant of the moment: $\mathcal{D}$, the perturbation of the energy density; $\mathcal{T}$, the divergence of the momentum density; $\mathcal{P}$, the perturbation of the pressure; and $\mathcal{A}$, the anisotropic stress. With the scale factors of the extra coordinates unperturbed, the Einstein tensor of the eight-dimensional metric gives, to first order,

$$2k^2\eta_{\text{sy}} - H_e\,h'_{\text{sy}} = -\mathcal{D}, \tag{127a}$$
$$2k^2\eta'_{\text{sy}} = \mathcal{T}, \tag{127b}$$
$$h''_{\text{sy}} + 2H_e\,h'_{\text{sy}} - 2k^2\eta_{\text{sy}} = -3\,\mathcal{P}, \tag{127c}$$
$$\alpha'_{\text{sy}} + 2H_e\,\alpha_{\text{sy}} - \eta_{\text{sy}} = -\tfrac{3}{2}\,\mathcal{A}/k^2, \tag{127d}$$

with

$$H_e = H_{\text{cf}} - \tfrac{1}{2}\,a_{\text{sp}}\,\text{gsum}. \tag{128}$$

They are the energy constraint, the momentum constraint, the trace of the space-space equations and their traceless part, the equation of the shear. An independent

program derives them by computer algebra for arbitrary rates and for both signatures (Appendix E). They are the four equations of general relativity [40] with two changes: Newton's constant is that of the moment, and $H_{\rm cf}$ is replaced by H_e in three of them. The second change can be read off Eq. (125). The time component of the Einstein tensor is the sum of the products of all pairs of expansion rates. If the rate of one ordinary direction is perturbed, the perturbation multiplies the sum of the rates of all *other* directions, which is $2H$ in four dimensions and $2H - \text{gsum}$ here.

The four equations cannot all hold with matter alone. In general relativity two of the four equations (127) imply the other two, because the energy and the momentum of matter are conserved. Here they do not. Let Eq. (127a) hold, and define three *gaps*, the amounts by which the other three fail when only matter and radiation stand on their right-hand sides:

$$\begin{aligned} \mathcal{Y}_M &= 2k^2\eta'_{\rm sy} - \mathcal{T}, \\ \mathcal{Y}_T &= h''_{\rm sy} + 2H_e\, h'_{\rm sy} - 2k^2\eta_{\rm sy} + 3\,\mathcal{P}, \\ \mathcal{Y}_S &= \alpha'_{\rm sy} + 2H_e\,\alpha_{\rm sy} - \eta_{\rm sy} + \tfrac{3}{2}\,\mathcal{A}/k^2. \end{aligned} \tag{129}$$

The conservation of the energy and of the momentum of matter in ordinary space gives two identities between them,

$$\mathcal{Y}_M = H_e\,\mathcal{Y}_T + \mathcal{Z}, \tag{130}$$

$$2k^2\,\mathcal{Y}_S = \mathcal{Y}_T + \frac{3}{k^2}\big(\mathcal{Y}'_M + 2H_e\,\mathcal{Y}_M\big), \tag{131}$$

with

$$\begin{aligned} \mathcal{Z} &= h'_{\rm sy}\,\mathcal{B} - \tfrac{1}{2}\,a_{\rm sp}\,\text{gsum}\,(\mathcal{D} - 3\,\mathcal{P}), \\ \mathcal{B} &= H'_e - H_e^2 + \tfrac{1}{2}(\mathcal{E} + \mathcal{E}_P). \end{aligned} \tag{132}$$

Here $\mathcal{E}$ and $\mathcal{E}_P$ are $8\pi G_N a_{\rm sp}^2$ times the density and the pressure of the unperturbed universe. By Eq. (125) the quantity $\mathcal{B}$ is proportional to gsum; in an era of matter it is $a_{\rm sp}\,\text{gsum}\,H_{\rm cf}/4$. Without deflation $\mathcal{B}$ and $\mathcal{Z}$ vanish, and Eqs. (130) and (131) say that if one gap vanishes at all times, all three do: that is general relativity. With deflation $\mathcal{Z} \neq 0$, and *the three gaps cannot all vanish.* This is not a defect of the calculation. It is the statement that the agent, which has a tension, takes part in the perturbations: the gaps are, up to the factor $8\pi G_N a_{\rm sp}^2$, the perturbations of its momentum density, of its pressure and of its anisotropic stress.

Closures. The two identities relate three gaps; one more relation closes the system, and the hypothesis offers no ground for a choice. Three natural choices set one gap to zero. In the closure "Momentum", $\mathcal{Y}_M = 0$: the agent carries no momentum density, and Eqs. (127a) and (127b) hold. In the closure "Pressure", $\mathcal{Y}_T = 0$: the agent carries no perturbation of its pressure, Eqs. (127a) and (127c) hold, and Eq. (127b) is replaced by $2k^2\eta'_{\rm sy} = \mathcal{T} + \mathcal{Z}$. In the closure "Shear", $\mathcal{Y}_S = 0$: the agent carries no anisotropic stress, as a perfect fluid or a scalar field would not. Eliminating $\mathcal{Y}_T$ from the two identities then gives an equation of motion for the momentum density of the agent,

$$\mathcal{Y}'_M = -\Big(\frac{k^2}{3H_e} + 2H_e\Big)\mathcal{Y}_M + \frac{k^2}{3H_e}\,\mathcal{Z}. \tag{133}$$

Far inside the horizon, $k \gg H_{\rm cf}$, the rate $k^2/3H_e$ is enormous, and $\mathcal{Y}_M$ relaxes at once to $\mathcal{Z}$, its value in the closure "Pressure". *Inside the horizon the closure "Shear" is the closure "Pressure"*, with corrections of the relative order $(H_{\rm cf}/k)^2$. We compute with the closures "Momentum" and "Pressure" and mark every result as depending on the closure or not.

All gaps are proportional to $a_{\rm sp}\,\text{gsum}/H_{\rm cf} = \text{gsum}/H$. At the largest rate that nucleosynthesis allows this ratio is 2.7×10^{-6} at recombination and 6.9×10^{-2} today. The closure therefore cannot matter for anything that is decided at recombination, and it must matter for the gravitational potentials and for the growth of structure of the recent universe.

F. The growth of structure

For pressureless matter with the density contrast δ_m and the density ρ_m each closure gives one equation. In the closure "Pressure", Eqs. (127a) and (127c) give

$$\frac{d^2\delta_m}{dx4^2} + \Big(2H - \frac{\text{gsum}}{2}\Big)\frac{d\delta_m}{dx4} = 4\pi\,G_N(x4)\,\rho_m\,\delta_m : \tag{134}$$

gravity is that of the moment, and the Hubble friction is reduced by gsum/2. In the closure "Momentum", the derivative of Eq. (127a) with Eq. (127b) gives, in conformal time,

$$\delta''_m + \frac{2H'_e + \mathcal{E}_m + \frac{4}{3}\mathcal{E}_r}{2H_e}\,\delta'_m = \frac{H_{\rm cf} - a_{\rm sp}\,\text{gsum}}{2H_e}\,\mathcal{E}_m\,\delta_m, \tag{135}$$

where $\mathcal{E}_m$ and $\mathcal{E}_r$ are the parts of $\mathcal{E}$ from matter and from radiation. Here the source is reduced as well, by the factor $(H - \text{gsum})/(H - \text{gsum}/2)$. (The term with $\mathcal{E}_r$ is the momentum density of radiation that streams freely: far inside the horizon its velocity divergence is $-h'_{\rm sy}/2$ [40].) Without deflation both equations reduce to the familiar $d^2\delta_m/dx4^2 + 2H\,d\delta_m/dx4 = 4\pi G_N\rho_m\delta_m$. With deflation that familiar equation, with $G_N(x4)$ inserted, is neither of the two. For matter alone and three equal rates the growth rate $d\ln\delta_m/d\ln a_{\rm sp}$ is, to the first order in gsum/H, $1 - \frac{9}{16}\,\text{gsum}/H$ in the closure "Pressure" and $1 - \frac{13}{16}\,\text{gsum}/H$ in the closure "Momentum".

Two effects oppose each other [Fig. 23(c), Table LVII]. At a *fixed* Hubble constant, weaker gravity in the past lowers the amplitude σ_8 of clustering today: $d\ln\sigma_8/d(\text{gsum}/H_0) = -0.443$ in the closure "Momentum" and -0.207 in the closure "Pressure". But the Hubble constant that keeps the angle of the sound horizon is lower, the cosmological term takes over later,

and matter clusters for longer. With that Hubble constant the net change is +0.003 and +0.239 per unit of gsum/H_0: +0.001 and +0.077 per cent at the bound of lunar laser ranging, and +0.01 and +1.60 per cent at the bound of nucleosynthesis. The growth rate today, $f_g = d\ln\delta_m/d\ln a_{\rm sp}$, rises in both closures, by the factors 1.0021 and 1.0035 at the first bound and 1.0463 and 1.0773 at the second. Common to both closures is that, at a fixed angle, the amplitude does not fall and the growth rate rises. How large the changes are depends on the closure: *the growth of structure on the deflating spacetime is not predicted until the perturbations of the agent are specified.*

G. The spectra of the microwave background

The program. The temperature and polarization spectra of the microwave background and the power spectrum of matter are computed with an Einstein-Boltzmann program written for this work. It integrates the equations of motion of photons, neutrinos, baryons and cold dark matter in the synchronous gauge [40] together with the equations of the metric perturbations, and obtains the spectra by integrating the sources along the line of sight [41]. Its version for static extra coordinates has Newton's constant built into its units. For the deflating spacetime four of its parts were copied and changed in a short, listed set of places: the expansion obeys Eq. (125); every source of the metric perturbations carries the Newton's constant of the moment; the energy constraint is Eq. (127a); and the sources of the temperature anisotropy contain the gap $\mathcal{Y}_S$ of the closure. The equations of motion of photons, neutrinos, baryons and dark matter are not changed: matter that is spread uniformly over the extra coordinates moves in ordinary space as it does in four dimensions, and Thomson scattering and recombination are atomic physics, which the model does not alter. The copies are written by a script from the original files, so that they cannot differ from them in any place that is not listed, and without deflation the two versions give the same spectra to the last binary digit.

Recombination and helium. Recombination is computed on the expansion history of Eq. (125), with the three-level atom and its standard corrections [42]. The helium fraction responds as well. At nucleosynthesis Newton's constant was smaller than today by the factor G_N ratio of Table XXXIV; a universe that expands more slowly by that factor in H^2 is equivalent, for nucleosynthesis, to one with $\frac{43}{7}(G_N \text{ ratio} - 1)$ additional neutrino species, a negative number, and the helium fraction follows from the standard fit [43]. (In a model with extra radiation, $\frac{43}{7}$ is replaced by $\frac{43}{7} + \Delta N$, where ΔN is the density of that radiation during nucleosynthesis in units of one neutrino species.) At the largest rate that nucleosynthesis allows, and without extra radiation, this is -0.379 species, and $Y_{\rm He}$ falls from 0.24540 to 0.24008. Table LVIII gives the thermal history of three models.

Spectra at a fixed angle. Table XXXVI lists the model without deflation and five models with three equal rates, each with the densities, the primordial amplitude and tilt and the optical depth of the standard model and with the Hubble constant that keeps $100\,\theta_* = 1.03965$. Keeping the angle keeps the acoustic peaks in place: the first five move by at most 0.32 in the multipole ℓ (Table LIX). What changes is their height and the damping tail [Fig. 25], for the reasons that apply to any change of the expansion rate at recombination: the ratio of the diffusion length to the sound horizon, the width of the last-scattering surface and the decay of the potentials shortly after recombination all change, and the lower helium fraction leaves more electrons per baryon.

At the rate that lunar laser ranging allows, the temperature spectrum differs from that of the model without deflation by at most 2.7×10^{-3} of itself, 0.134 of the cosmic variance of a single multipole, and $\Delta\chi^2 = 1.12$ against Planck-like mock data (Sec. XVI H) before any parameter is refitted. Such data cannot tell this rate from no deflation. At the rate that nucleosynthesis allows, Newton's constant at last scattering was 0.9383 of today's, the spectrum changes by up to 5.6 per cent, and $\Delta\chi^2 = 468$: the microwave background is then a stronger test than nucleosynthesis.

The closure. For $30 \le \ell \le 2500$ the spectra of the two closures differ by a thousandth of the effect of the deflation itself (1.4×10^{-3} and 1.2×10^{-3} of it at the two bounds), and $\Delta\chi^2$ is the same to five digits [Fig. 26(a)]. *The spectra at these multipoles follow from A1, A2 and A3, whatever the closure.* For $\ell < 30$, where the potentials of the recent universe act through the late integrated Sachs-Wolfe effect, the closures differ by more than the effect of the deflation in either of them, by a factor 2.1. The largest angular scales are not predicted; their cosmic variance is large, and at the rate of nucleosynthesis the closures differ by at most a third of it. The amplitude of matter is 0.9762 of its value without deflation in the closure "Momentum" and 0.9918 in the closure "Pressure" at the rate of nucleosynthesis, and 0.99892 and 0.99969 at the rate of lunar laser ranging (Table LX).

Figure 26(b) and Table LXI show what the agent must supply. In units of $a_{\rm sp}\,{\rm gsum}/H_{\rm cf}$ times the terms of each equation, the closure "Momentum" lacks 0.54 to 0.98 in Eq. (127c) and 0.31 to 0.49 in Eq. (127d) after recombination; the closure "Pressure" lacks 0.62 to 0.94 in Eq. (127b), and in Eq. (127d) an amount that falls as $(H_{\rm cf}/k)^2$ inside the horizon, as Eq. (133) requires. The two identities (130) and (131) hold along the numerical solutions to 8.6×10^{-4}, the accuracy of the numerical derivative with which they are tested. The complete calculation, with photons, neutrinos and baryons, reproduces the growth of Eqs. (134) and (135) to 1.3×10^{-4} [Fig. 26(c), Tables LXII and LXIII]; with the equation of the other closure it would disagree by a quarter of gsum/H_0.

TABLE XXXVI. Spectra of the microwave background on the deflating spacetime at a fixed angle of the sound horizon (assumptions A1, A2, A3; three equal rates). H_0 in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. $G_N(z_*)/G_N$: Newton's constant at last scattering divided by its value today. σ_8: amplitude of clustering for the same primordial amplitude, in the closures "Momentum" (M) and "Pressure" (P). Largest change: of the temperature spectrum D_ℓ^{TT} for $30 \le \ell \le 2500$ relative to the model without deflation. Ideal measurement: the root of the sum over those multipoles of the squared change divided by the variance $2/(2\ell+1)$ of one multipole, that is, the number of standard deviations by which a measurement of the temperature on the whole sky without noise would tell the model from no deflation if nothing else were unknown. $\Delta\chi^2$: against Planck-like mock data with all other parameters fixed.

gsum/H_0	H_0	$G_N(z_*)/G_N$	σ_8 (M)	σ_8 (P)	largest change	ideal measurement	$\Delta\chi^2$
0	67.360	1.0000	0.8228	0.8228	0	0.0	0.00
0.00324	67.149	0.9969	0.8219	0.8225	2.7×10^{-3}	3.1	1.12
0.0100	66.704	0.9905	0.8200	0.8219	8.3×10^{-3}	9.7	10.79
0.0300	65.365	0.9714	0.8142	0.8201	0.025	29.5	98.85
0.0644	62.976	0.9383	0.8032	0.8160	0.056	64.8	468.30
0.100	60.368	0.9037	0.7903	0.8104	0.090	103.5	1166.70

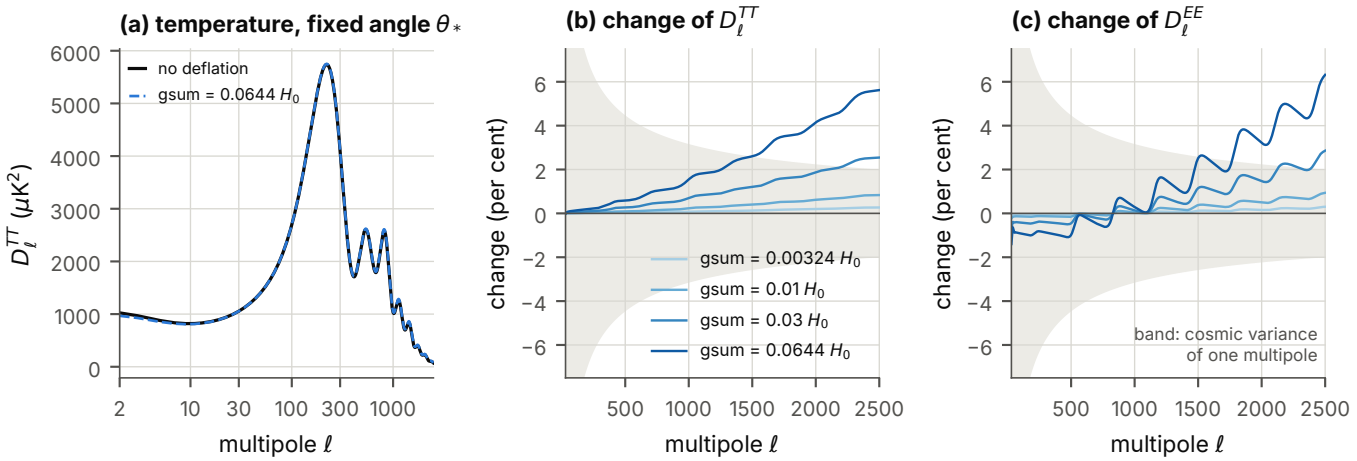


FIG. 25. The spectra of the microwave background at a fixed angle of the sound horizon. (a) The temperature spectrum without deflation and at the largest rate that nucleosynthesis allows. (b), (c) The change of the temperature and polarization spectra relative to the model without deflation for four rates (darker: faster); the band is the cosmic variance of a single multipole. Closure "Momentum"; at these multipoles the closure "Pressure" gives the same curves.

H. What mock data say about the rate, and about the Hubble constant

How far can the rate of deflation move the Hubble constant before the fit to the microwave background degrades? The question is answered here with mock data: the spectra of the standard model stand in for the measurements, with the noise and the sky coverage of the Planck satellite [43], and of Planck together with a deep survey of a quarter of the sky. The mock data contain no gravitational lensing, and the standard model fits them with $\Delta\chi^2 = 0$ by construction. The likelihood is the Gaussian likelihood of the temperature and polarization spectra with these noise levels; the six parameters of the standard model (the densities ω_b and ω_c, the Hubble constant, the amplitude A_s and the spectral index n_s of the primordial fluctuations, and the optical depth τ) are fitted again for every rate; the rate is a seventh parameter. Because the rates are positive, the derivative of the spectra with respect to gsum at zero rate is one-sided.

A bound, as a forecast. With the six parameters free, Planck-like data determine gsum/H_0 to 0.0196 (Table XXXVII), and the upper bound at 95 per cent is

$$\text{gsum} < 0.0385\,H_0 = 2.65 \times 10^{-12} \text{ per year}, \qquad (136)$$

or $0.0207\,H_0$ with the deep survey. With the six parameters *held* the error would be 0.00108: nearly all of the effect of the deflation on the spectra, above all the shift of the angle θ_*, can be absorbed by the other parameters [Fig. 27(c)]. The bound lies between those of Table XXXV: it is 11.9 times weaker than that of lunar laser ranging and 1.7 times stronger than that of nucleosynthesis. The closure changes it by half a per cent.

The Hubble constant falls in every fit. Along the degeneracy the best-fit Hubble constant changes by -84.35 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ per unit of gsum/H_0 for the Planck-like data and by -82.11 with the deep survey, that is, by -3.2 and -1.7 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at the respective bounds. This is steeper than the slope -65.24 at a

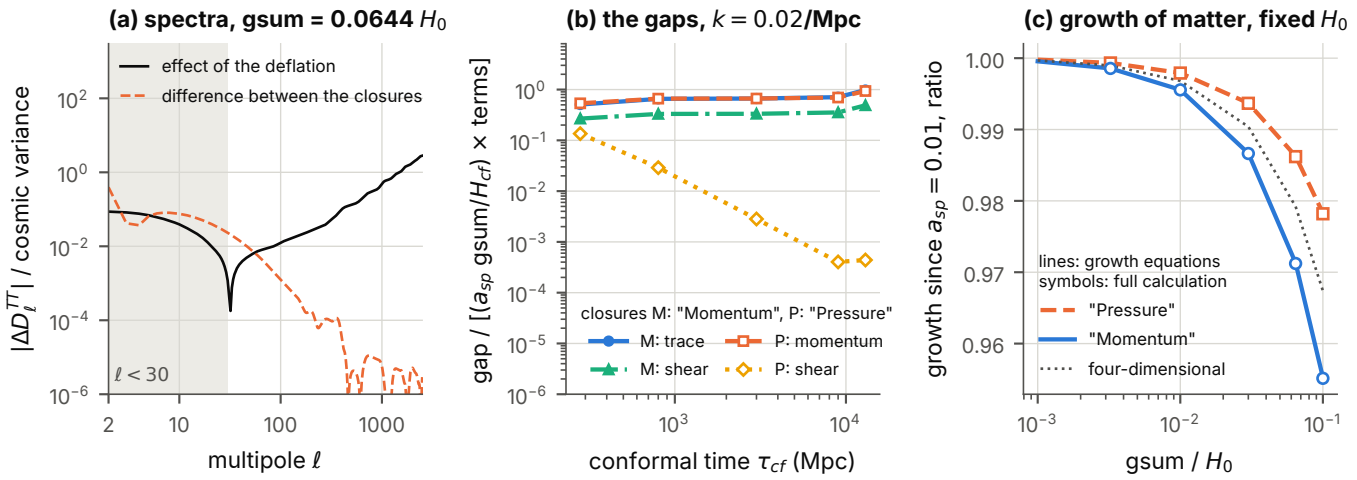


FIG. 26. The closure of the perturbation equations, at the largest rate that nucleosynthesis allows. (a) The difference between the temperature spectra of the closures "Momentum" and "Pressure", and the effect of the deflation itself, in units of the cosmic variance of a single multipole: the closure matters below $\ell = 30$ only. (b) What the agent supplies: the gaps of Eqs. (127b), (127c) and (127d) for one wavenumber, in units of $a_{\rm sp}\,{\rm gsum}/H_{\rm cf}$ times the terms of each equation. (c) The growth of the density contrast of matter since $a_{\rm sp} = 0.01$ relative to no deflation, at a fixed Hubble constant: Eqs. (134) and (135) (lines), the complete calculation (symbols), and the four-dimensional equation with the Newton's constant of the moment, which is neither closure.

TABLE XXXVII. Forecast for the rate of deflation from mock data of the microwave background (Fisher matrix; three equal rates; assumptions A1, A2, A3). M, P: the closures "Momentum" and "Pressure". $\sigma_{\rm rate}$: standard error of gsum$/H_0$ with the six parameters of the standard model free. Bound: the upper bound at 95 per cent, 1.96 standard errors, for a parameter that cannot be negative and whose best value is zero. Held: the standard error if the six parameters are held fixed. Slope: the change of the best-fit Hubble constant per unit of gsum$/H_0$, in ${\rm km\,s^{-1}\,Mpc^{-1}}$. Last two columns: the standard error of the Hubble constant in ${\rm km\,s^{-1}\,Mpc^{-1}}$ with the rate held at zero, and with the rate free.

mock data	closure	$\sigma_{\rm rate}$	bound	held	slope	error of H_0	rate free
Planck-like	M	0.01963	0.0385	0.00108	−84.35	0.561	1.748
Planck-like	P	0.01974	0.0387	0.00108	−84.40	0.561	1.758
with deep survey	M	0.01056	0.0207	0.00055	−82.11	0.372	0.943
with deep survey	P	0.01058	0.0207	0.00055	−82.13	0.372	0.945

fixed angle and fixed densities, because in the fit the density of dark matter rises and the tilt falls as well. Fits of the six parameters at fixed rates confirm the Fisher matrix [Fig. 27(a), (b); Table LXIV]: for rates up to $0.01\,H_0$ the minimum $\Delta\chi^2$ and the shift of the Hubble constant are 1.00 to 1.01 times its predictions, and the fits reach $\Delta\chi^2 = 3.84$ at gsum $= 0.0376\,H_0$ and, with the deep survey, at $0.0205\,H_0$. At the rate that lunar laser ranging allows the best fit has $\Delta\chi^2 = 0.027$ and $H_0 = 67.086\ {\rm km\,s^{-1}\,Mpc^{-1}}$. At the rate that nucleosynthesis allows it has $\Delta\chi^2 = 11.6$, a deviation of 3.4 standard deviations, and $H_0 = 61.70\ {\rm km\,s^{-1}\,Mpc^{-1}}$; with the deep survey $\Delta\chi^2 = 39.8$. *The largest rate that nucleosynthesis allows would be excluded by data of the quality of Planck.*

Deflation and dark radiation together. Extra radiation is the candidate of the model for the Hubble tension. Seven one-parameter families of extra radiation were fitted to the same mock data, each with a density N in units of one neutrino species: radiation that streams freely; a fluid; the *stepped fluid of the model*, at four transition redshifts z_t; and the stepped fluid of the Wess-Zumino model [44]. The stepped fluid of the model is the massless phase mode of the Vorton-Scalar in equilibrium with the sixteen components of the fermion field: when the fermions become non-relativistic, at the redshift z_t, they annihilate and heat the phase mode, so that the temperature of the fluid rises by the factor $15^{1/3}$ relative to the law of dilution (N is the density after this step; the Wess-Zumino fluid has the factor $(15/7)^{1/3}$). Every family can raise the Hubble constant, and every one at a price in $\Delta\chi^2$. Does the deflation lower the price? All fits were made with static extra coordinates and on the deflating spacetime at the largest rate that lunar laser ranging allows, and the Fisher matrix was computed with N and the rate free together (Table XXXVIII, Fig. 28; Tables LXV to LXVIII of Appendix D). Three results follow.

First, at that rate every best-fit Hubble constant of the static case falls by 0.274 to 0.279 ${\rm km\,s^{-1}\,Mpc^{-1}}$, which is

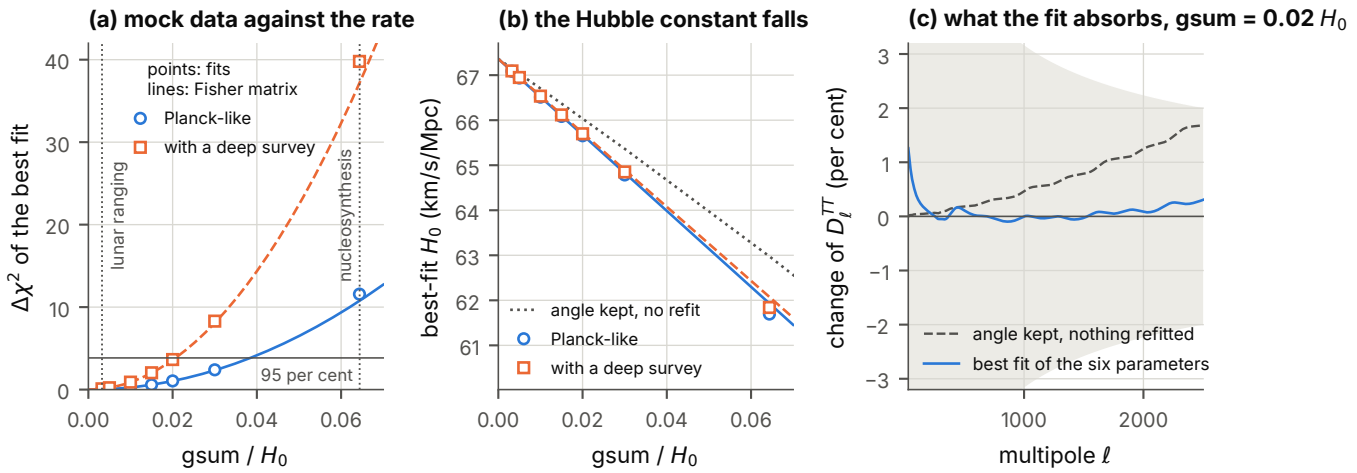


FIG. 27. Mock data of the microwave background against the rate of deflation. (a) The minimum $\Delta\chi^2$ of fits of the six parameters of the standard model at a fixed rate (symbols) and the parabola of the Fisher matrix (lines), for the two data sets; the rates that lunar laser ranging and nucleosynthesis allow are marked, and the horizontal line is $\Delta\chi^2 = 3.84$, the bound at 95 per cent. (b) The best-fit Hubble constant, with the slope of the Fisher matrix and, dotted, the Hubble constant that keeps the angle when nothing else is refitted. (c) The temperature spectrum at gsum $= 0.02\,H_0$ relative to the data: with the angle kept and nothing refitted, and at the best fit, which absorbs most of the difference; the band is the cosmic variance of a single multipole.

the shift of the model without extra radiation: the effects of the deflation and of the extra radiation add [Fig. 28(a), (b)].

Second, N and the rate are positively correlated, with correlation coefficients of 0.37 to 0.89. Extra radiation makes the early universe expand faster, the weaker gravity of the deflating spacetime makes it expand more slowly, and in the spectra the two compensate in part. Fits with extra radiation are therefore slightly *better* on the deflating spacetime than on the static one, slightly more extra radiation is allowed at a given $\Delta\chi^2$, and with both parameters free the error of each grows, for free-streaming radiation from 0.143 to 0.250 in N.

Third, the compensation does not help the Hubble constant. When the rate grows and N follows its best-fitting value, the best-fit Hubble constant still falls: by 35.3 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ per unit of gsum$/H_0$ for free-streaming radiation, against 84.35 without extra radiation, and by 22.1 to 76.9 for the seven families. The largest Hubble constant among the models with $N \geq 0$, gsum ≥ 0 and $\Delta\chi^2 \leq 3.84$ is therefore reached at zero rate, for every family and both data sets [Fig. 28(c)]. *The deflation lowers the Hubble constant faster than it makes room for extra radiation.*

I. Dark matter

What does not change. A superfluid needs particles that are degenerate (fermions) or condensed (bound pairs) at the mass density ρ and the velocity dispersion σ_v of observed halos. A gas of fermions with $\nu = 8$ internal states is degenerate if its Fermi velocity is at least σ_v, and bosons of the mass m_B are condensed if the number of particles in a cube with the edge of their thermal wavelength, at the temperature $k_B T = m_B \sigma_v^2$, is at least 2.612. Both conditions are upper limits on the mass,

$$m_{\mathrm{deg}} = \left[\frac{6\pi^2\hbar^3\rho}{\nu\,\sigma_v^3}\right]^{1/4}, \qquad m_{B,\mathrm{max}} = \left[\frac{(2\pi)^{3/2}\hbar^3\rho}{2.612\,\sigma_v^3}\right]^{1/4}, \tag{137}$$

the first being the bound of Tremaine and Gunn [45] for this model. They contain the densities and the velocities only, and the masses of the model do not change along the deflation (Table XXXIX): in the densest systems of dark matter that are observed, the cores of dwarf spheroidal galaxies, fermions are degenerate only if they are lighter than 0.276 keV and bound pairs are condensed only if they are lighter than 262 eV. The first limit conflicts with the bound of the Lyman-α forest derived below, which demands more than 1.5 keV; the second conflicts with the bound on the self-interaction of the pairs, which demands far heavier fermions. *The dark matter of galaxies is not a superfluid of these particles.* Heavy fermions or pairs can be cold dark matter of an ordinary kind. The deflation does not change this verdict.

Self-interaction, and the epoch of the resonance. The cross-section per mass of bound pairs must respect the bound of about $\kappa_{\mathrm{SI}} = 1\ \mathrm{cm^2/g}$ from colliding clusters [46, 47]. A pair bound by the Yukawa potential (69) has a scattering length of at least the range $1/m_s$; identical bosons with that scattering length scatter at low energy with the cross-section $8\pi/m_s^2$; and the pair exists only if $D_{\mathrm{VS}} = \alpha_{\mathrm{VS}}\, m/m_s > D_c$ (Sec. X). Together, for pairs of

TABLE XXXVIII. The seven families of dark radiation on the deflating spacetime (Planck-like mock data; assumptions A1, A2, A3). "Model fluid" is the stepped fluid of the model with the transition redshift z_t. Columns 2 to 5: the density N of extra radiation at which the best fit reaches $\Delta\chi^2 = 4$, and the Hubble constant there in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, with static extra coordinates and deflating at the largest rate that lunar laser ranging allows. Columns 6 to 10, from the Fisher matrix with N and the rate free together: the standard error of N with the rate held at zero and with the rate free; the correlation coefficient of N and the rate; the net slope, that is, the change of the best-fit Hubble constant in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ per unit of gsum/H_0 when N follows its best-fitting value; and the largest rise of the best-fit Hubble constant, in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, among the models with $N \geq 0$, gsum ≥ 0 and $\Delta\chi^2 \leq 3.84$, which is reached at zero rate for every family.

	N at $\Delta\chi^2 = 4$		H_0 there		error of N			net	largest
family	static	deflating	static	deflating	rate zero	rate free	correlation	slope	rise
free-streaming radiation	0.29	0.31	69.74	69.62	0.143	0.250	0.82	−35.3	2.30
fluid, no step	0.27	0.29	70.20	70.12	0.135	0.249	0.84	−23.9	2.77
model fluid, $z_t = 10^4$	0.25	0.26	68.75	68.53	0.122	0.145	0.54	−65.4	1.35
model fluid, $z_t = 2 \times 10^4$	0.17	0.18	68.16	67.90	0.085	0.091	0.37	−76.9	0.78
model fluid, $z_t = 5 \times 10^4$	0.12	0.13	68.48	68.27	0.060	0.076	0.61	−67.0	1.10
Wess-Zumino fluid, $z_t = 2 \times 10^4$	0.32	0.35	70.13	70.06	0.160	0.350	0.89	−22.1	2.69
model fluid, $z_t = 10^6$	0.30	0.32	70.39	70.32	0.150	0.250	0.80	−23.0	2.95

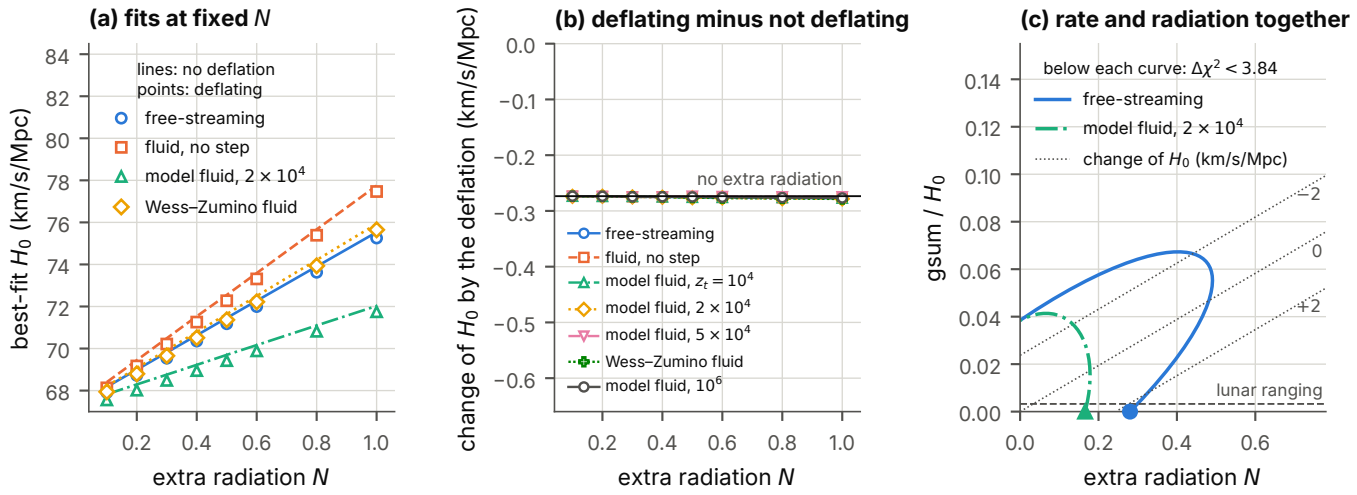


FIG. 28. The seven families of dark radiation on the deflating spacetime (Planck-like mock data). (a) The best-fit Hubble constant against the density N of extra radiation for four families: without deflation (lines) and at the largest rate that lunar laser ranging allows (symbols). (b) The difference for all seven families; the horizontal line is the shift of the model without extra radiation. (c) The rate and the extra radiation as free parameters together: the regions with $\Delta\chi^2 < 3.84$ of the Fisher matrix for two families, lines of constant change of the best-fit Hubble constant for free-streaming radiation (dotted), the point of the largest Hubble constant of each region (filled symbols, at zero rate), and the rate that lunar laser ranging allows (dashed).

TABLE XXXIX. Could the fermions of the model be degenerate, or their bound pairs condensed, in observed systems of dark matter? ρ: density; σ_v: velocity dispersion; m_{deg} and $m_{B,\mathrm{max}}$: the limits of Eq. (137).

system	ρ ($M_\odot$ pc^{-3})	σ_v (km/s)	m_{deg} (keV)	$m_{B,\mathrm{max}}$ (eV)
dwarf spheroidal galaxy, core	0.100	10	0.276	262
Milky Way at the Sun	0.0105	156	0.020	19.0
cluster of galaxies, core	0.00300	1000	0.004	3.45

the mass $2m$, these give a smallest fermion mass,

$$m \geq \left[\frac{4\pi\, D_c^2}{\alpha_{\mathrm{VS}}^2\, \kappa_{\mathrm{SI}}}\right]^{1/3}, \qquad (138)$$

which is 918 MeV for $\alpha_{\mathrm{VS}} = 0.1$ and falls as $\alpha_{\mathrm{VS}}^{-2/3}$, that is, as $e^{-2\tau_d/3}$ along the deflation. For a given particle the masses are constant and so is this cross-section. What the deflation changes is whether the pair *exists*: it is bound only after D_{VS} has grown to D_c. And at that moment the scattering length of two unbound fermions diverges (Sec. X), so that their cross-section rises to the limit that unitarity allows, which is larger the slower the

particles. For the example particle, $m = 918$ MeV and $m_s = 54.7$ MeV, at the relative velocities 10, 156 and 1000 km/s [Fig. 13(c)]: one unit of the clock before the binding the cross-section per mass is 0.84 cm^2/g at all three velocities; at the binding it peaks at 1.3×10^7, 5.2×10^4 and 1.3×10^3 cm^2/g; and it exceeds 100 cm^2/g during an interval of the clock of 0.27, 0.27 and 0.26 around the binding. *A deflating universe whose dark matter consists of these fermions passes through an epoch of strong, velocity-dependent self-interaction*, strongest in the smallest halos. Whether the epoch lies in the observable past depends on the rate. Under A2 and a constant rate it does so only if D_{VS} today exceeds D_c by less than a third of a per cent, a coincidence that nothing in the model explains.

Superfluid dark matter and small-scale structure. What becomes of the growth of structure if the dark matter is the superfluid itself, a fluid with first sound? Two limits are computed: a degenerate gas of the fermions with eight internal states, and a condensate of bound pairs with the sound speed c_{s0} today. The pressure of such a fluid suppresses structure below a wavenumber, and the bound of the Lyman-α forest on warm dark matter [48, 49] then limits the fermion mass from below and the sound speed from above. Computed again on the deflating background under A1 to A3, at the Hubble constant that keeps the angle, the wavenumber at which the power falls to a quarter of that of cold dark matter is lower by 0.15 to 0.16 per cent at the rate that lunar laser ranging allows and by 3.0 to 3.3 per cent at the rate that nucleosynthesis allows, for fifteen models (Fig. 29, Tables LXIX and LXX). The reason is the weaker gravity of the time at which the suppression is set: the Jeans wavenumber is proportional to the square root of Newton's constant. The limits move accordingly, from $m > 1.514$ keV and $c_{s0} < 1.645 \times 10^{-11}\, c$ without deflation to $m > 1.516$ keV and $c_{s0} < 1.641 \times 10^{-11}\, c$ at the first rate and $m > 1.555$ keV and $c_{s0} < 1.565 \times 10^{-11}\, c$ at the second. The two closures give the same wavenumbers to 1.5×10^{-5}. In this calculation the equation of state of the superfluid in ordinary space is that of static extra coordinates; whether the self-interaction of a condensate would itself follow the volume of the deflating coordinates is a question about the microscopic model that the calculation does not answer.

How fast the quantities drift. With $n = d\ln(\text{quantity})/d\tau_d$, the relative change of a quantity over the age of the universe is $n\,\mathrm{gsum}\,t_0$. The couplings themselves drift with $n = 1$ or $\frac{1}{2}$ (Table V). Quantities near a threshold drift much faster: the scattering length just below the binding, at $D_{\mathrm{VS}} = 1.5$, with $n = 9.1$; the binding energy just above it, at $D_{\mathrm{VS}} = 1.8$, with $n = 30$; the gap of a weakly paired state with $n = 1/\lambda$. These are the quantities in which a slow deflation would first be seen (Table LV).

J. Verdict on hypothesis H3

The Hubble tension. The candidate of the model, the phase mode of the Vorton-Scalar as dark radiation, is untouched at the allowed rates, since the couplings drift by parts in a thousand: with static extra coordinates it raises the Hubble constant by the amounts of the last column of Table XXXVIII at most, a fraction of the 6.1 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ that are needed, and the deflation does not improve this. The deflation itself contributes two effects. The first, through Newton's constant, has the wrong sign: at a fixed angle of the sound horizon the inferred Hubble constant falls by 0.2 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at the largest rate that lunar laser ranging allows, and in fits to mock data of the microwave background it falls by 84 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ per unit of gsum/H_0. Nor can the rate help the dark radiation: more extra radiation is allowed when the rate is free, but the largest Hubble constant that the mock data allow is reached at zero rate. The second effect, the loss of mass of bound pairs, has the right sign and is bounded, in the examples computed, to 0.5 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. *The hypothesis that the model resolves the Hubble tension is not supported by the deflation, and under A2 it is mildly disfavored.*

Dark matter. The mass ranges stated above hold unchanged, and the limits on superfluid dark matter move by two tenths of a per cent at the rate that lunar laser ranging allows. The deflation adds one qualitative prediction, the epoch of the resonance, and one quantitative one, a rise of the growth rate, whose size depends on the perturbations of the agent. Neither is in conflict with observation at the allowed rates, and neither is detectable at the rate that lunar laser ranging allows. *The hypothesis that the superfluid is the dark matter is neither strengthened nor weakened by the deflation.* It acquires a history, whose observable part is empty if gravity is eight-dimensional and the rate constant.

What rests on the additional assumptions. Everything in this section that concerns Newton's constant rests on A2. The spectra, the forecast and the growth of structure rest in addition on A3, and the growth of structure and the largest angular scales on a closure that the hypothesis of deflation does not supply. The forecasts are for mock data without gravitational lensing; no measured spectrum has been fitted.

XVII. THE HIGGS HYPOTHESIS

A. The hypothesis and the conditions

Hypothesis H2 says that the superfluid state, the condensate of the fermion pair along the hidden axis, *is the Higgs field* [50]. In the Standard Model the Higgs field is a doublet of the weak-isospin group with a vacuum value $v_{\mathrm{EW}} = 246$ GeV, which breaks the electroweak symmetry, gives the W and Z bosons and the quarks and charged leptons their masses, and leaves one scalar

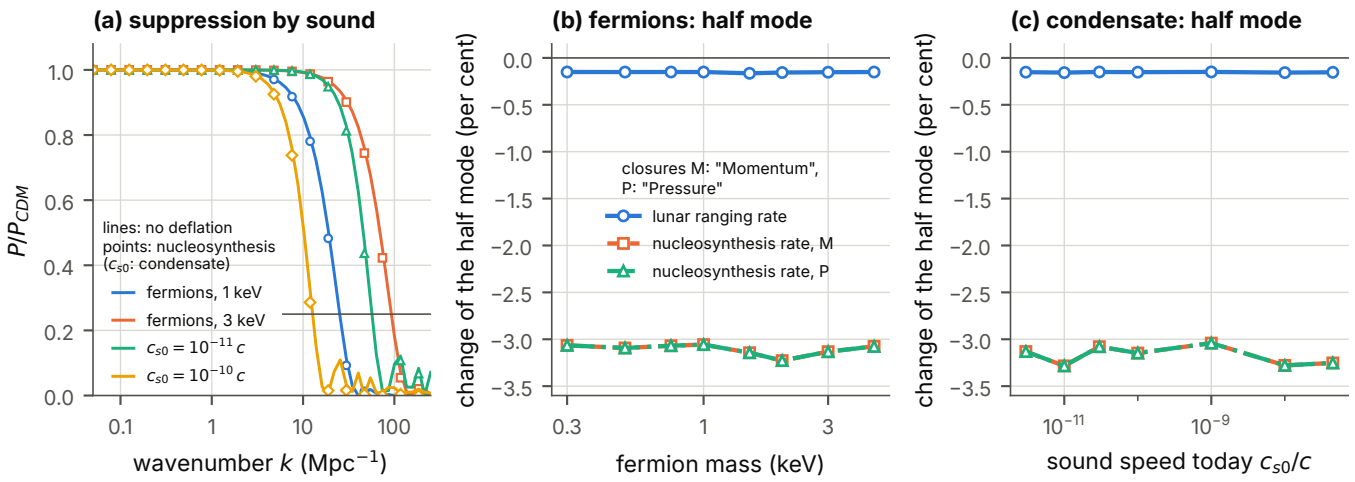


FIG. 29. Superfluid dark matter on the deflating spacetime. (a) The power spectrum of matter divided by that of cold dark matter, for a degenerate gas of the fermions of two masses and for a condensate of bound pairs with two sound speeds today: without deflation (lines) and at the rate that nucleosynthesis allows (symbols). The horizontal line marks one quarter, the value that defines the half mode. (b) The change of the wavenumber of the half mode for the degenerate gas, against the fermion mass, at the rates that lunar laser ranging and nucleosynthesis allow, for the closures "Momentum" (M) and "Pressure" (P). (c) The same for the condensate, against its sound speed today.

particle of 125 GeV. Its four real components form a vector of a group $SO(4)$, which the vacuum value breaks to $SO(3)$ [51]. From these facts follow seven conditions that a condensate must meet in order to *be* the Higgs field (Table XL). With static additional times the hypothesis has no geometric basis on the 4+4 spacetime, because the condensate is a singlet under the only non-Abelian symmetry that the geometry provides. This section examines the conditions on the deflating spacetime.

Two statements must be kept apart. H2 is the hypothesis that the *pair condensate* is the Higgs field; the electroweak scale is then the scale of the condensate. HV, used in one row of Table XXXV, is the different assumption that the vacuum value of the *Vorton-Scalar* sets the electroweak scale. HV is not examined here.

B. What the deflation leaves unchanged

The pair operators, their transformation properties and the internal symmetry are properties of the Dirac matrices of the frame, which the deflation does not change, and of the equations of the fields that do not depend on the extra coordinates, which for the weighted fields are the flat equations. By Sec. XIV C the four vector pair operators form the $(2,2)$ of $SU(2)_A \times SU(2)_B$, the representation of a Higgs field, and a condensate along one axis breaks $SO(4)$ to $SO(3)$. Hence, *within the sector*, C1 holds as on flat spacetime: the pair along the hidden axis is invariant under the rotations of $x1, x2, x3$, which remain symmetries of the spacetime, and under the boosts of ordinary spacetime as they act on the fields of the sector. (The boosts are no longer symmetries of the deflating spacetime as a whole; in the sector this shows only in the dependence of the couplings on the time.) C3 and C4 hold in the sector as well, in the sense that the pattern is right under its internal symmetry.

C. What the deflation changes

On the flat 4+4 spacetime the only non-Abelian internal symmetry that the geometry provides is $SO(3)_T$, the rotations of the three additional times, and under it the condensate along the hidden axis is a singlet, not a doublet: the pair operator along $x0$ commutes with the rotations of $x5, x6, x7$. By Sec. XIV C the deflating spacetime keeps $SO(3)_T$ only if the three rates are equal, and reduces it to one rotation, or to none, otherwise. The conclusion is the same where the rates are equal and holds a fortiori where they are not: *the hypothesis has no geometric basis on the* 4+4 *spacetime.* The pattern of the Higgs vacuum is present in the sector, under a hidden symmetry, and nowhere else. (On a spacetime with seven space-like dimensions and one time the six generators are rotations of the extra dimensions when these are static; a deflation of three of them removes that geometric basis as well.)

D. Condition C2: a condensate in empty space

With static extra coordinates C2 is open: a pair condensate forms in the vacuum only if $\alpha_{\rm VS}$ exceeds a critical value α_c of about 2 or more, depending on the cutoff (Sec. XII). On a deflating spacetime $\alpha_{\rm VS}$ grows as e^{τ_d} and reaches any such value at a definite time. Section XII showed what else has happened by then: the exchange of

TABLE XL. The conditions that a condensate must meet in order to be the Higgs field, and their status on the deflating 4+4 spacetime. The last row is a condition that only a deflating spacetime imposes. A dash marks a condition that is not examined for this spacetime, because C3 is not met there by any symmetry of the spacetime.

condition	status
C1: a scalar under the Lorentz group of ordinary spacetime	met in the sector of fields that do not depend on the extra coordinates
C2: non-zero in empty space	open; reached by the deflation only outside the range of validity of the model
C3: a doublet of the weak-isospin group with a neutral component	only under a hidden symmetry, which a contact self-interaction of form C breaks; a singlet under the rotations of the additional times
C4: breaks $SO(4)$ to $SO(3)$	met in the sector only, and only without a contact self-interaction of form C
C5: scale 246 GeV, with a scalar particle of 125 GeV with the observed couplings	—
C6: its couplings give the quarks and leptons their masses	—
C7: no further light or charged particles that would have been seen	—
new: the electroweak scale does not drift	requires gsum $< 1.16 \times 10^{-6}\, H_0/n_{\rm EW}$

one boson is no longer a controlled approximation, and the field at which the one-loop potential of the Vorton-Scalar falls below its vacuum value has moved to within a factor of two of the vacuum value. *C2 can be reached by waiting, but not within the range of validity of the model.* Its status remains open.

E. The self-interaction and the hidden symmetry

The internal symmetry of the sector, on which C3 and C4 rest, was established in Sec. XIV C for the model without self-interaction. The exchange of the Vorton-Scalar keeps the whole of it [it is a self-interaction of form Q in the scalar and the pseudoscalar channel with a finite range, Eq. (87)]. *A contact self-interaction of form C, the definition of the model, does not.* By Sec. IV F every self-interaction of form C that does not vanish breaks, explicitly, the three generators that mix the hidden axis with $x5, x6, x7$, and $U(1)_E$; of $SO(4)$ there remains $SO(3)_T$, the rotations of the additional times. The three generators that a condensate along the hidden axis would have to break spontaneously are then not symmetries of the model, so that the three flat directions of the potential of the order parameter that are rotations of its direction (Sec. XIV) are no longer protected by a symmetry; and the condensate along $x0$ is not degenerate with those along $x5, x6, x7$: with an attractive scalar coupling the pairs along the additional times have the larger gap (Sec. IX E). In the model as defined, the representation and the breaking pattern that C3 and C4 require are therefore present only if there is no contact self-interaction at all. In the alternative form Q they are present for any couplings: a scalar and a pseudoscalar contact coupling keep the whole internal symmetry, and a vector or an axial-vector coupling keeps the six generators of $SU(2)_A \times SU(2)_B$ and the phase symmetry and breaks only $U(1)_E$.

F. A new condition: the electroweak scale must not drift

Suppose that H2 holds. The electroweak scale $v_{\rm EW}$ is then the scale of the pair condensate in the vacuum, as in models in which a condensate of fermions replaces the elementary Higgs field [52]. That scale is determined by the parameters of the model: the masses m and m_s, the cutoff, and the coupling $\alpha_{\rm VS}$. On the deflating spacetime the masses are constant and $\alpha_{\rm VS} \propto e^{\tau_d}$. Therefore

$$v_{\rm EW} \;\propto\; e^{\,n_{\rm EW}\,\tau_d}, \qquad n_{\rm EW} = \frac{d\ln v_{\rm EW}}{d\ln \alpha_{\rm VS}}. \tag{139}$$

The exponent cannot vanish: below the critical coupling there is no condensate, so that the scale depends on the coupling, and most strongly just above the critical coupling.

Under H2 the masses of the electron and of the quarks are proportional to $v_{\rm EW}$ and change at the rate $n_{\rm EW}$ gsum. The mass of the proton is due mostly to the strong interaction and does not follow. The ratio m_p/m_e of the proton mass to the electron mass therefore changes at the rate $n_{\rm EW}$ gsum. Comparisons of atomic clocks find this ratio constant to 8.0×10^{-17} per year at two standard deviations [39]. Hence

$$\text{gsum} < \frac{8.0\times10^{-17}\ \text{per year}}{n_{\rm EW}} = \frac{1.16\times10^{-6}\,H_0}{n_{\rm EW}}. \tag{140}$$

The bound needs no assumption about gravity. For $n_{\rm EW} = 1$ it is 2787 times stronger than the bound that lunar laser ranging gives if gravity is eight-dimensional.

The size of the exponent. For the model itself, with massive fermions and a massive radial mode, the scale of the vacuum condensate has not been computed as a function of the coupling. It is known in closed form for the scale-invariant case of the ladder equation: massless fermions and massless bosons between an infrared scale

TABLE XLI. The exponent $n_{\rm EW}$ with which the scale of the vacuum condensate follows the coupling in the scale-invariant ladder equation, and the bound (140) on the present rate of deflation that follows from it under hypothesis H2.

$\alpha_{\rm VS}/\alpha_c$	ln(cutoff/scale)	$n_{\rm EW}$	gsum (per year)	gsum/H_0
1.05	12.082	147	5.4×10^{-19}	7.9×10^{-9}
1.10	7.998	54.0	1.5×10^{-18}	2.2×10^{-8}
1.25	4.429	15.1	5.3×10^{-18}	7.7×10^{-8}
1.50	2.702	6.05	1.3×10^{-17}	1.9×10^{-7}
2.00	1.571	2.57	3.1×10^{-17}	4.5×10^{-7}
3.00	0.870	1.15	6.9×10^{-17}	1.0×10^{-6}
5.00	0.464	0.540	1.5×10^{-16}	2.2×10^{-6}
10.0	0.215	0.230	3.5×10^{-16}	5.0×10^{-6}

and a cutoff [25]. There the scale of the condensate obeys

$$\varrho_c\,\ln\frac{\text{cutoff}}{\text{scale}}=\arctan\frac{1}{2\varrho_c},\qquad \varrho_c^2=\frac{1}{4}\Big(\frac{\alpha_{\rm VS}}{\alpha_c}-1\Big),\quad(141)$$

where ϱ_c is only an abbreviation for the square root that the second equation defines, from which

$$n_{\rm EW}=\frac{(1+4\varrho_c^2)\,\arctan(1/2\varrho_c)}{8\,\varrho_c^3}+\frac{1}{4\varrho_c^2}.\qquad(142)$$

The program verifies Eq. (142) against the numerical derivative to 3×10^{-8} and tabulates it with the bound (140) (Table XLI). The exponent diverges at the critical coupling, equals $1+\pi/2=2.57$ at twice the critical coupling, and exceeds 1 for $\alpha_{\rm VS}<3.27\,\alpha_c$. It falls below 1 only where the scale of the condensate approaches the cutoff, that is, where the four-dimensional description ends. Within the range in which a condensate of the kind that H2 needs can be described at all, $n_{\rm EW}$ is of order one or larger.

The conclusion. Under H2 the deflation clock can have advanced by at most $\text{gsum}\,t_0<1.1\times10^{-6}/n_{\rm EW}$ during the age of the universe at a constant rate. Every effect of the deflation described in Secs. VIII to XVI is then unobservably small. Conversely, a deflation at any rate at which those effects matter excludes H2. *The hypothesis H2 and a deflation of any consequence today exclude each other.* This conclusion does not depend on the unresolved conditions C2 and C5 to C7. It depends on two properties only: that the electroweak scale is set by the dynamics of the model, which is the content of H2, and that the coupling of the model follows the volume of the deflating coordinates, which is exact (Sec. VI).

G. Verdict on hypothesis H2

On the deflating spacetime the hypothesis that the singlet superfluid is the Higgs field is unproven and without geometric basis, as on the flat 4+4 spacetime, and it is incompatible with any deflation that is fast enough to matter today. Whether a mechanism exists that would make the electroweak scale insensitive to the coupling of the pair is not known; none is present in the model.

XVIII. WHAT HAS BEEN ESTABLISHED, AND WHAT HAS NOT

The sections above reported results of very different standing: identities verified in exact arithmetic, numerical results with a stated accuracy, consequences of additional assumptions, and estimates. This section sorts them. An item is called established if a machine check stands behind it (Appendix E); statements of interpretation deserve the reader's independent scrutiny more than the numbers do.

A. Established

Geometry (Sec. II). The curvature and the Einstein tensor of the deflating spacetime in closed form, for arbitrary rates and constants. The Killing vectors, 14, 15 or 20 in number, and the completeness of the lists. That the spacetime is not a solution of Einstein's equations without matter, with or without a cosmological constant; that with three equal rates it is an exact solution of Einstein-Gauss-Bonnet gravity without matter, for $\Lambda_8=3\,\mathrm{g}55^2$ and $\alpha_{\rm GB}=-1/(4\,\mathrm{g}55^2)$; that with unequal rates it is not; and that this solution is unstable (Sec. XV). The turning value of the ordinary time for a classical particle with momentum along a deflating time.

The Dirac field (Secs. III, V). The Dirac matrices in coordinate components, the spin connection, the covariant constancy of the matrices, and the curvature of the spinor derivative. The Lagrangian with the kinetic term, the mass term and the self-interaction; the field equation (24) with the self-interaction on its right-hand side; the weighted field; and the statement that fields which do not depend on $x5, x6, x7$ obey the flat equation with the couplings g_r/V_d of the moment. The exact solutions when one rate governs; that exactly eight of the sixteen solutions are bounded; that these are standing waves with vanishing charge; and the law of their extinction, Eqs. (58) to (60). All of this is confirmed numerically by two independent methods.

The self-interaction (Sec. IV). The Fierz matrix in closed form and from the 256 matrices. That for anticommuting fields the nine invariants obey five linear relations, so that only four couplings are independent, verified by exact integer arithmetic and on Grassmann-valued fields. The four effective couplings (47), the Hartree-Fock couplings (50), the pair couplings (54), and the conditions under which the self-interaction keeps the chiral symmetry and the symmetry $U(1)_E$. On the 4+4 spacetime: the exact relation (40) between the two forms of the self-interaction in the quantized sector, by which two particles of the same half of the field interact identically in the two forms and a particle of each half with opposite signs; the four mean-field couplings (49) of the two halves, the same four combinations in both forms with the roles (51); the pair couplings of Table III; and the conditions of Table IV, in particular that every self-interaction of form C

breaks the internal symmetry $SO(4) \times U(1)_E$ of fields in ordinary space to $SO(3)_T$.

Matter with the self-interaction (Sec. VII). The Kohn-Sham states of uniform matter and their thermodynamic consistency. The thresholds of Table VIII at which a gas becomes self-bound matter in the sector of fields that do not depend on the deflating coordinates, and the history of the saturated matter after that moment. That matter of form C polarizes and is self-bound with one half of the field filled, at two to three times the strength at which matter of form Q binds with both halves equally filled; and that a gas of fixed density polarizes before it binds.

The effective potential with the self-interaction (Sec. XIII). That the Kohn-Sham method and the CJT effective potential at the order of the Hartree and Fock diagrams give the same states, for one kind of particle and for the two halves of the field in both forms. The critical coupling (105) of the vacuum with the filled states of negative energy, and the moment at which the deflation crosses it. That above it a field without mass term acquires masses of opposite sign in the two halves in form C and of equal sign in form Q, with the same magnitude and at the same critical strength; and that with a bare mass form C has a sharp transition where form Q has a smooth one. The functional (108) with bare mass, self-interaction and Vorton-Scalar; that the exchange of the radial mode acts at low density as a scalar contact coupling; the shifts of the moment of binding of Table XXVII; and the mass (110) that a bare Dirac mass gives to the phase mode.

Pairing with the self-interaction (Sec. IX). The amplitudes (83) of the four contact channels in closed form; that the exchange of the Vorton-Scalar is a self-interaction of finite range; the ultraviolet criterion $U_{\rm uv} < 4\pi^2$ of the gap equation with the strength (92), below which the gap does not depend on the upper limit of the momentum integral; and the gaps of Table XV. The sign (84) with which the contact couplings of the two forms enter the two kinds of pairs, and the gaps of Table XVI: in form C a scalar contact attraction lowers the gap of the pair along the hidden axis and a vector repulsion raises it.

Gravity (Sec. XV). The Einstein-Lovelock tensors (119) and (120), derived in three ways. The reference evolution, reproduced by two programs. That the undisturbed deflation needs the constants (121) or a source with the equation of state (122); that the matter of the model cannot be that source; that matter accelerates the deflation and makes the flat coordinates expand; and that the undisturbed deflation is unstable in the tuned gravity.

The model along the clock (Secs. VI to XII). For fields that do not depend on the extra coordinates, the model on the deflating spacetime is the model on flat spacetime with the couplings of the moment, exactly. The masses are constant; $g_{\rm VS}$ grows as $e^{\tau_d/2}$; $\alpha_{\rm VS}$, $G_{\rm VS}$, $D_{\rm VS}$ and C grow as e^{τ_d}. The modes of the Vorton-Scalar grow with the weight only, and the potential between two fermions has the unchanged range and the strength of the moment. The exchange amplitudes are those of flat spacetime times the square of the coupling of the moment. In weak coupling the gap grows according to Eq. (81), and the critical temperature follows it. Two fermions become bound at a definite time, at a scattering resonance, with a binding energy that starts quadratically; the order of the other events depends on m_s/m alone. The paired state follows the deflation if and only if Ξ_d is below about 0.2 to 0.5; before that it lags and accumulates broken pairs. The solution of the gap equation is a minimum of the effective potential at every moment examined. Matter of fixed density loses its fermion mass at a critical value of C, is self-bound for $C > 3.7011$ and is more strongly bound later; the melting temperature of the fermion mass falls; the barrier of the one-loop potential of the vacuum moves toward the vacuum value.

Order parameter and symmetry (Sec. XIV). The potential of the order parameter prefers no direction among the four frame axes. The internal symmetry of the sector, the representation of the pair operators and the breaking pattern are unchanged by the deflation. Of the six generators that belong to the pairs of the four extra axes, the deflating spacetime keeps none, one or three as symmetries of the spacetime. A vortex line of unit winding carries four fermion zero modes, two for each direction along the line, at every moment; its energy per length falls as $e^{-\tau_d}$.

Cosmology (Sec. XVI), each item under the assumption named there. The equation of the expansion (125) and the fall of the inferred Hubble constant, with recombination computed on the deflating background and verified by an independent calculation. The bounds of Table XXXV, and that in the model as it stands the atomic clocks give none. The effect of the loss of mass of bound pairs. The pressure (126) of the agent. The equations (127) of the perturbations of ordinary space for unperturbed extra scale factors, the identities (130) and (131), and that the four equations cannot all hold with matter alone. The growth equations of the two closures, and that the closure "Shear" coincides with the closure "Pressure" inside the horizon. The spectra of the microwave background for $\ell \geq 30$ and their independence of the closure; the forecast for the rate from mock data; that the Hubble constant falls in every fit, with or without extra radiation; and that extra radiation and the rate of deflation compensate in part in the spectra, without any gain for the Hubble constant. The unchanged estimates of degeneracy and of condensation in halos, the limits on superfluid dark matter, the epoch of the resonance, and the drift of the quantities.

The Higgs hypothesis (Sec. XVII). Under H2 the electroweak scale follows the coupling with the exponent $n_{\rm EW}$, and atomic clocks bound the rate of deflation by Eq. (140). The exponent is given in closed form for the scale-invariant ladder equation.

B. What the mass term and the self-interaction change

Sections VIII to XIV were derived for the model in which the whole mass of the fermions comes from the Vorton-Scalar and the whole interaction from its exchange ($m_D = 0$, $g_r = 0$). Table XLII lists, result by result, what a bare Dirac mass and the self-interaction change.

C. The two methods as checks of each other and of the other results

The Kohn-Sham method and the CJT effective potential were used not only for new results but as checks, and each result of this paper that could be computed in two ways was (Table XLIII).

D. The two forms of the self-interaction: discussion

What each form is. Both forms keep what the quantum theory needs: the positive kinetic term, a real interaction, and the conservation of the number of particles. Both use the same nine couplings and the same four effective couplings, and they assign to these couplings different meanings. Form C takes the hypothesis literally: the quartic term is built from the invariants of the 4+4 spacetime. Form Q takes the particle content literally: the eight internal states of a particle are treated alike, as on the 7+1 spacetime.

For form C. (i) It is the quartic term of the hypothesis. Its invariants are invariant under all rotations and boosts of the 4+4 frame, and they can be written on the full deflating spacetime, for fields that depend on all eight coordinates, with the spin connection [Eq. (22)]; the quantized sector is a restriction of that covariant theory. Form Q is defined in the quantized sector only: its invariance group is not a symmetry group of the 4+4 spacetime, and it has no covariant extension to fields that depend on the additional times. (ii) Its dynamics respects the distinction that the metric makes between the space-like hidden coordinate and the additional times (Secs. IV E, IV F, IX E); form Q has an accidental symmetry that rotates them into one another. (iii) Each structure appears with its own adjoint: the mass that the interaction of form C generates in vacuum is the covariant mass term (Sec. XIII B).

Against form C, or for form Q. (i) Form C mixes the two adjoints: it combines the quartic term of the covariant Lagrangian with the kinetic and mass terms of the positive Lagrangian. The Lagrangian as a whole is invariant only under the thirteen generators that both parts share, the same thirteen as for form Q; the covariance of the quartic term is not a symmetry of the model. (ii) The bare mass term and the dynamical mass have different structures in form C, and with a bare mass the vacuum has a sharp transition at which the symmetry between the halves breaks spontaneously (Sec. XIII B). (iii) The results of the 7+1 spacetime do not carry over: in form Q every result for fields that do not depend on $x5, x6, x7$ is the same on the two spacetimes, and in form C the 4+4 model is a different model, with polarized matter, later binding, and pairs that are not degenerate. (iv) The pattern of the Higgs hypothesis is lost as soon as there is a contact self-interaction (Sec. XVII E), and with an attractive scalar coupling the self-interaction favors the pairs along the additional times over the pair along the hidden axis (Sec. IX E). (v) A scalar coupling of the sign that binds matter binds less: matter needs two to three times the strength (Table VIII), which moves it further outside the range in which the calculation without the filled states of negative energy is controlled.

What does not distinguish the forms. The reduction to four effective couplings and the Fierz matrix; the interaction between particles of the same half, hence everything about matter that occupies one half only; the pairs along $x5, x6, x7$; the exchange of the Vorton-Scalar, hence every result that does not involve the contact self-interaction; the equivalence of the Kohn-Sham and the CJT method; the conclusions on gravity; and the critical strength at which the vacuum of a massless field becomes unstable, with the magnitude of the mass that it acquires.

What could decide. Three kinds of consideration could decide between the forms; none is settled by the calculations of this paper. A principle: if the model is required to be the restriction of a covariant theory on the full 4+4 spacetime, form C follows; if it is required to be the theory of eight equivalent internal states, form Q follows. A consistent quantization of the covariant Lagrangian itself, with its indefinite sign: if one existed (with an indefinite metric on the space of states and a rule for the physical states), it would dictate the form of the interaction, and in general it would give neither form C nor form Q [Eq. (41)]; none is constructed here. And an observable consequence: the two forms differ in the internal states of self-bound matter (four or eight), in the pair that the self-interaction favors, and in the symmetry of the order parameter; a hypothesis that ties one of these to an observation would select a form.

E. Not established

The source of the deflation. The metric is prescribed. For equal rates it solves the equations of Einstein-Gauss-Bonnet gravity without matter, but that solution is unstable (Sec. XV); in Einstein gravity it needs a source with the equation of state (122), which the matter of the model cannot be. For unequal rates no field equation has been found that the metric solves. What holds the deflation, and how it absorbs the reaction of matter, is not known. All results on fields are results on a given background.

Which form of the self-interaction is the right one. The

TABLE XLII. What a bare Dirac mass m_D and the self-interaction (couplings g_r) change in the results obtained with the Vorton-Scalar alone.

result with the Vorton-Scalar alone	with a bare Dirac mass	with the self-interaction
fields that do not depend on $x5, x6, x7$ obey the flat equations with couplings of the moment (Secs. III, VI)	unchanged; m_D itself does not change with the clock	unchanged; every g_r grows as e^{τ_d}, in a fixed ratio to $g^2_{\rm VS}$
the chiral symmetry $U(1)_{\rm VS}$ is exact and the phase mode massless (Sec. VI)	broken explicitly; the phase mode has the mass of Eq. (110), which grows as $e^{\tau_d/2}$	kept only if $g_1^{\rm eff} - g_7^{\rm eff} = 0$ (form C; $g_0^{\rm eff} + g_8^{\rm eff} = 0$ in form Q)
the gap of the paired state and its growth (Sec. VIII)	smaller, because the massive phase mode binds less	in the pairs along $x5, x6, x7$, and in form Q: larger with a scalar contact attraction, smaller with a vector repulsion (Table XV); in the pair along the hidden axis of form C the reverse (Table XVI)
two fermions bind when $D_{\rm VS}$ reaches D_c (Sec. X)	the Yukawa potential of the radial mode is unchanged	a contact term $G_{\rm pair}$ is added to the potential; the threshold with it has not been computed
matter is self-bound for $C > 3\pi^2/8$ (Sec. XII)	earlier (Table XXVII)	shifted as in Table XXVII, differently in the two forms: less in form C, in which the couplings of the total densities are small
matter of fixed density loses its whole fermion mass (Sec. XII)	the particles keep the mass m_D	the masses of the two halves in matter are $m^*_\pm$ of Eq. (74)
the internal symmetry $[SU(2) \times SU(2)] \times U(1)_E$ of the sector (Sec. XIV)	unchanged	form C: broken to $SO(3)_T$ by every contact coupling; form Q: only $U(1)_E$ is broken, by vector and axial-vector couplings
cosmology and the bounds on the rate (Sec. XVI)	unchanged	unchanged

covariant Lagrangian has an indefinite kinetic term in the quantized sector, and the self-interaction can be completed in two ways (Sec. IV A). They differ in the sign of the interaction between particles of different halves of the field, and they give different matter, a different vacuum, different pairs and different symmetries. This paper takes form C, the covariant form, as the definition, by the decision of the author of the hypotheses; nothing in the calculations decides between the two (Sec. XVIII D).

States with coherence between the two halves, and other couplings. The density matrix (48) is diagonal in W_{1644}. Because the interaction connects the halves, states with $\langle \psi^\dagger_+ \cdots \psi_- \rangle \neq 0$ are possible in principle, in both forms; whether one of them has a lower energy than the states of Sec. VII has not been examined. Where the gap equations of form Q have no solution with non-negative masses (Sec. VII B), the continuation with a mass that has changed its sign has not been followed. The numbers are for the sets A and B; the exact formulas hold for all couplings. The potential of the order parameter with the contact couplings of form C has not been computed (Sec. XIV).

The cutoff. The self-interaction has couplings of negative mass dimension and defines an effective theory. The results without the filled states of negative energy presuppose a cutoff below the mass of the particles (Sec. XIII B); with the filled states the results depend on the cutoff, and the mass of the phase mode depends on it through the scalar density of the vacuum. The cutoff is a parameter of the model that nothing in the hypotheses fixes.

The couplings. The nine couplings g_r, of which four are independent, are free parameters. The examples use a scalar attraction and a vector repulsion in the ratio 10 to 1; no value is predicted.

Matter beyond the mean field. The Kohn-Sham states and the effective potential at the order of the Hartree and Fock diagrams neglect correlations. Without the filled states the energy has no lowest value as a function of an unrestricted trial mass; the states reported are solutions of the gap equation. Temperature was not included in the calculations with the self-interaction.

Pairing inside dense matter. The gaps of Table XV are computed for particles of the mass m, without regard to the polarization of the matter: in matter of form C with one filled half a pair along the hidden axis, which needs one particle of each half, cannot form at all, and this has not been taken into account. In self-bound matter the mass is $m^* < m$; the gap equation and the equation for m^* have not been solved together. The contact couplings enter the gap equation with a range $1/m_{\rm ct}$ that is a choice.

The theory of gravity and the assumption A2. If gravity is of the Einstein-Gauss-Bonnet kind, the relation (124) between Newton's constant and the volume of the deflating coordinates, and with it every result of Sec. XVI that is marked A2, would have to be derived anew.

TABLE XLIII. Results obtained in two or more independent ways, with the largest relative deviation between them.

result	obtained by	deviation
only four of the nine invariants of the self-interaction are independent	exact integer arithmetic on the antisymmetrized quartic form (program); Grassmann-valued fields (independent script)	9.4×10^{-16}
the Hartree-Fock couplings	exact traces (program); Wick's theorem with density matrices summed from numerical eigenvectors (independent script)	2.1×10^{-13}
the pair couplings of the 28 pairs at rest	exact diagonalization (program); numerical diagonalization for random couplings (independent script)	7.9×10^{-14}
saturation point of the reference example on flat 7+1 spacetime	a first calculation with a notebook; the program; the independent script	1.7×10^{-10}
states of uniform matter	Kohn-Sham method (gap equation, Gauss-Legendre integrals); CJT effective potential (minimization, Simpson integrals)	1.6×10^{-12}
matter with momentum along the deflating coordinates (7+1)	integral over the ellipsoid of occupied states (program); flat momenta integrated in closed form (independent script)	1.4×10^{-8}
the thresholds at which matter becomes self-bound	the program; a general-purpose minimizer (independent script)	1.0×10^{-9}
matter bound by the Vorton-Scalar (Sec. XII)	the functional of Sec. XIII C without bare mass and contact couplings; the functional of Sec. XII	2.1×10^{-8}
pairing amplitudes of the contact channels	closed forms; direct evaluation with the 256 matrices (program); projectors from numerical eigenvectors (independent script)	1.8×10^{-15}
the exchange of the radial mode as a contact coupling of finite range	the kernel of the exchange; the kernel of the scalar contact coupling with the range $1/m_s$	3.3×10^{-16}
gaps of the paired state without contact couplings	the gap equation of Sec. VIII; the gap equation of Sec. IX with vanishing contact couplings	1.3×10^{-15}
gaps with contact couplings	the program; plain iteration on another grid of momenta (independent script)	8.2×10^{-7}
the Einstein-Lovelock tensors of the orders 1, 2 and 3	sums over pair curvatures (program); the Riemann tensor of the metric and the Euler-Lagrange equations of the reduced action (independent script, computer algebra)	exact
the reference evolution with gravity	a first calculation with a notebook; the program (Runge-Kutta, order 4); the independent script (Runge-Kutta, order 8)	4.8×10^{-8}
the deflating spacetime with matter as a source	the program; a general-purpose integrator with an independent quadrature of the matter (independent script)	6.4×10^{-6}
the deflating spacetime as a solution of Einstein-Gauss-Bonnet gravity	the Lanczos tensor from its definition (Sec. II C); the Einstein-Lovelock polynomials of Sec. XV	exact

The agent. Nothing in the hypothesis says what keeps the three rates constant while ordinary space expands. Under A2 it must have the tension (126) and no energy density; whether a physical source with these properties exists has not been examined. Nor does the hypothesis say how the agent responds when ordinary space is perturbed. That the scale factors of the deflating coordinates are not perturbed (assumption A3) is an assumption; and even with it, a closure is needed that the hypothesis does not supply. The growth of structure, the amplitude σ_8 and the spectra of the microwave background at the largest angular scales are computed for two closures and differ between them at the first order in the rate. They are the quantities that a theory of the agent would have to predict.

The exclusion of the growing solutions. The eight growing solutions of each mode that depends on a deflating time are discarded because they are unbounded. That is a condition of admissibility; it is not derived from a principle. The theory has no positive conserved charge for such modes, and a consistent quantization exists only for the sector that does not depend on the additional times.

Unequal rates with simultaneous crossings. No closed form is known for a mode that depends on two deflating coordinates whose crossings coincide; the results are numerical.

The hidden coordinate. That the fields do not depend on $x0$ is assumed. Nothing in the deflation explains it.

The end of the history. The couplings grow without bound. The four-dimensional description with the exchange of one boson holds only while $\alpha_{\rm VS}$ is not large, and the eight-dimensional theory itself is an effective theory, with couplings of negative mass dimension, that must fail when the proper size of the deflating coordinates reaches its scale of validity. What happens then is outside the

model. Every history of Secs. VIII to XII ends in a regime that the two-loop truncation does not control; the numbers at the last moments of each table indicate the trend and are not predictions.

The energy. The energy of matter is not conserved while the couplings change. Where it goes, into whatever drives the deflation, is not described.

Dynamics. Sections VIII, X and XII give sequences of equilibrium states. Only for the paired state has the ability to follow been examined, in a model without dissipation; with dissipation the broken pairs would relax, and their lifetime is not known. The time-dependent two-body problem at the moment of binding, which decides how much of the population ends in the bound state, has not been solved. How a gas reaches the self-bound state, and whether the abnormal state forms by nucleation, are open. The equilibrium of a loop of vortex line with its current, and its stability, have not been computed.

The vacuum beyond one loop. The statements about the vacuum use the one-loop potential outside its range of reliability and draw only the conclusion stated in Sec. XII.

Cosmology. The forecasts of Sec. XVI H are for mock data without gravitational lensing; no measured spectrum of the microwave background has been fitted. All bounds that reach back to nucleosynthesis or to recombination assume a constant rate; a rate that was large early and is small today is neither excluded nor described. Only linear perturbations were computed. The consequences of the epoch of the resonance for the formation of halos have not been simulated.

The Higgs hypothesis. That a condensate forms in the vacuum, that any symmetry of the model is the electroweak gauge group, the scale and the mass of the scalar particle, and the masses of ordinary fermions: none of these is established. The exponent n_{EW} has not been computed for the model itself; the statement that it is of order one or larger rests on the scale-invariant case and on the general fact that the scale of the condensate vanishes at the critical coupling.

XIX. CONCLUSIONS

We asked what becomes of a 16-component Dirac field with a mass term and a self-interaction, and of the superfluid of its pairs, when three of the eight coordinates of a spacetime of signature 4+4 deflate exponentially.

The geometry is simple and rigid. The deflating spacetime is homogeneous, it has 14 to 20 symmetries instead of 36, and it is not a solution of Einstein's equations without matter. With three equal rates it is an exact solution of Einstein-Gauss-Bonnet gravity without matter, for one relation between the rate and the two constants of that theory; but the solution is unstable, the matter of the model cannot be the source that Einstein gravity would need, and matter that is present accelerates the deflation.

The Dirac field separates into two parts. Fields that do not depend on the deflating coordinates obey, after a change of normalization, exactly the flat equations. Fields that do depend on them are extinguished: every admissible mode is a standing wave that dies out as the exponential of an exponential of the time. The sector in which the 4+4 theory is not unitary is removed by the deflation, although the condition that selects the admissible modes is imposed and not derived.

The self-interaction has nine channels and, because the field anticommutes, four independent couplings. On the deflating spacetime they grow with the inverse volume of the deflating coordinates, in step with the coupling of the Vorton-Scalar. The Kohn-Sham method and the effective action of Cornwall, Jackiw and Tomboulis give the same matter, and the deflation carries it through definite events: a gas becomes self-bound, a massless field acquires a mass, and with a bare Dirac mass the phase mode of the Vorton-Scalar becomes heavy. On the 4+4 spacetime the quartic term has two forms in the sector that can be quantized. In the covariant form, which is the definition here, matter gathers in one half of the field and binds later, a massless field acquires masses of opposite sign in the two halves, and the internal symmetry of fields in ordinary space is broken to the rotations of the additional times; the alternative form reproduces the matter of the 7+1 spacetime.

The superfluid of hypothesis H1 exists on the deflating spacetime at every moment, with the couplings of that moment, and it acquires a history. The masses do not change; the coupling grows with the inverse volume of the deflating coordinates; the gap grows much faster than the coupling, in weak coupling as the exponential of an exponential; two fermions become bound at a scattering resonance; matter loses its fermion mass and becomes self-bound; and the paired state can follow the deflation only once its gap exceeds a few times the rate divided by the pairing strength. In the pairs along the additional times a scalar contact attraction raises the gap and a vector repulsion lowers it; in the pair along the hidden axis the covariant form reverses both. The gap equation has a solution that does not depend on the upper limit of its momentum integral only below an ultraviolet criterion, which the growing couplings reach at a definite moment. The direction of the order parameter is not selected by the deflation; a contact self-interaction of the covariant form selects it.

For hypotheses H2 and H3 the deflation brings no relief. The Higgs hypothesis had no geometric basis on the flat 4+4 spacetime and has none here, and a contact self-interaction of the covariant form removes even the hidden symmetry on which it rests. If the superfluid were the Higgs field, the electroweak scale would follow the growing coupling, and atomic clocks would then limit the rate of deflation to about a millionth of the Hubble rate. If gravity is the Einstein theory in eight dimensions, lunar laser ranging limits it to three thousandths of the Hubble rate, and the Hubble constant inferred from the

microwave background moves away from the local measurement, in the simplest estimate and in fits of all parameters to mock data alike. In either case nothing of the history just described can have happened within the age of the universe at a constant rate.

The perturbations of ordinary space show most plainly what the hypothesis of deflation leaves open. Whatever keeps the rates constant must carry a tension, and it must take part in the perturbations: with a Newton's constant that depends on the time, the Einstein equations of ordinary space cannot all hold with matter alone. The spectra of the microwave background at all but the largest angular scales do not depend on how the agent does so, and mock data of the quality of the Planck satellite would bound the rate to four per cent of the Hubble rate. The growth of structure does depend on it, at the first order in the rate.

These results are quantitative, and they come with the programs that produced them. They mark what a successful theory of this kind would have to supply: a source or a theory of gravity that sustains the deflation stably and that says how the source responds to matter and to perturbations, a rate that was large early and is small today, and a reason why the constants of ordinary matter do not follow the volume of the deflating coordinates.

DATA AVAILABILITY

The source code of all programs, the data files from which every figure and every computed number of this paper is generated, the instructions for compiling and running the programs (Appendix E), and the records of the provenance of each result are openly available [5].

Appendix A: Symbols

Tables XLIV to XLVIII list every symbol of this paper with its meaning and the place of its definition. No symbol has two meanings. Where one letter occurs with different indices or in different type, the tables say how the uses are told apart; a machine check of the registry from which these tables are taken verifies that no symbol and no name in a data file is defined twice (Appendix E).

Appendix B: The Dirac matrices

Table XLIX prints the eight Dirac matrices of the orthonormal frame and three matrices built from them. All are real, with entries 0, +1 and −1, and each Dirac matrix is a signed permutation matrix. The matrices $\gamma^0, \ldots, \gamma^3$ of the space-like coordinates are symmetric and square to +1; the matrices $\gamma^4, \ldots, \gamma^7$ of the time-like coordinates are antisymmetric and square to −1 [Eq. (15)]. Their product $\omega_{1644} = \gamma^0\gamma^1\cdots\gamma^7$ is the chirality matrix χ; it is real and diagonal, $\mathrm{diag}(-1_8, +1_8)$, so that the chiral projectors $P^{L,R} = \frac{1}{2}(1 \mp \chi)$ select the upper and the lower eight components. The matrix $C^- = \gamma^0\gamma^1\gamma^2\gamma^3$ is real and symmetric and turns each Dirac matrix into minus its transpose, $C^-\gamma^A(C^-)^{-1} = -(\gamma^A)^T$; it defines the covariant adjoint $\tilde{\psi} = \psi^\dagger C^-$. The matrix $C^+ = C^-\omega_{1644}$ turns each Dirac matrix into plus its transpose. Three further matrices follow by one multiplication: the Hermitian matrices $\beta = -i\gamma^4$ and $\alpha^j = -\gamma^4\gamma^j$ $(j = 0, 1, 2, 3)$ of the quantized sector, and $W_{1644} = i\,C^-\gamma^4$, which is Hermitian, squares to one, commutes with β and with the α^j, and satisfies $C^- = -W_{1644}\beta$. The program verifies every property stated here in exact integer arithmetic (Appendix E).

Appendix C: Curvature of the deflating metric

This appendix derives the Einstein tensor and the Lanczos tensor of Sec. II in a form that can be checked by hand, and gives the coordinates and the additional symmetries of the case of equal rates.

1. Curvatures on pairs of directions

In the orthonormal frame the Riemann tensor (7) of the deflating metric is diagonal on pairs of directions: for each pair of frame directions $A \neq B$ it has one independent component, $K_{AB} = K_{BA} = R^{AB}{}_{AB}$ (no summation), the sectional curvature of the plane that the two directions span. With the Riemann tensor (7) and the flat metric (2) the sectional curvatures are, on the 4+4 and on the 7+1 spacetime alike,

$$K_{4t} = \mathrm{rate}_t^2, \qquad K_{ts} = \mathrm{rate}_t\,\mathrm{rate}_s \quad (t \neq s), \tag{C1}$$

and $K_{AB} = 0$ for every pair that contains one of the flat directions $0, 1, 2, 3$. (A sectional curvature K_{AB} carries two subscripts; it is not the matrix K of Sec. III.) For a Riemann tensor of this kind the mixed components of the Einstein tensor and of the Lanczos tensor are diagonal, and

$$\mathcal{G}^A{}_A = -\sum_{\{B,C\}} K_{BC}, \tag{C2}$$

$$\mathcal{H}^A{}_A = -4\sum_{\{B,C\},\{B',C'\}} K_{BC}\,K_{B'C'}, \tag{C3}$$

without summation over A. The first sum runs over the pairs of directions that do not contain the direction A. The second runs over the unordered couples of two such pairs that have no direction in common; a set of four directions contributes its three divisions into two pairs. As a test, for a space of constant curvature in eight dimensions, where all K_{BC} equal one number, the first sum has 21 terms and the second 3×35, so that the two formulas give −21 times that number and −420 times its square, the known values $-\frac{1}{2}\cdot 7\cdot 6$ and $-\frac{1}{2}\cdot 7\cdot 6\cdot 5\cdot 4$.

TABLE XLIV. The symbols of this paper: geometry and the Dirac field.

symbol	meaning	defined in
$x0, \ldots, x7$	the eight coordinates: $x0$ hidden space-like coordinate; $x1, x2, x3$ ordinary space; $x4$ ordinary time; $x5, x6, x7$ the deflating coordinates, time-like	Sec. I
xt; xj; a	a deflating coordinate, $t \in \{5, 6, 7\}$; a flat coordinate other than the time, $j \in \{0, 1, 2, 3\}$; index of the four extra axes, $a \in \{0, 5, 6, 7\}$	Sec. I
g55, g66, g77; rate_t	the three rates of deflation (upright, without subscript; not components of the metric); the rate of xt	Eq. (1)
const5, const6, const7; const_t	the three positive constants of the metric: the scale factors at $x4 = 0$	Eq. (1)
gsum, gsq, gprod	the sum of the rates, the sum of their squares, the sum of their products in pairs	Sec. II
b_t	scale factor of the deflating coordinate xt	Eq. (3)
$\eta^{(4+4)}$; $g^{(4+4)}_{\mu\nu}$, $g^{\mu\nu}_{(4+4)}$; ${e_A}^\mu$	flat metric, metric of the deflating spacetime and its inverse, in the sign convention of the hypothesis (space-like components positive, time-like components negative; the superscript marks the spacetime, so that a component of the metric cannot be taken for a rate or a coupling); orthonormal frame	Eqs. (2), (4)
V_d	proper volume of a unit coordinate cell of the deflating coordinates	Eq. (5)
${\Gamma^\rho}_{\mu\nu}$; ${R^\rho}_{\sigma\mu\nu}$, ${R^\mu}_\nu$, R	Christoffel symbols; Riemann tensor [in Eq. (7) its components in the orthonormal frame], Ricci tensor, curvature scalar	Sec. II B
${\mathcal{G}^\mu}_\nu$; ${\mathcal{H}^\mu}_\nu$; Λ_8, α_{GB}	Einstein tensor; Lanczos tensor of the Gauss-Bonnet term; cosmological constant and coupling of Einstein-Gauss-Bonnet gravity	Sec. II C
$\mathcal{X}_{\mathrm{scale}}$, $\mathcal{X}_{(s)}$	the scaling Killing vector; the three special Killing vectors of the case of equal rates	Eqs. (12), (C8)
K_{AB}; $\bar{x}4$, $\bar{x}t$	sectional curvature of the plane of the frame directions A and B (two subscripts; K_{4i}, K_{ij} in Sec. XV); coordinates adapted to the case of equal rates	Appendix C
P_μ; ς; $x4_{\mathrm{turn}}$	conserved momenta of a classical particle; affine parameter; turning value of the ordinary time	Sec. II E
γ^A, γ^μ; Ω_μ, $\omega_{\mu AB}$	Dirac matrices of the frame (printed in Appendix B) and in coordinate components; spin connection and its coefficients	Sec. III
C^-; χ, $P^{L,R}$	the matrix that turns each Dirac matrix into minus its transpose; chirality matrix and the two chiral projectors	Secs. III, VI
ψ_{1644d}; $\hat{\psi}$	the 16-component field on the deflating spacetime; the weighted field $\sqrt{V_d}\,\psi$	Eqs. (23), (25)
K, $K_\perp$; γ^d; W_{1644}	generator of the evolution of a mode in the time, and its part without the scale factors; frame matrix along the wave vector in the deflating coordinates; matrix of the conserved charge	Eq. (27)
β, α^j; $\bar{\psi}$; $\tilde{\psi}$	the Hermitian matrices $-i\gamma^4$ and $-\gamma^4\gamma^j$ ($j = 0, 1, 2, 3$); the adjoint $\psi^\dagger\beta$ of the quantized sector; the covariant adjoint $\psi^\dagger C^-$	Secs. III B, IV A
k_j, k_t; $E_\perp$; k_d; u_d	wavenumbers along the flat and the deflating coordinates; $(m^2 + \sum_j k_j^2)^{1/2}$; physical wavenumber along the deflating coordinates; k_d/rate_t	Eqs. (28), (56)
w; S_{WKB}	one of the eight constant spinors from which the exact solutions are built; exponent of the extinction of a bounded solution	Eqs. (57), (59)

The Einstein tensor. With Eq. (C1), Eq. (C2) gives

$$\begin{aligned} {\mathcal{G}^4}_4 &= -\mathrm{gprod}, \qquad {\mathcal{G}^j}_j = -(\mathrm{gsq} + \mathrm{gprod}), \\ {\mathcal{G}^5}_5 &= -(\mathrm{g66}^2 + \mathrm{g77}^2 + \mathrm{g66\,g77}), \end{aligned} \tag{C4}$$

and the corresponding expressions for ${\mathcal{G}^6}_6$ and ${\mathcal{G}^7}_7$. Because $\mathrm{gsum}^2 = \mathrm{gsq} + 2\,\mathrm{gprod}$, these are Eq. (8).

The Lanczos tensor. A couple of pairs contributes to Eq. (C3) only if neither pair contains a flat direction, that is, only if the four directions are $4, 5, 6, 7$. For a flat direction $A = j$ these are allowed, and their three divisions into two pairs give

$$K_{45}K_{67} + K_{46}K_{57} + K_{47}K_{56} = \mathrm{g55\,g66\,g77\,gsum}, \tag{C5}$$

which is Eq. (10). For $A = 4, 5, 6, 7$ the direction A is itself among the four, and these components vanish. The Lovelock tensor of the third order is built in the same way from sets of six directions, and every such set contains a flat direction: it vanishes.

The field equations. In Eq. (9) the components along $x4$ and along the deflating coordinates read ${\mathcal{G}^4}_4 + \Lambda_8 = 0$ and ${\mathcal{G}^t}_t + \Lambda_8 = 0$. Since ${\mathcal{G}^5}_5 - {\mathcal{G}^6}_6 = (\mathrm{g55} - \mathrm{g66})\,\mathrm{gsum}$, they can hold together only if the three rates are equal, and then $\Lambda_8 = 3\,\mathrm{g55}^2$. The components along the flat coordinates read $-6\,\mathrm{g55}^2 + \Lambda_8 - 12\,\alpha_{\mathrm{GB}}\,\mathrm{g55}^4 = 0$, which gives $\alpha_{\mathrm{GB}} = -1/(4\,\mathrm{g55}^2)$: Eq. (11). The program verifies Eqs. (C2) and (C3) for the deflating metric in exact arithmetic, and an independent program obtains the same Lanczos tensor from its standard definition by computer algebra, for arbitrary rates and constants.

TABLE XLV. The symbols of this paper, continued: mass term and self-interaction, and its two forms.

symbol	meaning	defined in
$\mathcal{L}_{1644d}$; $\mathcal{L}_{\text{kin}}$, $\mathcal{L}_{\text{mass}}$, $\mathcal{L}_{\text{self}}$	Lagrangian of the fermion field, and its three parts: kinetic term, mass term, self-interaction	Eq. (19)
g_r; $g_0, \dots, g_8$	the nine couplings of the self-interaction, one for each rank (one digit as subscript: not a rate g55, not a component $g_{\mu\nu}^{(4+4)}$ of the metric, not the coupling g_{VS})	Eq. (22)
K, J; r, r', l	strictly increasing lists of indices out of $0, \dots, 7$; their ranks (numbers of indices), and a summation index (the matrix K of Sec. V is written without index or with the index $\perp$)	Eq. (21)
Θ^K, Θ_K; ζ_r; sgn_K	the 256 matrices of the self-interaction, $\Theta^K = \zeta_r \gamma^{A_1} \cdots \gamma^{A_r}$, and their inverses $\Theta_K = \text{sgn}_K \, \Theta^K$; the phase 1 or i; the sign ± 1	Eq. (21)
I_r	the nine quartic invariants	Eq. (22)
F, $\mathsf{F}_{rr'}$	the Fierz matrix (sans-serif; the amplitudes F_S, F_P, $F_0, \dots$ are in italic type)	Eq. (43)
g_0^{eff}, g_1^{eff}, g_7^{eff}, g_8^{eff}	the four independent couplings: scalar, vector, axial vector, pseudoscalar	Eq. (47)
n; n, n_S	density matrix of matter; its number density $\langle \psi^\dagger \psi \rangle$ and scalar density $\langle \bar\psi \psi \rangle$	Sec. IV D
G_S^{HF}, G_V^{HF}; G_S^{c}, G_V^{c}	four combinations of the effective couplings: the Hartree-Fock couplings of matter of one kind of particle with eight internal states, and the two further combinations that occur with the two halves of the field	Eq. (50)
G_{pair}	contact coupling of a pair of two particles at rest	Eq. (54)
form C, form Q; $\psi_\pm$	the two forms of the self-interaction in the quantized sector: the covariant form, which is the definition, and the quantized-sector form, which is the alternative; the two halves of the field, $W_{1644}\, \psi_\pm = \pm \psi_\pm$	Sec. IV A
γ_{44q}^A; Θ_Q^K, ζ_r^Q, sgn_K^Q	the matrices of form Q: its eight Dirac matrices, the 256 products of them, their phase 1 or i, and the sign ± 1 of their squares (the letter Q distinguishes them from Θ^K, ζ_r, sgn_K)	Eqs. (36), (37)
I_r^{C}, I_r^{Q}; $\mathcal{L}_{\text{self}}^{\text{C}}$, $\mathcal{L}_{\text{self}}^{\text{Q}}$	the nine invariants and the self-interaction in form C and in form Q (I_r^{C} is the invariant I_r of the covariant Lagrangian)	Eqs. (34), (38)
z_K; $\mathcal{J}_K^{++}$, $\mathcal{J}_K^{--}$, $\mathcal{J}_K^{+-}$, $\mathcal{J}_K^{-+}$	the phase in the relation between the matrices of the two forms; the bilinears of the two halves of the field	Eq. (40)
$n_\pm$, $n_{S\pm}$; n_W, n_{SW}; pol_W	number densities and scalar densities of the particles of the two halves $W_{1644} = \pm 1$ of the field; their differences; the polarization $(n_+ - n_-)/n$	Eqs. (48), (77)
G_S^{tot}, G_V^{tot}, G_S^{dif}, G_V^{dif}; G_S^{same}, G_V^{same}, G_S^{opp}, G_V^{opp}	the mean-field couplings of the two halves: of the squares of the total densities and of the squares of the differences between the halves; between two particles of the same half and between a particle of each half	Eqs. (49), (52)

2. Equal rates: a space of constant curvature

Let g55 = g66 = g77, and introduce in place of $x4, \dots, x7$ the coordinates

$$\bar{x}4 = \frac{e^{\text{g55}\, x4}}{\text{g55}}, \qquad \bar{x}t = \text{const}_t \, xt \quad (t = 5, 6, 7). \tag{C6}$$

The part of the line element that contains $x4, \dots, x7$ becomes

$$-dx4^2 - e^{-2\,\text{g55}\, x4} \sum_t d\bar{x}t^2 = \frac{-d\bar{x}4^2 - \sum_t d\bar{x}t^2}{(\text{g55}\, \bar{x}4)^2}. \tag{C7}$$

This is minus the line element of four-dimensional hyperbolic space of the radius 1/g55, in its upper-half-space form: the four time-like coordinates span a negative-definite space of constant curvature. The line element (C7) is a multiple of a flat one. The special conformal transformations of that flat metric along the three axes $\bar{x}5, \bar{x}6, \bar{x}7$ leave the factor $1/\bar{x}4^2$ unchanged and are therefore symmetries. Their generators are, for $s = 5, 6, 7$,

$$\begin{aligned} \mathcal{X}_{(s)} = 2\, \bar{x}s \left[\bar{x}4 \, \frac{\partial}{\partial \bar{x}4} + \sum_t \bar{x}t \, \frac{\partial}{\partial \bar{x}t} \right] \\ - \Big(\sum_t \bar{x}t^2 + \bar{x}4^2 \Big) \frac{\partial}{\partial \bar{x}s}, \end{aligned} \tag{C8}$$

with $\bar{x}4\, \partial/\partial \bar{x}4 = \text{g55}^{-1} \partial_4$. Together with the 14 Killing vectors of the general case and the three rotations of $x5, x6, x7$ they are the 20 Killing vectors of the case of equal rates.

Appendix D: Complete tables of the calculations

Several sections of this paper quote selected numbers from calculations whose complete results are tables. This appendix prints those tables, in the order of the sections to which they belong. Each caption defines every column; the numbers are written by a script from the data files of the program (Appendix E). M and P stand throughout for the closures "Momentum" and "Pressure" of the

TABLE XLVI. The symbols of this paper, continued: the Kohn-Sham and CJT calculations, pairing with the self-interaction, and gravity.

symbol	meaning	defined in
G_S, G_V	the scalar and the vector coupling of the mean-field description of uniform matter (attraction for $G_S < 0$, repulsion for $G_V > 0$)	Sec. VII A
$\mathsf{h}_{\rm KS}$; V_0; $E^*(k)$	Hamiltonian matrix of a particle in matter; the vector mean field $G_V\, n$; the energy $(k^2 + m^{*2})^{1/2}$	Eq. (70)
D; V_D, A_D; ν; $\langle f \rangle$	number of space dimensions in which the particles have momentum; volume and area of the unit ball in D dimensions; number of internal states of a particle, 8; average over the occupied states	Eq. (71)
ε; p_j, $p_{\rm int}$; μ	energy density of matter; its pressure along the coordinate xj and the interaction part of the pressure; chemical potential	Eqs. (72), (73)
$m^*_\pm$, $V_{0\pm}$, $E^*_\pm(k)$; $k_{F\pm}$, $k_{F,\rm tot}$	mass, energy shift and energy $(k^2 + m^{*2}_\pm)^{1/2}$ of a particle of the half $W_{1644} = \pm 1$ in matter; the Fermi momenta of the two halves, and the Fermi momentum that the total density would have with equal halves	Eq. (74)
g; set A, set B	the strength of the couplings of the Lagrangian (a plain g without index); the two sets of couplings for which numbers are given, $g_0 = -g$, and $g_0 = -g$ with $g_1 = +0.1\, g$	Sec. VII B
$\mathsf{R}_{\rm gap}$; $\Omega_{\rm half}$	the matrix of the derivatives of the right-hand sides of the two gap equations with respect to the two masses; the grand potential of free particles of one half	Sec. VII B, Eq. (102)
$\Gamma[S]$, Γ_2; S, S_0	the CJT effective action and its two-particle-irreducible part; full and free propagator of the fermions	Eq. (99)
$\Omega_{\rm CJT}$	the CJT effective potential of uniform matter with the self-interaction	Eq. (100)
Λ; G_c	cutoff of the momenta of the filled states of negative energy (without index or bar; Λ_8 and $\bar\Lambda$ are cosmological terms); critical coupling of the vacuum	Sec. XIII B
$m_{\rm vac}$, $m_{\rm bare}$; $\langle\bar\psi\psi\rangle_{\rm vac}$	with the filled states of negative energy: mass of a fermion in vacuum, mass parameter of the Lagrangian, scalar density of the vacuum	Eq. (104)
$n^{\rm vac}_{S\pm}$, $\Omega_{\rm vac}$; $g_{\rm crit}$; $\Lambda_{\rm max}$	with the filled states of negative energy: the scalar densities of the two halves in vacuum and the energy density of the vacuum; the strength at which the vacuum of a field without mass term becomes unstable; the largest cutoff for which the vacuum is stable at a given strength	Eq. (106)
x; f_D	expectation value of the Vorton-Scalar in units of its vacuum value, $\phi/v_{\rm VS}$; fraction m_D/m of the fermion mass that is a bare Dirac mass	Eq. (108)
$\vartheta_{\rm VS}$; m_ϑ	phase of the Vorton-Scalar; mass of its phase mode	Eq. (110)
F_0, F_1, F_7, F_8; Φ, $\Phi(\vec p)$, X	pairing amplitudes of the four contact channels; a state of a pair of two particles (an antisymmetric 16×16 matrix; without suffix, whereas the Vorton-Scalar is $\Phi_{\rm VS}$), the state of the pair along the hidden axis, and its matrix $X = C^-\gamma^0$	Secs. IV E, IX
$V_{\rm ct}$, $V_{\rm VS}$; $m_{\rm ct}$; q; Adj	pairing potentials of the contact interaction and of the exchange of the Vorton-Scalar; range parameter of the contact couplings; momentum transfer; the matrix of the adjoint in a vertex (β in form Q, C^- in form C)	Eqs. (85), (86)
$\Delta(k)$; $k_{\rm max}$; $w_\infty(t)$, s_Δ; $U_{\rm uv}$, u_S, u_V	the gap function; the upper limit of the momentum integral of the gap equation; the kernel at large momenta as a function of the ratio t of the two momenta, and the exponent with which the gap function falls; the ultraviolet strength of the kernel and its two numerical constants	Eqs. (88), (90)
$s_{\rm ct}$	the sign ± 1 with which the contact couplings enter the pairing potential of a pair	Eq. (84)
a, b; H_a, H_b; H_i	the two scale factors of Sec. XV (sans-serif), for $x0, \ldots, x3$ and for $x5, x6, x7$; their rates; the rate of the direction i	Eq. (115)
$E^{(k)\mu}{}_\nu$; α_1, α_2, α_3; $\bar\Lambda$; κ_8	Lovelock tensor of the order k (the order in parentheses; E_F, E_k, $E_\perp$, E^* are energies); couplings of the Lovelock terms (a digit as subscript; α^j are matrices); cosmological term; gravitational coupling in eight dimensions	Eq. (118)
$T^\mu{}_\nu$; loading; τ^0_d	energy-momentum tensor of matter; $\kappa_8\, \varepsilon(0)/(3\, \mathrm{g}55^2)$; the clock of the undisturbed deflation, $3\, \mathrm{g}55\, x4$	Sec. XV

perturbation equations (Sec. XVI E), and H_0 is given in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

1. The paired state, the bound pair, temperature and vacuum (Secs. X and XII)

Tables L and LI follow the paired state from zero temperature to its critical temperature at three moments of

TABLE XLVII. The symbols of this paper, continued: the model and the superfluid.

symbol	meaning	defined in
$\mathcal{L}^{\rm VS}_{1644d}$; $\Phi_{\rm VS}$, $\hat{\Phi}_{\rm VS}$	Lagrangian of the model; the Vorton-Scalar and its weighted form $\sqrt{V_d}\,\Phi_{\rm VS}$	Eq. (61)
$\lambda_{\rm VS}$, $v_{\rm VS}$, $y_{\rm VS}$	self-coupling of the Vorton-Scalar, the constant that fixes its vacuum value, its Yukawa coupling	Sec. VI
$U(1)_F$, $U(1)_{\rm VS}$; ϵ	the symmetries of fermion number and of chirality; parameter of a symmetry transformation	Sec. VI
m, m_D, m_Y; m_s; $g_{\rm VS}$, $\alpha_{\rm VS}$	mass of a fermion in vacuum, $m = m_D + m_Y$: the bare Dirac mass of the Lagrangian plus the part $m_Y = y_{\rm VS} v_{\rm VS}/\sqrt{2}$ from the Vorton-Scalar; mass of the radial mode; coupling $g_{\rm VS} = m_Y/v_{\rm VS}$ of both modes to the fermions, and $g^2_{\rm VS}/4\pi$	Eq. (62)
$D_{\rm VS}$, C; D_c	the two numbers that decide the state of matter; critical value of $D_{\rm VS}$ for a bound pair	Eq. (66)
τ_d; $\tau_{\rm pair}$	the deflation clock; the clock measured from the binding of the pair	Eqs. (65), (93)
$\varphi_{\rm bos}$, $m_{\rm bos}$, ϖ_b	a mode of the Vorton-Scalar, its mass, the frequency of its weighted form	Sec. VI D
F_S, F_P; $W(k,k')$	exchange amplitudes of the radial and of the phase mode; kernel of the gap equation (always written with its two arguments)	Sec. VIII
λ, λ_S, λ_P; Δ_F; T_c	pairing strength, and its parts from the radial and from the phase mode; gap at the Fermi momentum; critical temperature	Sec. VIII
k_F, E_F, v_F, $N(0)$	Fermi momentum, Fermi kinetic energy, Fermi velocity, density of states at the Fermi surface per internal state	Secs. VIII, XII
$a_{\rm sc}$, E_b	scattering length of two fermions; binding energy of the pair	Sec. X
Ξ_d; $\Delta_{\rm eq}$; $E_{\rm sh}$; $n_{\rm ex}$	the parameter that decides whether the paired state follows the deflation; equilibrium gap; half-width of the shell of states; density of broken pairs	Sec. XI
ξ_k, E_k, κ, n_k	single-particle energy, quasi-particle energy, pair correlation and occupation of a pair of states	Eq. (96)
$V[\Delta]$; ϕ; m^*; $T_{1/2}$	effective potential of the paired state; expectation value of the Vorton-Scalar, $\langle\Phi_{\rm VS}\rangle = \phi/\sqrt{2}$; mass of a fermion in matter; temperature at which the fermions have lost half of their mass	Secs. XII, VII
$a(T)$, b, $K_{\rm GL}$; $T_{\rm high}$, T_0	coefficients of the Ginzburg-Landau energy of the paired state near T_c (quadratic, quartic and gradient term); a closed estimate of the temperature above which the expectation value of the Vorton-Scalar vanishes, and the temperature at which it has fallen to 0.02 of its vacuum value	Appendix D
Δ^a, Δ_t; M; $d_\pm$	order parameter (frame components), and its coordinate components along the deflating coordinates; gap matrix; the two fourfold gaps of a state	Sec. XIV
$SU(2)_A$, $SU(2)_B$, $U(1)_E$; $SO(3)_T$	the factors of the internal symmetry of the fields that do not depend on the extra coordinates; the rotations of the three additional times $x5, x6, x7$	Sec. XIV C
P, P_c; w_z; θ_{12}	generator of $U(1)_E$ and the product of the five matrices of the sector; spinor of a zero mode; angle around a vortex line	Sec. XIV D

the deflation; they contain the universal ratios and the coefficients of the Ginzburg-Landau energy that Sec. XII quotes. Table LII compares two calculations of the critical value D_c at which two fermions bind. Table LIII lists the melting temperatures along the clock and Table LIV the fields at which the one-loop potential of the vacuum turns over.

2. The unperturbed universe (Sec. XVI)

Table LV gives the drift of the quantities of the model, Table LVI the examples of bound pairs that lose rest mass, and Table LVII the growth of structure from the two growth equations.

3. The microwave background (Secs. XVI E and XVI G)

Table LVIII gives the thermal history for three models, Table LIX the acoustic peaks, Table LX the difference between the two closures, Table LXI what the agent must supply in each closure, and Tables LXII and LXIII the growth of structure in the complete calculation.

4. Fits to mock data (Sec. XVI H)

Table LXIV lists the best fits of the six parameters of the standard model at fixed rates of deflation. Tables LXV to LXVIII give, for the seven families of extra radiation, the Fisher forecasts, the best fits at fixed density N of the extra radiation, their summary, and the forecasts with N and the rate free together. "Model fluid" is the stepped fluid of the model with the transi-

TABLE XLVIII. The symbols of this paper, continued: the cosmological part and the Higgs hypothesis.

symbol	meaning	defined in
$a_{\rm sp}$, H, H_0; G_N, G_8	scale factor of ordinary space, Hubble rate, Hubble constant; Newton's constant and the constant of gravitation in eight dimensions	Sec. XVI
ρ, ρ_m; ω_b, ω_c	energy density in ordinary space, and that of matter; the densities of baryons and of cold dark matter of the standard model of cosmology	Sec. XVI
θ_*, r_s; t_0; z_*, $Y_{\rm He}$	angle under which the sound horizon at last scattering is seen, and that sound horizon; age of the universe; redshift of last scattering, mass fraction of helium	Sec. XVI
δ_m, f_g, σ_8	density contrast of matter, its growth rate, amplitude of clustering	Sec. XVI
$p_{\rm agent}$	pressure, in ordinary space, of the agent: of whatever keeps the three rates of deflation constant	Eq. (126)
$\tau_{\rm cf}$; $H_{\rm cf}$, H_e	conformal time of ordinary space (a prime in Sec. XVI E is the derivative with respect to it); conformal Hubble rate, and $H_{\rm cf} - a_{\rm sp}{\rm gsum}/2$	Sec. XVI E
k (Sec. XVI); ℓ, D_ℓ^{TT}	comoving wavenumber of a perturbation of ordinary space; multipole, and the temperature spectrum of the microwave background	Secs. XVI E, XVI G
$h_{\rm sy}$, $\eta_{\rm sy}$; $\alpha_{\rm sy}$	the two functions of the perturbed metric of ordinary space in the synchronous gauge; $(h'_{\rm sy} + 6\eta'_{\rm sy})/2k^2$	Sec. XVI E
$\mathcal{D}$, $\mathcal{T}$, $\mathcal{P}$, $\mathcal{A}$	$8\pi G_N a_{\rm sp}^2$ times the perturbations of the energy density, of the divergence of the momentum density, of the pressure, and the anisotropic stress, of matter and radiation together	Eq. (127)
$\mathcal{E}$, $\mathcal{E}_P$, $\mathcal{E}_m$, $\mathcal{E}_r$; $\mathcal{B}$, $\mathcal{Z}$	$8\pi G_N a_{\rm sp}^2$ times the density and the pressure of the unperturbed universe, and the parts of the first from matter and from radiation; two combinations that vanish without deflation	Eq. (132)
$\mathcal{Y}_M$, $\mathcal{Y}_T$, $\mathcal{Y}_S$	the gaps: what the momentum constraint, the trace equation and the equation of the shear lack with matter and radiation alone	Eq. (129)
$\mathcal{U}$; $R_{\rm TF}$	$\eta_{\rm sy} + \alpha'_{\rm sy}$, the sum of the two gravitational potentials of the conformal Newtonian gauge; the radius of a core of condensed pairs that is supported by its own pressure	Appendix D
$\Delta\chi^2$; N, z_t; c_{s0}	difference of χ^2 from the best fit of the standard model to the mock data; density of extra radiation in units of that of one neutrino species, and the transition redshift of a stepped fluid; sound speed today of a condensate of bound pairs	Secs. XVI H, XVI I
σ_v; $m_{\rm deg}$, $m_{B,\rm max}$, m_B; $\kappa_{\rm SI}$	velocity dispersion of the dark matter of a halo; the largest masses at which fermions are degenerate and bosons of the mass m_B condensed there; the bound on the cross-section per mass of dark matter	Sec. XVI I
$v_{\rm EW}$, $n_{\rm EW}$; α_c, ϱ_c	electroweak scale and the exponent with which it would follow the coupling under H2; critical coupling of the vacuum condensate, and an abbreviation	Sec. XVII
m_p/m_e; $G_{\rm Fermi}$; $\alpha_{\rm em}$	ratio of the proton mass to the electron mass; Fermi constant; fine-structure constant	Sec. XVI
H1, H2, H3; A1, A2, A3; HV	the three hypotheses about the superfluid; the three assumptions of the cosmological part (A3, on the perturbations, with its closures "Momentum", "Pressure" and "Shear"); the assumption that the vacuum value of the Vorton-Scalar sets the electroweak scale	Secs. I, XVI, XVI E

tion redshift z_t; "with deep survey" means the Planck-like mock data together with a deep survey of a quarter of the sky.

5. Superfluid dark matter (Sec. XVI I)

Table LXIX lists the wavenumber of the half mode for the fifteen models of Sec. XVI I and Table LXX the limits that follow from the Lyman-α forest.

6. Symmetries (Secs. II and XIV C)

Table LXXI counts the rotations and boosts that remain symmetries of the deflating spacetime, and Table LXXII lists, item by item, the internal symmetry of the fields that depend on $x1, x2, x3, x4$ only, as the program computed it for the deflating spacetime.

TABLE XLIX. The real matrices of the 4+4 spacetime (+ stands for +1, − for −1, a dot for 0; the lines separate the upper and the lower eight components). First and second row: the eight Dirac matrices γ^A of the orthonormal frame. Third row: their product ω_{1644}, and the matrices C^- and C^+.

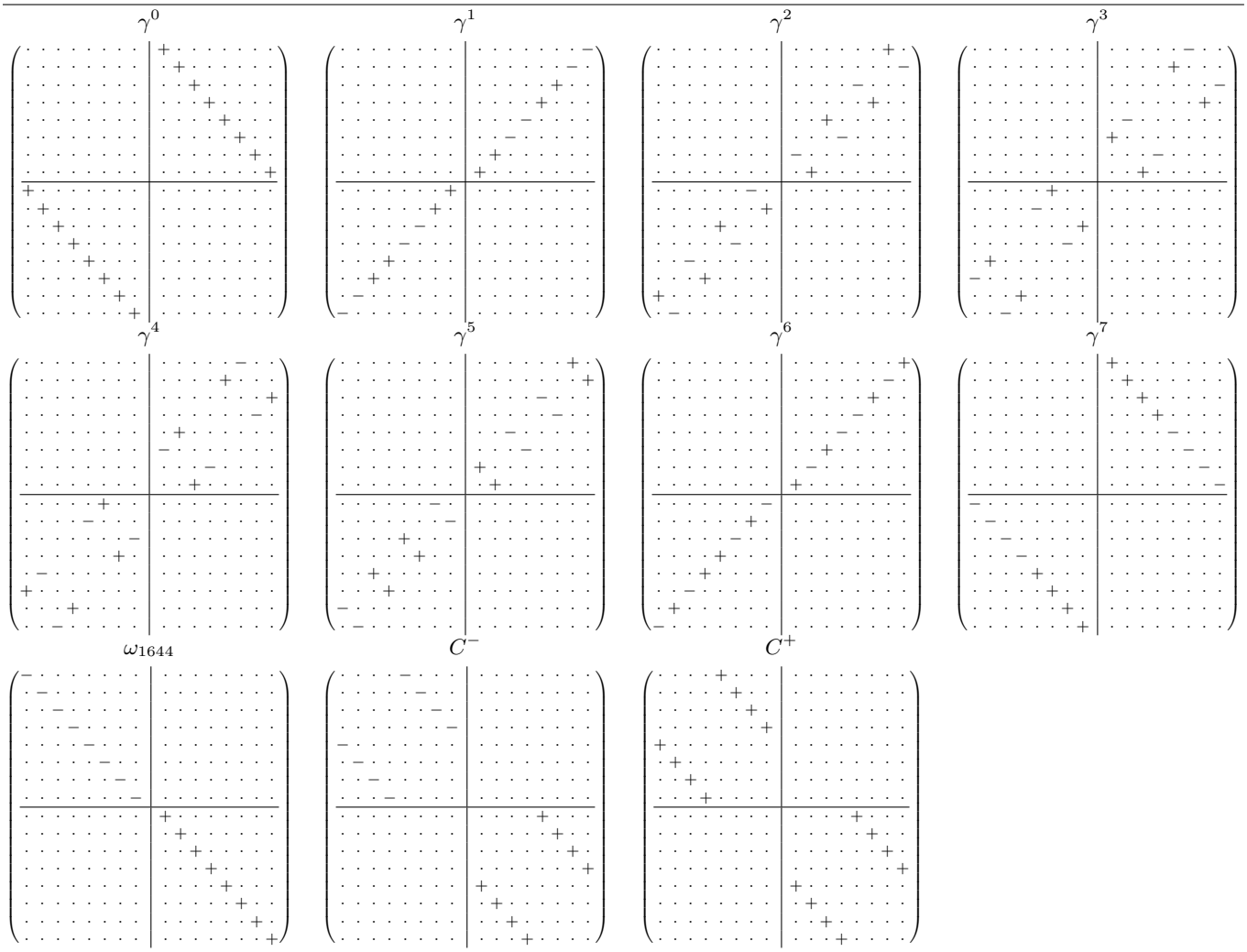


TABLE L. The paired state between zero temperature and the critical temperature T_c, at three moments of the deflation, from the CJT potential, for $m_s = m$ and $k_F = 0.5\,m$ (vector channel). λ: pairing strength. $\Delta_F(0)/T_c$: the gap at the Fermi momentum at zero temperature divided by T_c; its weak-coupling value is 1.7639. Jump: the heat capacity of the paired state minus that of the unpaired state, divided by the latter, at T_c; its weak-coupling value is 1.4261.

τ_d	$g_{\rm VS}$	λ	T_c/m	T_c/E_F	$\Delta_F(0)/T_c$	jump of the heat capacity
0.000	2.60	0.156	7.49×10^{-4}	0.0063	1.7640	1.4266
0.286	3.00	0.208	3.90×10^{-3}	0.0331	1.7668	1.4332
0.537	3.40	0.267	0.0123	0.1039	1.7829	1.4744

TABLE LI. The Ginzburg-Landau energy of the paired state at the three moments of Table L. Near T_c the CJT potential reduces to a quadratic term in the gap with the coefficient $a(T)$, a quartic term with the coefficient b and a gradient term with the coefficient $K_{\rm GL}$. Columns 4 to 6: the slope of a with respect to $\ln T$ at T_c, the coefficient b and the coefficient $K_{\rm GL}$, each divided by its value in the weak-coupling theory. Last two columns: the coherence length of the Ginzburg-Landau energy, $(K_{\rm GL}/|a|)^{1/2}$ at $|\ln(T/T_c)| = 1$, in units of $1/m$ and in units of v_F/T_c with the Fermi velocity v_F; in weak coupling the latter is 0.1333.

			ratio to weak coupling			coherence length	
τ_d	$g_{\rm VS}$	T_c/m	slope of $a(T)$	b	$K_{\rm GL}$	$\times m$	$\times T_c/v_F$
0.000	2.60	7.49×10^{-4}	1.0000	1.0001	1.0005	79.6	0.1333
0.286	3.00	3.90×10^{-3}	0.9993	0.9949	1.0120	15.4	0.1341
0.537	3.40	0.0123	0.9927	0.9639	1.0981	5.1	0.1402

TABLE LII. The critical value D_c of $D_{\rm VS} = \alpha_{\rm VS}\, m/m_s$ at which two fermions bind (Sec. X), from the linearized gap equation in vacuum, whose kernel is relativistic, and from the Schrödinger equation with the Yukawa potential. The two approach each other as the radial mode becomes light and the binding non-relativistic.

m_s/m	gap equation	Schrödinger equation	difference
0.30	1.7347	1.6798	3.3 %
0.10	1.6902	1.6798	0.6 %
0.030	1.6810	1.6798	0.1 %

TABLE LIII. Temperature along the clock: the melting of the expectation value of the Vorton-Scalar, for three ratios m_s/m and ten couplings each. $T_{\rm high}$: a closed estimate of the temperature above which the expectation value vanishes. $T_{1/2}$ and T_0: the temperatures at which the expectation value has fallen to one half and to 0.02 of its vacuum value, in the Hartree approximation. Last column: "no" marks the moments at which no melting temperature can be given, because the Hartree potential ceases to exist before the expectation value has fallen (Sec. XII).

m_s/m	$g_{\rm VS}$	C	$T_{\rm high}/m$	$T_{1/2}/m$	T_0/m	melting defined
0.30	0.30	1.00	0.939	0.857	0.941	yes
0.30	0.40	1.78	0.749	0.691	0.751	yes
0.30	0.53	3.16	0.617	0.573	0.618	yes
0.30	0.71	5.62	0.528	0.491	0.528	yes
0.30	0.95	10.00	0.471	0.435	0.471	yes
0.30	1.26	17.8	0.436	0.400	0.436	yes
0.30	1.69	31.6	0.415	—	—	no
0.30	2.25	56.2	0.402	—	—	no
0.30	3.00	100.0	0.395	—	—	no
0.30	4.00	178	0.391	—	—	no
1.0	0.30	0.0900	2.605	2.325	2.627	yes
1.0	0.40	0.160	1.967	1.776	1.991	yes
1.0	0.53	0.285	1.493	1.368	1.519	yes
1.0	0.71	0.506	1.143	1.065	1.170	yes
1.0	0.95	0.900	0.888	0.842	0.914	yes
1.0	1.26	1.60	0.705	0.679	0.726	yes
1.0	1.69	2.85	0.577	0.561	0.587	yes
1.0	2.25	5.06	0.490	0.476	0.490	yes
1.0	3.00	9.00	0.434	0.413	0.434	yes
1.0	4.00	16.0	0.399	0.366	0.399	yes
3.0	0.30	0.0100	4.809	4.536	4.899	yes
3.0	0.40	0.0178	3.609	3.492	3.719	yes
3.0	0.53	0.0316	2.710	2.707	2.842	yes
3.0	0.71	0.0562	2.037	2.115	2.190	yes
3.0	0.95	0.1000	1.535	1.668	1.706	yes
3.0	1.26	0.178	1.160	1.329	1.346	yes
3.0	1.69	0.316	0.881	1.070	1.075	yes
3.0	2.25	0.562	0.676	0.870	0.870	yes
3.0	3.00	1.000	0.527	0.713	0.713	yes
3.0	4.00	1.78	0.420	0.590	0.590	yes

TABLE LIV. The one-loop potential of the Vorton-Scalar in the vacuum along the clock ($C \propto e^{\tau_d}$): the field at which the potential has its maximum beyond the vacuum, and the field beyond which it lies below its value in the vacuum, both in units of the vacuum value $v_{\rm VS}$.

C	maximum	below the vacuum value
0.490	252	324
1.01	18.9	24.2
9.89	1.55	1.78
20.3	1.25	1.37

TABLE LV. How fast the quantities of the model drift. n: the logarithmic derivative of the quantity with respect to the clock τ_d. Last three columns: the relative change $n\,\mathrm{gsum}\,t_0$ of the quantity over the age t_0 of the universe at the largest total rate that each bound of Table XXXV allows: lunar laser ranging (assumption A2); nucleosynthesis through Newton's constant (A2); nucleosynthesis through the Fermi constant (assumption HV); with $H_0 t_0 = 0.9516$ of the model without deflation.

quantity	where evaluated	n	lunar ranging	nucleosynthesis through G_N	nucleosynthesis through G_{Fermi}
α_{VS}, G_{VS}, D_{VS}, C	any	1.000	3.08×10^{-3}	0.0612	0.223
g_{VS}	any	0.500	1.54×10^{-3}	0.0306	0.112
fermion mass m, mass m_s of the radial mode	any	0.000	0	0	0
scattering length of two fermions (no bound state)	$D_{\mathrm{VS}} = 0.5$	1.368	4.21×10^{-3}	0.0838	0.305
scattering length of two fermions (no bound state)	$D_{\mathrm{VS}} = 1$	2.336	7.20×10^{-3}	0.143	0.521
scattering length of two fermions (no bound state)	$D_{\mathrm{VS}} = 1.5$	9.092	0.0280	0.557	2.03
binding energy of the pair	$D_{\mathrm{VS}} = 1.8$	30.129	0.0928	1.85	6.72
binding energy of the pair	$D_{\mathrm{VS}} = 2$	12.657	0.0390	0.775	2.82
binding energy of the pair	$D_{\mathrm{VS}} = 3$	4.673	0.0144	0.286	1.04
binding energy of the pair	$D_{\mathrm{VS}} = 5$	3.101	9.55×10^{-3}	0.190	0.692
binding energy of the pair	$D_{\mathrm{VS}} = 10$	2.456	7.57×10^{-3}	0.150	0.548
rest mass of the bound pair	$D_{\mathrm{VS}} = 5$, $m_s/m = 0.2$	-0.162	-4.99×10^{-4}	-9.91×10^{-3}	-0.0361
rest mass of the bound pair	$D_{\mathrm{VS}} = 10$, $m_s/m = 0.1$	-0.218	-6.73×10^{-4}	-0.0134	-0.0488
gap and critical temperature of the paired state (weak coupling)	pairing strength $\lambda = 0.15$	6.667	0.0205	0.408	1.49
gap and critical temperature of the paired state (weak coupling)	pairing strength $\lambda = 0.2$	5.000	0.0154	0.306	1.12
gap and critical temperature of the paired state (weak coupling)	pairing strength $\lambda = 0.3$	3.333	0.0103	0.204	0.744
cross-section of two fermions at low energy (Born regime, $D_{\mathrm{VS}} \ll 1$)	$D_{\mathrm{VS}} = 0.05$	2.052	6.32×10^{-3}	0.126	0.458

TABLE LVI. Dark matter made of bound pairs, which lose rest mass as the binding energy grows (Sec. XVI): four examples of $D_{\rm VS}$, m_s/m and $\alpha_{\rm VS}$ today, each at six rates. "Gravity feels the deflation": yes, assumptions A1 and A2; no, Newton's constant is held fixed and the deflation acts on the dark sector alone. Mass ratio: the rest mass of a pair today divided by its rest mass in the early universe. H_0: the Hubble constant inferred with the density of dark matter at early times, the density of baryons and the angle θ_* held fixed; change: its difference from the value without deflation.

$D_{\rm VS}$ today	m_s/m	$\alpha_{\rm VS}$ today	gsum/H_0	gravity feels the deflation	mass ratio	H_0	change
2.50	0.20	0.50	0.00324	yes	0.99995	67.16	−0.20
2.50	0.20	0.50	0.0644	yes	0.99905	63.25	−4.11
2.50	0.20	0.50	0.0300	no	0.99953	67.37	0.01
2.50	0.20	0.50	0.100	no	0.99865	67.38	0.02
2.50	0.20	0.50	0.300	no	0.99736	67.38	0.02
2.50	0.20	0.50	1.00	no	0.99721	67.34	−0.02
5.00	0.20	1.00	0.00324	yes	0.99950	67.17	−0.19
5.00	0.20	1.00	0.0644	yes	0.99085	63.36	−4.00
5.00	0.20	1.00	0.0300	no	0.99557	67.42	0.06
5.00	0.20	1.00	0.100	no	0.98658	67.54	0.18
5.00	0.20	1.00	0.300	no	0.96882	67.68	0.32
5.00	0.20	1.00	1.00	no	0.95069	67.36	0.00
4.00	0.10	0.40	0.00324	yes	0.99994	67.16	−0.20
4.00	0.10	0.40	0.0644	yes	0.99880	63.25	−4.11
4.00	0.10	0.40	0.0300	no	0.99942	67.37	0.01
4.00	0.10	0.40	0.100	no	0.99824	67.38	0.02
4.00	0.10	0.40	0.300	no	0.99594	67.40	0.04
4.00	0.10	0.40	1.00	no	0.99408	67.35	−0.01
10.00	0.10	1.00	0.00324	yes	0.99933	67.17	−0.19
10.00	0.10	1.00	0.0644	yes	0.98756	63.40	−3.96
10.00	0.10	1.00	0.0300	no	0.99400	67.45	0.09
10.00	0.10	1.00	0.100	no	0.98168	67.61	0.25
10.00	0.10	1.00	0.300	no	0.95635	67.84	0.48
10.00	0.10	1.00	1.00	no	0.92342	67.58	0.22

TABLE LVII. Linear growth of pressureless matter from $a_{\rm sp} = 0.01$ to today, from the growth equations (135) of the closure M and (134) of the closure P. Columns 3 to 6, at the Hubble constant of column 2, which keeps the angle of the sound horizon: the amplitude today and the growth rate f_g today, each divided by its value without deflation. Columns 7 and 8: the amplitude ratio at the fixed Hubble constant 67.36 $\rm km\,s^{-1}\,Mpc^{-1}$. Last column: the same from the equation $d^2\delta_m/dx4^2 + 2H\,d\delta_m/dx4 = 4\pi G_N(x4)\rho_m\delta_m$, which is the equation of neither closure and is shown for comparison.

		amplitude		f_g		amplitude at fixed H_0		
gsum/H_0	H_0	M	P	M	P	M	P	four-dimensional
0	67.36	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
0.00100	67.29	1.0000	1.0002	1.0006	1.0011	0.9996	0.9998	0.9997
0.00324	67.15	1.0000	1.0008	1.0021	1.0035	0.9986	0.9993	0.9990
0.0100	66.70	1.0000	1.0024	1.0064	1.0109	0.9956	0.9979	0.9968
0.0300	65.36	1.0001	1.0073	1.0201	1.0339	0.9867	0.9937	0.9904
0.0644	62.98	1.0001	1.0160	1.0463	1.0773	0.9713	0.9863	0.9792
0.100	60.37	1.0001	1.0256	1.0780	1.1287	0.9553	0.9783	0.9674
0.300	41.94	1.0005	1.0907	1.4637	1.6751	0.8638	0.9288	0.8975

TABLE LVIII. Thermal history on the deflating spacetime (assumptions A1, A2; three equal rates) for three models: the standard model ("fiducial"); the same with a fluid of extra radiation of the density $N = 0.5$; and the same with the stepped fluid of the model, $N = 0.5$ and $z_t = 2 \times 10^4$. Held: the row keeps either the Hubble constant 67.36 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ or the angle θ_* of the same model without deflation. z_*: redshift of last scattering. r_s: sound horizon at last scattering. Y_{He}: mass fraction of helium. G_N ratio: Newton's constant at last scattering divided by its value today.

model	gsum/H_0	held	H_0	z_*	r_s (Mpc)	$100\,\theta_*$	Y_{He}	G_N ratio
fiducial	0	H_0	67.36	1089.92	144.45	1.03965	0.2454	1.0000
fiducial	0.00324	H_0	67.36	1089.86	144.68	1.04028	0.2451	0.9969
fiducial	0.00324	angle	67.15	1089.86	144.68	1.03965	0.2451	0.9969
fiducial	0.0644	H_0	67.36	1088.76	149.09	1.05252	0.2402	0.9400
fiducial	0.0644	angle	62.98	1088.72	149.23	1.03965	0.2401	0.9383
fluid, $N = 0.5$	0	H_0	67.36	1090.40	142.02	1.02248	0.2520	1.0000
fluid, $N = 0.5$	0.00324	H_0	67.36	1090.34	142.25	1.02310	0.2517	0.9969
fluid, $N = 0.5$	0.00324	angle	67.15	1090.34	142.25	1.02248	0.2517	0.9969
fluid, $N = 0.5$	0.0644	H_0	67.36	1089.22	146.59	1.03517	0.2468	0.9400
fluid, $N = 0.5$	0.0644	angle	62.97	1089.19	146.73	1.02248	0.2466	0.9383
stepped, $N = 0.5$	0	H_0	67.36	1090.25	142.55	1.02627	0.2481	1.0000
stepped, $N = 0.5$	0.00324	H_0	67.36	1090.19	142.77	1.02689	0.2479	0.9969
stepped, $N = 0.5$	0.00324	angle	67.15	1090.19	142.77	1.02627	0.2479	0.9969
stepped, $N = 0.5$	0.0644	H_0	67.36	1089.07	147.13	1.03900	0.2429	0.9400
stepped, $N = 0.5$	0.0644	angle	62.97	1089.04	147.27	1.02627	0.2428	0.9383

TABLE LIX. The first five acoustic peaks of the temperature spectrum at a fixed angle of the sound horizon (closure M; without gravitational lensing): the multipole ℓ and the height D_ℓ^{TT} in $\mu\mathrm{K}^2$, without deflation and at the largest rates that lunar laser ranging and nucleosynthesis allow.

	no deflation		lunar ranging rate		nucleosynthesis rate	
peak	ℓ	D_ℓ^{TT}	ℓ	D_ℓ^{TT}	ℓ	D_ℓ^{TT}
1	220.9	5740	220.9	5741	220.9	5752
2	537.4	2607	537.4	2608	537.5	2623
3	815.2	2584	815.2	2585	815.3	2612
4	1131.4	1264	1131.4	1265	1131.5	1287
5	1429.3	844	1429.3	845	1429.5	865

TABLE LX. The two closures compared, at a fixed angle of the sound horizon. Difference: the largest difference between the temperature spectra of the two closures in the range of multipoles named, in units of the cosmic variance of a single multipole. Share of the effect: the root of the sum of the squares of that difference over the range, divided by the same for the effect of the deflation itself. σ_8 ratio: the amplitude of clustering divided by its value without deflation. For $\ell \geq 30$ the closure does not matter; for $\ell < 30$ the closures differ by more than the effect of the deflation in either.

	$\ell \geq 30$		$\ell < 30$		σ_8 ratio	
gsum/H_0	difference	share of the effect	difference	share of the effect	M	P
0.00324	1.4×10^{-3}	1.4×10^{-3}	0.011	2.14	0.9989	0.9997
0.0100	4.3×10^{-3}	1.4×10^{-3}	0.037	2.15	0.9966	0.9990
0.0300	0.012	1.3×10^{-3}	0.129	2.16	0.9896	0.9967
0.0644	0.023	1.2×10^{-3}	0.330	2.13	0.9762	0.9918
0.100	0.031	9.9×10^{-4}	0.556	2.04	0.9605	0.9850

TABLE LXI. What the agent must supply in each closure, at the largest rate that nucleosynthesis allows, for the wavenumber $k = 0.02$ Mpc^{-1} at five conformal times. gsum$/H$: the ratio $a_{\rm sp}$gsum$/H_{\rm cf}$. $\mathcal{U} = \eta_{\rm sy} + \alpha'_{\rm sy}$: the sum of the two gravitational potentials of the conformal Newtonian gauge, the potential that light feels. The gaps $\mathcal{Y}_M$, $\mathcal{Y}_T$, $\mathcal{Y}_S$ of Eq. (129) are divided by gsum$/H$ times the size of the terms of their equations, $k^2 H_{\rm cf}|\mathcal{U}|$, $(k^2 + H_{\rm cf}^2)|\mathcal{U}|$ and $|\mathcal{U}|$, so that a number of order one means "of the first order in the rate". Last two columns: the residuals of the identities (131) and (130), relative to the size of their terms.

closure	$\tau_{\rm cf}$ (Mpc)	gsum$/H$	$\mathcal{U}$	$\mathcal{Y}_M$	$\mathcal{Y}_T$	$\mathcal{Y}_S$	residual of (131)	residual of (130)
M	280	2.7×10^{-6}	0.7373	0.000	-0.507	-0.2679	3×10^{-5}	5×10^{-5}
M	800	3.0×10^{-5}	0.5402	0.000	-0.656	-0.3314	4×10^{-5}	7×10^{-5}
M	3000	9.4×10^{-4}	0.5178	0.000	-0.666	-0.3333	4×10^{-4}	9×10^{-4}
M	9000	0.021	0.5083	0.000	-0.708	-0.3543	4×10^{-4}	9×10^{-4}
M	13000	0.055	0.4633	0.000	-0.980	-0.4900	4×10^{-4}	7×10^{-4}
P	280	2.7×10^{-6}	0.7373	0.536	0.000	0.1355	3×10^{-5}	5×10^{-5}
P	800	3.0×10^{-5}	0.5402	0.663	0.000	0.0289	4×10^{-5}	7×10^{-5}
P	3000	9.4×10^{-4}	0.5181	0.666	0.000	0.0028	4×10^{-4}	9×10^{-4}
P	9000	0.021	0.5139	0.697	0.000	0.0004	4×10^{-4}	9×10^{-4}
P	13000	0.055	0.4810	0.936	0.000	0.0004	4×10^{-4}	7×10^{-4}

TABLE LXII. Growth of the density contrast of cold dark matter and baryons at $k = 0.1$ Mpc^{-1} from $a_{\rm sp} = 0.01$ to today, divided by its value without deflation, at the fixed Hubble constant 67.36 km s^{-1} Mpc^{-1}: from the complete Einstein-Boltzmann calculation ("program") and from the growth equation of the same closure, Eq. (135) for M and Eq. (134) for P.

	closure M		closure P	
gsum/H_0	program	equation	program	equation
0	1.00000	1.00000	1.00000	1.00000
0.00324	0.99856	0.99857	0.99933	0.99933
0.0100	0.99556	0.99557	0.99791	0.99792
0.0300	0.98664	0.98668	0.99368	0.99371
0.0644	0.97123	0.97131	0.98620	0.98627
0.100	0.95513	0.95525	0.97819	0.97830

TABLE LXIII. The same growth in the complete calculation at the Hubble constant that keeps the angle of the sound horizon, and the amplitude σ_8 divided by its value without deflation.

		growth		σ_8 ratio	
gsum/H_0	H_0	M	P	M	P
0	67.360	1.00000	1.00000	1.00000	1.00000
0.00324	67.149	1.00001	1.00077	0.99892	0.99969
0.0100	66.704	1.00001	1.00239	0.99663	0.99901
0.0300	65.365	1.00003	1.00727	0.98955	0.99672
0.0644	62.976	1.00004	1.01597	0.97620	0.99177
0.100	60.368	1.00001	1.02545	0.96051	0.98497

TABLE LXIV. Best fits of the six parameters of the standard model to mock data of the microwave background at fixed total rates of deflation (three equal rates; assumptions A1, A2, A3). $\Delta\chi^2$: relative to the best fit without deflation, which is zero by construction. H_0, ω_c, n_s: the best-fitting Hubble constant, density of cold dark matter and spectral index. σ_8: the amplitude of clustering of the best fit. G_N ratio: Newton's constant at last scattering divided by its value today.

mock data	closure	gsum/H_0	$\Delta\chi^2$	H_0	ω_c	n_s	σ_8	G_N ratio
Planck-like	M	0.00324	0.027	67.086	0.12009	0.9638	0.8218	0.9969
Planck-like	M	0.00500	0.065	66.937	0.12014	0.9633	0.8213	0.9952
Planck-like	M	0.0100	0.261	66.512	0.12029	0.9616	0.8198	0.9905
Planck-like	P	0.0100	0.258	66.512	0.12029	0.9616	0.8218	0.9905
Planck-like	M	0.0150	0.589	66.084	0.12043	0.9599	0.8183	0.9857
Planck-like	M	0.0200	1.051	65.654	0.12058	0.9583	0.8168	0.9809
Planck-like	M	0.0300	2.398	64.784	0.12087	0.9549	0.8136	0.9713
Planck-like	M	0.0644	11.575	61.695	0.12193	0.9430	0.8016	0.9379
Planck-like	P	0.0644	11.467	61.695	0.12193	0.9430	0.8145	0.9379
with deep survey	M	0.00324	0.094	67.094	0.12008	0.9637	0.8218	0.9969
with deep survey	M	0.00500	0.225	66.948	0.12013	0.9631	0.8212	0.9952
with deep survey	M	0.0100	0.904	66.534	0.12025	0.9613	0.8197	0.9905
with deep survey	P	0.0100	0.900	66.534	0.12025	0.9613	0.8216	0.9905
with deep survey	M	0.0150	2.040	66.117	0.12038	0.9594	0.8181	0.9857
with deep survey	M	0.0200	3.644	65.699	0.12050	0.9576	0.8165	0.9809
with deep survey	M	0.0300	8.301	64.851	0.12076	0.9539	0.8131	0.9713
with deep survey	M	0.0644	39.791	61.842	0.12169	0.9408	0.8007	0.9380
with deep survey	P	0.0644	39.655	61.841	0.12170	0.9408	0.8136	0.9380

TABLE LXV. Fisher forecasts for the seven families of extra radiation, with static extra coordinates ("static") and on the deflating spacetime at the largest rate that lunar laser ranging allows ("deflating"): the standard error of the density N with the six parameters of the standard model free, and the change of the best-fit Hubble constant per unit of N.

		error of N		dH_0/dN	
mock data	family	static	deflating	static	deflating
Planck-like	free-streaming radiation	0.1431	0.1432	8.190	8.180
Planck-like	fluid, no step	0.1352	0.1352	10.441	10.435
Planck-like	model fluid, $z_t = 10^4$	0.1222	0.1222	5.631	5.624
Planck-like	model fluid, $z_t = 2 \times 10^4$	0.0846	0.0846	4.684	4.676
Planck-like	model fluid, $z_t = 5 \times 10^4$	0.0602	0.0602	9.283	9.273
Planck-like	Wess-Zumino fluid, $z_t = 2 \times 10^4$	0.1601	0.1598	8.578	8.573
Planck-like	model fluid, $z_t = 10^6$	0.1500	0.1500	10.042	10.038
with deep survey	free-streaming radiation	0.0770	0.0771	7.923	7.915
with deep survey	fluid, no step	0.0667	0.0666	9.801	9.797
with deep survey	model fluid, $z_t = 10^4$	0.0858	0.0858	6.275	6.271
with deep survey	model fluid, $z_t = 2 \times 10^4$	0.0591	0.0591	5.399	5.395
with deep survey	model fluid, $z_t = 5 \times 10^4$	0.0421	0.0421	6.928	6.919
with deep survey	Wess-Zumino fluid, $z_t = 2 \times 10^4$	0.0879	0.0876	8.246	8.241
with deep survey	model fluid, $z_t = 10^6$	0.0742	0.0741	9.445	9.442

TABLE LXVI. Best fits of the six parameters of the standard model at fixed density N of extra radiation, for the seven families and both sets of mock data, with static extra coordinates and on the deflating spacetime at the largest rate that lunar laser ranging allows. H_0: the best-fitting Hubble constant; shift: the difference of the two. $\Delta\chi^2$: relative to the best fit of the standard model with static extra coordinates.

			H_0			$\Delta\chi^2$	
mock data	family	N	static	deflating	shift	static	deflating
Planck-like	no extra radiation	0	67.360	67.086	−0.274	0	0.027
Planck-like	free-streaming radiation	0.2	68.996	68.722	−0.274	1.92	1.57
Planck-like	free-streaming radiation	0.5	71.455	71.179	−0.276	11.55	10.67
Planck-like	free-streaming radiation	1.0	75.533	75.255	−0.278	43.83	42.10
Planck-like	fluid, no step	0.2	69.445	69.171	−0.274	2.17	1.79
Planck-like	fluid, no step	0.5	72.548	72.273	−0.275	13.32	12.34
Planck-like	fluid, no step	1.0	77.746	77.469	−0.277	51.94	50.01
Planck-like	model fluid, $z_t = 10^4$	0.2	68.485	68.211	−0.274	2.64	2.38
Planck-like	model fluid, $z_t = 10^4$	0.5	70.174	69.898	−0.276	16.08	15.40
Planck-like	model fluid, $z_t = 10^4$	1.0	72.991	72.714	−0.277	61.23	59.89
Planck-like	model fluid, $z_t = 2 \times 10^4$	0.2	68.292	68.019	−0.274	5.47	5.21
Planck-like	model fluid, $z_t = 2 \times 10^4$	0.5	69.698	69.422	−0.276	32.98	32.32
Planck-like	model fluid, $z_t = 2 \times 10^4$	1.0	72.025	71.747	−0.278	124.24	122.93
Planck-like	model fluid, $z_t = 5 \times 10^4$	0.2	69.216	68.942	−0.274	10.81	10.18
Planck-like	model fluid, $z_t = 5 \times 10^4$	0.5	71.968	71.695	−0.274	65.29	63.71
Planck-like	model fluid, $z_t = 5 \times 10^4$	1.0	76.625	76.351	−0.275	247.48	244.38
Planck-like	Wess-Zumino fluid, $z_t = 2 \times 10^4$	0.2	69.072	68.798	−0.274	1.55	1.22
Planck-like	Wess-Zumino fluid, $z_t = 2 \times 10^4$	0.5	71.641	71.365	−0.276	9.38	8.52
Planck-like	Wess-Zumino fluid, $z_t = 2 \times 10^4$	1.0	75.926	75.647	−0.279	35.77	34.11
Planck-like	model fluid, $z_t = 10^6$	0.2	69.364	69.091	−0.274	1.76	1.44
Planck-like	model fluid, $z_t = 10^6$	0.5	72.345	72.071	−0.275	10.80	9.97
Planck-like	model fluid, $z_t = 10^6$	1.0	77.334	77.058	−0.277	42.16	40.49
with deep survey	no extra radiation	0	67.360	67.094	−0.266	0	0.094
with deep survey	free-streaming radiation	0.2	68.943	68.676	−0.267	6.63	5.36
with deep survey	free-streaming radiation	0.5	71.311	71.044	−0.268	40.18	36.94
with deep survey	free-streaming radiation	1.0	75.230	74.961	−0.269	153.44	147.06
with deep survey	fluid, no step	0.2	69.315	69.048	−0.267	8.89	7.29
with deep survey	fluid, no step	0.5	72.193	71.926	−0.267	54.51	50.44
with deep survey	fluid, no step	1.0	77.000	76.731	−0.269	211.85	203.79
with deep survey	model fluid, $z_t = 10^4$	0.2	68.614	68.347	−0.267	5.37	4.70
with deep survey	model fluid, $z_t = 10^4$	0.5	70.487	70.219	−0.268	32.60	30.85
with deep survey	model fluid, $z_t = 10^4$	1.0	73.616	73.346	−0.269	124.07	120.58
with deep survey	model fluid, $z_t = 2 \times 10^4$	0.2	68.436	68.169	−0.267	11.22	10.72
with deep survey	model fluid, $z_t = 2 \times 10^4$	0.5	70.035	69.766	−0.269	67.34	66.03
with deep survey	model fluid, $z_t = 2 \times 10^4$	1.0	72.694	72.423	−0.270	252.39	249.73
with deep survey	model fluid, $z_t = 5 \times 10^4$	0.2	68.742	68.475	−0.267	22.10	20.77
with deep survey	model fluid, $z_t = 5 \times 10^4$	0.5	70.759	70.493	−0.266	132.70	129.29
with deep survey	model fluid, $z_t = 5 \times 10^4$	1.0	74.148	73.879	−0.268	498.87	492.12
with deep survey	Wess-Zumino fluid, $z_t = 2 \times 10^4$	0.2	69.005	68.738	−0.267	5.17	4.00
with deep survey	Wess-Zumino fluid, $z_t = 2 \times 10^4$	0.5	71.456	71.189	−0.267	31.49	28.50
with deep survey	Wess-Zumino fluid, $z_t = 2 \times 10^4$	1.0	75.542	75.274	−0.268	121.43	115.53
with deep survey	model fluid, $z_t = 10^6$	0.2	69.243	68.976	−0.267	7.18	5.79
with deep survey	model fluid, $z_t = 10^6$	0.5	72.016	71.749	−0.267	44.03	40.45
with deep survey	model fluid, $z_t = 10^6$	1.0	76.641	76.373	−0.268	171.23	164.11

TABLE LXVII. Summary of the fits of Table LXVI, by interpolation in N: the density N at which the best fit reaches $\Delta\chi^2 = 4$; the Hubble constant there; and the $\Delta\chi^2$ of the best fit whose Hubble constant equals the local measurement $73.50\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. An asterisk marks a value obtained by extrapolation beyond the largest N computed.

	N at $\Delta\chi^2 = 4$		H_0 there		$\Delta\chi^2$ at $H_0 = 73.50$	
family	static	deflating	static	deflating	static	deflating
Planck-like mock data						
free-streaming radiation	0.29	0.31	69.74	69.62	25	26
fluid, no step	0.27	0.29	70.20	70.12	19	19
model fluid, $z_t = 10^4$	0.25	0.26	68.75	68.53	72*	77*
model fluid, $z_t = 2 \times 10^4$	0.17	0.18	68.16	67.90	211*	228*
model fluid, $z_t = 5 \times 10^4$	0.12	0.13	68.48	68.27	113	121
Wess-Zumino fluid, $z_t = 2 \times 10^4$	0.32	0.35	70.13	70.06	19	19
model fluid, $z_t = 10^6$	0.30	0.32	70.39	70.32	16	17
with deep survey						
free-streaming radiation	0.16	0.18	68.59	68.48	95	98
fluid, no step	0.13	0.15	68.67	68.59	87	90
model fluid, $z_t = 10^4$	0.17	0.19	68.44	68.25	120	126*
model fluid, $z_t = 2 \times 10^4$	0.12	0.12	68.00	67.75	330*	356*
model fluid, $z_t = 5 \times 10^4$	0.08	0.09	67.94	67.70	412	441
Wess-Zumino fluid, $z_t = 2 \times 10^4$	0.18	0.20	68.81	68.74	69	71
model fluid, $z_t = 10^6$	0.15	0.17	68.76	68.69	76	78

TABLE LXVIII. The density N of extra radiation and the total rate of deflation as free parameters together (Fisher matrices with eight parameters at the standard model). Columns 2 and 3: the standard error of N with the rate held at zero and with the rate free. Column 4: the standard error of gsum/H_0 with N free. Correlation: the correlation coefficient of N and the rate. N per unit rate: the change of the best-fitting N per unit of gsum/H_0. Net slope: the change of the best-fit Hubble constant per unit of gsum/H_0 when N follows its best-fitting value. Largest rise: the largest rise of the best-fit Hubble constant among the models with $N \geq 0$, $\mathrm{gsum} \geq 0$ and $\Delta\chi^2 \leq 3.84$, which is reached at zero rate.

	error of N		error of		N per	net	largest
family	rate zero	rate free	the rate	correlation	unit rate	slope	rise
Planck-like mock data							
free-streaming radiation	0.1431	0.2505	0.0344	0.821	5.98	−35.3	2.30
fluid, no step	0.1352	0.2490	0.0362	0.840	5.79	−23.9	2.77
model fluid, $z_t = 10^4$	0.1222	0.1451	0.0233	0.540	3.36	−65.4	1.35
model fluid, $z_t = 2 \times 10^4$	0.0846	0.0911	0.0211	0.370	1.60	−76.9	0.78
model fluid, $z_t = 5 \times 10^4$	0.0602	0.0760	0.0248	0.610	1.87	−67.0	1.10
Wess-Zumino fluid, $z_t = 2 \times 10^4$	0.1601	0.3503	0.0429	0.889	7.26	−22.1	2.69
model fluid, $z_t = 10^6$	0.1500	0.2496	0.0326	0.799	6.11	−23.0	2.95
with deep survey							
free-streaming radiation	0.0770	0.1540	0.0211	0.866	6.32	−32.1	1.20
fluid, no step	0.0667	0.1763	0.0279	0.926	5.85	−24.8	1.28
model fluid, $z_t = 10^4$	0.0858	0.1012	0.0125	0.530	4.31	−55.0	1.06
model fluid, $z_t = 2 \times 10^4$	0.0591	0.0617	0.0110	0.283	1.58	−73.6	0.63
model fluid, $z_t = 5 \times 10^4$	0.0421	0.0485	0.0121	0.495	1.97	−68.4	0.57
Wess-Zumino fluid, $z_t = 2 \times 10^4$	0.0879	0.2121	0.0255	0.910	7.57	−19.7	1.42
model fluid, $z_t = 10^6$	0.0742	0.1767	0.0251	0.908	6.38	−21.9	1.37

TABLE LXIX. Superfluid dark matter: the wavenumber of the half mode, at which the power spectrum of matter first falls to one quarter of that of cold dark matter, in Mpc^{-1}, for a degenerate gas of the fermions of eight masses and for a condensate of bound pairs with seven sound speeds c_{s0} today: without deflation, at the largest rate that lunar laser ranging allows, and at the largest rate that nucleosynthesis allows in the two closures. Last column: the change at the rate of nucleosynthesis relative to no deflation, in per cent.

	no	lunar ranging	nucleosynthesis rate		change
dark matter	deflation	rate	M	P	(per cent)
fermions, $m = 0.3$ keV	6.215	6.206	6.025	6.025	−3.07
fermions, $m = 0.5$ keV	11.222	11.205	10.875	10.875	−3.09
fermions, $m = 0.75$ keV	17.996	17.969	17.444	17.444	−3.07
fermions, $m = 1$ keV	25.196	25.159	24.426	24.426	−3.06
fermions, $m = 1.5$ keV	40.657	40.590	39.379	39.379	−3.14
fermions, $m = 2$ keV	57.249	57.159	55.401	55.401	−3.23
fermions, $m = 3$ keV	92.774	92.633	89.868	89.868	−3.13
fermions, $m = 4.5$ keV	150.645	150.419	146.013	146.010	−3.08
condensate, $c_{s0}/c = 3.0 \times 10^{-12}$	125.047	124.857	121.132	121.132	−3.13
condensate, $c_{s0}/c = 1.0 \times 10^{-11}$	56.870	56.781	55.003	55.003	−3.28
condensate, $c_{s0}/c = 3.0 \times 10^{-11}$	27.777	27.736	26.922	26.922	−3.08
condensate, $c_{s0}/c = 1.0 \times 10^{-10}$	12.734	12.714	12.333	12.333	−3.15
condensate, $c_{s0}/c = 1.0 \times 10^{-9}$	2.911	2.906	2.822	2.822	−3.04
condensate, $c_{s0}/c = 1.0 \times 10^{-8}$	0.693	0.692	0.670	0.670	−3.28
condensate, $c_{s0}/c = 4.5 \times 10^{-8}$	0.280	0.280	0.271	0.271	−3.25

TABLE LXX. Limits on superfluid dark matter from the requirement that its half mode lies at a wavenumber not smaller than that of thermal warm dark matter at two mass limits of the Lyman-α forest ("relic", the mass of the thermal relic). Half mode: the wavenumber required. Fermion mass: the smallest mass of the fermions of a degenerate gas that meets it. Largest c_{s0}/c: the largest sound speed today of a condensate of bound pairs that meets it, also in m/s. R_{TF}: the radius of a core of the condensate supported by its own pressure, at that sound speed.

deflation	relic (keV)	half mode (Mpc^{-1})	fermion mass (keV)	largest c_{s0}/c	in m/s	R_{TF} (pc)
none	3.5	25.94	1.0249	3.335×10^{-11}	0.01000	0.740
none	5.3	41.11	1.5140	1.645×10^{-11}	0.00493	0.365
lunar ranging rate	3.5	25.94	1.0262	3.327×10^{-11}	0.00997	0.739
lunar ranging rate	5.3	41.11	1.5161	1.641×10^{-11}	0.00492	0.364
nucleosynthesis rate, M	3.5	25.94	1.0523	3.178×10^{-11}	0.00953	0.705
nucleosynthesis rate, M	5.3	41.11	1.5554	1.565×10^{-11}	0.00469	0.347
nucleosynthesis rate, P	3.5	25.94	1.0523	3.178×10^{-11}	0.00953	0.705
nucleosynthesis rate, P	5.3	41.11	1.5554	1.565×10^{-11}	0.00469	0.347

TABLE LXXI. Of the 28 rotations and boosts of flat spacetime, the number that remain symmetries of the deflating spacetime for three sets of rates (g55, g66, g77); and of the six rotations of the extra axes $x0, x5, x6, x7$, whose generators act as internal symmetries on fields that do not depend on these coordinates, the number that remain symmetries of the spacetime.

	rotations	symmetries for the rates			internal	of these, symmetries for the rates		
spacetime	and boosts	$(1,2,3)$	$(2,2,5)$	$(1,1,1)$	generators	$(1,2,3)$	$(2,2,5)$	$(1,1,1)$
7+1	28	6	7	9	6	0	1	3
4+4	28	6	7	9	6	0	1	3

TABLE LXXII. The internal symmetry of the fields that depend on $x1, x2, x3, x4$ only (Sec. XIV C), item by item, computed with the matrices of the 4+4 spacetime; a number of the order of 10^{-16} is zero to the accuracy of the arithmetic. P: the traceless generator of $U(1)_E$. T_i: the generators of one of the two ideals $su(2)_A$, $su(2)_B$. The pair operators are the four vector pair operators of Eq. (78).

item	value	expected, and what it means
dimension of the algebra of the matrices that commute with $\alpha^1, \alpha^2, \alpha^3$, β and $i\beta\chi$	8	8
the same if the fields depend on $x0$ as well	4	4
the same if the fields depend on every space-like coordinate of the model	4	1 (7+1); 4 (4+4)
dimension of the centre of the algebra	2	2
$\|P^2 - 1\|$ for the traceless generator P of the centre	0	0
dimensions of the two simple ideals	3, 3	3, 3
structure constants $\|f_{123}\|$ of the two ideals, for generators T_i with $\mathrm{Tr}\, T_i T_j = 2\delta_{ij}$	1, 1	1, 1: each ideal is $su(2)$
$\|\sum_i T_i^2 - \frac{3}{4}$ projector$\|$ for the two ideals	2.2×10^{-16}, 2.2×10^{-16}	0, 0: doublets
number of states of positive energy with $P = +1$ and with $P = -1$ at a test momentum	4, 4	4, 4: the doublets $(2,1)$ and $(1,2)$
residual of the closure of the four pair operators $\psi^T C^- \gamma^a \psi$, $a \in \{0, 5, 6, 7\}$, under the algebra	3.1×10^{-16}	0
Casimir operator of each ideal on the four pair operators, minus $\frac{3}{4}$	5.6×10^{-17}, 5.6×10^{-17}	0, 0: the $(2,2)$
fermion number of the pair operators	2	2
charge of the pair operators under P	0	0
number of invariant quadratic forms on the four pair operators	1	1
the invariant quadratic form: its diagonal along the frame axes 0, 5, 6, 7 (first entry 1)	+1, −1, −1, −1	$+1, +1, +1, +1$ (7+1); $+1, -1, -1, -1$ (4+4): the signs of the metric of the four extra axes
largest off-diagonal entry of that form	2.2×10^{-16}	0
complex dimension of the generators that mix particles with antiparticles, without chemical potential	2	2: the symmetry is $O(4) \times O(4)$, of dimension 12
the same with a chemical potential	0	0: the symmetry is $U(2) \times U(2)$, of dimension 8
$\gamma^0\gamma^5$: character, and norm of its commutator with the five matrices	Hermitian; 0	anti-Hermitian (7+1: a rotation), Hermitian (4+4); 0
$\gamma^0\gamma^6$: character, and norm of its commutator with the five matrices	Hermitian; 0	anti-Hermitian (7+1: a rotation), Hermitian (4+4); 0
$\gamma^0\gamma^7$: character, and norm of its commutator with the five matrices	Hermitian; 0	anti-Hermitian (7+1: a rotation), Hermitian (4+4); 0
$\gamma^5\gamma^6$: character, and norm of its commutator with the five matrices	anti-Hermitian; 0	anti-Hermitian (a rotation) on both; 0
$\gamma^5\gamma^7$: character, and norm of its commutator with the five matrices	anti-Hermitian; 0	anti-Hermitian (a rotation) on both; 0
$\gamma^6\gamma^7$: character, and norm of its commutator with the five matrices	anti-Hermitian; 0	anti-Hermitian (a rotation) on both; 0

Appendix E: Programs, checks and provenance

This appendix says how every number, table and figure of this paper was computed, how the computation is verified, and how a reader repeats it.

1. The program and how to run it

All physics of this paper is computed by one program, `deflation_cjt_ks_c`, written in the Rust language (36 source files, 17774 lines with their comments, 77 unit tests). It uses no library other than a second Rust crate of the same repository, which supplies the Dirac matrices, exact integer arithmetic, quadrature, the gap equation and the Einstein-Boltzmann program for static extra coordinates. It is compiled, tested and run from the folder `Revision_Deflation_CJT_and_Kohn-Sham_definition_C` of the repository [5] with a Rust toolchain alone:

```
M=code/deflation-cjt-ks-c-rs/Cargo.toml
cargo test  --release --manifest-path $M
cargo build --release --manifest-path $M
B=code/deflation-cjt-ks-c-rs/target/release
$B/deflation_cjt_ks_c help
$B/deflation_cjt_ks_c kohnsham
```

The first and the fourth line only abbreviate two paths. The second line runs the unit tests, the third builds the program, the fifth lists its commands, and the sixth executes one of them. Each command writes its results as tables of numbers into the folder `data`, each table with a description of its content and of its columns at its head, together with a list of *checks*: statements that the command verified, each with its accuracy. A check marked "exact" is an identity verified in exact rational arithmetic; a check marked "numerical" states its largest deviation. Table LXXIII lists the commands. Then

```
python3 code/scripts/make_figures.py
python3 code/scripts/write_doc_tables.py
python3 code/scripts/build_manuscripts.py
```

draw the figures from the data files, write every table and every computed number of the text from the data files, and assemble and typeset this paper. No computed number in a table of this paper was typed by hand.

2. Independent verification

The central results are verified a second time by two scripts in the Python language that share no code with the program and use other methods: computer algebra with symbolic rates and constants instead of rational arithmetic; an implementation of the Grassmann algebra for the invariants of the self-interaction; general-purpose integrators, minimizers and quadratures; the Riemann tensor computed from the metric, and the Lovelock tensors derived as Euler-Lagrange equations of the reduced action, for the equations of gravity; and Peebles' three-level atom [53] for recombination. They are run by

```
S=code/scripts
python3 $S/verify_deflating_independent.py
python3 $S/verify_cjt_ks_independent.py
python3 $S/mutation_test_independent.py
```

The last line tests the two scripts themselves: it introduces deliberate errors into copies of them, one at a time (a wrong sign, a wrong factor, a wrong exponent), and records whether the checks notice. All 45 errors were detected: 29 in the script for the self-interaction, the Kohn-Sham and CJT calculations, pairing and gravity, 16 in the cosmological part of the other script.

3. Where each table and each figure comes from

Tables LXXIV to LXXVI give, for every table and every figure of this paper that contains computed numbers, the command that computed them, the source files of the program in which the calculation is written, and the data files from which the table or figure was made. The numbers quoted in the running text are taken from the same data files or from the statements of the checks, which the commands write to the files `data/*_checks.txt`; a script (`code/scripts/check_numbers.py`) verifies that every number of the text occurs there.

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TABLE LXXIII. The commands of the program: what each computes, and the number of its checks that pass. The checks of both deflating spacetimes are counted, since the program treats the two together. The last four rows are the verifications by other programs.

command	content	checks passed
`selfcoupling`	the nine channels of the self-interaction in exact arithmetic, on 7+1 and in the two forms of 4+4: Fierz matrix, reduction to four couplings, mean-field couplings, pair couplings, symmetries, relation between the forms	36 of 36
`kohnsham`	uniform matter by the Kohn-Sham method and by the CJT effective potential; the filled states of negative energy; bare mass, self-interaction and Vorton-Scalar together	13 of 13
`forms`	the two forms of the 4+4 model: matter of two halves, polarization, binding thresholds, saturated matter and gas along the clock, effective potential of two halves, vacuum, thresholds with the Vorton-Scalar	9 of 9
`pairing`	the gap with contact couplings and exchange of the Vorton-Scalar, for both signs of the contact couplings; the ultraviolet criterion and the upper limit of the momentum integral	7 of 7
`gravity`	the Einstein-Lovelock equations with matter as a source; stability of the deflation	6 of 6
`matrices`	the Dirac matrices of the frame	106 of 106
`matrices`	geometry, Killing vectors, Einstein and Lanczos tensors, geodesics; matrices in coordinate components, spin connection, Dirac operators	47 of 47
`modes`	solutions of the Dirac equation mode by mode: creation (7+1) and extinction (4+4)	7 of 7
`symmetry`	surviving rotations and boosts, internal symmetry of the sector, zero modes on a vortex line, exponents of the couplings	19 of 19
`channels`	exchange amplitudes in the pair channels; modes of the Vorton-Scalar	7 of 7
`history`	gap, bound pair, matter, vacuum and temperature along the clock; order of events	8 of 8
`quench`	dynamics of the paired state under a growing coupling	5 of 5
`paired`	the paired state and self-bound matter from the effective action	7 of 7
`cosmology`	inferred Hubble constant, bounds on the rate, the exponent n_{EW}	6 of 6
`matter`	dark-matter estimates, the resonance, growth of structure, drifts	6 of 6
`background`	recombination, sound horizons and helium on the deflating background	2 of 2
`spectrum`	spectra of the microwave background at a fixed angle; the two closures; the gaps and their identities; convergence; growth of structure	8 of 8
`forecast`	mock data against the rate of deflation: Fisher matrices and fits	5 of 5
`families`	the seven families of dark radiation on the deflating background, and together with the rate	4 of 4
`superfluid`	superfluid dark matter: suppression of small-scale structure	2 of 2
independent script 1	geometry, completeness of the Killing vectors, Lanczos tensor, spin connection and Dirac operators by computer algebra; exact solutions; vortex zero modes; Hubble constant; recombination; perturbation equations, identities, closures and growth	43 of 43
independent script 2	self-interaction on Grassmann-valued fields; Kohn-Sham and CJT calculations; pairing amplitudes and gap equation; the two forms of 4+4 and the ultraviolet criterion; Lovelock tensors from the metric and from the reduced action; the evolutions with gravity	45 of 45
deliberate errors	errors introduced one at a time into the two independent scripts, and detected by them	45 of 45
registry of symbols	no symbol and no name in a data file has two meanings	7 of 8

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TABLE LXXIV. Provenance of the tables and figures of this paper, first part. Source files are in `code/deflation-cjt-ks-c-rs/src`, data files in `data`.

item	command	source files	data files
Fig. 1	`symmetry`	`symmetry_defl.rs`, `cmd_symmetry.rs`	`isometries.csv`
Fig. 2	`matrices`	`metric.rs`, `geometry.rs`, `curved.rs`, `main.rs`	`geodesic_1644d.csv`, `geodesic_1671d.csv`
Table II	`selfcoupling`	`selfint.rs`, `cmd_selfcoupling.rs`	`selfcoupling_fierz.csv`
Table III	`selfcoupling`	`selfint.rs`, `cmd_selfcoupling.rs`	`selfcoupling_pairs.csv`
Table IV	`selfcoupling`	`selfint.rs`, `cmd_selfcoupling.rs`	`selfcoupling_internal_symmetry.csv`
Fig. 3	`channels`, `modes`	`scalar.rs`, `cmd_channels.rs`, `modes.rs`, `main.rs`	`boson_bounded.csv`, `extinction_profile.csv`
Table V	`symmetry`	`symmetry_defl.rs`, `cmd_symmetry.rs`	`coupling_exponents.csv`
Fig. 4	`history`	`history.rs`, `main.rs`	`events.csv`
Fig. 5	`channels`	`scalar.rs`, `cmd_channels.rs`	`scalar_zero_modes.csv`, `static_source.csv`
Table VI	`forms`	`halves.rs`, `cmd_forms.rs`, `cmd_forms_b.rs`	`forms_couplings.csv`
Table VII	`forms`	`halves.rs`, `cmd_forms.rs`, `cmd_forms_b.rs`	`forms_energy_vs_polarization.csv`
Fig. 6	`forms`	`halves.rs`, `cmd_forms.rs`, `cmd_forms_b.rs`	`forms_energy_vs_polarization.csv`
Table VIII	`forms`	`halves.rs`, `cmd_forms.rs`, `cmd_forms_b.rs`	`forms_thresholds.csv`
Table IX	`forms`	`halves.rs`, `cmd_forms.rs`, `cmd_forms_b.rs`	`forms_saturation.csv`
Fig. 7	`forms`	`halves.rs`, `cmd_forms.rs`, `cmd_forms_b.rs`	`forms_saturation.csv`, `forms_thresholds.csv`
Table X	`forms`	`halves.rs`, `cmd_forms.rs`, `cmd_forms_b.rs`	`forms_gas_events.csv`
Fig. 8	`forms`	`halves.rs`, `cmd_forms.rs`, `cmd_forms_b.rs`	`forms_fixed_density.csv`
Table XI	`kohnsham`	`kohn_sham.rs`, `cjt_contact.rs`, `cmd_kohnsham.rs`	`ks_reduced_thresholds.csv`
Fig. 9	`channels`	`scalar.rs`, `cmd_channels.rs`	`pair_channel_kernel.csv`
Table XII	`history`	`history.rs`, `main.rs`	`gap_history.csv`
Fig. 10	`history`	`history.rs`, `main.rs`	`gap_history.csv`
Table XIII	`pairing`	`pair_contact.rs`, `cmd_pairing.rs`	`pair_gap_history.csv`
Table XIV	`pairing`	`pair_contact.rs`, `cmd_pairing.rs`	`pairing_checks.txt`
Table XV	`pairing`	`pair_contact.rs`, `cmd_pairing.rs`	`pair_gap_history.csv`
Fig. 11	`pairing`	`pair_contact.rs`, `cmd_pairing.rs`	`pair_gap_history.csv`
Table XVI	`pairing`	`pair_contact.rs`, `cmd_pairing.rs`	`pair_gap_history.csv`
Fig. 12	`pairing`	`pair_contact.rs`, `cmd_pairing.rs`	`pair_gap_history.csv`
Table XVII	`pairing`	`pair_contact.rs`, `cmd_pairing.rs`	`pairing_checks.txt`
Table XVIII	`history`	`history.rs`, `main.rs`	`dimer_history.csv`
Fig. 13	`history`, `matter`	`history.rs`, `main.rs`, `cosmo_defl.rs`, `cmd_matter.rs`	`dimer_history.csv`, `dm_resonance.csv`
Table XIX	`history`	`history.rs`, `main.rs`	`events.csv`

TABLE LXXV. Provenance of the tables and figures of this paper, second part.

item	command	source files	data files
Table XX	quench	quench.rs, main.rs	quench_summary.csv
Fig. 14	quench	quench.rs, main.rs	quench_history.csv, quench_summary.csv
Table XXI	paired	cmd_paired.rs	cjt_history.csv
Fig. 15	paired	cmd_paired.rs	cjt_gl_history.csv, cjt_history.csv
Table XXII	paired	cmd_paired.rs	matter_saturation.csv
Fig. 16	history, paired	history.rs, main.rs, cmd_paired.rs	matter_saturation.csv, medium_history.csv, thermal_history.csv, vacuum_history.csv
Table XXIII	kohnsham	kohn_sham.rs, cjt_contact.rs, cmd_kohnsham.rs	cjt_vs_ks.csv, kohnsham_checks.txt
Table XXIV	kohnsham	kohn_sham.rs, cjt_contact.rs, cmd_kohnsham.rs	cjt_sea_vacuum.csv
Fig. 17	kohnsham	kohn_sham.rs, cjt_contact.rs, cmd_kohnsham.rs	cjt_sea_saturation.csv, cjt_sea_vacuum.csv
Table XXV	forms	halves.rs, cmd_forms.rs, cmd_forms_b.rs	forms_vacuum_critical.csv
Table XXVI	forms	halves.rs, cmd_forms.rs, cmd_forms_b.rs	forms_vacuum.csv
Fig. 18	forms	halves.rs, cmd_forms.rs, cmd_forms_b.rs	forms_vacuum.csv
Table XXVII	forms	halves.rs, cmd_forms.rs, cmd_forms_b.rs	forms_unified_thresholds.csv
Fig. 19	kohnsham	kohn_sham.rs, cjt_contact.rs, cmd_kohnsham.rs	unified_phase_mode.csv, unified_thresholds.csv
Table XXVIII	kohnsham	kohn_sham.rs, cjt_contact.rs, cmd_kohnsham.rs	unified_phase_mode.csv
Table XXIX	paired	cmd_paired.rs	order_parameter_states.csv
Table XXX	gravity	gravity.rs, cmd_gravity.rs	gravity_reference.csv
Fig. 20	gravity	gravity.rs, cmd_gravity.rs	gravity_reference_path.csv
Table XXXI	gravity	gravity.rs, cmd_gravity.rs	gravity_checks.txt
Table XXXII	gravity	gravity.rs, cmd_gravity.rs	gravity_deflating_summary.csv
Fig. 21	gravity	gravity.rs, cmd_gravity.rs	gravity_deflating.csv, gravity_vacuum_perturbation.csv
Fig. 22	gravity	gravity.rs, cmd_gravity.rs	gravity_deflating.csv, gravity_vacuum_perturbation.csv
Table XXXIII	gravity	gravity.rs, cmd_gravity.rs	gravity_vacuum_perturbation.csv
Table XXXIV	cosmology	cosmo_defl.rs, main.rs	hubble_deflation.csv
Fig. 23	cosmology, matter	cosmo_defl.rs, main.rs, cmd_matter.rs	growth_deflation.csv, hubble_deflation.csv, hubble_mass_loss.csv
Table XXXV	cosmology	cosmo_defl.rs, main.rs	deflation_bounds.csv
Fig. 24	cosmology, forecast	cosmo_defl.rs, main.rs, eb_fit.rs, eb_spectra.rs, cmd_spectra.rs	deflation_bounds.csv, deflation_fisher.csv
Table XXXVI	spectrum	spectra_defl.rs, eb_boltzmann.rs, eb_spectra.rs, cmd_spectra.rs	cmb_deflation_summary.csv
Fig. 25	spectrum	spectra_defl.rs, eb_boltzmann.rs, eb_spectra.rs, cmd_spectra.rs	cmb_deflation_cl.csv, cmb_deflation_summary.csv
Fig. 26	matter, spectrum	cosmo_defl.rs, cmd_matter.rs, spectra_defl.rs, eb_boltzmann.rs, eb_spectra.rs, cmd_spectra.rs	cmb_deflation_cl.csv, cmb_deflation_closure.csv, cmb_deflation_gaps.csv, cmb_deflation_growth.csv, cmb_deflation_summary.csv, growth_deflation.csv
Table XXXVII	forecast	eb_fit.rs, eb_spectra.rs, cmd_spectra.rs	deflation_fisher.csv

TABLE LXXVI. Provenance of the tables and figures of this paper, third part.

item	command	source files	data files
Fig. 27	cosmology, forecast	cosmo_defl.rs, main.rs, eb_fit.rs, eb_spectra.rs, cmd_spectra.rs	deflation_fisher.csv, deflation_profile.csv, deflation_residuals.csv, hubble_deflation.csv
Table XXXVIII	families	eb_fit.rs, eb_spectra.rs, cmd_spectra.rs	dr_deflation_joint.csv
Fig. 28	families	eb_fit.rs, eb_spectra.rs, cmd_spectra.rs	dr_deflation_joint.csv, dr_deflation_profiles.csv
Table XXXIX	matter	cosmo_defl.rs, cmd_matter.rs	dm_halo_estimates.csv
Fig. 29	superfluid	eb_boltzmann.rs, cmd_spectra.rs	sfdm_deflation_summary.csv, sfdm_deflation_transfer.csv
Table XLI	cosmology	cosmo_defl.rs, main.rs	higgs_scale_exponent.csv
Table L	paired	cmd_paired.rs	cjt_thermo_history.csv
Table LI	paired	cmd_paired.rs	cjt_gl_history.csv
Table LII	paired	cmd_paired.rs	cjt_threshold.csv
Table LIII	history	history.rs, main.rs	thermal_history.csv
Table LIV	history	history.rs, main.rs	vacuum_history.csv
Table LV	matter	cosmo_defl.rs, cmd_matter.rs	dm_drift.csv
Table LVI	cosmology	cosmo_defl.rs, main.rs	hubble_mass_loss.csv
Table LVII	matter	cosmo_defl.rs, cmd_matter.rs	growth_deflation.csv
Table LVIII	background	spectra_defl.rs, eb_cosmo.rs, cmd_spectra.rs	background_deflation.csv
Table LIX	spectrum	spectra_defl.rs, eb_boltzmann.rs, eb_spectra.rs, cmd_spectra.rs	cmb_deflation_peaks.csv
Table LX	spectrum	spectra_defl.rs, eb_boltzmann.rs, eb_spectra.rs, cmd_spectra.rs	cmb_deflation_summary.csv
Table LXI	spectrum	spectra_defl.rs, eb_boltzmann.rs, eb_spectra.rs, cmd_spectra.rs	cmb_deflation_gaps.csv
Table LXII	spectrum	spectra_defl.rs, eb_boltzmann.rs, eb_spectra.rs, cmd_spectra.rs	cmb_deflation_growth.csv
Table LXIII	spectrum	spectra_defl.rs, eb_boltzmann.rs, eb_spectra.rs, cmd_spectra.rs	cmb_deflation_growth.csv
Table LXIV	forecast	eb_fit.rs, eb_spectra.rs, cmd_spectra.rs	deflation_profile.csv
Table LXV	families	eb_fit.rs, eb_spectra.rs, cmd_spectra.rs	dr_deflation_fisher.csv
Table LXVI	families	eb_fit.rs, eb_spectra.rs, cmd_spectra.rs	dr_deflation_profiles.csv
Table LXVII	families	eb_fit.rs, eb_spectra.rs, cmd_spectra.rs	dr_deflation_joint.csv
Table LXVIII	families	eb_fit.rs, eb_spectra.rs, cmd_spectra.rs	dr_deflation_joint.csv
Table LXIX	superfluid	eb_boltzmann.rs, cmd_spectra.rs	sfdm_deflation_summary.csv
Table LXX	superfluid	eb_boltzmann.rs, cmd_spectra.rs	sfdm_deflation_bounds.csv
Table LXXI	symmetry	symmetry_defl.rs, cmd_symmetry.rs	isometries.csv
Table LXXII	symmetry	symmetry_defl.rs, cmd_symmetry.rs	reduced_symmetry.csv

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